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Properties and uses of the Wilson flow in lattice QCD

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Theoretical and numerical studies of the Wilson flow in lattice QCD suggest that the gauge field obtained at flow time $t > 0$ is a smooth renormalized field. The expectation values of local gauge-invariant expressions in this field are thus well-defined physical quantities that probe the theory at length scales on the order of $\sqrt{t}$. Moreover, by transforming the QCD functional integral to an integral over the gauge field at a specified flow time, the emergence of the topological (instanton) sectors in the continuum limit becomes transparent and is seen to be caused by a dynamical effect that rapidly separates the sectors when the lattice spacing is reduced from 0.1 fm to smaller values.

1. Introduction

Flows in field space are an interesting tool that may allow new insights to be gained into the physical mechanisms described by highly non-linear quantum field theories such as QCD. The flow $B_\mu(t, x)$ of $SU(3)$ gauge fields studied in this paper is defined by the equations

$$\dot{B}_\mu = D_\nu G_{\nu\mu}, \quad B_\mu|_{t=0} = A_\mu, \quad (1.1)$$

$$G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu], \quad D_\mu = \partial_\mu + [B_\mu, \cdot], \quad (1.2)$$

where A_μ is the fundamental gauge field in QCD (see Appendix A for unexplained notation; differentiation with respect to the “flow time” t is abbreviated by a dot). Evidently, for increasing t and as long as no singularities develop, the flow equation (1.1) drives the gauge field along the direction of steepest descent towards the stationary points of the Yang-Mills action.

In lattice QCD, the simplest choice of the action of the gauge field $U(x, \mu)$ is the Wilson action [1]

$$S_w(U) = \frac{1}{g_0^2} \sum_p \text{Re tr} \{1 - U(p)\}, \quad (1.3)$$

where g_0 is the bare coupling, p runs over all oriented plaquettes on the lattice and $U(p)$ denotes the product of the link variables around p . The associated flow $V_t(x, \mu)$ of lattice gauge fields (the “Wilson flow”) is defined by the equations

$$\dot{V}_t(x, \mu) = -g_0^2 \{\partial_{x, \mu} S_w(V_t)\} V_t(x, \mu), \quad V_t(x, \mu)|_{t=0} = U(x, \mu), \quad (1.4)$$

in which $\partial_{x, \mu}$ stands for the natural $\mathfrak{su}(3)$ -valued differential operator with respect to the link variable $V_t(x, \mu)$ (see Appendix A). The existence, uniqueness and smoothness of the Wilson flow at all positive and negative times t is rigorously guaranteed on a finite lattice [2]. Moreover, from eq. (1.4) one immediately concludes that the action $S_w(V_t)$ is a monotonically decreasing function of t . The flow therefore tends to have a smoothing effect on the field and it is, in fact, generated by infinitesimal “stout link smearing” steps [3].

The Wilson flow previously appeared in ref. [4] in the context of trivializing maps of field space. Familiarity with this paper is not assumed, but some mathematical results obtained there will be used here again. An important goal in the following is to find out whether the expectation values of local observables constructed from the gauge field at positive flow time can be expected to have a well-defined continuum limit. Evidence for the existence of the limit is provided by performing a sample calculation to one-loop order of perturbation theory directly in the continuum theory, using dimensional regularization, and through a numerical study of the $\text{SU}(3)$ gauge theory at three values of the lattice spacing. Two applications of the Wilson flow are then discussed, one concerning the scale-setting in lattice QCD and the other the question of how exactly the topological (instanton) sectors emerge when the lattice spacing goes to zero.

2. Properties of the Wilson flow at small coupling

The aim in this section partly is to show that the Wilson flow can be studied straightforwardly in perturbation theory and partly to check that the expectation values of

local gauge-invariant observables calculated at positive flow time are renormalized quantities.

For simplicity the perturbation expansion is discussed in the continuum theory using dimensional regularization. The gauge group is taken to be $SU(N)$ and it is assumed that there are N_f flavours of massless quarks. As a representative case, the observable

$$E = \frac{1}{4} G_{\mu\nu}^a G_{\mu\nu}^a \quad (2.1)$$

is considered and its expectation value is worked out to next-to-leading order in the gauge coupling.

2.1 Gauge fixing

The flow equation (1.1) is invariant under t -independent gauge transformations. E is therefore a gauge-invariant function of the fundamental field A_μ and its expectation value can consequently be calculated in any gauge.

In perturbation theory, the gauge invariance of the flow equation leads to some technical complications that are better avoided by considering the modified equation

$$\dot{B}_\mu = D_\nu G_{\nu\mu} + \lambda D_\mu \partial_\nu B_\nu. \quad (2.2)$$

For any given value of the gauge parameter λ , the solution of eq. (2.2) is related to the one at $\lambda = 0$ through

$$B_\mu = \Lambda B_\mu|_{\lambda=0} \Lambda^{-1} + \Lambda \partial_\mu \Lambda^{-1}, \quad (2.3)$$

where the gauge transformation $\Lambda(t, x)$ is determined by

$$\dot{\Lambda} = -\lambda \partial_\nu B_\nu \Lambda, \quad \Lambda|_{t=0} = 1. \quad (2.4)$$

The expectation value of E can thus be computed using the modified flow equation. Moreover, one is free to set $\lambda = 1$, which turns out to be a particularly convenient choice.

Note that the use of the modified flow equation does not interfere with the fixing of the gauge of the fundamental field, since E is unchanged and therefore remains a gauge-invariant function of the latter.

2.2 Solution of the modified flow equation

In perturbation theory the gauge potential is scaled by the bare coupling,

$$A_\mu \rightarrow g_0 A_\mu, \quad (2.5)$$

and the functional integral is then expanded in powers of g_0 . The flow $B_\mu(t, x)$ thus becomes a function of the coupling with an asymptotic expansion of the form

$$B_\mu = \sum_{k=1}^{\infty} g_0^k B_{\mu,k}, \quad B_{\mu,k}|_{t=0} = \delta_{k1} A_\mu. \quad (2.6)$$

When this series is inserted in eq. (2.2), and if one sets $\lambda = 1$, a tower of equations

$$\dot{B}_{\mu,k} - \partial_\nu \partial_\nu B_{\mu,k} = R_{\mu,k}, \quad k = 1, 2, \dots, \quad (2.7)$$

is obtained, where the expressions on the right are given by

$$R_{\mu,1} = 0, \quad (2.8)$$

$$R_{\mu,2} = 2[B_{\nu,1}, \partial_\nu B_{\mu,1}] - [B_{\nu,1}, \partial_\mu B_{\nu,1}], \quad (2.9)$$

$$R_{\mu,3} = 2[B_{\nu,2}, \partial_\nu B_{\mu,1}] + 2[B_{\nu,1}, \partial_\nu B_{\mu,2}] \\ - [B_{\nu,2}, \partial_\mu B_{\nu,1}] - [B_{\nu,1}, \partial_\mu B_{\nu,2}] + [B_{\nu,1}, [B_{\nu,1}, B_{\mu,1}]], \quad (2.10)$$

and so on. In particular, in D dimensions the leading-order equation implies

$$B_{\mu,1}(t, x) = \int d^D y K_t(x - y) A_\mu(y), \quad (2.11)$$

$$K_t(z) = \int \frac{d^D p}{(2\pi)^D} e^{ipz} e^{-t p^2} = \frac{e^{-z^2/4t}}{(4\pi t)^{D/2}}, \quad (2.12)$$

which shows explicitly that the flow is a smoothing operation. More precisely, the gauge potential is averaged over a spherical range in space whose mean-square radius in four dimensions is equal to $\sqrt{8t}$.

The higher-order equations (2.7) can be solved one after another by noting that

$$B_{\mu,k}(t, x) = \int_0^t ds \int d^D y K_{t-s}(x - y) R_{\mu,k}(s, y). \quad (2.13)$$

Recalling eqs. (2.9),(2.10), it is clear that this formula generates tree-like expressions, where the fundamental field A_μ is attached to the endpoints of the trees.

2.3 Expansion of $\langle E \rangle$

When the series (2.6) is inserted in

$$\begin{aligned} \langle E \rangle = & \frac{1}{2} \langle \partial_\mu B_\nu^a \partial_\mu B_\nu^a - \partial_\mu B_\nu^a \partial_\nu B_\mu^a \rangle \\ & + f^{abc} \langle \partial_\mu B_\nu^a B_\mu^b B_\nu^c \rangle + \frac{1}{4} f^{abc} f^{cde} \langle B_\mu^a B_\nu^b B_\mu^c B_\nu^d \rangle, \end{aligned} \quad (2.14)$$

a sequence of terms of increasing order in g_0 is obtained. The lowest-order term is

$$\mathcal{E}_0 = \frac{1}{2} g_0^2 \langle \partial_\mu B_{\nu,1}^a \partial_\mu B_{\nu,1}^a - \partial_\mu B_{\nu,1}^a \partial_\nu B_{\mu,1}^a \rangle \quad (2.15)$$

and the terms at the next order are

$$\mathcal{E}_1 = g_0^3 f^{abc} \langle \partial_\mu B_{\nu,1}^a B_{\mu,1}^b B_{\nu,1}^c \rangle, \quad (2.16)$$

$$\mathcal{E}_2 = g_0^3 \langle \partial_\mu B_{\nu,2}^a \partial_\mu B_{\nu,1}^a - \partial_\mu B_{\nu,2}^a \partial_\nu B_{\mu,1}^a \rangle. \quad (2.17)$$

Each of these terms is a power series in the gauge coupling, which may be worked out by expressing the coefficients $B_{\mu,k}(t, x)$ through the fundamental field $A_\mu(x)$ and by expanding the correlation functions of the latter using the standard Feynman rules.

In practice it is advantageous to pass to momentum space by inserting the Fourier representations

$$B_{\mu,1}(t, x) = \int_p e^{ipx} e^{-tp^2} \tilde{A}_\mu^a(p), \quad (2.18)$$

$$\begin{aligned} B_{\mu,2}^a(t, x) = & i f^{abc} \int_0^t ds \int_{q,r} e^{i(q+r)x} e^{-s(q^2+r^2) - (t-s)(q+r)^2} \\ & \times \{ \delta_{\mu\lambda} r_\sigma - \delta_{\mu\sigma} q_\lambda + \frac{1}{2} \delta_{\sigma\lambda} (q-r)_\mu \} \tilde{A}_\nu^b(q) \tilde{A}_\lambda^c(r), \end{aligned} \quad (2.19)$$

and the corresponding expressions for the higher-order fields $B_{\mu,k}(t, x)$. The short-hand

$$\int_p = \int \frac{d^D p}{(2\pi)^D} \quad (2.20)$$

has been introduced here in order to simplify the notation.

Note that the flow-time integral in eq. (2.19) is similar to a Feynman parameter integral. In particular, if Feynman parameters are used for the diagrams contributing to the gluon correlation functions, one ends up with gaussian momentum integrals that can be easily evaluated in any dimension D . The integrals over the parameters then look very much the same as the ones usually encountered, except for the fact that the flow-time parameters are integrated up to t rather than infinity.

2.4 Computation of $\mathcal{E}_0$

In the case of the lowest-order term (2.15), the steps sketched in the previous subsection lead to the formula

$$\mathcal{E}_0 = \frac{1}{2}g_0^2(N^2 - 1) \int_p e^{-2tp^2} (p^2 \delta_{\mu\nu} - p_\mu p_\nu) D(p)_{\mu\nu}, \quad (2.21)$$

where $D(p)_{\mu\nu}$ denotes the unrenormalized full gluon propagator. Setting $D = 4 - 2\epsilon$ and choosing the Feynman gauge, the propagator assumes the form

$$D(p)_{\mu\nu} = \frac{1}{(p^2)^2} \left\{ (p^2 \delta_{\mu\nu} - p_\mu p_\nu) (1 - \omega(p))^{-1} + p_\mu p_\nu \right\}, \quad (2.22)$$

$$\omega(p) = \sum_{k=1}^{\infty} g_0^{2k} (p^2)^{-k\epsilon} \omega_k, \quad (2.23)$$

from which one infers that

$$\mathcal{E}_0 = \frac{1}{2}g_0^2 \frac{N^2 - 1}{(8\pi t)^{D/2}} (D - 1) \left\{ 1 + g_0^2 (2t)^\epsilon \frac{\Gamma(2 - 2\epsilon)}{\Gamma(2 - \epsilon)} \omega_1 + \dots \right\}. \quad (2.24)$$

To this order, the computation is then easily completed by quoting the known result

$$\omega_1 = \frac{1}{16\pi^2} (4\pi e^{-\gamma_E})^\epsilon \left\{ N \left(\frac{5}{3\epsilon} + \frac{31}{9} \right) - N_f \left(\frac{2}{3\epsilon} + \frac{10}{9} \right) + O(\epsilon) \right\} \quad (2.25)$$

for the gluon self-energy, $\gamma_E = 0.577 \dots$ being Euler's constant.

2.5 Computation of $\langle E \rangle$ to order g_0^4

At the next-to-leading order, the expectation value of E receives contributions from $\mathcal{E}_0$ and the terms $\mathcal{E}_1, \mathcal{E}_2, \dots$ up to order g_0^6 generated by the expansion of the flow

equation (cf. subsect. 2.3). Some of the latter involve the 3-point gluon vertex, but in all cases only the leading-order gluon correlation functions are required.

In the case of the term $\mathcal{E}_2$, for example, the purely algebraic part of the calculation leads to the integral

$$\mathcal{E}_2 = N(N^2 - 1)g_0^4 \int_0^t ds \int_{q,r} \frac{e^{-(2t-s)p^2 - s(q^2 + r^2)}}{p^2 q^2 r^2} \times \{(D-1)p^2(p^2 + q^2 + r^2) + 2(D-2)(q^2 r^2 - (qr)^2)\} + O(g_0^6) \quad (2.26)$$

where $p = q + r$. This integral appears to be quite complicated, but the polynomial in the numerator of the integrand allows the expression to be simplified and eventually to be evaluated analytically (see Appendix B for further details). The result of the computation,

$$\mathcal{E}_2 = N(N^2 - 1) \frac{g_0^4}{(4\pi)^D (2t)^{D-2}} \left(\frac{D}{2\epsilon} - \frac{D}{2} + 45 \ln 2 - \frac{D}{2} \ln 3 + O(\epsilon) \right) + O(g_0^6), \quad (2.27)$$

shows that these contributions are not free of ultra-violet singularities.

The computation of the other terms follows the same pattern and does not present any additional difficulties. Collecting all contributions, the result

$$\langle E \rangle = \frac{1}{2} g_0^2 \frac{N^2 - 1}{(8\pi t)^{D/2}} (D-1) \{ 1 + c_1 g_0^2 + O(g_0^4) \}, \quad (2.28)$$

$$c_1 = \frac{1}{16\pi^2} (4\pi)^{\epsilon} (8t)^{\epsilon} \left\{ N \left(\frac{11}{3\epsilon} + \frac{52}{9} - 3 \ln 3 \right) - N_f \left(\frac{2}{3\epsilon} + \frac{4}{9} - \frac{4}{3} \ln 2 \right) + O(\epsilon) \right\}, \quad (2.29)$$

is then obtained.

2.6 Renormalization

The bare coupling g_0 is related to the renormalized coupling g in the $\overline{\text{MS}}$ scheme and the associated normalization mass μ according to [5]

$$g_0^2 = g^2 \mu^{2\epsilon} (4\pi e^{-\gamma_E})^{-\epsilon} \left\{ 1 - \frac{b_0}{\epsilon} g^2 + O(g^4) \right\}, \quad (2.30)$$

$$b_0 = \frac{1}{16\pi^2} \left\{ \frac{11}{3} N - \frac{2}{3} N_f \right\}. \quad (2.31)$$

Now if eq. (2.28) is written as an expansion in the renormalized coupling, the terms proportional to $1/\epsilon$ cancel and one obtains

$$\langle E \rangle = \frac{3(N^2 - 1)g^2}{128\pi^2 t^2} \{1 + \bar{c}_1 g^2 + O(g^4)\}, \quad (2.32)$$

$$\bar{c}_1 = \frac{1}{16\pi^2} \left\{ N \left(\frac{11}{3}L + \frac{52}{9} - 3 \ln 3 \right) - N_f \left(\frac{2}{3}L + \frac{4}{9} - \frac{4}{3} \ln 2 \right) \right\}, \quad (2.33)$$

at $\epsilon = 0$, where $L = \ln(8\mu^2 t) + \gamma_E$. To this order in the gauge coupling, $\langle E \rangle$ thus turns out to be a quantity that does not need to be renormalized and which therefore encodes some physical property of the theory.

In terms of the running coupling $\alpha(q)$ at scale $q = (8t)^{-1/2}$, the expansion (2.32) assumes the form

$$\langle E \rangle = \frac{3(N^2 - 1)}{32\pi t^2} \alpha(q) \{1 + k_1 \alpha(q) + O(\alpha^2)\}, \quad (2.34)$$

$$k_1 = \frac{1}{4\pi} \left\{ N \left(\frac{11}{3}\gamma_E + \frac{52}{9} - 3 \ln 3 \right) - N_f \left(\frac{2}{3}\gamma_E + \frac{4}{9} - \frac{4}{3} \ln 2 \right) \right\}. \quad (2.35)$$

In particular, for $N = 3$ one obtains

$$\langle E \rangle = \frac{3}{4\pi t^2} \alpha(q) \{1 + k_1 \alpha(q) + O(\alpha^2)\}, \quad k_1 = 1.0978 + 0.0075 \times N_f. \quad (2.36)$$

The next-to-leading order correction is reasonably small in this case and corresponds to a change in the momentum q by a factor 2 or so if $N_f \leq 3$.

3. Lattice studies of the Wilson flow

In QCD the perturbation expansion (2.36) is expected to be applicable at small flow times only, where the smoothing range $\sqrt{8t}$ is at most 0.3 fm or so. The properties of the Wilson flow at larger values of t can however be studied straightforwardly using the lattice formulation of the theory and numerical simulations.

The simulations of the SU(3) gauge theory reported in this section mainly serve to clarify whether $\langle E \rangle$ scales to the continuum limit as suggested by perturbation

Table 1. Lattice parameters, statistics and reference flow time

Lattice	β	a [fm]	N_{conf}	t_0/a^2
48×24^3	5.96	0.0999(4)	100	2.7854(62)
64×32^3	6.17	0.0710(3)	100	5.489(14)
96×48^3	6.42	0.0498(3)	100	11.241(23)

theory. Along the way, a new scale-setting method will be proposed based on the observed properties of $\langle E \rangle$.

3.1 Simulation parameters

Ensembles of representative gauge fields were generated on three lattices, using the Wilson gauge action and a combination of the well-known link-update algorithms. The values of the lattice spacing quoted in table 1 derive from the results in lattice units for the Sommer reference scale $r_0 = 0.5$ fm [6] published by Guagnelli et al. [7]. At the chosen couplings $\beta = 6/g_0^2$, the spacings of the three lattices thus decrease from roughly 0.1 to 0.05 fm by factors of $1/\sqrt{2}$. Moreover, the lattice sizes in physical units are approximately constant.

In order to safely suppress any residual statistical correlations of the N_{conf} generated fields, the separation in simulation time of the fields was taken to be at least 10 times the integrated autocorrelation time of the topological charge. The definition of the latter on the lattice is ambiguous to some extent, but its autocorrelation time is largely independent of the choices one makes and is known to increase very rapidly when the lattice spacing is reduced [8,9]. In particular, already at $a = 0.07$ fm all other usual quantities of interest tend to be far less correlated in simulation time.

3.2 Observables

For any given gauge field configuration $U(x, \mu)$, the flow equation (1.4) can be integrated numerically up to the desired flow time and one may then construct gauge-invariant local observables from the gauge field at this time. In particular,

$$E = 2 \sum_{p \in P_x} \text{Re tr} \{ 1 - V_t(p) \} \quad (3.1)$$

is a possible definition of the density E on the lattice, P_x being the set of unoriented plaquettes with lower-left corner x .

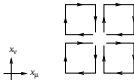


Fig. 1. The anti-hermitian traceless part of the average of the four plaquette Wilson loops shown in this figure can be taken as the definition of the field tensor $a^2 G_{\mu\nu}(x)$ on the lattice. All loops are in the (μ, ν) -plane, have the same orientation and start and end at the point x .

Another more symmetric definition of E is obtained by introducing a lattice version of the field tensor $G_{\mu\nu}(x)$ (see fig. 1). E is then simply given by the continuum formula (2.1). Both definitions are equally acceptable at this point, since they respect all formal requirements (locality and gauge invariance in particular) and since they converge to the correct expression in the classical continuum limit.

For the numerical integration of the flow equation (1.4), any of the widely known integration schemes can in principle be used (see ref. [10], for example). The Euler integrators previously discussed in ref. [4] are particularly simple but also the least efficient ones. Evidently, the integration errors should be much smaller than the statistical errors of the calculated quantities. A fairly simple and numerically stable integrator that allows this condition to be easily met is described in Appendix C.

3.3 Time dependence of $\langle E \rangle$

To leading-order perturbation theory, the dimensionless combination $t^2 \langle E \rangle$ is a constant proportional to the gauge coupling. At the next order, the scale invariance of the theory is broken and $t^2 \langle E \rangle$ develops a non-trivial dependence on the flow time t . Asymptotic freedom actually implies that the combination slowly goes to zero in the limit $t \rightarrow 0$.

Recalling the result

$$\Lambda|_{N_f=0} = 0.602(48)/r_0 \quad (3.2)$$

for the Λ -parameter in the $\overline{\text{MS}}$ scheme obtained by the ALPHA collaboration [11], and using the four-loop evolution equation for the running coupling $\alpha(g)$ [12], the perturbation series (2.36) can be evaluated at any value of the flow time given in

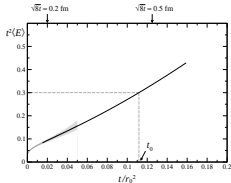


Fig. 2. Simulation data obtained at $a = 0.05$ fm for $t^2(E)$ as a function of the flow time t (black line). Statistical errors are smaller than 0.3% and therefore invisible on the scale of the figure. The curve predicted by the perturbation series (2.36) and the known value (3.2) of the Λ -parameter is also shown (grey band).

units of r_0 . The curve obtained in this way is shown in fig. 2 together with the error band that derives from the error of Λ quoted in eq. (3.2).

Perhaps somewhat fortuitously, the simulation results obtained at $a = 0.05$ fm accurately match the perturbative curve over a significant range of t . The symmetric definition of E has here been used and for clarity the data from only one lattice are shown (as discussed below, the lattice-spacing effects are small and the data from the other lattices would therefore lie nearly on top of the line plotted in fig. 2).

Beyond the perturbative regime, $t^2(E)$ grows roughly linearly with t , at least so within the range covered by the simulation data. The slowdown of the density $\langle E \rangle$ from the perturbative $1/t^2$ to a smoother $1/t$ behaviour may perhaps be explained by noting that the Wilson flow tends to drive the gauge field towards the stationary points of the gauge action. In the vicinity of these points of field space, the right-hand side of the flow equation (1.4) is small and E consequently changes only little with time.

3.4 Lattice-spacing effects

Perturbation theory suggests that the density $\langle E \rangle$ scales to the continuum limit like

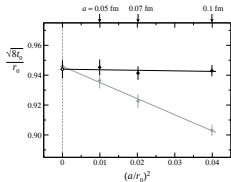


Fig. 3. Extrapolation of the dimensionless ratio $\sqrt{8t_0}/r_0$ of reference scales to the continuum limit (open data points). The black data points were obtained using the symmetric definition of E and the grey ones using the expression (3.1).

a physical quantity of dimension 4. The scaling behaviour of $\langle E \rangle$ can be checked by introducing a reference scale t_0 through the implicit equation

$$\{t^2 \langle E \rangle\}_{t=t_0} = 0.3 \quad (3.3)$$

(see fig. 2). If $\langle E \rangle$ is physical, the dimensionless ratio t_0/r_0^2 must be independent of the lattice spacing, up to corrections vanishing proportionally to a power of a .

The data plotted in fig. 3 clearly show that the ratio of reference scales smoothly extrapolates to the continuum limit. As may have been suspected, the lattice effects appear to be of order a^2 . They are at most a few percent on the lattices considered and particularly small if the symmetric definition of E is employed. At other points in time, the scaling violations behave similarly, but tend to increase towards small t , where the smoothing range $\sqrt{8t}$ is only 2 or 3 times larger than the lattice spacing.

3.5 Synthesis

(a) *Continuum limit.* The numerical studies of the Wilson flow strongly support the conjecture that the gauge fields generated at positive flow time are smooth renormalized fields (except for their gauge degrees of freedom). In the case of QCD with a non-zero number of sea quarks, similar studies however still need to be performed.

Additional confirmation in perturbation theory and through the consideration of further observables is evidently desirable.

(b) *Scale setting.* The time t_0 defined through eq. (3.3) may serve as a reference scale similar to the Sommer radius r_0 . With respect to the latter, t_0 has the advantage that its computation does not require any fits or extrapolations. Moreover, an ensemble of only 100 independent representative field configurations allows t_0 to be obtained with a statistical precision of a small fraction of a percent (for illustration, the values of t_0/a^2 computed using the symmetric definition of E are listed in table 1).

(c) *Universality.* On the lattice the definition of local gauge-invariant quantities like E is not unique, but the results obtained in this section indicate that the differences become irrelevant in the continuum limit (see fig. 3). This kind of universality is a consequence of the fact, further elucidated in sect. 4, that the gauge fields generated by the Wilson flow are smooth on the scale of the lattice spacing. The universality classes of the local fields composed from the gauge field at fixed t/t_0 are therefore determined by the asymptotic behaviour of the fields in the classical continuum limit.

4. Functional integral and topological sectors

In lattice gauge theory, the space of gauge fields is connected and the concept of a topological sector has therefore no a priori well-defined meaning. However, one also knows since a long time that any classical continuous gauge field (including multi-instanton configurations) can be approximated arbitrarily well by lattice fields. The sectors are hence included in the field space but are not separated from one another.

An understanding of how exactly the sectors get divided when the lattice spacing is taken to zero can now be achieved using the Wilson flow. The existence of the topological sectors thus turns out to be a dynamical property of the theory rather than being a consequence of an assumed or imposed continuity of the fields.

4.1 Field transformation

On a finite lattice, however large, the transformation

$$U \rightarrow V = V_{t_0} \quad (4.1)$$

is invertible and actually a diffeomorphism of the field space [4]. One can therefore perform a change of integration variables in the QCD functional integral from the

fundamental field to the field at flow time t_0 . As already noted in ref. [4], the associated Jacobian can be worked out analytically and be expressed through the Wilson action along the flow. The expectation value of any observable $\mathcal{O}(V)$ is then given by

$$\langle \mathcal{O} \rangle = \frac{1}{\mathcal{Z}} \int \mathcal{D}[V] \mathcal{O}(V) e^{-\tilde{S}(V)}, \quad (4.2)$$

$$\tilde{S}(V) = S(U) + \frac{16g_0^2}{3a^2} \int_0^{t_0} dt S_w(V_t), \quad \mathcal{D}[V] = \prod_{x,\mu} dV(x,\mu), \quad (4.3)$$

where $S(U)$ denotes the total action (including the quark determinants) of the theory before the transformation.

Evidently, the functional integral may be rewritten in this way as an integral over the gauge field at any flow time t . Setting t to the reference time t_0 is however an interesting and natural choice. In particular, the fields that dominate the transformed integral then have a characteristic wavelength on the order of the fundamental low-energy scales of the theory.

4.2 Smoothness of the dominant fields

A quantitative measure for the smoothness of a given lattice gauge field V is

$$h = \max_p s_p, \quad s_p = \text{Re tr} \{1 - V(p)\}, \quad (4.4)$$

the maximum being taken over all plaquettes p (as before, $V(p)$ denotes the product of the link variables around p). In the functional integral (4.2), all possible values of h occur, but large values are expected to be unlikely in view of the fact that the expectation value $\langle s_p \rangle$ scales proportionally to a^4 .

Since all terms in the action (4.3) have the same sign and favour smooth configurations, it is certainly plausible that large plaquette values s_p are strongly suppressed. The probability for s_p on a given plaquette p to be larger than some specified value s in fact decreases roughly like a^{10} when the lattice spacing is reduced (see fig. 4). In a finite volume of fixed physical size, the statistical weight of the field configurations with values of h beyond a given threshold is therefore rapidly going to zero in the continuum limit, while the fields that dominate the transformed functional integral become uniformly smooth in space.

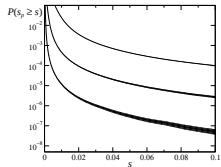


Fig. 4. Plot of the probability for s_p [eq. (4.4)] on a given plaquette p to be above some specified value s . From top to bottom, the three curves shown correspond to the values $a = 0.1, 0.07$ and 0.05 fm of the lattice spacing. The calculation was performed in the $SU(3)$ gauge theory using the ensembles of fields described in subsect. 3.1.

4.3 Dynamical separation of the topological sectors

Many years ago, the space of lattice gauge fields was shown to divide into disconnected sectors if only fields satisfying a certain smoothness condition are admitted [13,14]. The proof is based on a geometrical construction of a local lattice expression for the topological charge which assumes integer values and which has the correct classical continuum limit.

In the case of QCD, the fields satisfying †

$$h < 0.067 \quad (4.5)$$

are included in the subspace covered by the geometrical construction and thus fall into topological sectors very much like the continuous gauge fields in the continuum theory. The space of fields “between the sectors” may accordingly be characterized by the inequality $h \geq 0.067$.

† The theorems proved by Phillips and Stone [14] assume a simplicial lattice. To be able to apply them in the present context, the hypercubic cells of the lattice must be divided into simplices. An interpolation of the gauge field to the added links is then required and can easily be accomplished using the same interpolation methods as the ones applied elsewhere in refs. [13,14].

Recalling the discussion in subsect. 4.2 of the smoothness properties of the fields that dominate the functional integral (4.2), the topological sectors are now seen to emerge as a consequence of the fact that the fields with large values of h have a rapidly decreasing weight in the integral when the lattice spacing is taken zero. In the case of the simulations reported in sect. 3, for example, the fraction of representative fields satisfying the bound (4.5) increases from 0% at $a = 0.1$ fm to 8% and 70% at $a = 0.07$ and 0.05 fm, respectively. The asymptotic behaviour of the weight of the fields between the sectors cannot be safely determined from these data, but the curves shown in fig. 4 suggest that the probability of finding such a configuration on lattices of a given physical size goes to zero proportionally to a^6 .

4.4 Topological susceptibility

Once the functional integral (4.2) splits into a sum of integrals over effectively disconnected sectors, the assignment of the topological charge Q to the field configurations in a representative ensemble of fields becomes unambiguous. The geometrical construction mentioned before could be used to compute the charge, but in view of the universality property discussed at the end of sect. 3, a straightforward discretization of the topological density, using the symmetric lattice expression for the field tensor $G_{\mu\nu}$, is expected to give the same results for the moments $\langle Q^n \rangle$ in the continuum limit.

On the lattices listed in table 1, the second moment $\langle Q^2 \rangle$ computed along these lines is about 50 and the topological susceptibility χ_t turns out to be independent of the lattice spacing within statistical errors. A fit by a constant then gives the result

$$\chi_t^{1/4} = 187.4(3.9) \text{ MeV}, \quad (4.6)$$

which differs by less than two standard deviations from the value $194.5(2.4)$ MeV [16] obtained at flow time $t = 0$ using a chiral lattice Dirac operator and the index theorem [15]. Moreover, as shown in a forthcoming publication, the ratios of the fourth root of the second moments computed here and through a variant [18] of the universal formula proposed in [17] extrapolate to 1 in the continuum limit to within a statistical uncertainty of 3%.

All these empirical results support the conjecture that the moments of the charge distribution in the functional integral (4.2) coincide with those obtained from the index theorem [15–18] and thus the ones appearing in the chiral Ward identities [19,20]. However, while highly plausible, the equality remains to be theoretically established.

5. Concluding remarks

The results reported in this paper shed some new light on the nature of non-abelian gauge theories and the continuum limit in lattice QCD. They are obviously incomplete in several respects and give rise to some interesting questions. In particular, an all-order analysis of the Wilson flow in perturbation theory will probably be required in order to achieve a structural understanding of its renormalization properties.

The one-loop calculation in sect. 2 suggests that the fields obtained at positive flow time are renormalized fields for any number of quark flavours. So far the question was studied numerically only in the pure gauge theory, but it may be encouraging to note that the conjecture is true in QED, for any charged matter multiplet, as a consequence of the gauge Ward identity.

An intriguing aspect of the transformed functional integral (4.2) is the fact that it is dominated by smooth fields. Since the field transformation (4.1) is invertible, the integral nevertheless encodes all the physics described by the theory. Some properties (the division into topological sectors, for example) however become particularly transparent in this formulation of the theory, while its behaviour at high energies is better discussed in terms of the fundamental gauge field.

I wish to thank Peter Weisz for a critical reading of the paper and for checking some of the formulae in sect. 2. All numerical simulations were performed on a dedicated PC cluster at CERN. I am grateful to the CERN management for providing the required funds and to the CERN IT Department for technical support.

Appendix A. Notational conventions

The Lie algebra $\mathfrak{su}(N)$ of $SU(N)$ may be identified with the linear space of all anti-hermitian traceless $N \times N$ matrices. With respect to a basis T^a , $a = 1, \dots, N^2 - 1$, of such matrices, the elements $X \in \mathfrak{su}(N)$ are given by $X = X^a T^a$ with real components X^a (repeated group indices are automatically summed over). The structure constants f^{abc} in the commutator relation

$$[T^a, T^b] = f^{abc} T^c \quad (\text{A.1})$$

are real and totally anti-symmetric in the indices if the normalization condition

$$\text{tr}\{T^a T^b\} = -\frac{1}{2} \delta^{ab} \quad (\text{A.2})$$

is imposed. Moreover, $f^{acd}f^{bcd} = N\delta^{ab}$.

Gauge fields in the continuum theory take values in the Lie algebra of the gauge group. Their normalization is usually such that gauge transformations and covariant derivatives do not involve the gauge coupling (see, however, subsect. 2.2). Lorentz indices $\mu, \nu, \dots$ in D dimensions range from 0 to $D-1$ and are automatically summed over when they occur in matching pairs. The space-time metric is assumed to be Euclidian. In particular, $p^2 = p_\mu p_\mu$ for any momentum p .

The lattice theories considered in this paper are set up as usual on a four-dimensional hyper-cubic lattice with spacing a . In particular, lattice gauge fields $U(x, \mu)$ reside on the links (x, μ) of the lattice and take values in the gauge group. The link differential operators acting on functions $f(U)$ of the gauge field are

$$\partial_{x,\mu}^\alpha f(U) = \left. \frac{d}{ds} f(e^{sX} U) \right|_{s=0}, \quad X(y, \nu) = \begin{cases} T^\alpha & \text{if } (y, \nu) = (x, \mu), \\ 0 & \text{otherwise.} \end{cases} \quad (\text{A.3})$$

While these depend on the choice of the generators T^α , the combination

$$\partial_{x,\mu} f(U) = T^\alpha \partial_{x,\mu}^\alpha f(U) \quad (\text{A.4})$$

can be shown to be basis-independent.

Appendix B. One-loop integrals

The Feynman integrals other than $\mathcal{E}_0$ which contribute to $\langle E \rangle$ at order g_0^4 involve an integration over two momenta, q and r , and over at most two flow-time parameters. In all cases the integrand is of the form

$$e^{-sq^2 - ur^2 - v(q+r)^2} \frac{P(q, r)}{q^2 r^2 (q+r)^2} \quad (\text{B.1})$$

where $P(q, r)$ is a Lorentz-invariant polynomial and s, u, v are linear combinations of the flow-time parameters.

As already mentioned, these integrals can in principle be evaluated by substituting the Feynman parameter representation for the propagators $1/q^2$, $1/r^2$ and $1/(q+r)^2$. A more economic computation is, however, always possible using the basic integrals

$$\int_q e^{-sq^2} (q^2)^{-\alpha} = \frac{s^\alpha}{(4\pi s)^{D/2}} \frac{\Gamma(D/2 - \alpha)}{\Gamma(D/2)}, \quad (\text{B.2})$$

$$\int_{q,r} \frac{e^{-sq^2 - ur^2 - v(q+r)^2}}{q^2} = \frac{2}{(4\pi)^D(D-2)(u+v)}(su + uv + vs)^{1-D/2}. \quad (\text{B.3})$$

The second formula, for example, allows the integral

$$\int_{q,r} \frac{e^{-t(q^2+r^2+(q+r)^2)}}{q^2 r^2} = \frac{1}{(4\pi)^4(2t)^2} \{8 \ln 2 - 4 \ln 3 + \text{O}(\epsilon)\} \quad (\text{B.4})$$

to be quickly evaluated once the propagator $1/r^2$ is replaced by its Feynman parameter representation.

A special case are integrals like

$$\begin{aligned} \int_0^t ds \int_{q,r} \frac{e^{-(t+s)(q^2+r^2)-(t-s)(q+r)^2}}{q^2 r^2} qr \\ = \frac{1}{(4\pi)^4(2t)^2} \left\{ \frac{1}{2} - 4 \ln 2 + 2 \ln 3 + \text{O}(\epsilon) \right\}, \end{aligned} \quad (\text{B.5})$$

whose integrand is proportional to qr . Noting

$$qr = \frac{1}{2} \{ (q+r)^2 - q^2 - r^2 \}, \quad (\text{B.6})$$

the factor can be traded for a differentiation with respect to the flow-time parameters. Most of the time, this allows one integral over these parameters to be performed right away and thus leads to simpler integrals without factors of qr .

Appendix C. Numerical integration of the Wilson flow

On a finite lattice, the space $\mathcal{G}$ of all gauge fields is a finite power of the gauge group and thus itself a Lie group. The associated Lie algebra $\mathfrak{g}$ coincides with the linear space of all link fields with values in the Lie algebra of the gauge group. From this abstract point of view, the flow equation (1.4) is an ordinary first-order differential equation of the form

$$\dot{V}_t = Z(V_t)V_t, \quad (\text{C.1})$$

where $V_t \in \mathcal{G}$ and $Z(V_t) \in \mathfrak{g}$.

The Runge-Kutta scheme described in this appendix obtains the solution of the flow equation at times $t = n\epsilon$, $n = 1, 2, 3, \dots$, recursively, starting from the initial configuration at $t = 0$. The rule for the integration from time t to $t + \epsilon$ is

$$\begin{aligned} W_0 &= V_t, \\ W_1 &= \exp\left\{\frac{1}{4}Z_0\right\}W_0, \\ W_2 &= \exp\left\{\frac{8}{9}Z_1 - \frac{17}{36}Z_0\right\}W_1, \\ V_{t+\epsilon} &= \exp\left\{\frac{2}{3}Z_2 - \frac{8}{9}Z_1 + \frac{17}{36}Z_0\right\}W_2, \end{aligned} \tag{C.2}$$

where

$$Z_i = \epsilon Z(W_i), \quad i = 0, 1, 2. \tag{C.3}$$

Note that this rule is fully explicit. Moreover, since the gauge field can be overwritten from one equation to the next, and since Z_0 can be overwritten by $\frac{8}{9}Z_1 - \frac{17}{36}Z_0$, intermediate storage space for only one of these latter fields is required.

A straightforward calculation shows that the integration scheme (C.2) is accurate up to errors of order ϵ^4 per step. The total error of the integration up to a specified flow time thus scales like ϵ^3 . Empirically one finds that the integration is numerically stable in the direction of positive flow time, the integration errors in the link variables being on the order of 10^{-6} if $\epsilon = 0.01$.

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P08：梯度流的微扰分析

主题归属	梯度流、涂抹与流时算符
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Perturbative analysis of the gradient flow in non-abelian gauge theories

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Abstract

The gradient flow in non-abelian gauge theories on $\mathbb{R}^4$ is defined by a local diffusion equation that evolves the gauge field as a function of the flow time in a gauge-covariant manner. Similarly to the case of the Langevin equation, the correlation functions of the time-dependent field can be expanded in perturbation theory, the Feynman rules being those of a renormalizable field theory on $\mathbb{R}^4 \times [0, \infty)$. For any matter multiplet and to all loop orders, we show that the correlation functions are finite, i.e. do not require additional renormalization, once the theory in four dimensions is renormalized in the usual way. The flow thus maps the gauge field to a one-parameter family of smooth renormalized fields.

1. Introduction

The physics described by non-abelian gauge theories can be studied in many ways. Depending on the context and the questions to be answered, the desired information is extracted from correlation functions of local fields, expectation values of Wilson loops or the Schrödinger functional, for example.

As explained in refs. [1, 2], the gradient flow provides further opportunities to probe these theories and allows some of their otherwise elusive properties to be understood (in lattice gauge theory, the gradient flow is referred to as the Wilson flow, while in the mathematical literature it is commonly known as the Yang–Mills gradient flow).

Evidently, physically meaningful probes must be safe of ultra-violet divergences or must be such that these can be canceled by a well-defined renormalization procedure.

In the case of the gradient flow, there is some evidence that the gauge field generated by the flow does not require renormalization [1,2], but a formal proof of the absence of ultra-violet divergences at positive flow time has not been given so far. The aim of the present paper is to fill this gap through an all-order analysis of the flow in perturbation theory. With respect to the closely related case of the renormalization of the Langevin equation discussed by Zinn-Justin and Zwanziger [3,4], there are two important differences, one being the absence of the noise term in the flow equation and the other the fact that the initial distribution of the gauge field (which is given by the functional integral of the theory considered) is not ignored.

The perturbative analysis presented in this paper applies to renormalizable gauge theories with any compact simple gauge group and any matter multiplet. However, in order to simplify the discussion as much as possible, only the pure $SU(N)$ gauge theory with dimensional regularization will be considered, the generalization to other cases being straightforward (see sect. 8).

In the following three sections, the Feynman rules for the correlation functions of the gauge field generated by the gradient flow are derived and are shown to be those of a local field theory with an extra dimension (the flow time). The finiteness of the correlation functions at positive flow time can then be established using power-counting and the BRS symmetry (sects. 6,7), but for illustration the divergent parts of a set of one-loop diagrams are first worked out in sect. 5.

2. Iterative solution of the flow equation

In this section, we introduce the gradient flow and derive the Feynman rules for the associated flow-line diagrams. The pure $SU(N)$ gauge theory in $D = 4 - 2\epsilon$ euclidean dimensions is considered and the fundamental gauge field $A_\mu(x)$ is normalized so that its action at bare coupling g_0 is given by

$$S = -\frac{1}{2g_0^2} \int d^D x \operatorname{tr} \{F_{\mu\nu}(x) F_{\mu\nu}(x)\}, \quad (2.1)$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \quad (2.2)$$

(see appendix A for unexplained notation).

2.1 Definition of the gradient flow

The gradient flow evolves the gauge field as a function of a parameter $t \geq 0$ that is referred to as the flow time. Starting from the fundamental gauge field,

$$B_\mu|_{t=0} = A_\mu, \quad (2.3)$$

the time-dependent field $B_\mu(t, x)$ is determined by the differential equation

$$\partial_t B_\mu = D_\nu G_{\nu\mu} + \alpha_0 D_\mu \partial_\nu B_\nu, \quad (2.4)$$

$$G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu], \quad D_\mu = \partial_\mu + [B_\mu, \cdot]. \quad (2.5)$$

The name “gradient flow” derives from the fact that first term on the right of eq. (2.4) is proportional to the gradient of the gauge action along the flow. Note that neither the initial condition (2.3) nor the flow equation (2.4) involve the gauge coupling.

The second term on the right of eq. (2.4) is included in order to damp the evolution of the gauge degrees of freedom of the field. As in the case of the Langevin equation [4], some technicalities in the perturbative analysis of the flow can be avoided in this way without affecting the evolution of the gauge-invariant observables. The latter are in fact independent of the parameter α_0 , since the solutions of eq. (2.4) obtained at different values of α_0 are related by a (time-dependent) gauge transformation [1].

2.2 Expansion in powers of the fundamental gauge field

Equation (2.4) may be split into a linear and remainder part according to

$$\partial_t B_\mu = \partial_\nu \partial_\nu B_\mu + (\alpha_0 - 1) \partial_\mu \partial_\nu B_\nu + R_\mu, \quad (2.6)$$

$$R_\mu = 2[B_\nu, \partial_\nu B_\mu] - [B_\nu, \partial_\mu B_\nu] + (\alpha_0 - 1)[B_\nu, \partial_\nu B_\nu] + [B_\nu, [B_\nu, B_\mu]]. \quad (2.7)$$

The linearized equation can be solved using the heat kernel

$$K_t(z)_{\mu\nu} = \int_p \frac{e^{ipz}}{p^2} \{ (\delta_{\mu\nu} p^2 - p_\mu p_\nu) e^{-t p^2} + p_\mu p_\nu e^{-\alpha_0 t p^2} \}, \quad (2.8)$$

where

$$\int_p = \int \frac{d^D p}{(2\pi)^D}. \quad (2.9)$$

Taking the boundary condition (2.3) into account, the flow equation may then be cast in the integral form

$$B_\mu(t, x) = \int d^D y \left\{ K_t(x - y)_{\mu\nu} A_\nu(y) + \int_0^t ds K_{t-s}(x - y)_{\mu\nu} R_\nu(s, y) \right\}. \quad (2.10)$$

From this representation the retarded character of the equation is evident and it is also quite clear that the sensitivity to the initial value of the field dies away as t increases, although only slowly so at small momenta (α_0 is assumed to be positive).

When passing to momentum space,

$$B_\mu(t, x) = \int_p e^{ipx} \hat{B}_\mu(t, p), \quad (2.11)$$

the integral equation (2.10) becomes

$$\hat{B}_\mu(t, p) = \hat{K}_t(p)_{\mu\nu} \hat{A}_\nu(p) + \int_0^t ds \hat{K}_{t-s}(p)_{\mu\nu} \hat{R}_\nu(s, p). \quad (2.12)$$

It is helpful at this point to introduce the vertices $X^{(2,0)}$ and $X^{(3,0)}$ through

$$\begin{aligned} \hat{R}_\mu^a(t, p) &= \sum_{n=2}^3 \frac{1}{n!} \int_{q_1} \dots \int_{q_n} (2\pi)^D \delta(p + q_1 + \dots + q_n) \\ &\quad \times X^{(n,0)}(p, q_1, \dots, q_n)^{ab_1 \dots b_n} \hat{B}_{\nu_1}^{b_1}(t, -q_1) \dots \hat{B}_{\nu_n}^{b_n}(t, -q_n) \end{aligned} \quad (2.13)$$

and the requirement that they are totally symmetric in the momentum-index combinations $(q_1, \nu_1, a_1), \dots, (q_n, \nu_n, a_n)$. The solution of the integral equation in powers of the fundamental gauge field,

$$\begin{aligned} \hat{B}_\mu^a(t, p) &= \hat{K}_t(p)_{\mu\nu} \hat{A}_\nu^a(p) + \frac{1}{2} \int_0^t ds \hat{K}_{t-s}(p)_{\mu\nu} \int_{q,r} (2\pi)^D \delta(p - q - r) \\ &\quad \times X^{(2,0)}(p, -q, -r)^{abc} \hat{K}_s(q)_{\rho\delta} \hat{K}_s(r)_{\sigma\tau} \hat{A}_\delta^b(q) \hat{A}_\tau^c(r) + \dots, \end{aligned} \quad (2.14)$$

is then obtained through iteration, i.e. by recursively inserting the equation on the right of itself.

The vertices $X^{(2,0)}$ and $X^{(3,0)}$ are given explicitly in appendix B. Note that the momentum-index combination (p, μ, a) plays a special rôle in eq. (2.13). In particular, the vertices are symmetric only in their other arguments.



Fig. 1. The diagrams contributing to $\tilde{B}_\mu^a(t, p)$ are directed tree graphs with a single external line (a little square is drawn at the end of this line). Flow lines always start from a one-point vertex (circle with a cross) or a flow vertex (circle) and end at another flow vertex if the line is not the external one. Each flow vertex has one outgoing flow line that corresponds to the momentum-index combination (p, μ, a) in eq. (2.13).

2.3 Flow-line diagrams

The terms contributing to the expansion (2.14) in powers of the fundamental gauge field can be graphically represented by Feynman diagrams (see fig. 1). There are two kinds of vertices in these diagrams, the flow vertices $X^{(2,0)}$ and $X^{(3,0)}$ introduced in the previous subsection and the one-point vertex

$$\begin{array}{c} \mu, a \\ \rightarrow \\ \text{---} \otimes \end{array} = \hat{A}_\mu^a(p). \quad (2.15)$$

Each flow vertex is inserted at some flow time that is integrated from zero to infinity, while the one-point vertices reside at time zero. The vertices are connected through the flow lines

$$\begin{array}{c} t, \mu, a \\ \xrightarrow{\quad} \\ \text{---} \end{array} \begin{array}{c} s, \nu, b \\ \xrightarrow{\quad} \end{array} = \delta^{ab} \theta(t-s) \hat{K}_{t-s}(p)_{\mu\nu}, \quad (2.16)$$

where t and s are the flow times at the endpoints of the line. In view of the retarded nature of the propagator (2.16), the flow time increases from zero at the one-point vertices to time t at the outer end of the external line as one follows the arrows in the diagram. In particular, the times associated with the vertices are effectively integrated only up to t (rather than infinity).

As far as the momenta, the index contractions and the symmetry factors are concerned, the Feynman rules are the usual ones. It is then not difficult to show that the sum of all diagrams solves the flow equation (2.12) order by order in the fundamental gauge field.

3. Perturbation theory

The n -point correlation functions of the field $B_\mu^a(t, x)$ can be computed in perturbation theory by expanding the field in powers of the fundamental field, as explained sect. 2, and by calculating the correlation functions of the latter in the $SU(N)$ gauge theory as usual. We now show that the correlation functions can be directly obtained from a set of Feynman rules in $D + 1$ dimensions.

3.1 Gauge fixing

The flow equation (2.4) is invariant under the infinitesimal transformation

$$\delta B_\mu = D_\mu \omega, \quad (3.1)$$

provided the time-dependence of $\omega(t, x) \in \mathfrak{su}(N)$ is such that

$$\partial_t \omega = \alpha_0 D_\mu \partial_\mu \omega. \quad (3.2)$$

Since the initial value of ω is unconstrained, these transformations generate the full gauge group at flow time $t = 0$ and thus extend the gauge symmetry of the $SU(N)$ gauge theory to all flow times.

The symmetry can be fixed as usual by including the gauge-fixing term

$$S_{gf} = -\frac{\lambda_0}{g_0^2} \int d^D x \operatorname{tr} \{ \partial_\mu A_\mu(x) \partial_\nu A_\nu(x) \} \quad (3.3)$$

and the associated ghost action

$$S_{ce} = -\frac{2}{g_0^2} \int d^D x \operatorname{tr} \{ \partial_\mu \bar{c}(x) D_\mu c(x) \} \quad (3.4)$$

in the total action of the theory, where c and $\bar{c}$ are the Faddeev–Popov ghost fields. As far as the n -point correlation functions of the fundamental gauge field and the ghosts are concerned, the Feynman rules are then the standard ones. Evidently, since the transformation (3.1) is an infinitesimal gauge variation, the expectation values of gauge-invariant expressions in the field generated by the flow are independent of the gauge parameter λ_0 .

3.2 Gauge-field propagator

The way in which the flow-line diagrams combine with the Feynman diagrams of the underlying theory is best explained by considering the two-point function of the time-dependent gauge field. To leading order in the gauge coupling, the flow-line diagram


(3.5)

is the only one that contributes to the correlation function. The contraction of the gauge fields at the one-point vertices then shows that

$$\langle \hat{B}_\mu^\alpha(t, p) \hat{B}_\nu^\beta(s, q) \rangle = (2\pi)^D \delta(p+q) \delta^{\alpha\beta} g_0^2 \hat{D}_{t+s}(p)_{\mu\nu} + \mathcal{O}(g_0^4), \quad (3.6)$$

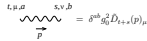
$$\hat{D}_t(p)_{\mu\nu} = \frac{1}{(p^2)^2} \{ (\delta_{\mu\nu} p^2 - p_\mu p_\nu) e^{-tp^2} + \lambda_0^{-1} p_\mu p_\nu e^{-\alpha_0 tp^2} \}. \quad (3.7)$$

This formula includes the mixed propagator

$$\langle \hat{A}_\mu^\alpha(p) \hat{B}_\nu^\beta(s, q) \rangle = \langle \hat{B}_\mu^\alpha(0, p) \hat{B}_\nu^\beta(s, q) \rangle \quad (3.8)$$

as well as the two-point function of the gauge field at flow time zero.

Since all three propagators are given by the same analytic expression, the same graphical symbol


(3.9)

may be used for them. Note that the contraction of the one-point vertices always has the effect of converting the terminal flow lines to gauge-field lines. If one starts from the flow-line diagram


(3.10)

instead of the diagram (3.5), for example, the leading-order contribution is obtained by substituting the tree-level diagram for the correlation function of the gauge fields

at the three one-point vertices. This leads to a diagram



$$(3.11)$$

that has an ordinary three-point vertex (filled circle) and a flow vertex. Two of the lines attached to the latter are gauge-field lines instead of flow lines, but the expression for the vertex is the same as before.

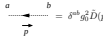
3.3 Feynman rules in $D + 1$ dimensions

The flow time will now be interpreted as an additional space-time coordinate. Since only non-negative times are considered, the $D + 1$ dimensional space is a half-space with a D dimensional boundary at flow time zero. The $SU(N)$ gauge theory lives at the boundary, while the field generated by the gradient flow extends to the extra dimension.

From this point of view, the ordinary and the flow vertices represent boundary and bulk interaction terms, respectively, while the propagation of the fields in $D + 1$ dimensions is described by the gauge-field propagator (3.9) and the flow propagator (2.16). The ghost-field propagator

$$\langle \tilde{c}^a(p) \tilde{c}^b(q) \rangle = (2\pi)^D \delta(p + q) \delta^{ab} g_0^2 \tilde{D}(p) + O(g_0^4), \quad \tilde{D}(p) = \frac{1}{p^2}, \quad (3.12)$$

on the other hand, is defined at flow time zero only. Graphically it is represented through a dotted line,



$$(3.13)$$

where the arrow is drawn in the direction from $\tilde{c}$ to c .

It should be quite clear at this point that the n -point functions of the field generated by the gradient flow are given by Feynman diagrams in $D + 1$ dimensions with the graphical elements listed in the caption of fig. 2. Some special features of the Feynman rules are worth pointing out:

- (a) Each flow vertex has exactly one outward-directed flow line corresponding to the first momentum-index combination in eq. (2.13). The other lines attached to these vertices may be gauge-field lines or ingoing flow lines.



Fig. 2. Examples of diagrams contributing to the two- and three-point functions of the time-dependent gauge field. All diagrams are built from flow propagators (directed solid lines), gauge-field propagators (wiggly lines), ghost-field propagators (directed dotted lines), flow vertices (open circles) and ordinary vertices at flow time zero (filled circles). The little square at the end of the external lines indicates that they are not amputated.

(b) Flow lines must start at a flow vertex and are either external or end at another flow vertex. The gauge-field lines, on the other hand, can start and end at both the flow vertices and the ordinary vertices.

(c) The flow vertices are inserted at some flow time which is integrated from zero to infinity. All other vertices are at flow time zero. The flow times on which the propagators depend are the ones at the endpoints of the corresponding lines.

(d) Diagrams with closed flow-line loops are set to zero. This rule derives from the fact that flow-line diagrams are tree diagrams and that the contraction of the gauge fields at the one-point vertices never leads to new flow lines.

4. Field theory $D+1$ dimensions

The Feynman rules for the correlation functions of the time-dependent gauge field obtained in the previous section are those of a local field theory in $D+1$ dimensions [4]. It is possible to show this through somewhat formal functional-integral manipulations, but one can also adopt a purely algebraic point of view, where the action in $D+1$ dimensions merely serves as a generating function for the Feynman rules.

4.1 Action

Except for the boundary condition (2.3), the field $B_\mu(t, x)$ will now be considered to be an independent field. One also needs to introduce a Lagrange-multiplier field $L_\mu(t, x) = L_\mu^a(t, x)T^a$ with purely imaginary components. The action of the theory

in $D + 1$ dimensions is then given by [4]

$$S_{\text{tot}} = S + S_{\text{gf}} + S_{\text{cc}} + S_{\text{fl}}, \quad (4.1)$$

$$S_{\text{fl}} = -2 \int_0^\infty dt \int d^D x \text{tr} \{ L_\mu(t, x) (\partial_t B_\mu - D_\nu G_{\nu\mu} - \alpha_0 D_\mu \partial_\nu B_\nu)(t, x) \}. \quad (4.2)$$

In this framework, the flow equation (2.4) coincides with the field equation obtained by varying the action with respect to the Lagrange-multiplier field. Note that the latter is not required to satisfy any particular boundary conditions.

4.2 Propagators

The quadratic part of the action is the sum of

$$\int_0^\infty dt \int d^D x L_\mu^\alpha(t, x) (\partial_t B_\mu^\alpha - \partial_\nu \partial_\nu B_\mu^\alpha - (\alpha_0 - 1) \partial_\mu \partial_\nu B_\nu^\alpha)(t, x) \quad (4.3)$$

and the quadratic part of the action in D dimensions. The bulk fields B_μ and L_μ thus couple to each other only, but there is also an implicit coupling to the fundamental gauge field through the boundary condition (2.3). This complication can easily be overcome by substituting

$$B_\mu(t, x) = \int d^D y K_t(x - y)_{\mu\nu} A_\nu(y) + b_\mu(t, x). \quad (4.4)$$

The field b_μ then satisfies homogenous boundary conditions, while the first term in eq. (4.4) solves the linearized flow equation and thus drops out in the action (4.3).

It follows from these remarks that the propagators of the fundamental gauge field and the ghost fields coincide with the expressions given in the previous section. All other two-point functions vanish except for

$$\langle b_\mu^\alpha(t, x) L_\nu^b(s, y) \rangle \big|_{\text{leading order}} = \delta^{\alpha b} H(t, x; s, y)_{\mu\nu}, \quad (4.5)$$

which is determined by the field equation

$$\{ \delta_{\mu\rho} \partial_t - \delta_{\mu\rho} \partial_\sigma \partial_\sigma - (\alpha_0 - 1) \partial_\mu \partial_\rho \} H(t, x; s, y)_{\mu\nu} = \delta_{\mu\nu} \delta(t - s) \delta(x - y) \quad (4.6)$$

and the boundary condition $H_{\mu\nu}(t, x; s, y)|_{t=0, s>0} = 0$. The unique solution of these equations is

$$H(t, x; s, y)_{\mu\nu} = \theta(t - s) K_{t-s}(x - y)_{\mu\nu} \quad (4.7)$$



Fig. 3. Examples of diagrams with flow-line loops. All such diagrams vanish as a consequence of the retarded nature of the flow propagator and the rules of dimensional regularization [4].

and the bL propagator is thus seen to coincide with the flow propagator.

The two-point functions involving the B field may finally be calculated by combining eq. (4.4) with the results obtained so far. Since the AL and the bb propagators vanish, one quickly finds that the BL propagator is equal to the flow propagator and that the BB and the AB propagators are equal to the expressions given in the previous section.

4.3 Vertices

The theory described by the action (4.1) has the vertices of the theory in D dimensions plus those deriving from the bulk action S_B . Recalling eqs. (2.6) and (2.13), it is straightforward to show that the LB^2 and LB^3 vertices generated by the interaction part of the latter,

$$S_B|_{\text{interaction}} = 2 \int_0^\infty dt \int d^D x \text{tr} \{ L_\mu(t, x) R_\mu(t, x) \}, \quad (4.8)$$

coincide with the vertices $X^{(2,0)}$ and $X^{(3,0)}$.

All vertices and propagators of the Feynman rules of the previous sections are thus recovered. A somewhat unusual feature of these rules is the presence of a field (the Lagrange multiplier) that propagates only through its mixing with the other fields. The algebraic structure of the Feynman rules is however entirely standard and does not involve any prescriptions apart from the ones deriving from the expansion of the action in powers of the fields.

4.4 Flow lines and flow-line loops

Flow lines represent the BL propagator. They can start and end at the flow vertices but are never attached to an ordinary vertex. Outward- and inward-directed external flow lines are external B and L lines, respectively.

Diagrams with closed flow lines comply with these rules, but should be absent if a complete matching with the Feynman rules of the previous section is to be achieved (cf. point (d) at the end of sect. 3). Such diagrams are in fact equal to zero [4]. The diagrams 1 and 2 in fig. 3, for example, vanish because dimensional regularization sets the momentum integral to zero. If there are two or more vertices in the loop, as in diagram 3, the time integrations vanish, because the flow propagator is retarded and thus forces the flow times at the vertices to be squeezed to a range of measure zero (singularities in the time coordinates are excluded in view of the regularization of the momentum integral).

If a lattice regularization is used, the vanishing of diagram 2 may not be guaranteed, but one can always include a ghost field in the action that cancels the flow-line loops algebraically, i.e. at the level of the Feynman integrands [3]. With dimensional regularization, however, this device is not needed and is therefore omitted here.

5. Sample calculation at one-loop order

For illustration, the divergent parts of the one-loop diagrams that contribute to the gauge-field two-point functions are worked out in this section. The parameter and field renormalization required at flow time zero is then seen to cancel the singularities at all flow times.

5.1 Computation of self-energy diagrams

There are only few self-energy diagrams at one-loop order (see fig. 4). The diagrams 1 – 3 coincide with the ones contributing to the two-point function of the gauge field at flow time zero. We therefore merely quote the known result

$$\Gamma_{AA}^{(1)}(p)_{\mu\nu} \Big|_{\text{pole}} = \delta^{ab} (\delta_{\mu\nu} p^2 - p_\mu p_\nu) \frac{N}{16\pi^2 \epsilon} \left(\frac{13}{6} - \frac{1}{2\lambda_0} \right) \quad (5.1)$$

for the sum of their singular parts.

In the case of diagram 4, the Feynman integrand is a linear combination of terms of the form

$$\delta^{ab} \frac{Q(k, p)_{\mu\nu}}{\{k^2(k+p)^2\}^2} e^{-t\{uk^2+v(k+p)^2\}}, \quad u, v \in \{1, \alpha_0\}, \quad (5.2)$$

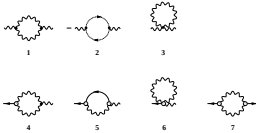


Fig. 4. One-loop diagrams contributing to the AA (1–3), LA (4), LB (5, 6) and LL (7) vertex functions. The time-momentum-index combinations at the external legs on the left and right are (t, p, μ, a) and (s, q, ν, b) , respectively (the flow times are absent in the case of the A legs).

where k is the loop momentum and $Q(k, p)_{\mu\nu}$ a homogeneous polynomial in k and p of degree 6 (the momentum-conservation factor $(2\pi)^D \delta(p+q)$ is now always suppressed). This diagram is finite at all $t > 0$ and in any dimension D , but eventually it will be multiplied by the external propagators and must then be integrated over t from 0 to infinity. The behaviour of the integral small t thus matters and can give rise to singularities at $D = 4$.

If $f(t)$ is any smooth test function (an external propagator, for example), the time integral

$$I(k, p) = \int_0^\infty dt f(t) e^{-t\{uk^2 + v(k+p)^2\}} \quad (5.3)$$

can be worked out in an asymptotic series at large k , the leading term being

$$I(k, p) = \frac{f(0)}{(u+v)k^2} + O(k^{-3}). \quad (5.4)$$

The integration over the flow time t thus improves the degree of divergence of the momentum integral. In particular, the first term in the decomposition

$$I(k, p) = \left\{ I(k, p) - \frac{f(0)}{(u+v)(k+r)^2} \right\} + \frac{f(0)}{(u+v)(k+r)^2} \quad (5.5)$$

(where r is an arbitrary external momentum) makes the integral convergent at $D = 4$. One is then left with a logarithmic divergence and therefore a pole singularity equal to the one of the ordinary Feynman integral

$$\delta^{ab}\delta(t)\frac{1}{u+v}\int_k\frac{Q(k,0)_{\mu\nu}}{(k^2)^4(k+r)^2}, \quad (5.6)$$

which is easily evaluated using standard techniques.

Proceeding in this way, one obtains

$$\Gamma_{LA}^{(1)}(t,p)_{\mu\nu}^{ab}\Big|_{\text{pole}} = g_0^2\delta^{ab}\delta(t)\delta_{\mu\nu}\frac{N}{16\pi^2\epsilon}\left(\frac{3}{4} + \frac{3}{4\lambda_0}\right), \quad (5.7)$$

$$\Gamma_{LB}^{(1)}(t,s,p)_{\mu\nu}^{ab}\Big|_{\text{pole}} = g_0^2\delta^{ab}\delta(t)\delta(s)\delta_{\mu\nu}\frac{N}{16\pi^2\epsilon}\left(-\frac{1}{2\lambda_0}\right), \quad (5.8)$$

for the divergent parts of the LA and LB vertex functions. The diagram 7 and thus the LL vertex function are finite at $D = 4$.

5.2 Renormalization

The bare coupling and gauge-fixing parameter are related to the renormalized parameters g and λ through

$$g_0^2 = \mu^{2\epsilon}g^2Z, \quad \lambda_0 = \lambda Z_3^{-1}, \quad (5.9)$$

where μ is the normalization mass. To one-loop order, the renormalization constants Z and Z_3 are given by

$$Z = 1 - \frac{b_0}{\epsilon}g^2 + O(g^4), \quad b_0 = \frac{N}{16\pi^2}\frac{11}{3}, \quad (5.10)$$

$$Z_3 = 1 + \frac{c_0}{\epsilon}g^2 + O(g^4), \quad c_0 = \frac{N}{16\pi^2}\left(\frac{13}{6} - \frac{1}{2\lambda}\right), \quad (5.11)$$

up to scheme-dependent finite terms.

The renormalization of the fundamental gauge field,

$$A_\mu^a = Z^{1/2}Z_3^{1/2}(A_R)_\mu^a, \quad (5.12)$$

involves both renormalization constants as a result of the unconventional normalization conventions adopted in this paper. For reasons explained in sect. 7, the ghost fields are renormalized asymmetrically according to

$$c^a = \hat{Z}_3 Z^{1/2} Z_3^{1/2} (c_R)^a, \quad \bar{c}^a = Z^{1/2} Z_3^{-1/2} (\bar{c}_R)^a, \quad (5.13)$$

where $\hat{Z}_3$ is the usual ghost renormalization constant. Note that eq. (5.13) is equivalent to the standard renormalization prescription for the correlation functions at flow time zero, because the fields c and $\bar{c}$ always occur in pairs and only the product of their renormalization factors matters.

5.3 Do the bulk fields require renormalization?

To one-loop order of perturbation theory, the question may be answered by explicitly calculating the divergent parts (if any) of bulk-field correlation functions. We first consider the two-point function of the B field and define its Lorentz-invariant parts $\mathcal{A}$ and $\mathcal{B}$ through

$$\begin{aligned} \langle \hat{B}_\mu^a(t, p) \hat{B}_\nu^b(s, q) \rangle &= (2\pi)^D \delta(p + q) \frac{\delta^{ab}}{(p^2)^2} \\ &\times \{ (\delta_{\mu\nu} p^2 - p_\mu p_\nu) \mathcal{A}(t, s, p^2) + p_\mu p_\nu \mathcal{B}(t, s, p^2) \}. \end{aligned} \quad (5.14)$$

All self-energy diagrams drawn in fig. 4 contribute to $\mathcal{A}$ and $\mathcal{B}$. The diagrams 4 – 6 actually make two contributions, because their external legs are different.

In the renormalized perturbation expansion

$$\mathcal{X} = \mu^{2\epsilon} \sum_{l=0}^{\infty} g^{2l+2} \mathcal{X}^{(l)}, \quad \mathcal{X} = \mathcal{A} \text{ or } \mathcal{B}, \quad (5.15)$$

the leading-order coefficients are

$$\mathcal{A}^{(0)} = e^{-(t+s)p^2}, \quad \mathcal{B}^{(0)} = \lambda^{-1} e^{-\alpha_0(t+s)p^2}. \quad (5.16)$$

At the next order, the residues of the poles that derive from the renormalization of the coupling and the gauge-fixing parameter are thus given by

$$\text{res}\{\mathcal{A}^{(1)}\} \Big|_{Z \text{ factors}} = -b_0 \mathcal{A}^{(0)}, \quad (5.17)$$

$$\text{res}\{\mathcal{B}^{(1)}\} \Big|_{Z \text{ factors}} = (c_0 - b_0) \mathcal{B}^{(0)}. \quad (5.18)$$

Recalling the results obtained in subsect. 5.1, it is now straightforward to verify that these poles are canceled by the divergent parts of the one-loop diagrams. Up to this order of perturbation theory, the two-point function of the time-dependent gauge field is thus finite and does not require further renormalization.

In the case of the BL , BA_R , LA_R and LL correlation functions, the cancellation of the singularities at $D = 4$ can be shown in the same way. There is actually little to prove in the last two instances, because these two-point functions vanish to all orders of perturbation theory (there are no diagrams with ingoing and no outgoing flow lines). All calculations reported in this section thus support the conjecture that the B and the L field do not need to be renormalized.

6. BRS symmetry

We now proceed with the general discussion that will lead to the proof of finiteness of the correlation functions of the bulk fields to all orders in the gauge coupling. As a first step, the BRS symmetry of the theory in $D + 1$ dimensions is reviewed in this section.

The gauge fixing discussed in sect. 3.1 follows the standard procedure and therefore leads to a theory with a BRS symmetry. This symmetry acts on the boundary fields in the usual way, but the transformation of the bulk fields requires the solution of the diffusion equation (3.2) and consequently tends to be non-local. In their work on the renormalization of the Langevin equation, Zinn-Justin and Zwanziger [4] however showed that the locality of the transformation can be restored by introducing additional ghost fields.

6.1 BRS transformation of the boundary fields

The BRS variation of the unrenormalized fields A_μ , $\bar{c}$ and c is defined by [5,6]

$$\delta A_\mu = D_\mu c, \quad (6.1)$$

$$\delta c = -c^2, \quad (6.2)$$

$$\delta \bar{c} = \lambda_0 \partial_\mu A_\mu. \quad (6.3)$$

Note that the components c^a and $\bar{c}^a$ of the ghost fields and the operator δ anti-commute with one another. The product on the right of eq. (6.2), for example, is

given by

$$c^2 = c^a c^b T^a T^b = \frac{1}{2} c^a c^b f^{abc} T^c, \quad (6.4)$$

and the Leibniz rule for the operator δ must take its anti-commuting character into account. The action $S + S_{gf} + S_{\varepsilon\varepsilon}$ and the measure in the functional integral of the theory in D dimensions are then easily shown to be BRS invariant.

6.2 Bulk ghost fields

The additional ghost fields $d(t, x)$ and $\bar{d}(t, x)$ mentioned above live in $D + 1$ dimensions but are otherwise of the same kind as the Faddeev–Popov ghosts. Their action is

$$S_{d\bar{d}} = -2 \int_0^\infty dt \int d^D x \operatorname{tr} \{ \bar{d}(t, x) (\partial_t d - \alpha_0 D_\mu \partial_\mu d)(t, x) \} \quad (6.5)$$

and the d field is required to satisfy the boundary condition

$$\bar{d}|_{t=0} = c. \quad (6.6)$$

No boundary condition is imposed on the $\bar{d}$ field, which, in many respects, plays a rôle similar to the Lagrange-multiplier field L_μ in the case of the gauge field.

The propagator of the ghost fields,

$$\langle \bar{d}^a(t, p) d^b(s, q) \rangle = (2\pi)^D \delta(p + q) \delta^{ab} \theta(t - s) \tilde{K}_{t-s}(p) + O(g_0^2), \quad (6.7)$$

$$\tilde{K}_t(p) = e^{-\alpha_0 t p^2}, \quad (6.8)$$

can be worked out following the steps taken in subsect. 4.2. Graphically the propagator is represented by a dashed line,

$$\begin{array}{c} t,a \qquad \qquad s,b \\ \cdots \cdots \cdots \cdots \cdots \cdots \\ \xrightarrow{p} \end{array} = \delta^{ab} \theta(t - s) \tilde{K}_t(p), \quad (6.9)$$

where the arrow is drawn in the direction from $\bar{d}$ to d . There is a mixed propagator,

$$\langle \bar{d}^a(t, p) \bar{c}^b(q) \rangle = (2\pi)^D \delta(p + q) \delta^{ab} g_0^2 \tilde{D}_t(p) + O(g_0^4), \quad (6.10)$$

$$\tilde{D}_t(p) = \frac{1}{p^2} e^{-\alpha_0 t p^2}, \quad (6.11)$$

as well, which coincides with the $c\bar{c}$ propagator at $t = 0$ and is therefore represented by the same graphical symbol (a directed dotted line). The cd two-point function, on the other hand, vanishes to all orders.

The action

$$S_{dd}|_{\text{interaction}} = - \int_0^\infty dt \int_{p,q,r} (2\pi)^D \delta(p+q+r) \\ \times X^{(1,1)}(p,q,r)_{\mu}^{abc} \tilde{B}_\mu^a(t,-p) \tilde{d}(t,-q)^b \tilde{d}(t,-r)^c, \quad (6.12)$$

also gives rise to a new flow vertex $X^{(1,1)}$. In the Feynman diagrams, it is represented by an open circle as the other flow vertices. The explicit expression for the vertex is given in appendix B.

As in the case of the flow-line loops discussed in subsect. 4.4, it is possible to show that diagrams with $d\bar{d}$ loops vanish. Note that ghost loops with mixed propagators do not exist, because there is no $c\bar{d}$ propagator and the vertices only couple c to $\bar{c}$ or d to $\bar{d}$. The only non-zero ghost loops are thus the usual ones of the theory at flow time zero.

6.3 BRS variation of the fields in the bulk

The BRS symmetry acts on the bulk fields according to [4]

$$\delta B_\mu = D_\mu d, \quad (6.13)$$

$$\delta L_\mu = [L_\mu, d], \quad (6.14)$$

$$\delta d = -d^2, \quad (6.15)$$

$$\delta \bar{d} = D_\mu L_\mu - \{d, \bar{d}\}. \quad (6.16)$$

Note that $\delta B_\mu = \delta A_\mu$ and $\delta d = \delta c$ at the boundary, as must be the case in view of the boundary conditions (2.3) and (6.6).

In order to show that the BRS variation of the bulk action $S_B + S_{dd}$ vanishes, it is helpful to introduce the fields

$$E_\mu = \partial_\nu B_\mu - D_\nu G_{\nu\mu} - \alpha_0 D_\mu \partial_\nu B_\nu, \quad (6.17)$$

$$e = \partial_t d - \alpha_0 D_\mu \partial_\mu d. \quad (6.18)$$

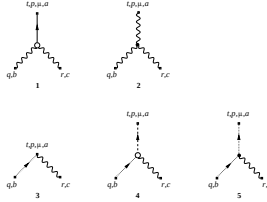


Fig. 5. Tree diagrams contributing to the correlation function on the left of eq. (6.23) (diagrams 1 and 2) and to the one on the right of the equation (diagrams 3, 4 and 5). The momenta are ingoing and satisfy $p + q + r = 0$. In diagram 3, the momentum p flows into a vertex representing the insertion of the field $[B_\mu, d]$, while in the case of the diagrams 4 and 5 the external lines with momentum p are multiplied by ip_μ (thus representing the field $\partial_\mu d$).

After some algebra, one finds that

$$\delta E_\mu = [E_\mu, d] + D_\mu e, \quad (6.19)$$

$$\delta e = -\{e, d\}, \quad (6.20)$$

and the invariance of the bulk action is then easily established.

6.4 Examples of BRS identities

Since the functional integration measure is also invariant, it follows that $\langle \delta \mathcal{O} \rangle = 0$ for any product $\mathcal{O}$ of the basic fields. This leads to identities such as

$$\lambda_0 \langle \partial_\mu A_\mu^a(x) \partial_\nu A_\nu^b(y) \rangle = g_0^2 \delta^{ab} \delta(x - y), \quad (6.21)$$

$$\lambda_0 \langle B_\mu^a(t, x) \partial_\nu A_\nu^b(y) \rangle = -\langle (D_\mu d)^a(t, x) \bar{c}^b(y) \rangle, \quad (6.22)$$

$$\lambda_0 \langle B_\mu^a(t, x) \partial_\nu A_\nu^b(y) \partial_\rho A_\rho^c(z) \rangle = -\langle (D_\mu d)^a(t, x) \bar{c}^b(y) \partial_\rho A_\rho^c(z) \rangle, \quad (6.23)$$

the first of them being the familiar gauge Ward identity in the theory in D dimensions, which determines the longitudinal part of the gauge-field two-point function.

It is instructive to check eqs. (6.22) and (6.23) at tree level of perturbation theory. The first equation relates the longitudinal part of the mixed gauge-field propagator to the mixed ghost propagator, an identity that is immediate from the expressions for the propagators. The other equation involves the 3-point vertices (see fig. 5) and a non-trivial cancellation among various contributions (appendix C).

7. Finiteness of the renormalized perturbation expansion

The renormalization constants Z , Z_3 and $\bar{Z}_3$ are now assumed to be such that the singularities of the correlation functions of the fields $(A_\mu)_R$, c_R and $\bar{c}_R$ at $D = 4$ cancel to all orders of the renormalized coupling. Our aim in this section is to show that the correlation functions involving the bulk fields at positive flow time are then finite too and thus do not require further renormalization.

7.1 Renormalized perturbation theory

We first need to reorganize the perturbation expansion in terms of the renormalized parameters and fields. In this form, the expansion is generated by an action $S_0 + \Delta S$ of the renormalized fields, which includes the counterterms ΔS required to cancel the poles of the Feynman diagrams at $D = 4$.

Explicitly S_0 and ΔS are obtained by expressing the bare parameters and fields in the total action

$$S_{\text{tot}} = S + S_{\text{gf}} + S_{\text{cc}} + S_R + S_{\text{d}\bar{d}} \quad (7.1)$$

through the renormalized ones and by setting

$$S_0 = S_{\text{tot}}|_{Z=Z_3=\bar{Z}_3=1}, \quad (7.2)$$

$$\Delta S = S_{\text{tot}} - S_0 + \Delta S_{\text{bc}}, \quad (7.3)$$

where

$$\begin{aligned} \Delta S_{\text{bc}} = 2 \int d^D x \text{tr} \{ & (Z^{1/2} Z_3^{1/2} - 1) L_\mu(0, x) (A_R)_\mu(x) \\ & + (\hat{Z}_3 Z^{1/2} Z_3^{1/2} - 1) \bar{d}(0, x) c_R(x) \}. \end{aligned} \quad (7.4)$$

The boundary conditions are then

$$B_\mu|_{t=0} = (A_R)_\mu, \quad \bar{d}|_{t=0} = c_R, \quad (7.5)$$

and the Feynman rules derived from the action S_0 thus coincide with those discussed in sect. 3 (apart from the fact that the bare parameters g_0 and λ_0 are replaced by $\mu^* g$ and λ , respectively).

Note that the boundary conditions (7.5) differ from the ones imposed on the bare fields (a product of renormalization factors is missing). The counterterm ΔS_{bc} must be included in the action of the renormalized fields to correct for this. It amounts to adding two-point vertices to the Feynman rules, whose effect on the gauge-field and ghost propagators is equivalent to a change of the boundary conditions by the missing renormalization factors.

7.2 Absence of bulk counterterms

In the renormalized perturbation expansion, singularities at $D = 4$ (if any) appear for the first time at some loop order $l \geq 1$. Since the interactions are local, and since all propagators are tempered distributions in position space with singularities only at coinciding arguments, the divergent parts of the correlation functions at this loop order are expected to be such that they can be canceled by local counterterms. Similarly to the case of an ordinary field theory on a half-space studied by Symanzik [7], the counterterms can be localized either in the bulk or at the boundary of the half-space.

In the theory considered here, bulk counterterms can be excluded from the outset. In order to show this, first note that the correlation functions of the bulk fields B_μ , L_μ , d and $\bar{d}$ are, at large flow times, given by Feynman diagrams built from flow lines and flow vertices only. All diagrams of this kind are directed trees or products such trees, each tree ending at one of the B and d fields in the correlation function considered. Moreover, the flow lines at the other ends of the trees must start from the L and $\bar{d}$ fields in the correlation function.

Since there are no loop diagrams, the correlation functions of the bulk fields are non-singular at large flow times and do not require renormalization. Divergent bulk

counterterms are therefore excluded. Note, incidentally, that correlation functions of local fields composed from B and d fields are finite too, because no loop diagrams are generated when the arguments of some of these fields coalesce. The field $D_\mu d$ that appears in the BRS identity (6.23), for example, is of this kind and consequently does not need to be renormalized.

7.3 Boundary counterterms

The discussion in the previous subsection shows that the structure of any divergent parts of the correlation functions must correspond to the insertion of a counterterm localized at flow time zero. Since the correlation functions of the renormalized fields $(A_n)_\mu$, c_n and $\bar{c}_n$ are finite to all orders, divergences can only arise from diagrams with at least one flow vertex and thus at least one such vertex with an external flow line. The possible counterterms are therefore proportional to L_μ or $\bar{d}$.

Since the gauge coupling is dimensionless, the counterterm may not involve composite fields of dimension larger than 4 (such terms would be irrelevant). Moreover, Lorentz symmetry, global gauge invariance and the ghost number conservation must be respected. Since L_μ and $\bar{d}$ have dimension 3 and all other fields dimension 1, it follows that the possible counterterms at l -loop order are of the form

$$2g^{2l} \int d^D x \operatorname{tr} \{ z_1 L_\mu(0, x) (A_n)_\mu(x) + z_2 \bar{d}(0, x) c_n(x) \} \quad (7.6)$$

with some (singular) coefficients z_1 and z_2 . An LB and a $\bar{d}d$ term should in principle be included here, but in view of the boundary conditions (7.5), these terms are not independent and their inclusion would be equivalent to a change of the coefficients z_1 and z_2 .

7.4 Consequences of the BRS symmetry

We now show that the boundary counterterm (7.6) is excluded by the BRS symmetry of the theory. In terms of the renormalized fields and parameters, the BRS identity (6.23) reads

$$\lambda \langle B_\mu^a(t, x) \partial_\nu (A_n)^b(y) \partial_\rho (A_n)^c_\rho(z) \rangle = - \langle (D_\mu d)^a(t, x) (\bar{c}_n)^b(y) \partial_\rho (A_n)^c_\rho(z) \rangle. \quad (7.7)$$

The unusual asymmetric renormalization (5.13) of the ghost fields c and $\bar{c}$ was chosen to ensure that the renormalization factors drop out in this equation. From the point of view of the $SU(N)$ gauge theory, the standard and the asymmetric renormalization of the ghost fields are equivalent, but the situation is different in the theory in $D+1$ dimensions, because only c couples to the bulk fields.

Equation (7.7) holds as long as no extra counterterms need to be added and thus up to loop order l inclusive. Note that all fields in the correlation functions in this equation, including the composite field $D_\mu d$, are renormalized fields that cannot be additionally renormalized (cf. subsect. 7.2). If the correlation functions are singular at l -loop order, one must therefore be able to cancel the singularities by a counterterm of the form (7.6).

The contribution of the counterterm (7.6) to the correlation functions is obtained by inserting the corresponding two-point vertices in the tree diagrams in fig. 5. The insertions have the effect of multiplying the values of the diagrams 1 – 5 by g^2 times

$$2z_1, z_1, z_1 + z_2, z_1 + z_2 \text{ and } z_2 \quad (7.8)$$

respectively. However, recalling the values of the diagrams quoted in appendix C, the sum of the diagrams weighted by the factors (7.8) turns out to violate the BRS identity (7.7) unless $z_1 = z_2 = 0$.

The correlation functions in eq. (7.7) must therefore be finite at l -loop order and all other (renormalized) correlation functions must be non-singular too, because the addition to the action of a counterterm of the form (7.6) is excluded. We have thus shown that the theory in $D+1$ dimensions does not require further renormalization.

8. Miscellaneous remarks

8.1 Behaviour of the gauge field near flow time zero

In the regularized theory, the time-dependent field $B_\mu(t, x)$ satisfies the boundary condition (2.3) and its correlation functions thus converge to those of the bare gauge field $A_\mu(x)$ when t goes to zero. However, since the latter requires renormalization by a divergent constant, the correlation functions tend to become singular at $t = 0$ after renormalization and removal of the regularization. Their asymptotic behaviour for $t \rightarrow 0$ is then described by an expansion

$$B_\mu(t, x) = c_B(t)(A_R)_\mu(x) + O(t) \quad (8.1)$$

in local renormalized fields with singular coefficients [7]. At one-loop order of perturbation theory, for example, the coefficient $c_B(t)$ is found to diverge logarithmically at $t = 0$.

It is straightforward to show that the renormalization group equation

$$\left\{ \mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} - 2\gamma \lambda \frac{\partial}{\partial \lambda} + \gamma + \beta/g \right\} c_B(t) = \mathcal{O}(t) \quad (8.2)$$

holds, where

$$\beta = -b_0 g^3 + \mathcal{O}(g^5), \quad \gamma = c_0 g^2 + \mathcal{O}(g^4), \quad (8.3)$$

are the beta function and the anomalous dimension of the gauge field. In the Landau gauge, for example, the equation can be easily integrated and one finds that

$$B_\mu(t, x) \underset{t \rightarrow 0}{\sim} (2b_0 \bar{g}(q)^2)^{1/2 - \alpha/2b_0} R(g)(A_R)_\mu(x), \quad (8.4)$$

$\bar{g}(q)$ being the running coupling at momentum $q = (8t)^{-1/2}$ and $R(g)$ the factor that relates the renormalized to the renormalization-group-invariant gauge field.

Equation (8.4) is a remnant of the boundary condition satisfied by the gauge field in the regularized theory. It shows that the field generated by the flow equation is connected to the fundamental field in a universal manner, i.e. there is no room for finite renormalizations here, the reason being that any such renormalization would violate the BRS symmetry.

8.2 Gauge-invariant composite fields

Wilson loops and gauge-invariant local fields are independent of the parameter a_0 in the flow equation and do not require renormalization at positive flow time. At small flow times, their asymptotic behaviour is determined by the scaling properties of the corresponding renormalized fields in the $SU(N)$ gauge theory and thus reflects the singular nature of the latter.

For illustration, consider the density

$$E = \frac{1}{4} G_{\mu\nu}^a G_{\mu\nu}^a \quad (8.5)$$

that has previously been studied in ref. [1]. Since E can mix with the unit field, the dominant term at small flow times,

$$E(t, x) = \langle E(t, x) \rangle + c_E(t) \left\{ \frac{1}{4} F_{\mu\nu}^a F_{\mu\nu}^a \right\}_R(x) + \mathcal{O}(t), \quad (8.6)$$

is its expectation value, while the first subleading term is proportional to the renormalized action density of the fundamental field. The associated coefficients, $\langle E(t, x) \rangle$

and $c_E(t)$, satisfy a renormalization group equation that allows their exact asymptotic behaviour at small t to be worked out analytically.

8.3 Theories with matter fields

In the presence of matter fields, the gradient flow is defined by the same equations as in the pure gauge theory (subsect. 2.1). The interesting but rather more complicated case where the matter fields are included in the time evolution is not considered here.

In a renormalizable theory, matter fields are scalar fields of dimension 1 or fermion fields of dimension 3/2. Provided the regularization preserves the gauge symmetry, our argumentation in sects. 6 and 7 then carries over literally. In particular, additional boundary counterterms involving the matter fields are excluded by the global symmetries and the requirement that their dimension must be less than or equal to 4. The finiteness of the correlation functions at positive flow times is therefore again guaranteed.

QCD is a prominent example of a gauge theory that has all the required properties to ensure finiteness, but the same applies to many more theories of interest, including the SU(2) Higgs model, supersymmetric versions of QCD and technicolour theories.

8.4 Lattice regularization

While the Feynman rules are more complicated than with dimensional regularization, the perturbative analysis of the gradient flow on the lattice is not expected to run into fundamental difficulties.

A possible choice of the gradient term in the lattice flow equation is the gradient of the Wilson action [1]. There is however no reason to choose this particular action or to require that it coincides with the gauge action of the theory. As long as the term has the correct form and normalization in the classical continuum limit (and thus at tree-level of perturbation theory), the correlation functions of the time-dependent gauge field will not depend on the exact choices one makes, except for lattice effects vanishing proportionally to a positive power of the lattice spacing a .

The continuous Lorentz symmetry is broken on the lattice, but the remaining exact symmetries are sufficient to exclude counterterms that have not already appeared in the continuum theory. We therefore expect that the correlation functions of the time-dependent fields do not require renormalization and that their values in the continuum limit are independent of the regularization (up to finite renormalizations of the coupling and the gauge-fixing parameter). Moreover, if the lattice theory is $O(a)$ improved, the correlation functions are automatically improved too, because there are no candidate $O(a)$ counterterms with all the required properties.

9. Conclusions

The fact that the gauge field generated by the gradient flow does not require renormalization is a consequence of the locality, the symmetries and the parabolic nature of the flow equation. In particular, in the associated field theory in $D+1$ dimensions, bulk counterterms are excluded simply because the evolution of the bulk fields is retarded at large times and thus described by tree diagrams. The absence of boundary counterterms other than those needed for the renormalization of the theory at flow time zero is however non-trivial and can only be shown using power-counting and the BRS symmetry.

In presence of matter fields, the situation is essentially unchanged as long as only the gauge field is evolved in time. The finiteness of the correlation functions of the field at positive flow time is therefore still guaranteed. Including all or some of the matter fields in the flow is however an interesting option that remains to be explored.

We thank Jean Zinn-Justin for helpful discussions on the renormalization of the Langevin equation and for encouraging us to proceed directly with the flow equation rather than considering the latter to be a limit of the Langevin equation.

Appendix A. Notational conventions

The Lie algebra $\mathfrak{su}(N)$ of $SU(N)$ may be identified with the linear space of all anti-hermitian traceless $N \times N$ matrices. With respect to a basis T^a , $a = 1, \dots, N^2 - 1$, of such matrices, the elements $X \in \mathfrak{su}(N)$ are given by $X = X^a T^a$ with real components X^a (repeated group indices are automatically summed over). The structure constants f^{abc} in the commutator relation

$$[T^a, T^b] = f^{abc} T^c \quad (\text{A.1})$$

are real and totally anti-symmetric in the indices if the normalization condition

$$\text{tr}\{T^a T^b\} = -\frac{1}{2} \delta^{ab} \quad (\text{A.2})$$

is imposed. Moreover, $f^{acd} f^{bcd} = N \delta^{ab}$.

Gauge fields in the continuum theory take values in the Lie algebra of the gauge group. Lorentz indices $\mu, \nu, \dots$ are automatically summed over when they occur in

matching pairs. The space-time metric is assumed to be euclidean. In particular, $p^2 = p_\mu p_\mu$ for any momentum p and $\delta_{\mu\mu} = D$.

Appendix B. Flow vertices

The flow vertices $X^{(2,0)}$, $X^{(3,0)}$ and $X^{(1,1)}$ are defined through eqs. (2.13) and (6.12), respectively. They are explicitly given by

$$\begin{aligned} X^{(2,0)}(p, q, r)_{\mu\nu\rho}^{abc} = & if^{abc} \{ (r - q)_\mu \delta_{\nu\rho} + 2q_\rho \delta_{\mu\nu} - 2r_\nu \delta_{\mu\rho} \\ & + (\alpha_0 - 1)(q_\nu \delta_{\mu\rho} - r_\rho \delta_{\mu\nu}) \}, \end{aligned} \quad (\text{B.1})$$

$$\begin{aligned} X^{(3,0)}(p, q, r, s)_{\mu\nu\rho\sigma}^{abcd} = & f^{abe} f^{cde} (\delta_{\mu\sigma} \delta_{\nu\rho} - \delta_{\mu\rho} \delta_{\sigma\nu}) \\ & + f^{ade} f^{bce} (\delta_{\mu\rho} \delta_{\sigma\nu} - \delta_{\mu\nu} \delta_{\rho\sigma}) + f^{ace} f^{dbe} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}), \end{aligned} \quad (\text{B.2})$$

$$X^{(1,1)}(p, q, r)_\mu^{abc} = \alpha_0 i f^{abc} r_\mu. \quad (\text{B.3})$$

In the Feynman diagrams, the momenta in these formulae are identified with the ingoing line momenta.

Appendix C. Evaluation of the diagrams in fig. 5

The contributions of the diagrams in fig. 5 to the Fourier transform of the correlation functions in the BRS identity (6.23) are of the form

$$-\frac{g_0^4 i f^{abc}}{\lambda_0 p^2 q^2 r^2} C_{\mu,n}, \quad (\text{C.1})$$

where n is the diagram number. Explicitly, one obtains

$$C_{\mu,1} = \frac{1}{2} p^2 (q - r)_\mu e^{-i\alpha_0(q^2 + r^2)} + \frac{1}{2} (q^2 - r^2) p_\mu e^{-i\alpha_0 p^2} - C_{\mu,2}, \quad (\text{C.2})$$

$$C_{\mu,2} = \frac{1}{2} \{ p^2 (q - r)_\mu + (q^2 - r^2) p_\mu \} e^{-ip^2}, \quad (\text{C.3})$$

for the diagrams contributing to the left-hand side of the equation and

$$C_{\mu,3} = -p^2 r_\mu e^{-\tan(q^2+r^2)}, \quad (C.4)$$

$$C_{\mu,4} = -\frac{1}{2}p^2 p_\mu \{e^{-\tan(q^2+r^2)} - e^{-\tan p^2}\}, \quad (C.5)$$

$$C_{\mu,5} = -(qr + r^2)p_\mu e^{-\tan p^2}, \quad (C.6)$$

for the diagrams on the right-hand side. The BRS symmetry requires

$$C_{\mu,1} + C_{\mu,2} = C_{\mu,3} + C_{\mu,4} + C_{\mu,5}, \quad (C.7)$$

which is indeed the case.

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Chiral symmetry and the Yang-Mills gradient flow

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Abstract

In the last few years, the Yang-Mills gradient flow was shown to be an attractive tool for non-perturbative studies of non-Abelian gauge theories. Here a simple extension of the flow to the quark fields in QCD is considered. As in the case of the pure-gauge gradient flow, the renormalizability of correlation functions involving local fields at positive flow times can be established using a representation through a local field theory in $4+1$ dimensions. Applications of the extended flow in lattice QCD include non-perturbative renormalization and $O(a)$ improvement as well as accurate calculations of the chiral condensate and of the pseudo-scalar decay constant in the chiral limit.

1. Introduction

In view of its renormalization properties [1,2], and since its application in lattice gauge theory is technically straightforward, the Yang-Mills gradient flow allows the dynamics of non-Abelian gauge theories to be probed in many interesting ways. The flow can be used for accurate scale setting, for example, and it provides an understanding of how exactly the topological (instanton) sectors emerge in the continuum limit of lattice QCD [1]†. Moreover, observables at positive flow time are natural quantities to consider for non-perturbative renormalization and step scaling [3–5].

Matter fields may or may not be included in the flow equations. In this paper, a fairly trivial extension of the flow to the quark fields in QCD is considered, where the flow equation for the gauge field is unchanged, while the evolution of the quark

† In lattice gauge theory, where the right-hand side of the flow equation coincides with the gradient of the Wilson plaquette action, the flow is often referred to as the “Wilson flow”. The term “gradient flow” is used here both in the continuum theory and on the lattice.

fields as a function of the flow time is determined by a gauge-covariant heat equation (see sect. 2). The theoretical analysis and practical implementation of the flow is not significantly complicated by the inclusion of the quark fields. In particular, following ref. [2], the renormalization of correlation functions of gauge-invariant local fields at positive flow times can be shown to require no more than a multiplicative renormalization of the time-dependent quark fields once the parameters of QCD are renormalized as usual.

For illustration, two applications of the extended flow are worked out in this paper, one being a new strategy for the calculation of the axial-current renormalization constant in lattice QCD and the other a computation of the chiral condensate essentially through the evaluation of the expectation value of the scalar quark density at positive flow time. In both cases, the use of the flow proves to be technically attractive. The chiral condensate, for example, is easily obtained with high precision, because no additive renormalization is required.

In the next two sections, the extension of the gradient flow to the quark fields is discussed in the continuum theory. Since the flow equations respect chiral symmetry, the correlation functions of the time-dependent fields satisfy simple chiral Ward identities. As explained in sect. 4, these allow the correlation functions to be related to the physics of the light pseudo-scalar mesons. Then follows a more technical part of the paper (sects. 5-7), where the flow is set up in the framework of lattice QCD. In particular, the associated field theory in 4+1 dimensions is shown to admit a well-defined local lattice regularization. The viability of the proposed applications of the flow in numerical lattice QCD is finally demonstrated through a sample calculation in 2+1 flavour QCD (sects. 8,9).

2. The gradient flow in QCD

The theory considered in this paper is QCD with gauge group $SU(N)$ and a multiplet of two or more massive quarks in the fundamental representation of the gauge group. Many results are however of a fairly general nature and not limited to QCD. The theory is set up in Euclidean space and quantized through the functional integral as usual. In this and the following section, dimensional regularization is employed. The notational conventions are summarized in appendix A.

2.1 Flow equations

Let $A_\mu(x)$ be the $SU(N)$ gauge potential, $\psi(x)$ the quark field and $\bar{\psi}(x)$ the antiquark field integrated over in the functional integral. The latter carry Dirac, colour and flavour indices, which are usually suppressed for simplicity.

The Yang-Mills gradient flow evolves the gauge field as a function of a parameter $t \geq 0$ that is referred to as the flow time. Starting from the fundamental gauge field,

$$B_\mu|_{t=0} = A_\mu, \quad (2.1)$$

the time-dependent field $B_\mu(t, x)$ is determined by the differential equation

$$\partial_t B_\mu = D_\nu G_{\nu\mu}, \quad (2.2)$$

$$G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu], \quad D_\mu = \partial_\mu + [B_\mu, \cdot] \quad (2.3)$$

(see ref. [1] for an introduction to the subject).

As already mentioned in sect. 1, the extension of the flow to the quark fields considered in this paper is a minimal one, where the evolution of the gauge field is left unchanged. The gauge field however appears in the flow equations[†]

$$\partial_t \chi = \Delta \chi, \quad \partial_t \bar{\chi} = \bar{\chi} \overleftarrow{\Delta}, \quad (2.4)$$

$$\Delta = D_\mu D_\mu, \quad D_\mu = \partial_\mu + B_\mu, \quad (2.5)$$

which, together with the initial conditions

$$\chi|_{t=0} = \psi, \quad \bar{\chi}|_{t=0} = \bar{\psi}, \quad (2.6)$$

define the time-dependent quark and antiquark fields $\chi(t, x)$ and $\bar{\chi}(t, x)$. In the flow equations (2.4), the gauge-covariant Laplacian Δ could be replaced by the square of the Dirac operator $\not{D}$, for example, or be multiplied by a proportionality factor, but such modifications do not appear to offer any advantages.

The flow of quark fields introduced in this section is similar to the source smoothing operations proposed many years ago in refs. [6,7]. With respect to these popular methods, there are, however, a few important differences, one of them being the fact

[†] The left-action of any differential operator Δ (or of a difference operator in lattice field theory) is defined through the requirement that the relation $\eta^\dagger \overleftarrow{\Delta} = (\Delta \eta)^\dagger$ holds for all complex-valued fields η .

that the flow operates in four dimensions rather than on the fields on an equal-time hyper-plane. Moreover, the flow time varies continuously and the gauge field is not set to the fundamental gauge field, but is evolved together with the quark field.

2.2 Local composite fields

Since the flow equations are gauge-covariant, the time-dependent fields transform in the same way under gauge transformations as the fundamental fields. Examples of gauge-invariant composite fields that are local both in space-time and in flow time are the densities

$$E_t(x) = -\frac{1}{2}\text{tr}\{G_{\mu\nu}(t,x)G_{\mu\nu}(t,x)\}, \quad (2.7)$$

$$S_t^{rs}(x) = \overline{\chi}_r(t,x)\chi_s(t,x), \quad P_t^{rs}(x) = \overline{\chi}_r(t,x)\gamma_5\chi_s(t,x), \quad (2.8)$$

where r, s are flavour indices.

Through the initial conditions (2.1),(2.6), the time-dependent fields depend on the fundamental fields. From the point of view of the QCD functional integral, composite fields like the densities (2.7),(2.8) are therefore observables, similar to Wilson loops or the ordinary pseudo-scalar densities, for example. In the following, the quantities of interest are the correlation functions of these fields, i.e. the expectation values of products of local fields composed from the basic fields at any flow time.

2.3 Perturbation theory

As explained in refs. [1,2], such correlation functions can be expanded in powers of the gauge coupling in a straightforward manner. First the flow equations are solved in powers of the fundamental fields. The substitution of these expansions in the local fields then leads to linear combinations of correlation functions of the fundamental fields that can be computed using the standard QCD Feynman rules.

In perturbation theory, the flow equations (2.2) and (2.4) are replaced by

$$\partial_t B_\mu = D_\nu G_{\nu\mu} + \alpha_0 D_\mu \partial_\nu B_\nu, \quad (2.9)$$

$$\partial_t \chi = \Delta \chi - \alpha_0 \partial_\nu B_\nu \chi, \quad \partial_t \overline{\chi} = \overline{\chi} \overline{\Delta} + \alpha_0 \overline{\chi} \partial_\nu B_\nu, \quad (2.10)$$

in order to avoid some technical subtleties. The terms proportional to the parameter $\alpha_0 > 0$ serve to damp the gauge modes and could be removed by a time-dependent gauge transformation [1]. Correlation functions of gauge-invariant fields are therefore not affected by these extra terms.

The expansion of the time-dependent quark field $\chi(t, x)$ in powers of the fundamental fields is obtained by iterating the quark flow equation in its integral form,

$$\chi(t, x) = \int d^D y \left\{ K_t(x - y) \psi(y) + \int_0^t ds K_{t-s}(x - y) \Delta' \chi(s, y) \right\}, \quad (2.11)$$

$$\Delta' = (1 - \alpha_0) \partial_\nu B_\nu + 2B_\nu \partial_\nu + B_\nu B_\nu, \quad (2.12)$$

and the corresponding integral equation for the time-dependent gauge field [1], where

$$K_t(z) = \frac{e^{-z^2/4t}}{(4\pi t)^{D/2}} \quad (2.13)$$

denotes the heat kernel of the Laplacian in D dimensions. At the lowest order in the gauge coupling, the interaction term in eq. (2.11) can be dropped and the expression

$$\langle \chi(t, x) \bar{\chi}(s, y) \rangle = \int \frac{d^D p}{(2\pi)^D} e^{ip(x-y)} \frac{e^{-(t+s)p^2}}{M_0 + ip^2} + \mathcal{O}(g_0^2) \quad (2.14)$$

is then obtained for the two-point function of the time-dependent quark field, where g_0 and M_0 are the bare coupling and (diagonal) quark mass matrix. The smoothing character of the quark flow is evident from these equations. Moreover, the smoothing range at leading order, $\sqrt{8t}$, is seen to be the same as in the case of the gauge field.

There are 8 self-energy diagrams that contribute to the two-point function (2.14) at one-loop order of perturbation theory. The vertices in these diagrams derive from the QCD action and from the iteration of the integral equation (2.11). Following the lines of ref. [2], it is straightforward to compute the ultraviolet divergent parts of the diagrams and one then finds that the two-point function can be renormalized by renormalizing the quark masses as usual and the fields according to

$$\chi = Z_\chi^{-1/2} \chi_R, \quad \bar{\chi} = \bar{\chi}_R Z_\chi^{-1/2}. \quad (2.15)$$

In the $\overline{\text{MS}}$ scheme in $D = 4 - 2\epsilon$ dimensions, the calculation yields

$$Z_\chi = 1 + \frac{3C_F}{16\pi^2\epsilon} g^2 + \mathcal{O}(g^4), \quad C_F = \frac{N^2 - 1}{2N}, \quad (2.16)$$

for the field renormalization constant, g being the renormalized coupling,

2.4 Quark condensate at non-zero flow time

At non-zero flow times, the large momenta in the integral (2.14) are exponentially suppressed and the quark two-point function consequently has no singularities as $(t, x) \rightarrow (s, y)$. The absence of short-distance singularities is a general feature of the correlation functions at positive flow times, which derives from the smoothing property of the flow equations. In particular, the “time-dependent quark condensates”

$$\Sigma_t^{rr} = -\langle S_t^{rr}(x) \rangle \quad (2.17)$$

do not require additive renormalization.

In perturbation theory, the renormalized condensates

$$\Sigma_{R,t}^{rr} = Z_\chi \Sigma_t^{rr} \quad (2.18)$$

can be worked out in powers of the renormalized coupling g , with coefficients that depend on the renormalized quark masses $m_{R,r}, m_{R,s}, \dots$ and the flow time t . The leading-order term in four dimensions is given by

$$\Sigma_{R,t}^{rr}|_{g=0} = \frac{2Nm_{R,r}}{(4\pi)^2 t} \int_0^\infty dv \frac{e^{-vz}}{(1+v)^2}, \quad z = 2tm_{R,r}^2, \quad (2.19)$$

and thus vanishes in the chiral limit, consistently with the fact that chiral symmetry can only be spontaneously broken at the non-perturbative level.

3. Field theory in $D+1$ dimensions

The discussion in the following sections heavily builds on the fact that the correlation functions of the time-dependent fields coincide with the correlation functions in a local field theory in $D+1$ dimensions, the extra dimension being the flow time. The observation dates back to the seminal work of Zinn-Justin and Zwanziger [8,9] on the Langevin equation and recently allowed the renormalizability of the (pure-gauge) gradient flow to be established to all orders of perturbation theory [2].

As will become clear below, the inclusion of the quark fields in the theory in $D+1$ dimensions is straightforward. The general setup is exactly the same as in ref. [2], where an introduction to the subject and further details can be found.

3.1 Fields and action

In addition to the fundamental and the time-dependent fields already encountered, the theory in $D+1$ dimensions involves the Lagrange-multiplier fields $L_\mu(t, x)$, $\lambda(t, x)$ and $\bar{\lambda}(t, x)$. The latter are fermion fields with the same indices as the quark fields, while $iL_\mu(t, x)$ is a vector field that takes values in the Lie algebra of $SU(N)$.

As explained in ref. [2], gauge fixing in perturbation theory requires the introduction of the Faddeev–Popov ghost fields $c(x)$, $\bar{c}(x)$ and the associated time-dependent fields $d(t, x)$, $\bar{d}(t, x)$, where

$$d|_{t=0} = c. \quad (3.1)$$

No boundary condition is imposed on the field $\bar{d}(t, x)$, which, in many ways, plays a rôle similar to the Lagrange-multiplier fields. Apart from the boundary conditions (2.1), (2.6) and (3.1), all fields are assumed to be unconstrained at this point.

The total action of the theory,

$$S_{\text{tot}} = S + S_{G,\beta} + S_{FP,\beta} + S_{F,\beta}, \quad (3.2)$$

includes the gauge-fixed QCD action S of the fields at flow time zero and the bulk actions

$$S_{G,\beta} = -2 \int_0^\infty dt \int d^D x \operatorname{tr} \{ L_\mu(t, x) (\partial_t B_\mu - D_\nu G_{\nu\mu} - \alpha_0 D_\mu \partial_\nu B_\nu)(t, x) \}, \quad (3.3)$$

$$S_{FP,\beta} = -2 \int_0^\infty dt \int d^D x \operatorname{tr} \{ \bar{d}(t, x) (\partial_t d - \alpha_0 D_\mu \partial_\mu d)(t, x) \}, \quad (3.4)$$

$$S_{F,\beta} = \int_0^\infty dt \int d^D x \{ \bar{\lambda}(t, x) (\partial_t - \Delta + \alpha_0 \partial_\nu B_\nu) \chi(t, x) \\ + \bar{\chi}(t, x) (\overleftarrow{\partial}_t - \overleftarrow{\Delta} - \alpha_0 \partial_\nu B_\nu) \lambda(t, x) \}. \quad (3.5)$$

With respect to the pure-gauge flow studied in ref. [2], the bulk quark action $S_{F,\beta}$ is the only new term in the total action.

3.2 Correlation functions

The n -point correlation functions of the gauge, quark, ghost and Lagrange-multiplier fields are formally defined through the functional integral over all fields with “Boltzmann factor” $\exp\{-S_{\text{tot}}\}$.

In the case of the correlation functions of the gauge and quark fields, the functional integral can be worked out, to some extent, by performing the integral over the Lagrange-multiplier fields. Since the action is linear in these fields, the integration yields a product of functional Dirac δ -functions. The bulk actions were chosen such that the δ -functions amount to imposing the flow equations on the time-dependent gauge and quark fields. Up to mathematical subtleties and possibly non-trivial Jacobian factors, the correlation functions thus coincide with the ones calculated directly, as in sect. 2, by solving the flow equations with the initial conditions (2.1),(2.6) and treating the calculated time-dependent fields as QCD observables.

The equivalence of the theory in $D+1$ dimensions and the direct computation of the correlation functions can be rigorously shown in perturbation theory [2] and in lattice QCD (sect. 5). In particular, all correlation functions that do not involve the time-dependent quark fields or the associated Lagrange-multiplier fields coincide with the ones studied in ref. [2], where the quark fields were not evolved in flow time. This “reduction property” of the theory in $D+1$ dimensions strongly constrains the form of the possible counterterms required for renormalization (see subsect. 3.4).

3.3 Fermion integral

Since the action in $D+1$ dimensions is quadratic in the fermion fields, the functional integral over these fields can be evaluated (at fixed bosonic fields) by applying Wick’s theorem. The basic Wick contractions are worth being given explicitly, because the expressions provide some insight into the structure of the theory in $D+1$ dimensions and will, in any case, be frequently needed later.

The Wick contractions of the basic fields are determined by the action and the boundary conditions (2.6). In particular, the propagator of the quark fields at flow time zero,

$$\overline{\psi(x)}\psi(y) = S(x, y), \quad (3.6)$$

$$(\not{D} + M_0)S(x, y) = \delta(x - y), \quad (3.7)$$

coincides with the one in D dimensions. Apart from the quark propagator $S(x, y)$, the contractions of the time-dependent fermion fields,

$$\overline{\lambda(t, x)}\lambda(s, y) = 0, \quad (3.8)$$

$$\overline{\chi(t, x)}\lambda(s, y) = \theta(t - s)K(t, x; s, y), \quad (3.9)$$

$$\overline{\lambda(t, x) \chi(s, y)} = \theta(s - t) K(s, y; t, x)^\dagger, \quad (3.10)$$

$$\overline{\chi(t, x) \chi(s, y)} = \int d^D v d^D w K(t, x; 0, v) S(v, w) K(s, y; 0, w)^\dagger, \quad (3.11)$$

involve the fundamental solution

$$\{\partial_t - \Delta + \alpha_0 \partial_\nu B_\nu\} K(t, x; s, y) = 0 \quad \text{if } t \geq s, \quad (3.12)$$

$$\lim_{t \rightarrow s} K(t, x; s, y) = \delta(x - y), \quad (3.13)$$

of the quark flow equation. Note that the kernel $K(t, x; s, y)$ is flavour-independent, proportional to the unit Dirac matrix and an $N \times N$ complex matrix in colour space (the adjoint of the kernel in eqs. (3.10) and (3.11) is taken in index space only).

Consistently with the boundary conditions at vanishing flow time, the contractions involving the fundamental quark and antiquark fields,

$$\overline{\chi(t, x) \psi(y)} = \int d^D v K(t, x; 0, v) S(v, y), \quad (3.14)$$

$$\overline{\psi(x) \chi(s, y)} = \int d^D w S(x, w) K(s, y; 0, w)^\dagger, \quad (3.15)$$

$$\overline{\lambda(t, x) \psi(y)} = \overline{\psi(x) \lambda(s, y)} = 0, \quad (3.16)$$

are included in eqs. (3.9)–(3.11) as special cases. Note, however, that the behaviour of correlation functions near space-time boundaries tends to be complicated by short-distance singularities [10]. The limits

$$\lim_{s \rightarrow 0} \overline{\lambda(0, x) \chi(s, y)} = \lim_{t \rightarrow 0} \overline{\chi(t, x) \lambda(0, y)} = \delta(x - y), \quad (3.17)$$

for example, differ from the contractions (3.16) at $t = s = 0$ by a contact term.

3.4 Renormalization

In perturbation theory, the renormalization properties of the theory in $D+1$ dimensions can be determined using power counting and symmetry arguments. The task of figuring out all possible local counterterms to the action is simplified by the fact that the renormalization of the theory where only the gauge field is evolved in flow

time is already known to require no more than the usual QCD parameter and field renormalizations [2]. In particular, at flow time $t > 0$, the fields $B_\mu(t, x)$ and $L_\mu(t, x)$ need not be renormalized.

When the time-dependent quark fields are included in the theory, further counterterms may need to be added to the total action. The reduction property mentioned at the end of subsect. 3.2 however implies that any such term must be proportional to the quark fields or the associated Lagrange-multiplier fields. Moreover, as explained in subsect. 7.2 of ref. [2], bulk counterterms (i.e. local terms integrated over all $D+1$ coordinates) are excluded, because the correlation functions at large flow times are given by tree diagrams.

Taking the symmetries of the unrenormalized theory into account, and the fact that the quark and Lagrange-multiplier fields have power-counting dimension $3/2$ and $5/2$, respectively, a moment of thought reveals that the only possible additional counterterm is proportional to

$$\int d^D x \left\{ \bar{\chi}(0, x) \psi(x) + \bar{\psi}(x) \lambda(0, x) \right\}. \quad (3.18)$$

The insertion of this term in the Feynman diagrams is equivalent to a renormalization

$$\chi = Z_\chi^{-1/2} \chi_{\text{R}}, \quad \lambda = Z_\lambda^{1/2} \lambda_{\text{R}}, \quad (3.19)$$

of the fermion and, similarly, the anti-fermion fields at positive flow time. Apart from the usual renormalization of the QCD parameters and fields, the renormalization of the theory in $D+1$ dimensions thus requires the time-dependent quark and associated Lagrange multiplier fields to be renormalized as specified above.

4. Chiral symmetry relations

Since the quark flow equations (2.4) preserve chiral symmetry, the time-dependent composite fields like the densities (2.8) transform in a simple way under chiral rotations and thus allow the spontaneous breaking of the symmetry to be studied in useful new ways. In this section, the principal goal is to relate the time-dependent condensates (2.17) to the ordinary chiral condensate and the pseudo-scalar decay constant in the chiral limit.

4.1 Quark field equation and PCAC relation

The space-time dimension is now set to 4 and the need for regularization and renormalization is ignored for the moment, i.e. the statements made in this subsection are, strictly speaking, only valid at tree level of perturbation theory.

Perhaps somewhat unexpectedly, the quark field equation in the theory in 4+1 dimensions,

$$\{(\not{D} + M_0)\psi(x) - \lambda(0, x)\} \phi_1(t_1, x_1) \dots \phi_n(t_n, x_n)\rangle = \text{contact terms}, \quad (4.1)$$

differs from the one in QCD by a term that couples to the quark fields at non-zero flow time. Equation (4.1) holds for any local fields $\phi_1, \dots, \phi_n$ and the contact terms vanish if all points $(t_1, x_1), \dots, (t_n, x_n)$ in 4+1 dimensions are different from $(0, x)$. The validity of the field equation (4.1) follows from the identities

$$(\not{D} + M_0)\overbrace{\psi(x)\bar{\psi}(y)} = \overbrace{\lambda(0, x)\bar{\psi}(y)} + \delta(x - y), \quad (4.2)$$

$$(\not{D} + M_0)\overbrace{\psi(x)\bar{\chi}(s, y)} = \overbrace{\lambda(0, x)\bar{\chi}(s, y)}, \quad (4.3)$$

$$(\not{D} + M_0)\overbrace{\psi(x)\bar{\lambda}(s, y)} = \overbrace{\lambda(0, x)\bar{\lambda}(s, y)}, \quad (4.4)$$

which are a direct consequence of the formulae quoted in subsect. 3.3 for the Wick contractions of the fermion fields. Equations (4.2)–(4.4) moreover show that contact terms only arise from contractions of the fundamental quark fields.

Now let

$$A_\mu^{rs}(x) = \bar{\psi}_r(x)\gamma_\mu\gamma_5\psi_s(x), \quad \hat{P}^{rs}(x) = \bar{\psi}_r(x)\gamma_5\psi_s(x), \quad (4.5)$$

be the axial currents and densities in QCD. For $r \neq s$, the quark field equation then implies the PCAC relation

$$\begin{aligned} & \{[\partial_\mu A_\mu^{rs}(x) - (m_{0,r} + m_{0,s})\hat{P}^{rs}(x) + \hat{P}^{rs}(x)] \phi_1(t_1, x_1) \dots \phi_n(t_n, x_n)\} \\ & = \text{contact terms}, \end{aligned} \quad (4.6)$$

where $m_{0,r}, m_{0,s}$ are the bare masses of the quarks with flavour labels r, s and

$$\hat{P}^{rs}(x) = \bar{\lambda}_r(0, x)\gamma_5\psi_s(x) + \bar{\psi}_r(x)\gamma_5\lambda_s(0, x). \quad (4.7)$$

In particular, the identity

$$\langle \{ \partial_\mu A_\mu^{\tau s}(x) - (m_{0,\tau} + m_{0,s}) P^{\tau s}(x) + \hat{P}^{\tau s}(x) \} P_t^{\tau s}(y) \rangle = 0 \quad (4.8)$$

holds at all flow times $t > 0$ and all points x including $x = y$. There are no contact terms in this case, because the fermion propagators that contribute to the correlation function on the left of eq. (4.8) all go from $(0, x)$ to (t, y) in 4+1 dimensions or from (t, y) to $(0, x)$. The contraction (4.2) thus does not occur. In view of the smoothing character of the flow equations, all parts of the correlation function are in fact regular functions of x .

Another purely algebraic consequence of the structure of the Wick contractions is the relation

$$\int d^4x \langle \hat{P}^{\tau s}(x) P_t^{\tau s}(y) \rangle = \langle S_t^{\tau \tau}(y) \rangle + \langle S_t^{\tau s}(y) \rangle, \quad (4.9)$$

which, when combined with the PCAC relation (4.8), leads to an identity

$$(m_{0,\tau} + m_{0,s}) \int d^4x \langle P^{\tau s}(x) P_t^{\tau s}(y) \rangle = -\Sigma_t^{\tau \tau} - \Sigma_t^{\tau s} \quad (4.10)$$

that allows the time-dependent quark condensates to be related to the physics of the pseudo-scalar mesons (see subsect. 4.3). In the derivation of this equation, both the quark masses and the flow time were assumed to be positive, as otherwise the absence of boundary terms at large x and non-integrable singularities at $x = y$ would not be guaranteed.

4.2 Renormalized PCAC relation

Beyond tree level of perturbation theory, the fields and quark masses in eqs. (4.8)–(4.10) require renormalization. The equations then become relations among renormalized correlation functions, whose exact form depends on the chosen normalization conventions. There are some natural choices one can make here and it is the goal in the following lines to specify these. The discussion implicitly assumes a sensible regularization of the theory, such as dimensional regularization or Wilson's lattice formulation [11], where most symmetries are preserved. As before, only the flavour non-diagonal channels are considered.

The non-singlet axial currents and densities renormalize multiplicatively,

$$A_{R,\mu}^{\tau s} = Z_A A_\mu^{\tau s}, \quad P_R^{\tau s} = Z_P P^{\tau s}, \quad (4.11)$$

and the same is the case for the time-dependent densities

$$S_{n,t}^{\tau s} = Z_\chi S_t^{\tau s}, \quad P_{n,t}^{\tau s} = Z_\chi P_t^{\tau s}. \quad (4.12)$$

As already mentioned, since there are no short-distance singularities at positive flow time, these fields (including the flavour diagonal ones) renormalize according to their quark content, i.e. like the product of a time-dependent quark and antiquark field at non-zero distance in 4+1 dimensions.

In the case of the field $\hat{P}^{\tau s}$, the multiplicative renormalizability,

$$\hat{P}_n^{\tau s} = \hat{Z}_P \hat{P}^{\tau s}, \quad (4.13)$$

is less obvious, because the field has dimension 4 and could mix with other fields of dimension 3 and 4. The Wick contractions involving the Lagrange-multiplier fields λ and $\bar{\lambda}$ are however such that, up to contact terms, the two-point correlation functions of $\hat{P}^{\tau s}$ and any local field composed from the gauge and quark fields at flow time zero vanish. Divergent additive renormalizations by such fields are therefore excluded. One is then left with fields involving the Lagrange-multiplier fields, but since $\hat{P}^{\tau s}$ is the only one among these with dimension 4 and the required symmetry properties, additive renormalizations are again not possible and the field must consequently renormalize multiplicatively.

The normalizations of the renormalized fields and the renormalized quark masses $m_{n,\tau}$ may now be chosen such that the PCAC relation assumes the form

$$\langle \{ \partial_\mu A_{n,\mu}^{\tau s}(x) - (m_{n,\tau} + m_{n,s}) P_n^{\tau s}(x) + \hat{P}_n^{\tau s}(x) \} P_{n,t}^{\tau s}(y) \rangle = 0. \quad (4.14)$$

Note that this convention only fixes the relative normalization of the axial current $A_{n,\mu}^{\tau s}$, the product $(m_{n,\tau} + m_{n,s}) P_n^{\tau s}$ and the density $\hat{P}_n^{\tau s}$. The normalization of the latter may however be fixed by requiring the equation

$$\int d^4x \langle \hat{P}_n^{\tau s}(x) P_{n,t}^{\tau s}(y) \rangle = \langle S_{n,t}^{\tau \tau}(y) \rangle + \langle S_{n,t}^{\tau s}(y) \rangle \quad (4.15)$$

and thus the identity

$$(m_{n,\tau} + m_{n,s}) \int d^4x \langle P_n^{\tau s}(x) P_{n,t}^{\tau s}(y) \rangle = -\Sigma_{n,t}^{\tau \tau} - \Sigma_{n,t}^{\tau s} \quad (4.16)$$

to hold (where $\Sigma_{n,t}^{\tau \tau} = Z_\chi \Sigma_t^{\tau \tau}$). Further normalization conditions then still need to be imposed on the renormalized quark masses and the time-dependent densities, but for

the masses one can adopt one of the standard conventions, while the normalization of the time-dependent fields tends to cancel in the equations of interest and is left unspecified.

The normalization of the axial currents implicitly determined by eqs. (4.14),(4.15) coincides with the one usually chosen, where the axial charges assume integer values. There are different ways to show this, the perhaps most straightforward one starting from a formulation of lattice QCD that preserves chiral symmetry [12–19]. Using chiral Ward identities [20], the bare fields can be shown to satisfy eqs. (4.8),(4.9) in this case, up to terms vanishing proportionally to the lattice spacing. Equations (4.14),(4.15) then imply $Z_A = \tilde{Z}_P = 1$ and thus the canonical normalization of the renormalized axial currents. In view of the universality of the continuum limit, the canonical normalization is therefore guaranteed by these equations independently of the chosen regularization of the theory.

4.3 Pseudo-scalar matrix elements and the chiral condensate

The quark multiplet is now assumed to contain two light quarks, referred to as the u and d quarks, with the same renormalized mass m_n . If there are not too many further quarks, chiral symmetry in the (u, d) -channel is expected to be spontaneously broken and there is a triplet of pseudo-scalar mesons, the “pions”, whose mass vanishes as m_n goes to zero.

At large distances, the light-quark pseudo-scalar correlation functions are dominated by the intermediate one-pion states. In particular, as $x_0 \rightarrow \infty$

$$\int d^3\mathbf{x} \langle P_n^{ad}(x) P_n^{da}(0) \rangle = -\frac{G_\pi^2}{M_\pi} e^{-M_\pi x_0} \{1 + O(e^{-\Delta E x_0})\}, \quad (4.17)$$

$$\int d^3\mathbf{x} \langle A_{n,0}^{ad}(x) P_n^{da}(0) \rangle = F_\pi G_\pi e^{-M_\pi x_0} \{1 + O(e^{-\Delta E x_0})\}, \quad (4.18)$$

where ΔE denotes the energy gap to the higher states and M_π , F_π and G_π are the pion mass, decay constant and vacuum-to-pion matrix element of the axial density.

From the point of view of the theory in 4 dimensions, $P_{n,t}^{da}(x)$ is not a local field, but it is still a well localized expression in the fundamental fields since the smoothing kernel $K(t, x; 0, y)$ falls off approximately like a Gaussian at large distances $|x - y|$. At times x_0 much larger than the smoothing range $\sqrt{8t}$, the two-point function

$$\int d^3\mathbf{x} \langle P_n^{ad}(x) P_{n,t}^{da}(0) \rangle = -\frac{G_\pi G_{\pi,t}}{M_\pi} e^{-M_\pi x_0} \{1 + O(e^{-\Delta E x_0})\} \quad (4.19)$$

therefore decays in the same way as the ordinary pseudo-scalar correlation functions. The coefficient $G_{\pi,t}$ coincides with the vacuum-to-pion matrix element of a certain operator, but is here considered to be defined through eq. (4.19).

The low-energy effective constants characterizing the (two-flavour) chiral limit,

$$\Sigma = \lim_{m_R \rightarrow 0} F_\pi G_\pi, \quad (4.20)$$

$$F = \lim_{m_R \rightarrow 0} F_\pi, \quad (4.21)$$

are the chiral condensate Σ and the value F of the pion decay constant at $m_R = 0$. As already mentioned, these can be related to the time-dependent light-quark condensate $\Sigma_{R,t} = \Sigma_{R,t}^{\text{qu}}$ via the chiral Ward identity (4.16). Using the PCAC relation

$$2m_R G_\pi = M_\pi^2 F_\pi, \quad (4.22)$$

the latter may be written in the form

$$\Sigma_{R,t} = -\frac{M_\pi^2 F_\pi}{2G_\pi} \int d^4x \langle P_R^{ud}(x) P_R^{du}(0) \rangle. \quad (4.23)$$

The integral on the right of this equation diverges when $m_R \rightarrow 0$ and its asymptotic behaviour is determined by the pion pole of the correlation function in momentum space. Recalling eq. (4.19), one is then led to conclude that

$$\Sigma = \lim_{m_R \rightarrow 0} \frac{\Sigma_{R,t} G_\pi}{G_{\pi,t}}, \quad (4.24)$$

$$F = \lim_{m_R \rightarrow 0} \frac{\Sigma_{R,t}}{G_{\pi,t}}. \quad (4.25)$$

In the chiral limit, the time-dependent condensate $\Sigma_{R,t}$ thus provides an order parameter for the spontaneous breaking of chiral symmetry, which coincides with the standard condensate Σ up to a proportionality constant.

A remarkable feature of eqs. (4.24),(4.25) is the fact that there is no reference to the axial currents, not even implicitly via the normalization conditions, and that the equations hold for any (positive) flow time t . Moreover, the renormalization constant Z_χ drops out so that the decay constant F , for example, is directly given through the bare time-dependent condensate and pseudo-scalar vacuum-to-pion matrix element.

4.4 Chiral perturbation theory

Close to the chiral limit, the asymptotic behaviour of many quantities in QCD can be described by chiral perturbation theory. If the Compton wavelength of the pion is much larger than the smoothing range of the gradient flow, i.e. if

$$8tM_\pi^2 \ll 1, \quad (4.26)$$

the time-dependent densities $S_{n,t}^{rs}$ and $P_{n,t}^{rs}$ are local fields from the point of view of pion physics, with the same symmetry properties as the densities at vanishing flow time. At low energies, the correlation functions of the time-dependent fields and the ordinary axial currents and densities are therefore expected to be dominated by the pion intermediate states and should hence be accessible to a quantitative description in the framework of chiral perturbation theory.

The time-dependent densities can in fact easily be included in chiral perturbation theory by adding the appropriate source terms to the chiral Lagrangian [21]. At each order of the chiral expansion, these terms require a number of effective couplings to be introduced, which will, in general, depend on the flow time. The expansion of the chiral condensate $\Sigma_{n,t}$ and the matrix element $G_{\pi,t}$, for example, can then be worked out and provides insight into how exactly the chiral limits (4.24),(4.25) are reached.

5. Lattice regularization

The starting point in this section is a formulation of lattice QCD, where the quark fields are put on the lattice as proposed by Wilson [11]. No particular assumptions are made on the lattice Dirac operator or the gauge action, but both should be local and respect all symmetries preserved by the standard Wilson theory. For notational convenience, the lattice spacing a is set to unity. The lattice is taken to be infinitely extended in all directions unless specified otherwise.

5.1 Flow equations

On the lattice, the time-dependent gauge field $V(t, x, \mu)$ is defined by

$$V|_{t=0} = U, \quad (5.1)$$

$$\partial_t V V^{-1} = -g_0^2 \partial S_w, \quad (5.2)$$

where $U(x, \mu)$ is the fundamental lattice gauge field and $(\partial S_w)(V, x, \mu)$ the gradient of the Wilson plaquette action (see appendix A and ref. [1] for further explanations).

In the case of the quark fields, the boundary conditions and the form of the flow equations are the same as in the continuum theory, but the operator Δ in eq. (2.4) gets replaced by the lattice Laplacian

$$\Delta = \nabla_\mu^* \nabla_\mu, \quad (5.3)$$

∇_μ and ∇_μ^* being the gauge-covariant forward and backward difference operators in presence of the time-dependent gauge field.

As already noted in ref. [1], the global existence, uniqueness and smoothness of the solution of the flow equation (5.2) is rigorously guaranteed. Since Δ is a bounded operator, and since it depends smoothly on the flow time, the same can be shown to be true in the case of quark flow equations [22]. Local fields at non-zero flow time, such as those considered in the previous sections, are thus completely well defined on the lattice.

5.2 Lattice theory in $4+1$ dimensions

Similarly to the continuum theory, the lattice QCD correlation functions of the time-dependent local fields can be obtained from a lattice field theory in $4+1$ dimensions. To be able to fully control this construction, the latter is first set up for a discretized version of the flow equations, where the flow time is restricted to a finite set

$$0, \epsilon, 2\epsilon, 3\epsilon, \dots, T \quad (5.4)$$

of values separated by a time step $\epsilon > 0$.

The integration of the discretized evolution equation

$$V(t + \epsilon, x, \mu) V(t, x, \mu)^{-1} = e^{-\epsilon g_s^2 (\partial S_w)(V, x, \mu)} \quad (5.5)$$

for the gauge field amounts to an approximate integration of the continuous equation (5.2) using the Euler method (or, equivalently, through the application of a sequence of "stout smearing" steps [23]). In the case of the quark fields, the discretized flow equation has the same form as the continuous one, but the time derivative of the quark field is replaced by the forward difference

$$\partial_t \chi(t, x) = \frac{1}{\epsilon} \{ \chi(t + \epsilon, x) - \chi(t, x) \}. \quad (5.6)$$

Note that the discrete equations are such that they can be solved in steps, starting from the fundamental fields at time $t = 0$ and recursively proceeding from time t to $t + \epsilon$. In particular, the solutions are uniquely determined by the initial values of the fields.

In the lattice theory in 4+1 dimensions, all fields integrated over in the functional integral are defined at the times (5.4) only. Since gauge fixing is not needed in the present context, there are no ghost fields and no gauge terms in the action, but the boundary conditions (2.6) and (5.1) are imposed as before. The components of the Lagrange-multiplier fields at time T decouple from the other fields and can therefore be set to zero (or be omitted from the beginning).

A possible choice of the bulk actions is then

$$S_{G,\beta} = -2 \sum_{t=0}^{T-\epsilon} \sum_{x,\mu} \text{tr} \left\{ L(t, x, \mu) \right. \\ \left. \times \mathcal{P} \left(V(t + \epsilon, x, \mu) V(t, x, \mu)^{-1} - e^{-\epsilon g_0^2 (\partial S_\mu)(V, x, \mu)} \right) \right\}, \quad (5.7)$$

$$S_{F,\beta} = \epsilon \sum_{t=0}^{T-\epsilon} \sum_x \left\{ \overline{\lambda}(t, x) (\partial_t - \Delta) \chi(t, x) + \overline{\lambda}(t, x) (\tilde{\partial}_t - \tilde{\Delta}) \lambda(t, x) \right\}, \quad (5.8)$$

where the function

$$M \rightarrow \mathcal{P}(M) = \frac{1}{2}(M - M^\dagger) - \frac{1}{2N} \text{tr}(M - M^\dagger) \quad (5.9)$$

maps any complex $N \times N$ matrix to an element of the Lie algebra of $\text{SU}(N)$. Apart from these actions and the lattice QCD action, the measure term

$$S_{ms} = \sum_{t=0}^{T-\epsilon} \sum_{x,\mu} \mathcal{L}_{ms}(V(t + \epsilon, x, \mu) V(t, x, \mu)^{-1}), \quad (5.10)$$

$$\mathcal{L}_{ms}(W) = \begin{cases} -\ln \det(J(W)) & \text{if } W \in \mathfrak{U}, \\ \infty & \text{otherwise,} \end{cases} \quad (5.11)$$

must be included in the total action of the lattice theory (see appendix B for the definition of the Jacobian matrix $J(W)^{ab}$ of the mapping (5.9) and the neighbourhood $\mathfrak{U} \subset \text{SU}(N)$ of unity).

Having specified the fields and their action, the functional integral of the lattice theory in 4+1 dimensions is defined in the standard manner. Note that the measure term (5.10),(5.11) effectively restricts the link variables in the functional integral to the domain where

$$V(t + \epsilon, x, \mu)V(t, x, \mu)^{-1} \in \mathfrak{V} \quad \text{for all } t, x, \mu. \quad (5.12)$$

Since the action $S_{G, \mathfrak{A}}$ is purely imaginary, the integral over the Lagrange-multiplier field $L(t, x, \mu)$ is not absolutely convergent, but it is understood that the integration ambiguity this may entail is removed through the addition of an infinitesimal quadratic term in the Lagrange-multiplier field to the action.

5.3 Recovering the theory in four dimensions

The correlation functions defined in this way of local fields composed from the fields V , χ and $\overline{\chi}$ can be evaluated, to some extent, by first integrating over the Lagrange-multiplier fields. Since the total action is linear in the latter, the integration yields a product

$$\begin{aligned} & \prod_{t=0}^{T-\epsilon} \prod_{x, \mu} \delta \left\{ \mathcal{P} \left(V(t + \epsilon, x, \mu)V(t, x, \mu)^{-1} \right) - \mathcal{P} \left(e^{-\epsilon g_0^2 (\partial S_\mu)(V, x, \mu)} \right) \right\} \\ & \times \prod_{t=0}^{T-\epsilon} \prod_x \delta \left\{ \partial_t \chi(t, x) - \Delta \chi(t, x) \right\} \delta \left\{ \overline{\chi}(t, x) \tilde{\partial}_t - \overline{\chi}(t, x) \tilde{\Delta} \right\} \end{aligned} \quad (5.13)$$

of Dirac δ -functions. The arguments of these δ -functions vanish if V , χ and $\overline{\chi}$ satisfy the discrete flow equations, and for sufficiently small step sizes ϵ there is in fact no other way the arguments can be made to vanish[†]. In the case of the gauge field, the important point to note here is that the mapping $\mathcal{P}$ in the δ -functions operates on $SU(N)$ matrices in a small neighbourhood of unity. The properties of the mapping quoted in appendix B then imply that the constraints imposed by the δ -functions are equivalent to imposing the flow equation (5.5) on the gauge field.

The integral over the field variables at all flow times $t > 0$ can now be performed using the δ -functions. Starting with the fields at time $t = T$ and proceeding from there to the smaller times, the δ -functions allow all fields to be eliminated recursively until the only integration variables left are the fundamental fields U , ψ and $\overline{\psi}$. In

[†] Uniqueness is guaranteed if $\exp\{\epsilon g_0^2 (\partial S_\mu)(V, x, \mu)\} \in \mathfrak{V}$ on all links of the lattice and at all flow times. Some rigorous estimates show that this condition is satisfied if $\epsilon \leq (24\sqrt{2}N)^{-1}$.

this process, the integrals over the δ -functions for the gauge field variables give rise to a product of Jacobians of the mapping $\mathcal{P}$, but these factors are canceled by the measure term in the action (see appendix B).

The calculation thus shows that the correlation functions in 4+1 dimensions of local fields that do not involve the Lagrange-multiplier fields are exactly equal to the correlation functions of the same fields defined in 4 dimensions through the discrete flow equations and the QCD functional integral.

5.4 Continuous time limit

At this point, the time-dependent fields in the correlation functions are still the ones obtained by the Euler integration of the flow equations. Since the associated integration errors are of order ϵ , the continuous-time correlation functions are recovered when $\epsilon \rightarrow 0$ (and, trivially, $T \rightarrow \infty$). Moreover, the measure term

$$S_{\text{meas}} = \sum_{t=0}^{T-\epsilon} \sum_{x,\mu} \ln \det J \left(e^{-\epsilon g_0^2 (\partial S_\mu)(V, x, \mu)} \right) = O(\epsilon) \quad (5.14)$$

vanishes in this limit.

In the following, the continuous-time formulation of the lattice theory will normally be considered, but the functional integral in 4+1 dimensions is always assumed to be defined through the limit $\epsilon \rightarrow 0$ of the discrete-time functional integral.

6. On-shell $O(a)$ improvement

The lattice-spacing dependence of correlation functions involving the fields at non-zero flow time is best discussed in the theory in 4+1 dimensions. In this framework, the standard argumentation based on locality, power counting and symmetry can be applied. The relevant Symanzik local effective theory [24,25] then coincides with the continuum theory in 4+1 dimensions, with $O(a)$ and higher-order terms in the lattice spacing a added to the action and the local fields. Moreover, on-shell improvement [26] means that the cancellation of lattice-spacing effects is limited to correlation functions at non-zero distances in 4+1 dimensions.

Although the theoretical discussion is more widely applicable, the $O(a)$ -improved Wilson formulation of lattice QCD [27,28] is assumed from now on. Where possible, the same conventions and notation as in ref. [28] are used.

6.1 Effective action

Since the flow equations in the lattice theory are classically $O(a)$ improved, and since loop diagrams do not contribute to the correlation functions at asymptotically large flow times, all $O(a)$ terms in the Symanzik effective action must be boundary terms, i.e. local terms at flow time zero. Furthermore, if the effective theory is to describe on-shell quantities only, as is the case here, many terms can be eliminated using the quark field equations, the flow equations and the boundary conditions (2.1),(2.6). In the chiral limit, a possible choice of the $O(a)$ term in the effective action is then

$$a \int d^4x \left\{ c_1 \bar{\psi}(x) \frac{1}{4} \sigma_{\mu\nu} F_{\mu\nu}(x) \psi(x) + c_2 \bar{\lambda}(0, x) \lambda(0, x) \right\}, \quad (6.1)$$

where $F_{\mu\nu}(x)$ is the field tensor of the fundamental gauge potential (here and below, $c_1, c_2, \dots$ denote coefficients of the effective theory).

Quite many more terms are required to describe the $O(a)$ lattice-spacing effects at non-zero quark masses. In the theory in 4 dimensions, these were completely classified by Bhattacharya et al. [29]. The Symanzik effective action in 4+1 dimensions in addition includes a term

$$a \int d^4x \left\{ \bar{\lambda}(0, x) (c_3 M + c_4 \text{tr} M) \psi(x) + \bar{\psi}(x) (c_3 M + c_4 \text{tr} M) \lambda(0, x) \right\} \quad (6.2)$$

that contributes to correlation functions involving the bulk fields, M being the quark mass matrix in the effective theory. Its effect on the correlation functions is equivalent to a renormalization

$$\chi(t, x) \rightarrow Z \chi(t, x), \quad \lambda(t, x) \rightarrow Z^{-1} \lambda(t, x), \quad (6.3)$$

$$Z = 1 + a c_3 M + a c_4 \text{tr} M, \quad (6.4)$$

of the bulk fermion (and anti-fermion) fields.

6.2 Effective local fields

The form of the effective fields representing the local lattice fields in the Symanzik theory is constrained by the same general principles that determine the form of the effective action. In particular, the effective fields representing fields at positive flow time have no $O(a)$ terms.

Such terms however occur in the case of the local fields at flow time zero. Considering again the theory with only massless quarks, the effective fields representing

the axial currents with flavour indices $r \neq s$, for example, are given by

$$\begin{aligned} (A_{\text{eff}})^{rs}(x) &= \bar{\psi}_r(x) \gamma_\mu \gamma_5 \psi_s(x) + ac_5 \partial_\mu \{ \bar{\psi}_r(x) \gamma_5 \psi_s(x) \} \\ &\quad + ac_6 \{ \bar{\lambda}_r(0, x) \gamma_\mu \gamma_5 \psi_s(x) + \bar{\psi}_r(x) \gamma_\mu \gamma_5 \lambda_s(0, x) \} + \dots, \end{aligned} \quad (6.5)$$

where the ellipsis stands for terms of order a^2 . The associated effective pseudo-scalar densities,

$$\begin{aligned} (P_{\text{eff}})^{rs}(x) &= \bar{\psi}_r(x) \gamma_5 \psi_s(x) \\ &\quad + ac_7 \{ \bar{\lambda}_r(0, x) \gamma_5 \psi_s(x) + \bar{\psi}_r(x) \gamma_5 \lambda_s(0, x) \} + \dots, \end{aligned} \quad (6.6)$$

have a similar expansion.

The mass-dependent $O(a)$ corrections to these fields coincide with the ones in the theory in 4 dimensions [29]. In general, the determination of the $O(a)$ terms however requires a careful consideration of a possible mixing of fields and may lead to fairly complicated expressions.

6.3 $O(a)$ improved lattice action

Following common practice, the mass-dependent $O(a)$ counterterms are omitted in the improved action and are instead included in the parameter and field renormalization factors [28]. The $O(a)$ counterterms in 4+1 dimensions are then given by

$$\delta S_{\text{tot}} = \sum_x \left\{ c_{\text{sw}} \bar{\psi}(x) \frac{1}{4} \sigma_{\mu\nu} \hat{F}_{\mu\nu}(x) \psi(x) + c_{\text{B}} \bar{\lambda}(0, x) \lambda(0, x) \right\}, \quad (6.7)$$

where $\hat{F}_{\mu\nu}(x)$ denotes the standard clover expression for the gauge field tensor $F_{\mu\nu}(x)$ on the lattice. While the first term in eq. (6.7) (the Sheikholeslami-Wohlert term [27, 28]) is already required for the improvement of the theory in 4 dimensions, the one proportional to the improvement coefficient $c_{\text{B}}(g_0)$ is needed to cancel the $O(a)$ lattice effects in correlation functions involving the bulk fields.

Having specified the action, the Wick contractions of the basic fermion fields in the $O(a)$ improved theory can be worked out straightforwardly (see appendix C). In the continuous time limit, the contractions are practically the same as in the continuum theory with the obvious modifications (integrals over the space-time coordinates are replaced by sums over lattice points, the quark propagator $S(x, y)$ by the inverse of the massive $O(a)$ improved lattice Dirac operator and the kernel $K(t, x; s, y)$ by the

fundamental solution of the quark flow equation on the lattice). A special case is the contraction

$$\overbrace{\chi(t, x) \bar{\chi}(s, y)} = \sum_{v, w} K(t, x; 0, v) (S(v, w) - c_B \delta_{v, w}) K(s, y; 0, w)^\dagger \quad (6.8)$$

of the time-dependent quark fields, which is the only one depending on the improvement coefficient c_B . In particular, the contraction and thus the correlation functions of the fields ψ and $\bar{\psi}$ are independent of c_B , as has to be the case, since the theory in 4 dimensions is already improved through the inclusion of the Sheikholeslami-Wohlert term.

A perhaps puzzling aspect of eq. (6.8) is the fact that the limit

$$\lim_{t \rightarrow 0} \overbrace{\chi(t, x) \bar{\chi}(s, y)} = \overbrace{\psi(x) \bar{\chi}(s, y)} - c_B K(s, y; 0, x)^\dagger \quad (6.9)$$

differs from $\overbrace{\psi(x) \bar{\chi}(s, y)}$ by a term of order a , which is not just a contact term. The subtractions required for $O(a)$ improvement thus depend on whether the quark fields in the correlation functions are at zero and non-zero flow time. This is actually not too surprising since these fields also renormalize differently. Both differences merely reflect the fact that the continuum limit of correlation functions involving the time-dependent fields must be taken at fixed flow times given in physical units and that the Symanzik effective theory in 4+1 dimensions correctly describes the deviation of the lattice theory from its continuum limit only when the latter is approached in this way.

6.4 Renormalized improved fields

As usual, the additively renormalized bare quark masses $m_{q,r}$ are defined by

$$m_{q,r} = m_{0,r} - m_c, \quad (6.10)$$

where m_c denotes the critical bare mass in the theory with mass-degenerate quarks. It is also helpful to introduce the associated subtracted bare mass matrix M_q and the combinations

$$m_{q,rs} = \frac{1}{2} (m_{q,r} + m_{q,s}) \quad (6.11)$$

of quark masses.

The lattice fermion fields at positive flow time require multiplicative renormalization and $O(a)$ improvement by a mass-dependent factor (cf. subsect. 6.1). Quark and antiquark fields are scaled with the same factor,

$$\chi_{n,r} = \{Z_\chi(1 + b_\chi m_{q,r} + \bar{b}_\chi \text{tr} M_q)\}^{1/2} \chi_r, \quad (6.12)$$

while the Lagrange-multiplier fields $\lambda, \bar{\lambda}$ are renormalized with the inverse of the same factor[†]. Time-dependent composite fields like the pseudo-scalar density

$$P_{n,t}^{rs} = Z_\chi(1 + b_\chi m_{q,r} + \bar{b}_\chi \text{tr} M_q) P_t^{rs} \quad (6.13)$$

then renormalize according to their quark content.

Starting from the bare fields

$$A_\mu^{rs}(x) = \bar{\psi}_r(x) \gamma_\mu \gamma_5 \psi_s(x), \quad (6.14)$$

$$\hat{A}_\mu^{rs}(x) = \bar{\lambda}_r(0, x) \gamma_\mu \gamma_5 \psi_s(x) + \bar{\psi}_r(x) \gamma_\mu \gamma_5 \lambda_s(0, x), \quad (6.15)$$

$$P^{rs}(x) = \bar{\psi}_r(x) \gamma_5 \psi_s(x), \quad (6.16)$$

$$\hat{P}^{rs}(x) = \bar{\lambda}_r(0, x) \gamma_5 \psi_s(x) + \bar{\psi}_r(x) \gamma_5 \lambda_s(0, x), \quad (6.17)$$

the improved flavour non-singlet axial current and density are given by

$$A_{1,\mu}^{rs} = A_\mu^{rs} + c_A \hat{\partial}_\mu P^{rs} + \tilde{c}_A \hat{A}_\mu^{rs}, \quad (6.18)$$

$$P_1^{rs} = P^{rs} + \tilde{c}_P \hat{P}^{rs}, \quad (6.19)$$

and these are renormalized according to [28,29]

$$A_{n,\mu}^{rs} = Z_A(1 + b_A m_{q,r} + \bar{b}_A \text{tr} M_q) A_{1,\mu}^{rs}, \quad (6.20)$$

$$P_n^{rs} = Z_P(1 + b_P m_{q,r} + \bar{b}_P \text{tr} M_q) P_1^{rs}. \quad (6.21)$$

The new improvement coefficients, $\tilde{c}_A$ and $\tilde{c}_P$, are required for the $O(a)$ improvement of correlation functions involving the fermion fields at non-zero flow time, but all

[†] Following refs. [28,29], b_χ and $\bar{b}_\chi$ generically denote improvement coefficients multiplying mass-dependent $O(a)$ counterterms.

other coefficients already occur in the theory in four dimensions [27–29]. At tree-level of perturbation theory,

$$c_B = \frac{1}{2}, \quad \tilde{c}_A = \tilde{c}_P = -\frac{1}{2}, \quad \tilde{b}_\chi = 1. \quad (6.22)$$

All coefficients $\tilde{b}_\chi$ are, incidentally, of order g_0^4 [29].

6.5 Improvement and renormalization of the field $\tilde{P}^{rs}$

In the chiral symmetry relations derived in sect. 4, the field $\tilde{P}^{rs}$ plays a prominent rôle. The improvement and renormalization is a bit complicated in this case and is therefore discussed separately from the other fields.

As already noted in subsect. 4.2, the field can only mix with local composite fields that include at least one Lagrange-multiplier field. Charge conjugation, the lattice symmetries and the flavour symmetry then imply that the field is multiplicatively renormalizable. There are, however, several fields that can mix with the density at order a , among them two fields

$$\tilde{P}^{rs}(x) = \overline{\lambda}_r(0, x) \gamma_5 \lambda_s(0, x), \quad (6.23)$$

$$Q^{rs}(x) = \overline{\lambda}_r(0, x) \gamma_5 \psi_s(x) - \overline{\psi}_r(x) \gamma_5 \lambda_s(0, x), \quad (6.24)$$

that have not appeared before. Inspection then shows that a possible choice of the improved and renormalized densities is

$$\tilde{P}_i^{rs} = \tilde{P}^{rs} + \tilde{c}_A \hat{\partial}_\mu \tilde{A}_\mu^{rs} + \tilde{c}_P \tilde{P}^{rs}, \quad (6.25)$$

$$\tilde{P}_B^{rs} = \tilde{Z}_P \{ (1 + b_P m_{q,rs} + \tilde{b}_P \text{tr} M_q) \tilde{P}_i^{rs} + \tilde{b}_P (m_{q,r} - m_{q,s}) Q^{rs} \}, \quad (6.26)$$

where, as usual, the field equations were used to reduce the number of terms. The fields on the right of these equations are distinguished by their symmetry properties or content in Lagrange-multiplier fields. They are all multiplicatively renormalizable and the contributions of the counterterms proportional to $\tilde{A}_\mu^{rs}$, $\tilde{P}^{rs}$ and Q^{rs} in on-shell correlation functions are therefore of order a .

The expression (6.25), (6.26) for the renormalized improved density can be simplified by noting that the relations

$$\sum_x \langle \tilde{P}^{rs}(x) P_i^{sr}(y) \rangle = \langle S_i^{rr}(y) \rangle + \langle S_i^{ss}(y) \rangle + 2c_B \sum_x \langle \tilde{P}^{rs}(x) P_i^{sr}(y) \rangle, \quad (6.27)$$

$$\sum_x \langle Q^{rs}(x) P_t^{sr}(y) \rangle = \langle S_t^{rs}(y) \rangle - \langle S_t^{sr}(y) \rangle, \quad (6.28)$$

hold exactly, for any $t > 0$, as a consequence of the form of the Wick contractions of the fermion fields. On the other hand, as discussed in subsect. 4.2, the renormalization constants can be (and are to be) chosen so that the normalization condition (4.15) is satisfied in the continuum limit. Since the correlation functions in eq. (4.15) converge to the continuum limit with a rate proportional to a^2 , the comparison with the unrenormalized identities, eqs. (6.27),(6.28), then shows that

$$\hat{Z}_P = 1, \quad (6.29)$$

$$\hat{c}_P = -2c_B, \quad b_{\hat{P}} = \bar{b}_{\hat{P}} = 0, \quad \hat{b}_{\hat{P}} = -\frac{1}{2}b_\chi. \quad (6.30)$$

The renormalized density

$$\hat{P}_N^{rs} = \hat{P}^{rs} + \hat{c}_A \hat{\partial}_\mu \hat{A}_\mu^{rs} - 2c_B \hat{P}^{rs} - b_\chi \frac{1}{2}(m_{q,r} - m_{q,s}) Q^{rs} \quad (6.31)$$

thus assumes a fairly simple form, in which $\hat{c}_A$ is the only new improvement coefficient.

6.6 PCAC relation on the lattice

In the continuum limit, the renormalized PCAC relation (4.14) holds provided the renormalization constants Z_A, Z_P and the renormalized quark masses $m_{N,r}$ are chosen appropriately. If also all improvement coefficients are properly tuned, this implies

$$\begin{aligned} & \langle \partial A_N^{rs}(x) P_{N,t}^{sr}(y) \rangle \\ &= (m_{N,r} + m_{N,s}) \langle \hat{P}_N^{rs}(x) P_{N,t}^{sr}(y) \rangle - \langle \hat{P}_N^{rs}(x) P_{N,t}^{sr}(y) \rangle + O(a^2), \end{aligned} \quad (6.32)$$

where, following the tradition, the divergence of the improved axial current is taken to be

$$\partial A_N^{rs} = Z_A (1 + b_A m_{q,rs} + \bar{b}_A \text{tr} M_q) \partial A_1^{rs}, \quad (6.33)$$

$$\partial A_1^{rs} = \hat{\partial}_\mu \{ A_\mu^{rs} + \hat{c}_A \hat{A}_\mu^{rs} \} + c_A \partial_\mu^* \partial_\mu P^{rs}. \quad (6.34)$$

A technical detail worth emphasizing here again is the fact that the PCAC relation (6.32) holds at all x , including $x = y$, as long as the flow time t is set to a positive value given in physical units.

7. Calculation of correlation functions

In numerical lattice QCD, correlation functions involving fields at non-zero flow time can be computed following the steps usually taken in the case of ordinary hadronic correlation functions. The fact that the flow equations must be solved at some point nevertheless represents a complication that needs to be carefully considered. For illustration, the details are worked out in this section for two representative cases.

7.1 Pseudo-scalar two-point function

In practice one is interested in the correlation function

$$\langle P^{rs}(x)P^{sr}(y) \rangle = - \sum_{v,w} (\text{tr} \{ [K(t, y; 0, v)S(v, x)_{ss}]^\dagger K(t, y; 0, w)S(w, x)_{rr} \}) \quad (7.1)$$

at vanishing spatial momentum and computes its average over a lattice Γ of source points, using random source fields [30], in order to reduce the statistical errors.

If x is taken to be the source point, the averaging amounts to the substitution

$$P^{rs}(x) \rightarrow \frac{1}{N_\Gamma N_s} \sum_{k=1}^{N_s} (\bar{\psi}_r, \eta_k)(\eta_k, \gamma_5 \psi_s), \quad (7.2)$$

where N_Γ is the number of points in Γ and

$$\eta_k(x), \quad k = 1, \dots, N_s, \quad (7.3)$$

are randomly chosen complex spinor fields on Γ with mean zero and variance

$$\langle \eta_k(v) \eta(w)^\dagger \rangle = \begin{cases} \delta_{kv} \delta_{vw} & \text{if } v, w \in \Gamma, \\ 0 & \text{otherwise.} \end{cases} \quad (7.4)$$

In a QCD simulation, the computation of the averaged correlation function at zero spatial momentum,

$$\frac{1}{N_\Gamma} \sum_{x \in \Gamma} \sum_y \langle P^{rs}(x)P^{sr}(y) \rangle = - \frac{1}{N_\Gamma N_s} \sum_{k=1}^{N_s} \sum_y \langle \phi_{k,s}(t, y)^\dagger \phi_{k,r}(t, y) \rangle, \quad (7.5)$$

then amounts to calculating the functions

$$\phi_{k,h}(t, y) = \sum_{w,x} K(t, y; 0, w)S(w, x)_{hh} \eta_k(x) \quad (7.6)$$

for a representative ensemble of gauge fields, all $k = 1, \dots, N_s$ and $h = r, s$.

For a given gauge field, the computation of $\phi_{k,h}(t, y)$ proceeds in two steps, where one first calculates $\phi_{k,h}(0, w)$ by solving the lattice Dirac equation with mass $m_{0,h}$ and source field $\eta_k(x)$. The calculated field must then be evolved in flow time from time 0 to time t . In the direction of increasing flow time, the flow equations are numerically stable and the solution can easily be obtained, with negligible integration errors, using a Runge-Kutta integrator (appendix D).

7.2 Chiral condensate

In the case of the time-dependent quark condensate,

$$\langle S_t^{rr}(x) \rangle = - \sum_{v,w} \langle \text{tr} \{ K(t, x; 0, v) [S(v, w)_{rr} - c_{\text{H}} \delta_{vw}] K(t, x; 0, w)^\dagger \} \rangle, \quad (7.7)$$

an averaging over the position x is again desirable. Introducing random source fields as above, one is then led to the representation

$$\frac{1}{N_\Gamma} \sum_{x \in \Gamma} \langle S_t^{rr}(x) \rangle = - \frac{1}{N_\Gamma} \sum_{v,w} \langle \xi_k(t; 0, v)^\dagger [S(v, w)_{rr} - c_{\text{H}} \delta_{vw}] \xi_k(t; 0, w) \rangle \quad (7.8)$$

in terms of the fields

$$\xi_k(t; s, w) = \sum_x K(t; x; s, w)^\dagger \eta_k(x) \quad (7.9)$$

at flow time $s = 0$.

The computation of these fields requires the *adjoint flow equation*

$$(\partial_s + \Delta) \xi_k(t; s, w) = 0 \quad (7.10)$$

to be solved from time $s = t$ (where ξ_k coincides with η_k) to $s = 0$. A straightforward Runge-Kutta integration should not be used here, because the Laplacian Δ is the one in presence of the gauge field determined by the gradient flow at time s , which would therefore have to be evolved backward in time, i.e. in the unstable direction. As explained in appendix E, this difficulty can be overcome through a hierarchical scheme that avoids the backward integration of the gauge field.

8. Strategy for the computation of Z_A and c_{B}

The PCAC relation (6.32) holds provided the relevant renormalization constants and improvement coefficients are chosen appropriately. As explained below, the constant Z_A and the coefficient c_{B} can conversely be determined, up to terms of order a^2 and a , respectively, by requiring the relation to be satisfied exactly at two points in the space of kinematical parameters.

The computation through numerical simulation of Z_A and c_{B} along these lines assumes that the improvement coefficients c_{sw} and c_A have already been calculated. Periodic boundary conditions should be imposed in the space directions, but apart from this no further assumptions are made. In particular, the boundary conditions in time do not need to be specified, since the computed values of Z_A and c_{B} depend on them only at the level of the lattice-spacing effects.

8.1 Integrated form of the PCAC relation

The $\mathcal{O}(a)$ improvement unfortunately leads to an inflation of terms when the renormalized PCAC relation is expanded in the correlation functions of the unimproved bare fields. A slight simplification can however be achieved by summing the equation over an interval of time x_0 around y_0 and over all points $\vec{x}$ on the spatial lattice. One of the bare correlation functions entering the relation is then

$$C_P^{rs}(t, d) = \sum_{x=y_0-d}^{y_0+d} \sum_{\vec{x}} \langle P^{rs}(x) P_t^{sr}(y) \rangle, \quad (8.1)$$

and there are 6 further such correlation functions in which P^{rs} is replaced by

$$\tilde{P}^{rs}, \tilde{P}^{rs}, Q^{rs}, \hat{\partial}_\mu A_\mu^{rs}, \hat{\partial}_\mu \hat{A}_\mu^{rs} \text{ or } \partial_\mu^* \partial_\mu P^{rs}. \quad (8.2)$$

In the case of the last three fields in this list, the sum over x_0 reduces to a sum of terms at the boundary of the summation range as in

$$C_{\partial A}^{rs}(t, d) = \frac{1}{2} \sum_{\vec{x}} \left\{ \langle (A_0^{rs}(x)|_{x_0=y_0+d+1} + A_0^{rs}(x)|_{x_0=y_0+d}) P_t^{sr}(y) \rangle \right. \\ \left. - \langle (A_0^{rs}(x)|_{x_0=y_0-d-1} + A_0^{rs}(x)|_{x_0=y_0-d}) P_t^{sr}(y) \rangle \right\}, \quad (8.3)$$

for example.

For notational convenience, it is now helpful to introduce the abbreviations

$$\hat{Z}_A^{rs} = Z_A (1 + b_A m_{q,rs} + \bar{b}_A \text{tr} M_q), \quad (8.4)$$

$$\hat{Z}_P^{rs} = Z_P (1 + b_P m_{q,rs} + \bar{b}_P \text{tr} M_q). \quad (8.5)$$

Defining the bare current-quark mass sums through

$$m_{rs} = \frac{\hat{Z}_P^{rs}}{\hat{Z}_A^{rs}} (m_{\text{R},r} + m_{\text{R},s}), \quad (8.6)$$

the integrated PCAC relation then becomes

$$\begin{aligned} & \hat{Z}_A^{rs} \{ C_{\partial\bar{A}}^{rs} + \bar{c}_A C_{\partial\bar{A}}^{rs} + c_A C_{\partial\partial P}^{rs} - m_{rs} (C_P^{rs} + \bar{c}_P C_P^{rs}) \} \\ & + C_{\bar{P}}^{rs} + \bar{c}_A C_{\partial\bar{A}}^{rs} - 2c_{\text{H}} C_{\bar{P}}^{rs} - b_{\text{A}} \frac{1}{2} (m_{\text{q},r} - m_{\text{q},s}) C_Q^{rs} = \mathcal{O}(a^2). \end{aligned} \quad (8.7)$$

This equation still looks fairly complicated, but one should keep in mind that it is a linear relation among computable correlation functions, which must hold for all flow times $t > 0$, time ranges $d \geq 0$ and quark flavours $r \neq s$. The unknown coefficients are thus strongly constrained by the identity.

At values of d larger than the smoothing range $\sqrt{8t}$ of the gradient flow, $C_{\partial\bar{A}}^{rs}$ falls off like a Gaussian and rapidly becomes negligible with respect to the other terms in eq. (8.7). Moreover, the sum of the terms in the curly bracket as well as all other terms in the equation are then practically independent of d . Without further notice, only this range of d is considered in the following and the two terms proportional to $C_{\partial\bar{A}}^{rs}$ are dropped.

8.2 Up-down flavour channel

In QCD with mass-degenerate up and down quarks, the PCAC relation in the up-down channel assumes the form

$$\hat{Z}_A^{ud} \{ C_{\partial\bar{A}}^{ud} + c_A C_{\partial\partial P}^{ud} - m_{ud} C_P^{ud} \} - 2c_{\text{H}} C_{\bar{P}}^{ud} = -(1 - \hat{Z}_A^{ud} \bar{c}_P m_{ud}) C_{\bar{P}}^{ud} \quad (8.8)$$

(for simplicity, the $\mathcal{O}(a^2)$ error term is dropped from now on). The bare mass combination m_{ud} can be determined as usual through the PCAC relation at vanishing flow time and may be assumed to be known at this point. By computing the correlation functions in eq. (8.8) at two values of the flow time t , one then obtains two linear

equations for the ratios

$$\frac{\hat{Z}_A^{ud}}{1 - \hat{Z}_A^{ud} \tilde{c}_P m_{ud}} \quad \text{and} \quad \frac{c_{\mathbb{B}}}{1 - \hat{Z}_A^{ud} \tilde{c}_P m_{ud}}. \quad (8.9)$$

Clearly, as long as the chosen flow times are in the scaling range and are held fixed in physical units, the calculated ratios do not depend on them, up to terms of order a^2 and a , respectively.

At quark masses close to their physical values, and lattice spacings $a \leq 0.1$ fm, the factor $1 - \hat{Z}_A^{ud} \tilde{c}_P m_{ud}$ differs from unity by a fraction of a percent. An estimate of the size of this correction term can be obtained by inserting the known value of m_{ud} and the tree-level value (6.22) of $\tilde{c}_P$. $\hat{Z}_A^{ud}$ and $c_{\mathbb{B}}$ can thus be computed up to this small correction factor. Note that the bare pion decay constant determined through the vacuum-to-pion matrix element of the improved axial current is to be renormalized with the mass-dependent factor $\hat{Z}_A^{ud}$ rather than Z_A . The difference is however small and can be estimated using the tree-level value $b_A = 1$.

8.3 Up-strange flavour channel

The additional term that appears in the PCAC relation

$$\begin{aligned} & \hat{Z}_A^{us} \{ C_{\partial A}^{us} + c_A C_{\partial\partial P}^{us} - m_{us} C_P^{us} \} - 2c_{\mathbb{B}} C_P^{us} - b_A \frac{1}{2} (m_{q,u} - m_{q,s}) C_Q^{us} \\ &= - (1 - \hat{Z}_A^{us} \tilde{c}_P m_{us}) C_P^{us} \end{aligned} \quad (8.10)$$

in this channel is proportional to the square of the mass difference $m_{q,u} - m_{q,s}$ and is therefore likely to be negligible in practice. To be able to determine the ratios (8.9) (with ud replaced by us), eq. (8.10) must otherwise be evaluated at three values of the flow time t .

For $a \leq 0.1$ fm and physical quark masses, the correction factor $1 - \hat{Z}_A^{us} \tilde{c}_P m_{us}$ may deviate from unity by up to one percent or so. The comparison with the results obtained in the ud channel,

$$\frac{\hat{Z}_A^{us} (1 - \hat{Z}_A^{ud} \tilde{c}_P m_{ud})}{\hat{Z}_A^{ud} (1 - \hat{Z}_A^{us} \tilde{c}_P m_{us})} = 1 + b_A \frac{1}{2} (m_{q,s} - m_{q,d}) + Z_A \tilde{c}_P (m_{us} - m_{ud}), \quad (8.11)$$

unfortunately only allows a certain linear combination of the coefficients b_A and $\tilde{c}_P$ to be estimated.

9. Sample calculation in 2+1 flavour QCD

The aim in the following paragraphs is to demonstrate the feasibility of the proposed strategies for the computation of the renormalization constant Z_A , the improvement coefficient c_R and the low-energy constants Σ and F . A single simulation run suffices for this test and no attempt is made to actually compute the low-energy constants, as this would require the data to be extrapolated to the infinite-volume, chiral and continuum limit.

9.1 Simulation details

The calculations reported below are based on a recent simulation of QCD with 2+1 flavours of quarks [31]. In this run (labeled I_1 in ref. [31]), a 64×32^3 lattice with open boundary conditions in time [32] was simulated at a point in parameter space, where the lattice spacing is estimated to be 0.090 fm [33,34] and where the pion and kaon masses are about 203 and 520 MeV, respectively [31]. The physical size of the lattice is thus such that the finite-volume effects on the calculated meson masses and matrix elements are expected to be small.

For the Iwasaki gauge action [35] and the standard $O(a)$ -improved Wilson–Dirac operator [27,28] used in the simulation, the coefficients c_{sw} and c_A are known non-perturbatively [36,37]. A representative ensemble of 150 gauge-field configurations, separated by 9.6 units of molecular-dynamics time, was produced in the run. In order to suppress any residual autocorrelation effects, the statistical errors quoted in this section were estimated by dividing the measurement data series for the primary observables into bins of 5 consecutive measurements and by applying the jackknife method to the binned data.

9.2 Computation of $\hat{Z}_A^*$ and c_R

The values 2.47, 3.86, 5.56 and 7.56 of the flow time t in lattice units, where the correlation functions entering the PCAC relation (8.7) were computed, correspond to smoothing ranges of the gradient flow equal to 0.4, 0.5, 0.6 and 0.7 fm, respectively. In order to keep away from the boundaries of the lattice in the time direction, the time-dependent field $P_t^*(y)$ was inserted in the middle of the lattice. The correlation functions were then averaged over space translations using 12 random source fields at time y_0 .

All correlation functions in the PCAC relation can easily be obtained with small statistical errors over the whole range of the summation window d . As expected, the sum of correlation functions multiplying the axial-current renormalization constant

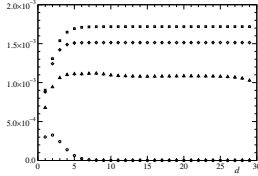


Fig. 1. Combinations of correlation functions appearing in the PCAC relation (8.8), plotted as a function of the summation range d . The linear combinations of correlation functions shown are the ones multiplying the renormalization constant $\hat{Z}_A^{ud}$ (triangles), the improvement coefficient c_R (diamonds) and the factor $1 - \hat{Z}_A^{ud} \tilde{c}_P m_{ud}$ (squares) at flow time $t = 3.86$. For $d \geq 9$, the function C_{alt}^{ud} (circles) is many orders of magnitude smaller than the other functions and can be safely neglected (cf. subsect. 8.1). All quantities are given in lattice units.

and the other correlation functions reach a plateau at values of d larger than 2 or 3 times the smoothing range of the gradient flow (see fig. 1 for a representative case; the slight bending down of the function multiplying $\hat{Z}_A^{ud}$ close to the boundary of the lattice is a lattice-spacing effect).

Following the strategy outlined in sect. 8, the ratios (8.9) are now determined by requiring the PCAC relation (8.8) to hold exactly at two values t_1, t_2 of t . The choice of the summation window d has very little influence on the calculated ratios as long as one stays within the plateau range of the correlation functions. Setting $d = 18$ then leads to the results quoted in the second and third column of table 1, where the factor $1 - \hat{Z}_A^{ud} \tilde{c}_P m_{ud}$ was estimated using the tree-level value (6.22) of the coefficient $\tilde{c}_P$. As anticipated, this correction is very small (about 0.2%).

In the up-strange flavour channel, there are two sources of $O(a)$ mass corrections. Proceeding as above, the correction coming from the factor $1 - \hat{Z}_A^{us} \tilde{c}_P m_{us}$ is estimated to be approximately 1.3%, while the one deriving from the term in eq. (8.10) proportional to b_χ changes the results by about 0.2%. As can be seen from table 1,

Table 1. Simulation results for $\hat{Z}_A^{rs}$ and c_{Π}

t_1, t_2	$\hat{Z}_A^{sd}$	c_{Π}	$\hat{Z}_A^{us}$	c_{Π}
2.47, 3.86	0.7929(87)	0.5703(11)	0.8096(41)	0.56867(60)
2.47, 5.56	0.7904(85)	0.5712(14)	0.8073(40)	0.56990(72)
2.47, 7.56	0.7895(85)	0.5716(16)	0.8067(39)	0.57025(83)
3.86, 5.56	0.7885(85)	0.5734(22)	0.8054(40)	0.5726(10)
3.86, 7.56	0.7883(85)	0.5735(26)	0.8055(39)	0.5724(12)
5.56, 7.56	0.7882(86)	0.5738(36)	0.8056(38)	0.5723(16)

The values of c_{Π} in the 3rd and 5th column were obtained in the ud and us channel, respectively

there is a remarkable consistency among the values of c_{Π} obtained in the two flavour channels. There is also little dependence on the flow times and the difference of the renormalization constants $\hat{Z}_A^{ud}$ and $\hat{Z}_A^{us}$ might very well be explained by the mass-dependent factor in eq. (8.4). The $O(a)$ improvement thus works out very well, with residual lattice-spacing effects at the level of a fraction of a percent in the case of the renormalization constant.

The results for the renormalization constant $\hat{Z}_A^{sd}$ quoted in table 1 are in agreement with the value $Z_A = 0.781(20)$ previously obtained in ref. [38] using a method based on the Schrödinger functional [39, 40]. Note that the improvement coefficient c_{Π} turns out to be close to its tree-level value (6.22), although there is no very good reason for this to be so at the gauge coupling considered.

9.3 Computation of the chiral condensate

Having determined the improvement coefficient c_{Π} , the unrenormalized time-dependent condensate Σ_t^{uu} can be computed straightforwardly as described in subsect. 7.2. The source time x_0 is taken to be in the middle of the lattice in order to minimize the excited-states contributions to the expectation value of the scalar density. Using 12 random source fields, and evaluating c_{Π} at $t_1, t_2 = 3.86, 5.56$, the calculation yields the results quoted in the second column of table 2. In the range of flow time considered, the time-dependent condensate is thus seen to decrease, roughly like $1/t$, when t increases. The statistical error follows this evolution and is, in any case, encouragingly small.

To be able to relate the time-dependent condensate to the quark condensate Σ in

Table 2. Simulation results for Σ_t^{uu} and G_t^{ud} in lattice units

t	Σ_t^{uu}	G_t^{ud}	$\Sigma_t^{uu} G_t^{ud} / G_t^{ud}$	Σ_t^{uu} / G_t^{ud}
2.47	0.0006277(95)	0.01484(23)	0.003962(61)	0.04230(85)
3.86	0.0004251(58)	0.01028(16)	0.003872(55)	0.04134(81)
5.56	0.0002911(36)	0.00720(11)	0.003785(51)	0.04040(78)
7.56	0.0002017(23)	0.005092(80)	0.003711(48)	0.03962(76)

the chiral limit, the vacuum-to-pion matrix elements

$$G_\pi = Z_P(1 + b_P m_{q,ud} + \bar{b}_P \text{tr} M_q) G^{ud}, \quad (9.1)$$

$$G_{\pi,t} = Z_\chi(1 + b_\chi m_{q,ud} + \bar{b}_\chi \text{tr} M_q) G_t^{ud}, \quad (9.2)$$

need to be computed as well. Actually, since only the ratio

$$\frac{\Sigma_{\pi,t}}{G_{\pi,t}} = \frac{\Sigma_t^{uu}}{G_t^{ud}} \quad (9.3)$$

appears in eqs. (4.24),(4.25), it suffices to calculate G_π and the unrenormalized matrix element G_t^{ud} . The latter can be determined from the pseudo-scalar correlation function (7.1) at large time separations $|x_0 - y_0|$ in the same way as the bare matrix element G^{ud} at vanishing flow time is usually extracted from the ordinary pseudo-scalar two-point function. Setting x_0 to a value next to the boundaries of the lattice, there is in all cases a wide range in time y_0 , where the one-pion intermediate states dominate and the desired matrix elements can be easily determined.

Similarly to the time-dependent condensate, the matrix element G_t^{ud} is a monotonically decreasing function of t . The ratios listed in the last two columns of table 2 however vary much less with t , consistently with the fact that the flow-time dependence of the ratios must disappear in the chiral limit.

9.4 Remarks

Conversion to physical units. The spacing of the simulated lattice, $a = 0.08995(40)$ fm, was determined by PACS-CS through a computation of the mass of the Ω baryon [34]. Using step scaling and the Schrödinger functional, PACS-CS also calculated the renormalization constant $Z_P = 0.580(21)$ [38] that relates the bare matrix element

G^{ud} to G_π in the $\overline{\text{MS}}$ scheme at 2 GeV † . At $t = 3.86$, for example, the results

$$\frac{\Sigma_{0,t} G_\pi}{G_{\pi,t}} = [287(4) \text{ MeV}]^3, \quad (9.4)$$

$$\frac{\Sigma_{0,t}}{G_{\pi,t}} = 91(2) \text{ MeV}, \quad (9.5)$$

are then obtained, where the $\mathcal{O}(a)$ mass correction in eq. (9.1) was neglected (the error in eq. (9.4) is anyway dominated by the error of Z_P). Clearly, while these ratios may be close to the condensate Σ and decay constant F in the chiral limit, simulations of a range of lattices will be required to be able to determine Σ and F with controlled systematic errors.

Pseudo-scalar decay constants. Since the calculation of the axial-current renormalization constant does not require separate simulations, the renormalized pseudo-scalar meson decay constants become directly accessible on each simulated lattice. Depending on the flavour channel and the precision that is to be attained, the mass-dependent $\mathcal{O}(a)$ corrections in the PCAC relation used to determine the renormalization constant Z_A^P may however need to be estimated non-perturbatively.

Scaling to the continuum limit. The results of the computations reported in this section depend on a choice of flow times. When lattices with different spacings are simulated, these parameters should be held fixed in units of a suitable reference scale such as the reference flow time t_0 [1]. Only then can the calculated renormalized quantities be expected to converge to the continuum limit with a rate proportional to a^2 .

10. Concluding remarks

Through its extension to the quark fields, the gradient flow becomes a powerful tool for studies of the spontaneous breaking of chiral symmetry in QCD and the physics of the light pseudo-scalar mesons. The focus in this paper has been on the theoretical framework and its implementation in one of the widely used formulations of lattice QCD. In view of the renormalizability properties of the flow, the choice of

† A combination of table entries quoted in ref. [38], with errors added in quadrature, was required to obtain the value of Z_P given here.

the lattice regularization is however expected to be irrelevant in the continuum limit as long as the locality and gauge invariance of the lattice theory are guaranteed. The theoretical analysis obviously applies to field theories with other gauge groups and fermion multiplets as well.

In the cases worked out so far, the application of the gradient flow in numerical lattice QCD offers important technical advantages with respect to the established computational strategies. Apart from the ones already mentioned in sect. 1, there are potentially many further uses of the flow. The flavour singlet channel, for example, has not been considered yet and the time-dependent condensate is likely to be an easily accessible chiral order parameter in QCD at non-zero temperatures. Moreover, the application of the flow in QCD-like theories near the conformal window may conceivably lead to interesting qualitative insights.

All calculations reported in sect. 9 were performed on a dedicated PC cluster at CERN. I am grateful to the CERN management for providing the required funds and to the CERN IT Department for technical support.

Appendix A. Notational conventions

A.1 Gauge group

The Lie algebra of $SU(N)$ may be identified with the linear space of all complex, anti-hermitian traceless $N \times N$ matrices. With respect to a basis T^a , $a = 1, \dots, N^2 - 1$, of such matrices, the general element X of the Lie algebra is given by $X = X^a T^a$ with real components X^a (repeated group indices are automatically summed over). The generators are assumed to satisfy

$$\text{tr}\{T^a T^b\} = -\frac{1}{2}\delta^{ab}, \quad (\text{A.1})$$

but are otherwise left unspecified.

Gauge fields are normalized such that gauge transformations and covariant derivatives do not involve the gauge coupling. Lorentz indices $\mu, \nu, \dots$ in D dimensions range from 0 to $D - 1$ and are summed over when they occur in matching pairs. The space-time metric is assumed to be Euclidean. In particular, $p^2 = p_\mu p_\mu$ for any momentum p and $\delta_{\mu\mu} = D$.

A.2 Dirac matrices

The Dirac matrices γ_μ satisfy

$$\gamma_\mu^\dagger = \gamma_\mu, \quad \{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}. \quad (\text{A.2})$$

Scalar products $\gamma_\mu p_\mu$ are abbreviated by $\not{p}$, as usual, and $\sigma_{\mu\nu} = \frac{1}{2}[\gamma_\mu, \gamma_\nu]$. In any dimension D close to 4,

$$\gamma_5 = \gamma_0\gamma_1\gamma_2\gamma_3, \quad \text{tr}\{1\} = 4. \quad (\text{A.3})$$

Note that the axial currents require renormalization if dimensional regularization is used with these conventions for the Dirac matrices.

A.3 Lattice theory

The lattice theories considered in this paper are set up as usual on a four-dimensional hypercubic lattice with spacing $a = 1$. For the quark fields, Wilson's formulation is chosen, where the quark fields reside on the sites of the lattice and carry the same indices as the quark fields in the continuum theory. The gauge covariant forward and backward difference operators acting on a quark field $\psi(x)$ in presence of the gauge field $U(x, \mu)$ are defined by

$$\nabla_\mu \psi(x) = U(x, \mu) \psi(x + \hat{\mu}) - \psi(x), \quad (\text{A.4})$$

$$\nabla_\mu^* \psi(x) = \psi(x) - U(x - \hat{\mu}, \mu)^{-1} \psi(x - \hat{\mu}), \quad (\text{A.5})$$

where $\hat{\mu}$ denotes the unit vector in direction μ . On the lattice, ∂_μ and ∂_μ^* stand for the ordinary forward and backward difference operators and

$$\hat{\partial}_\mu = \frac{1}{2}(\partial_\mu + \partial_\mu^*) \quad (\text{A.6})$$

for their symmetric combination.

The differential operators $\partial_{x,\mu}^*$ act on differentiable functions $f(U)$ of the gauge field U according to

$$\partial_{x,\mu}^* f(U) = \frac{d}{ds} f(e^{sX} U) \Big|_{s=0}, \quad X(y, \nu) = \begin{cases} T^a & \text{if } (y, \nu) = (x, \mu), \\ 0 & \text{otherwise.} \end{cases} \quad (\text{A.7})$$

While these operators depend on the choice of the generators T^a , the gradient field

$$(\partial f)(U, x, \mu) = T^a \partial_{x,\mu}^* f(U) \quad (\text{A.8})$$

can be shown to be basis-independent.

Appendix B. Properties of the mapping (5.9)

The mapping (5.9) is differentiable and satisfies

$$\mathcal{P}(e^{sX}) = sX + \mathcal{O}(s^2) \quad (\text{B.1})$$

for all matrices X in the Lie algebra of $\text{SU}(N)$. These properties imply that there exists an open neighbourhood of unity in $\text{SU}(N)$, where $\mathcal{P}$ is one-to-one and in which the Jacobian matrix

$$J(U)^{ab} = -2\text{tr}\{T^a \mathcal{P}(T^b U)\} \quad (\text{B.2})$$

is nowhere singular.

Using simple norm estimates, the interior of the domain

$$\mathfrak{U} = \{U \in \text{SU}(N) \mid \text{Re tr}\{1 - U\} \leq (16N)^{-1}\} \quad (\text{B.3})$$

can be shown to be such a neighbourhood of unity. Moreover, the Jacobian $\det J(U)$ is strictly positive on $\mathfrak{U}$. As a consequence, the identity

$$\int_{\mathfrak{U}} dU \delta(\mathcal{P}(U) - \mathcal{P}(V)) f(U) = k \det J(V)^{-1} f(V) \quad (\text{B.4})$$

holds for all integrable functions $f(U)$ and all $V \in \mathfrak{U}$, the Dirac δ -function being the one associated to the $\text{SU}(N)$ invariant measure on the Lie algebra. The proportionality constant k in this equation is positive and could be worked out explicitly, but its value is not needed in this paper.

Appendix C. Wick contractions

The Wick contractions of the basic fermion fields in the $\mathcal{O}(a)$ improved lattice theory are first worked out for the case of the discretized flow equations with time step ϵ . Time-ordering ambiguities are avoided in this way and the limit $\epsilon \rightarrow 0$ can then easily be taken at the end of the calculation.

C.1 Fermion action and functional integral

To get started, it may be helpful to recall that the fermion part of the total action of the $O(a)$ improved lattice theory in 4+1 dimensions is the sum of a boundary term,

$$S_F = \sum_x \{ \bar{\psi}(x) D \psi(x) + c_B \bar{\lambda}(0, x) \lambda(0, x) \}, \quad (C.1)$$

and of the bulk action (5.8). The Wilson-Dirac operator D in eq. (C.1) includes the Sheikholeslami-Wohlert term [27,28] and the mass term with bare mass matrix M_0 .

In the bulk action (5.8), it is understood that the time-dependent fields satisfy the boundary conditions (2.6). The unconstrained fermion (and similarly the anti-fermion) fields integrated over in the functional integral can thus be taken to be

$$\psi(x), \quad \chi(\epsilon, x), \dots, \chi(T, x), \quad \lambda(0, x), \dots, \lambda(T - \epsilon, x). \quad (C.2)$$

Since the action is quadratic in these fields, the fermion integral is given through Wick's theorem.

C.2 Field transformation

In the following, an important rôle is played by the kernel $K_\epsilon(t, x; s, y)$ that satisfies

$$K_\epsilon(t, x; t, y) = \delta_{xy} \quad (C.3)$$

at all t in the range $0 \leq t \leq T$ and the discretized quark flow equation

$$(\partial_t - \Delta) K_\epsilon(t, x; s, y) = 0 \quad (C.4)$$

if $0 \leq s \leq t < T$. Clearly, since the flow equation (C.4) can be solved step by step starting from time $t = s$, these equations uniquely determine the kernel at all $t \geq s$.

The calculation of the fermion propagators can now be simplified by performing a field transformation, where $\chi(t, x)$ is expressed through $\psi(x)$ and another field $\phi(t, x)$, defined at $t = \epsilon, \dots, T$, according to

$$\chi(t, x) = \sum_y K_\epsilon(t, x; 0, y) \psi(y) + \epsilon \sum_{s=\epsilon}^t \sum_y K_\epsilon(t, x; s, y) \phi(s, y). \quad (C.5)$$

For any given field $\psi(x)$, the mapping from the field variables $\chi(\epsilon, x), \dots, \chi(T, x)$ to

the new variables $\phi(\epsilon, x), \dots, \phi(T, x)$ is invertible in view of the identity

$$\phi(t + \epsilon, x) = \begin{cases} (\partial_t - \Delta) \chi(t, x) & \text{if } 0 < t < T, \\ \frac{1}{\epsilon}(\chi(\epsilon, x) - \psi(x)) - \Delta \psi(x) & \text{if } t = 0. \end{cases} \quad (\text{C.6})$$

Moreover, the associated Jacobian matrix is lower triangular and its determinant does not depend on the gauge field.

After the field transformation and the corresponding transformation of the anti-fermion fields, the fermion action assumes the form

$$\begin{aligned} S_F + S_{F,\mathbb{A}} = & \sum_x \{ \overline{\psi}(x) D \psi(x) + c_{\mathbb{A}} \overline{\lambda}(0, x) \lambda(0, x) \} \\ & + \epsilon \sum_{t=0}^{T-\epsilon} \sum_x \{ \overline{\lambda}(t, x) \phi(t + \epsilon, x) + \overline{\phi}(t + \epsilon, x) \lambda(t, x) \}. \end{aligned} \quad (\text{C.7})$$

The transformation thus achieves a nearly perfect decoupling of the field variables.

C.3 Contractions

Since the fundamental quark fields $\psi, \overline{\psi}$ are completely decoupled from the other fields, the contraction of these fields

$$\overbrace{\psi(x) \overline{\psi}(y)} = S(x, y), \quad (\text{C.8})$$

$$DS(x, y) = \delta_{xy}, \quad (\text{C.9})$$

coincides with the quark propagator in the theory in 4 dimensions as expected. The other non-vanishing contractions deriving from the action (C.7) are

$$\overbrace{\phi(t + \epsilon, x) \overline{\lambda}(t, y)} = \overbrace{\lambda(t, x) \overline{\phi}(t + \epsilon, y)} = \frac{1}{\epsilon} \delta_{xy}, \quad t = 0, \epsilon, \dots, T - \epsilon, \quad (\text{C.10})$$

$$\overbrace{\phi(\epsilon, x) \overline{\phi}(\epsilon, y)} = -\frac{c_{\mathbb{A}}}{\epsilon^2} \delta_{xy}. \quad (\text{C.11})$$

For the contractions involving the fields χ and $\overline{\chi}$ one then obtains

$$\overbrace{\chi(t, x) \overline{\chi}(s, y)} = \theta(t > s) K_{\epsilon}(t, x; s + \epsilon, y), \quad (\text{C.12})$$

$$\overline{\lambda(t, x) \chi(s, y)} = \theta(t < s) K_\epsilon(s, y; t + \epsilon, x)^\dagger, \quad (\text{C.13})$$

$$\overline{\chi(t, x) \psi(y)} = \sum_v K_\epsilon(t, x; 0, v) S(v, y), \quad (\text{C.14})$$

$$\overline{\psi(x) \chi(s, y)} = \sum_w S(x, w) K_\epsilon(s, y; 0, w)^\dagger, \quad (\text{C.15})$$

$$\begin{aligned} \overline{\chi(t, x) \chi(s, y)} &= \sum_{v, w} K_\epsilon(t, x; 0, v) S(v, w) K_\epsilon(s, y; 0, w)^\dagger \\ &\quad - c_B \sum_v K_\epsilon(t, x; \epsilon, v) K_\epsilon(s, y; \epsilon, v)^\dagger. \end{aligned} \quad (\text{C.16})$$

In all these equations, it is understood that the flow-time arguments of χ and $\bar{\chi}$ are in the range $(0, T]$ and those of λ and $\bar{\lambda}$ in the range $[0, T)$.

C.4 Contractions in the continuous time limit

When the time step ϵ is taken to zero, the kernel $K_\epsilon(t, x; s, y)$ smoothly converges to the fundamental solution $K(t, x; s, y)$ of the quark flow equation in the continuous time formulation of the lattice theory. The complete set of the non-zero contractions of the fermion fields, given by eqs. (C.8) and (C.12)–(C.16), therefore has a well-defined limit.

Appendix D. Integration of the lattice flow equations

The numerical integration of the pure-gauge gradient flow has already been discussed in appendix C of ref. [1]. Here the 3rd order Runge–Kutta integrator described there is extended to the quark flow equation.

D.1 Integration of the gradient flow

Following ref. [1], the flow equation (5.2) is written in an abstract form

$$\partial_t V_t = Z(V_t) V_t, \quad (\text{D.1})$$

where the gauge field V_t is considered to be an element of a (high-dimensional) Lie group and the gradient $Z(V_t)$ of the Wilson action an element of the associated Lie algebra. The Runge-Kutta integrator mentioned above proceeds in time steps of size ϵ . Assuming V_t is known at some flow time t , an approximation to the exact solution $V_{t+\epsilon}$ at time $t + \epsilon$ is obtained by computing the fields

$$\begin{aligned} W_0 &= V_t, \\ W_1 &= \exp\left\{\frac{1}{4}Z_0\right\}W_0, \\ W_2 &= \exp\left\{\frac{8}{9}Z_1 - \frac{4\pi}{9\epsilon}Z_0\right\}W_1, \\ W_3 &= \exp\left\{\frac{3}{4}Z_2 - \frac{8}{9}Z_1 + \frac{4\pi}{9\epsilon}Z_0\right\}W_2, \end{aligned} \quad (\text{D.2})$$

where

$$Z_i = \epsilon Z(W_i), \quad i = 0, 1, 2 \quad (\text{D.3})$$

These rules are such that

$$W_3 = V_{t+\epsilon} + \mathcal{O}(\epsilon^4) \quad (\text{D.4})$$

and W_3 thus provides the desired approximation to $V_{t+\epsilon}$.

D.2 Integration of the quark flow equation

It is again helpful to rewrite the flow equation in an abstract form,

$$\partial_t \chi_t = \Delta(V_t) \chi_t, \quad (\text{D.5})$$

where the quark field χ_t is considered to be an element of a complex vector space. Given V_t and χ_t at some flow time t , and assuming the gauge fields W_0 , W_1 and W_2 are as above, the quark fields

$$\begin{aligned} \phi_0 &= \chi_t, \\ \phi_1 &= \phi_0 + \frac{1}{4}\Delta_0\phi_0, \\ \phi_2 &= \phi_0 + \frac{8}{9}\Delta_1\phi_1 - \frac{2}{9}\Delta_0\phi_0, \\ \phi_3 &= \phi_1 + \frac{3}{4}\Delta_2\phi_2, \end{aligned} \quad (\text{D.6})$$

may then be calculated, where

$$\Delta_i = \epsilon \Delta(W_i), \quad i = 0, 1, 2. \quad (\text{D.7})$$

It is not difficult to show that

$$\phi_3 = \chi_{t+\epsilon} + O(\epsilon^4), \quad (\text{D.8})$$

and ϕ_3 thus approximates $\chi_{t+\epsilon}$ to an accuracy that matches the one attained by the integrator of the pure-gauge flow.

Appendix E. Integration of the adjoint flow equation (7.10)

The numerical integration of the adjoint flow equation is complicated by the fact that the backward integration of the flow equation for the gauge field is exponentially unstable. In this appendix, the problem is first reformulated and its solution is then discussed. For simplicity, the index k in eq. (7.10) is omitted.

E.1 Adjoint Runge-Kutta integration

For a given step size $\epsilon > 0$ and $t = n\epsilon$, $n = 0, 1, 2, \dots$, let

$$V_t^*(x, \mu) \quad \text{and} \quad \chi_t^*(x) \quad (\text{E.1})$$

be the gauge and quark fields generated by the Runge-Kutta integrator defined in appendix D. There exists another sequence of quark fields

$$\xi_s^*(x), \quad s = t, t - \epsilon, t - 2\epsilon, \dots, 0, \quad (\text{E.2})$$

with initial value

$$\xi_t^*(x) = \eta(x), \quad (\text{E.3})$$

such that the identity

$$(\xi_s^*, \chi_s^*) = (\eta, \chi_t^*) \quad (\text{E.4})$$

holds exactly for all s . This property ensures that ξ_s^ϵ is an approximate solution of the adjoint flow equation, accurate to order ϵ^3 , with the required initial value.

The Runge–Kutta steps that lead from χ_s^ϵ to $\chi_{s+\epsilon}^\epsilon$ amount to applying the operator

$$R_s^\epsilon = 1 + \frac{1}{4}\Delta_0 + \frac{3}{4}\Delta_2 \left\{ 1 - \frac{2}{9}\Delta_0 + \frac{8}{9}\Delta_1 \left(1 + \frac{1}{4}\Delta_0 \right) \right\} \quad (\text{E.5})$$

to χ_s^ϵ , where Δ_i is given by eq. (D.7) and the fields W_i that appear there by eq. (D.2) with $W_0 = V_s^\epsilon$. From eq. (E.4) one then infers that

$$\xi_s^\epsilon(x) = (R_s^{\epsilon\dagger} \xi_{s+\epsilon}^\epsilon)(x) \quad (\text{E.6})$$

$$R_s^{\epsilon\dagger} = 1 + \frac{1}{4}\Delta_0 + \left\{ 1 - \frac{2}{9}\Delta_0 + \left(1 + \frac{1}{4}\Delta_0 \right) \frac{8}{9}\Delta_1 \right\} \frac{3}{4}\Delta_2, \quad (\text{E.7})$$

and ξ_s^ϵ is thus obtained from $\xi_{s+\epsilon}^\epsilon$ by calculating

$$\begin{aligned} \lambda_3 &= \xi_{s+\epsilon}^\epsilon, \\ \lambda_2 &= \frac{3}{4}\Delta_2 \lambda_3, \\ \lambda_1 &= \lambda_3 + \frac{8}{9}\Delta_1 \lambda_2, \\ \lambda_0 &= \lambda_1 + \lambda_2 + \frac{1}{4}\Delta_0 (\lambda_1 - \frac{8}{9}\lambda_2), \end{aligned} \quad (\text{E.8})$$

and setting $\xi_s^\epsilon = \lambda_0$.

E.2 Stability issue

The adjoint recursion (E.8) is a smoothing operation and is therefore numerically stable. To be able to apply the recursion at time $s + \epsilon$, the gauge field V_s^ϵ at flow time s must however be known. In principle, the sequence of gauge fields from $s = t$ to $s = 0$ could be reconstructed through backward integration of the gradient flow, but such strategies are not practical, because the flow is exponentially unstable in this direction.

Alternatively, V_s^ϵ can be computed through forward integration of the flow starting from the fundamental field $V_0^\epsilon = U$. If the field is recomputed for every s , the total number of gauge update steps required in the course of the integration scales like the square of the number $m = t/\epsilon$ of time steps. This method is numerically safe and the calculated quark fields satisfy the identity (E.4) practically to machine precision. The required computational effort can however be prohibitively large.

E.3 Improved scheme

The adjoint flow equation can be integrated more efficiently by dividing the sequence $0, \epsilon, \dots, t$ of flow times into n_b consecutive blocks of about equal size. One may then first compute the field V'_t at the time t at the beginning of the last block by forward integration from time 0 and keep this field in the memory of the computer. After that the adjoint flow equation is integrated from time t to r by applying the update step (E.8) and using the safe reconstruction method described above for the gauge field, where the reconstruction now starts from the stored field rather than the field at time 0.

In the next step the gauge field is calculated at the first time r in the next-to-last block and stored in memory. The integration of the adjoint flow equation can then proceed to time r as before. Clearly, the blocks can be processed in this way one after another until the integration reaches time 0.

The total number of gauge update steps in this scheme is minimized if $m = n_b^2$. While m will rarely be the square of an integer, one can always set $n_b = \lceil m^{1/2} \rceil$ and divide m into blocks of size $m_b = \lfloor m/n_b \rfloor$ and $m_b + 1$. The computational effort then scales approximately like $m^{3/2}$ and is much smaller than the one required for the safe scheme described in the previous subsection already at low values of m .

E.4 Hierarchical scheme

A further acceleration of the computation can be achieved by dividing each block into n_b smaller blocks, and these into n_b even smaller blocks, and so on, with a saved gauge field at each block level. Given m and the number n_s of saved gauge fields, a nearly optimal choice of the block number n_b is

$$n_b = \lceil m^{1/(n_s+1)} \rceil. \quad (\text{E.9})$$

For $m \leq 10^3$ and $n_s = 4$, for example, the total number of gauge-field updates that need to be performed is then not more than about $5m$. Moreover, when several quark fields are evolved simultaneously, the intermediate gauge fields need to be computed only once.

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P10: 杨米尔斯梯度流提取能动量张量

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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

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Energy-momentum tensor from the Yang-Mills gradient flow

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The product of gauge fields generated by the Yang-Mills gradient flow for positive flow times does not exhibit the coincidence-point singularity and a local product is thus independent of the regularization. Such a local product can furthermore be expanded by renormalized local operators at zero flow time with finite coefficients that are governed by renormalization group equations. Using these facts, we derive a formula that relates the small flow-time behavior of certain gauge-invariant local products and the correctly-normalized conserved energy-momentum tensor in the Yang-Mills theory. Our formula provides a possible method to compute the correlation functions of a well-defined energy-momentum tensor by using lattice regularization and Monte Carlo simulation.

Subject Index B01, B31, B32, B38

1. Introduction

Although lattice regularization provides a very powerful non-perturbative formulation of field theories, it is unfortunately incompatible with fundamental global symmetries quite often. The most well-known example is chiral symmetry [1, 2]; supersymmetry is another infamous example [3], as is, needless to say, translational invariance. When a regularization is not invariant under a symmetry, it is not straightforward to construct the corresponding Noether current that is conserved and generates the symmetry transformation through Ward-Takahashi (WT) relations. This makes the measurement of physical quantities related to the Noether current in a solid basis very difficult. To solve this problem, one can imagine at least three possible approaches.

The first approach is an ideal one: One finds a lattice formulation that realizes (a lattice-modified form of) the desired symmetry. If such a formulation comes to hand, the corresponding Noether current can easily be obtained by the standard Noether method. The best successful example of this sort is the lattice chiral symmetry [4–11], which can be defined with a lattice Dirac operator that satisfies the Ginsparg-Wilson relation [12]. Although this is certainly an ideal approach, it appears that such an ideal formulation does not always come to hand, especially for spacetime symmetries (see, e.g., Ref. [13] for a no-go theorem for supersymmetry).

The second approach is to construct the Noether current by tuning coefficients in the linear combination of operators that can mix with the Noether current under lattice symmetries.¹ For example, for the energy-momentum tensor—the Noether current associated with the

¹Here, we assume that fine tuning of bare parameters to the target (symmetric) theory is done.

translational invariance and rotational and conformal symmetries [14, 15]—one can construct a conserved lattice energy-momentum tensor by adjusting coefficients in the linear combination of dimension 4 operators [16, 17]²; the overall normalization of the energy-momentum tensor has to be fixed in some other way.³ Although this method is in principle sufficient when the energy-momentum tensor is in “isolation”, i.e., when the energy-momentum tensor is separated from other composite operators, as in the on-shell matrix elements, it is not obvious a priori whether one can control the ambiguity of possible higher-dimensional operators that may contribute when the energy-momentum tensor coincides with other composite operators in position space. This implies that it is not obvious whether the energy-momentum tensor constructed in the above method generates correctly-normalized translations (and rotational and conformal transformations) on operators through WT relations. (If the energy-momentum tensor generates correctly-normalized translations, it is ensured [21] (see also Sect. 7.3 of Ref. [22]) that the trace or conformal anomaly [23, 24] is proportional to the renormalization group functions [25–27].)

The third possible approach is to utilize some ultraviolet (UV) finite quantity. Since such a quantity must be independent of the regularization adopted (in the limit in which the regulator is removed), there emerges a possibility that one can relate the lattice regularization and some other regularization that preserves the desired symmetry. This methodology can be found e.g. in Ref. [28] (see also Ref. [29]), where an ultraviolet finite representation of the topological susceptibility is derived. Although the derivation of the representation itself relies on a lattice regularization that preserves the chiral symmetry [4–11], one can use any regularization (e.g., the Wilson fermion [30]) to compute the representation because it must be independent of the regularization.

In the present paper, we consider the above third approach for the energy-momentum tensor, by taking the pure Yang-Mills theory as an example. For this, we utilize the so-called Yang-Mills gradient flow (or the Wilson flow in the context of lattice gauge theory) whose usefulness in lattice gauge theory has recently been revealed [31–39]. A salient feature of the Yang-Mills gradient flow is its robust UV finiteness [33]. More precisely, any product of gauge fields generated by the gradient flow for a positive flow time t is UV finite under standard renormalization. Such a product, moreover, does *not* exhibit any singularities even if some positions of gauge fields coincide. The basic mechanism for this UV finiteness is that the flow equation is a type of the diffusion equation and the evolution operator in the momentum space $\sim e^{-tk^2}$ acts as an UV regulator for $t > 0$. This property of the gradient flow implies that the definition of a local product of gauge fields for positive flow times is independent of the regularization. In our present context, there is a hope of relating quantities obtained by the lattice regularization and the dimensional regularization with which the translational invariance *is* manifest.

As noted in Ref. [33], on the other hand, a local product of gauge fields for a positive flow time can be expanded by renormalized local operators of the original gauge theory with finite coefficients. Those coefficients satisfy certain renormalization group equations that, combined with the dimensional analysis, provide information on the coefficients as a function of the

² A somewhat different approach on the basis of the $\mathcal{N} = 1$ supersymmetry has been given in Refs. [18, 19].

³ It might be possible to employ “current algebra” for this, as for the axial current [20].

flow time. Because of the asymptotic freedom, one can then use the perturbation theory to find the asymptotic behavior of the coefficients for small flow times.

By using the above properties of the gradient flow, one can obtain a formula that relates the small flow-time behavior of certain gauge-invariant local products and the energy-momentum tensor defined by the dimensional regularization. Since the former can be computed by using the Wilson flow with lattice regularization [31–39] and the latter is conserved and generates correctly-normalized translations on composite operators, our formula provides a possible method to compute the correlation functions of a correctly-normalized conserved energy-momentum tensor by using Monte Carlo simulation.

In the present paper, we follow the notational convention of Ref. [33] unless otherwise stated.

2. Yang-Mills theory and the energy-momentum tensor

2.1. The energy-momentum tensor with dimensional regularization

In the present paper, we consider the $SU(N)$ Yang-Mills theory defined in a D dimensional Euclidean space. The action is given by

$$S = \frac{1}{4g_0^2} \int d^D x F_{\mu\nu}^a(x) F_{\mu\nu}^a(x), \quad (2.1)$$

from the Yang-Mills field strength

$$F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) + [A_\mu(x), A_\nu(x)]. \quad (2.2)$$

We set

$$D = 4 - 2\epsilon, \quad (2.3)$$

and then the mass dimension of the bare gauge coupling g_0 is ϵ .

Assuming that the theory is regularized by the dimensional regularization (for a very nice exposition, see Ref. [40]), one can define the energy-momentum tensor for the system (2.1) simply by (see, e.g., Ref. [41])

$$T_{\mu\nu}(x) = \frac{1}{g_0^2} \left[F_{\mu\rho}^a(x) F_{\nu\rho}^a(x) - \frac{1}{4} \delta_{\mu\nu} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x) \right], \quad (2.4)$$

up to terms attributed to the gauge fixing and the Faddeev-Popov ghost fields, which are irrelevant in correlation functions of gauge-invariant operators. Note that the mass dimension of the energy-momentum tensor is D .

The advantage of dimensional regularization is its translational invariance. Because of this property, the energy-momentum tensor naively constructed from bare quantities, Eq. (2.4), is conserved and generates correctly-normalized translations through a WT relation,

$$\int d^D x \langle \partial_\mu T_{\mu\nu}(x) \mathcal{O} \rangle = - \langle \partial_\nu \mathcal{O} \rangle, \quad (2.5)$$

where it is understood that the derivative on the right-hand side is acting all positions in a gauge-invariant operator $\mathcal{O}$. Used in combination with dimensional counting and gauge invariance, this WT relation implies that the energy-momentum tensor $T_{\mu\nu}(x)$ is finite [26,

42] and thus, in the minimal subtraction (MS) scheme,⁴

$$T_{\mu\nu}(x) - \langle T_{\mu\nu}(x) \rangle = \{T_{\mu\nu}\}_R(x). \quad (2.6)$$

The finiteness of the energy-momentum tensor (2.4) provides further useful information on the renormalization of dimension 4 gauge-invariant operators. The gauge coupling renormalisation with dimensional regularization is defined by

$$g_0^2 \equiv \mu^{2\epsilon} g^2 Z, \quad (2.7)$$

where μ is the renormalization scale and Z is the renormalization factor. In the MS scheme,

$$Z = 1 - \frac{1}{\epsilon} \left[b_0 g^2 + \frac{1}{2} b_1 g^4 + O(g^6) \right] + O\left(\frac{1}{\epsilon^2}\right), \quad (2.8)$$

and

$$b_0 = \frac{11N}{48\pi^2}, \quad b_1 = \frac{17N^2}{384\pi^4}. \quad (2.9)$$

From the rotational invariance that the dimensional regularization keeps, we see that the operator-renormalization possesses the following structures:⁵

$$\begin{aligned} & F_{\mu\rho}^a(x) F_{\nu\rho}^a(x) - \langle F_{\mu\rho}^a(x) F_{\nu\rho}^a(x) \rangle \\ &= Z_T \{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) + Z_M \delta_{\mu\nu} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x), \end{aligned} \quad (2.10)$$

and

$$F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x) - \langle F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x) \rangle = Z_S \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x). \quad (2.11)$$

Substituting the above relations into Eqs. (2.4) and (2.6), we have

$$\begin{aligned} & \{T_{\mu\nu}\}_R(x) \\ &= \frac{1}{g^2} \mu^{-2\epsilon} Z^{-1} \left[Z_T \{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) - \frac{1}{4} (Z_S - 4Z_M) \delta_{\mu\nu} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) \right]. \end{aligned} \quad (2.12)$$

Since the left-hand side is finite for $\epsilon \rightarrow 0$, in the MS scheme in which only pole terms are subtracted, we infer (by considering the cases, $\mu \neq \nu$ and $\mu = \nu$) that

$$Z_T = Z = 1 - b_0 g^2 \frac{1}{\epsilon} + O(g^4) \quad (2.13)$$

and

$$Z_S - 4Z_M = Z. \quad (2.14)$$

⁴Here, we define the renormalized operator by subtracting its vacuum expectation value. In the perturbation theory using dimensional regularization, this subtraction is automatic.

⁵Here again, we define renormalized operators by subtracting their vacuum expectation values.

2.2. Implications of the trace anomaly

Another important property of the energy-momentum tensor (2.4) is the trace anomaly [25–27],

$$\delta_{\mu\nu} \{T_{\mu\nu}\}_R(x) = -\frac{\beta}{2g^4} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x). \quad (2.15)$$

By Eq. (2.6), this relation is equivalent to

$$\begin{aligned} \delta_{\mu\nu} [T_{\mu\nu}(x) - \langle T_{\mu\nu}(x) \rangle] &= \epsilon \frac{1}{2g_0^4} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x) - \left\langle \epsilon \frac{1}{2g_0^4} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x) \right\rangle \\ &\xrightarrow{\epsilon \rightarrow 0} -\frac{\beta}{2g^4} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x). \end{aligned} \quad (2.16)$$

In Eqs. (2.15) and (2.16), β denotes the β function for $D = 4$, defined by

$$\beta \equiv \left(\mu \frac{\partial}{\partial \mu} \right)_0 g = -\frac{1}{2} g \left(\mu \frac{\partial}{\partial \mu} \right)_0 \ln Z, \quad (2.17)$$

where the subscript 0 implies that the derivative is taken while the bare quantities are kept fixed. Equations (2.8) and (2.7) yield

$$\beta = -b_0 g^3 - b_1 g^5 + O(g^7). \quad (2.18)$$

Then, substituting Eqs. (2.12) into Eq. (2.15) and using Eqs. (2.13) and (2.14), we observe that

$$\delta_{\rho\lambda} \{F_{\rho\sigma}^a F_{\lambda\sigma}^a\}_R(x) = \left(1 - \frac{\beta}{2g}\right) \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x), \quad (2.19)$$

i.e., the contraction with the metric and the minimal subtraction, the subtraction of $1/\epsilon$ poles, do not commute; this is a peculiar but legitimate property of the dimensional regularization [40].

Also, substituting Eqs. (2.7) and (2.11) into Eq. (2.16), we see

$$\epsilon \frac{Z_S}{Z} \xrightarrow{\epsilon \rightarrow 0} -\frac{\beta}{g}. \quad (2.20)$$

In the $\overline{\text{MS}}$ scheme in which only pole terms are subtracted, this implies

$$Z_S = \left(1 - \frac{\beta}{g} \frac{1}{\epsilon}\right) Z = 1 + O(g^4), \quad (2.21)$$

and Eq. (2.14) then shows

$$Z_M = -\frac{\beta}{4g} Z \frac{1}{\epsilon} = \frac{b_0}{4} g^2 \frac{1}{\epsilon} + O(g^4). \quad (2.22)$$

We thus observe that all the renormalization constants in Eqs. (2.10) and (2.11), Z_T , Z_M and Z_S , in the $\overline{\text{MS}}$ scheme can eventually be expressed by the gauge coupling renormalization constant Z in Eq. (2.7).

3. Yang–Mills gradient flow and the small flow-time expansion

The Yang–Mills gradient flow defines a $D + 1$ dimensional gauge potential $B(t, x)$ along a fictitious time t , according to the flow equation

$$\partial_t B_\mu(t, x) = D_\nu G_{\nu\mu}(t, x) + \alpha_0 D_\mu \partial_\nu B_\nu(t, x), \quad (3.1)$$

where the $D + 1$ dimensional field strength and the covariant derivative are defined by

$$G_{\mu\nu}(t, x) = \partial_\mu B_\nu(t, x) - \partial_\nu B_\mu(t, x) + [B_\mu(t, x), B_\nu(t, x)] \quad (3.2)$$

and

$$D_\mu = \partial_\mu + [B_\mu, \cdot], \quad (3.3)$$

respectively. The initial condition for the flow is given by the D dimensional gauge potential in the previous section:

$$B_\mu(t = 0, x) = A_\mu(x). \quad (3.4)$$

In Eq. (3.1), the last term is introduced to suppress the evolution of the field along the direction of gauge degrees of freedom. Although this term breaks the gauge symmetry, it does not affect the evolution of any gauge-invariant operators [31]. Note that the mass dimension of the flow time t is -2 .

Now, from the field strength extended to the $D + 1$ dimension (3.2), we define a $D + 1$ dimensional analogue of the energy-momentum tensor by

$$U_{\mu\nu}(t, x) \equiv G_{\mu\rho}^a(t, x) G_{\nu\rho}^a(t, x) - \frac{1}{4} \delta_{\mu\nu} G_{\rho\sigma}^a(t, x) G_{\rho\sigma}^a(t, x). \quad (3.5)$$

Although this is similar in form to the original energy-momentum tensor (2.4), it is not obvious a priori how this $D + 1$ dimensional object and Eq. (2.4) are related (or not). To find the relationship between them is the principal task of the present paper. We also use the density operator studied in Ref. [31]:

$$E(t, x) \equiv \frac{1}{4} G_{\mu\nu}^a(t, x) G_{\mu\nu}^a(t, x). \quad (3.6)$$

Now, as shown in Ref. [33], for $t > 0$, any correlation function of $B_\mu(t, x)$ is UV finite after standard renormalization in the D dimensional Yang–Mills theory. This property holds even for any local products of $B_\mu(t, x)$ such as Eqs. (3.5) and (3.6). Also, for small flow times, a local product of $B_\mu(t, x)$ can be regarded as a local field in the D dimensional sense because the flow equation (3.1) is basically the diffusion equation along the time t and the diffusion length in x is $\sqrt{8t}$. These properties allow us to express, as explained in Sect. 8 of Ref. [33], $U_{\mu\nu}(t, x)$ and $E(t, x)$ as an asymptotic series of D dimensional renormalized local operators with finite coefficients. Considering the gauge invariance and the index structure, for $D = 4$, we can write

$$U_{\mu\nu}(t, x) = c_T(t) \{T_{\mu\nu}\}_R(x) + c_S(t) \delta_{\mu\nu} \left\{ \frac{1}{4} F_{\rho\sigma}^a F_{\rho\sigma}^a \right\}_R(x) + O(t), \quad (3.7)$$

where abbreviated terms are the contributions of operators with a mass dimension higher than or equal to 6. For Eq. (3.6), we similarly have

$$E(t, x) = \langle E(t, x) \rangle + c_E(t) \left\{ \frac{1}{4} F_{\rho\sigma}^a F_{\rho\sigma}^a \right\}_R(x) + O(t). \quad (3.8)$$

We note that, when the renormalized gauge coupling is fixed, $U_{\mu\nu}(t, x)$ (3.5) is traceless for $D = 4$,

$$\delta_{\mu\nu} U_{\mu\nu}(t, x) = 2cE(t, x) \xrightarrow{\epsilon \rightarrow 0} 0, \quad (3.9)$$

because $E(t, x)$ (3.6) is finite [31] and does not produce a $1/\epsilon$ singularity (this explains why there is no c number expectation value term in Eq. (3.7)). Thus, considering the trace part of Eq. (3.7), we see that the coefficients $c_T(t)$ and $c_S(t)$ are not independent and are related by, for $D = 4$,

$$c_S(t) = \frac{\beta}{2g^2} c_T(t), \quad (3.10)$$

because of the trace anomaly (2.15).

By eliminating the renormalized action density from Eqs. (3.7) and (3.8), we have

$$\{T_{\mu\nu}\}_R(x) = \frac{1}{c_T(t)} U_{\mu\nu}(t, x) - \frac{c_S(t)}{c_T(t)c_E(t)} \delta_{\mu\nu} [E(t, x) - \langle E(t, x) \rangle] + O(t). \quad (3.11)$$

This expression relates the energy-momentum tensor (2.6) and the short flow-time behavior of gauge-invariant local products defined by the gradient flow. Thus, once the coefficients are known, one can extract the energy-momentum tensor from the $t \rightarrow 0$ behavior of the combination on the right-hand side.

4. Renormalization group equation and the asymptotic formula

4.1. Renormalization group equation for the coefficients

We now operate

$$\left(\mu \frac{\partial}{\partial \mu} \right)_0, \quad (4.1)$$

on both sides of Eq. (3.7). Since the left-hand side of Eq. (3.7), i.e., Eq. (3.5), is entirely expressed by bare quantities through the flow equation (3.1) and the initial condition (3.4), the action of (4.1) on the left-hand side identically vanishes. On the right-hand side, this vanishing must hold in each power of t . Thus we infer that

$$\left(\mu \frac{\partial}{\partial \mu} \right)_0 c_T(t) \{T_{\mu\nu}\}_R(x) = 0, \quad (4.2)$$

$$\left(\mu \frac{\partial}{\partial \mu} \right)_0 c_S(t) \left\{ \frac{1}{4} F_{\mu\nu}^a F_{\mu\nu}^a \right\}_R(x) = 0. \quad (4.3)$$

For the first relation (4.2), we recall that the energy-momentum tensor is not renormalized as Eq. (2.6). Then, by expressing the operation (4.1) in terms of renormalized quantities, we have

$$\left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} \right) c_T(t) = 0. \quad (4.4)$$

For Eq. (4.3), on the other hand, from Eq. (2.11),

$$\left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} + \gamma_S \right) c_S(t) = 0, \quad (4.5)$$

where

$$\gamma_S \equiv - \left(\mu \frac{\partial}{\partial \mu} \right)_0 \ln Z_S. \quad (4.6)$$

Equations (2.21), (2.7), and (2.17) yield

$$\gamma_S = -g^3 \frac{d}{dg} \left(\frac{\beta}{g^2} \right) = 2b_1 g^4 + O(g^6). \quad (4.7)$$

Similarly, for Eq. (3.8), we have

$$\left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} \right) \langle E(t, x) \rangle = 0, \quad (4.8)$$

$$\left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} + \gamma_S \right) c_E(t) = 0, \quad (4.9)$$

and thus

$$\left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} \right) \frac{c_S(t)}{c_E(t)} = 0. \quad (4.10)$$

By the standard argument and from the fact that dimensionless quantities can depend on the renormalization scale μ only through the dimensionless combination $\sqrt{8t}\mu$, the above renormalization group equations imply that

$$c_T(t)(g; \mu) = c_T(t_0)(\bar{g}(-\xi); \mu_0), \quad (4.11)$$

$$c_S(t)(g; \mu) = \exp \left[\int_0^{-\xi} d\xi' \gamma_S(\bar{g}(\xi')) \right] c_S(t_0)(\bar{g}(-\xi); \mu_0), \quad (4.12)$$

$$t^2 \langle E(t, x) \rangle(g; \mu) = t_0^2 \langle E(t_0, x) \rangle(\bar{g}(-\xi); \mu_0), \quad (4.13)$$

$$\frac{c_S(t)}{c_E(t)}(g; \mu) = \frac{c_S(t_0)}{c_E(t_0)}(\bar{g}(-\xi); \mu_0), \quad (4.14)$$

where the dependence on the renormalized gauge coupling and on the renormalization scale has been explicitly written. In these expressions, the running coupling $\bar{g}(\xi)$ is defined by

$$\frac{d\bar{g}(\xi)}{d\xi} = \beta(\bar{g}(\xi)), \quad \bar{g}(0) = g, \quad (4.15)$$

and we introduce a variable

$$\xi \equiv \ln \frac{\sqrt{8t}\mu}{\sqrt{8t_0}\mu_0}. \quad (4.16)$$

In the one-loop order, the running coupling (4.15) is given by

$$\bar{g}(-\xi)^2 = \frac{1}{2b_0 - \xi + 1/(2b_0 g^2)} = \frac{1}{2b_0 - \ln(\sqrt{8t}\Lambda) + \ln(\sqrt{8t_0}\mu_0)}, \quad (4.17)$$

where Λ is the Λ parameter in the one-loop level,

$$\Lambda = \mu e^{-1/(2b_0 g^2)}, \quad (4.18)$$

and the integral appearing in Eqs. (4.12) is

$$\int_0^{-\xi} d\xi' \gamma_S(\bar{g}(\xi')) = \frac{b_1}{b_0} [g^2 - \bar{g}(-\xi)^2]. \quad (4.19)$$

In the small flow-time limit $t \rightarrow 0$, $-\xi \rightarrow +\infty$ and the running coupling $\bar{g}(-\xi)$ (4.17) becomes very small thanks to the asymptotic freedom. Thus, the right-hand sides of Eqs. (4.11)–(4.14) allow us to compute the small flow-time behavior of the coefficients by using the perturbation theory.

4.2. Lowest-order approximation and the asymptotic formula

By substituting the solution of the flow equation (3.1) (see Ref. [33]) in the tree-level approximation to Eq. (3.7), we have

$$c_T(t) = g_0^2, \quad (4.20)$$

simply because our energy-momentum tensor (2.4) is proportional to $1/g_0^2$. If we apply the right-hand side of Eq. (4.11) to this expression by substituting Eq. (4.17), however, it depends on $\sqrt{8t}\mu_0$ while the left-hand side of Eq. (4.11) does not. This shows that $c_T(t)$ should depend on g^2 and $\sqrt{8t}\mu$ through a particular combination as (for $D = 4$)

$$c_T(t) = g^2 \left\{ 1 + 2b_0 g^2 \left[\ln(\sqrt{8t}\mu) + c_1 \right] + O(g^4) \right\}, \quad (4.21)$$

where c_1 is a constant. Similarly, since the lowest-order approximation in Eqs. (3.7) and (3.8) yields

$$c_E(t) = 1, \quad c_S(t) = -\frac{b_0}{2} g_0^2 \mu^{-2s}, \quad (4.22)$$

where the latter follows from Eq. (3.10), from Eq. (4.14) we have

$$\frac{c_S(t)}{c_E(t)} = -\frac{b_0}{2} g^2 \left\{ 1 + 2b_0 g^2 \left[\ln(\sqrt{8t}\mu) + c_2 \right] + O(g^4) \right\}, \quad (4.23)$$

where c_2 is another constant.⁶

Applying Eqs. (4.11) and (4.14) to above expressions and using Eq. (4.17), we finally have the asymptotic behaviors of the coefficients in Eq. (3.11),

$$\frac{1}{c_T(t)} \stackrel{t \rightarrow 0^+}{\sim} -2b_0 \left[\ln(\sqrt{8t}\Lambda) + c_1 \right] \quad (4.26)$$

and

$$\frac{c_S(t)}{c_E(t)} \stackrel{t \rightarrow 0^+}{\sim} -\frac{b_0}{2} \frac{1}{-2b_0 \left[\ln(\sqrt{8t}\Lambda) + c_2 \right]}, \quad (4.27)$$

and hence

$$\frac{c_S(t)}{c_T(t)c_E(t)} \stackrel{t \rightarrow 0^+}{\sim} -\frac{b_0}{2} \left[1 - \frac{c_1 - c_2}{-\ln(\sqrt{8t}\Lambda)} \right]. \quad (4.28)$$

That is,

$$\begin{aligned} \{T_{\mu\nu}\}_R(x) &\stackrel{t \rightarrow 0^+}{\sim} \left\{ -2b_0 \left[\ln(\sqrt{8t}\Lambda) + c_1 \right] U_{\mu\nu}(t, x) \right. \\ &\quad \left. + \frac{b_0}{2} \left[1 - \frac{c_1 - c_2}{-\ln(\sqrt{8t}\Lambda)} \right] \delta_{\mu\nu} [E(t, x) - \langle E(t, x) \rangle] \right\}. \end{aligned} \quad (4.29)$$

This is the relation that we were seeking: One can obtain the correctly-normalized conserved energy-momentum tensor from the small flow-time behavior of gauge-invariant products

⁶ Using Eq. (4.21) in Eq. (3.10), we have

$$c_S(t) = -\frac{b_0}{2} g^2 \left\{ 1 + 2b_0 g^2 \left[\ln(\sqrt{8t}\mu) + c_1 + \frac{b_1}{2b_0^2} \right] + O(g^4) \right\}, \quad (4.24)$$

and then using Eq. (4.23),

$$c_E(t) = 1 + 2b_0 g^2 \left(c_1 - c_2 + \frac{b_1}{2b_0^2} \right) + O(g^4). \quad (4.25)$$

given by the Yang–Mills gradient flow. It is interesting to note that the leading $t \rightarrow 0$ behavior is completely independent of the detailed definition of the gradient flow; the structure and coefficients follow solely from the finiteness of the local products and the renormalizability of the Yang–Mills theory. The sub-leading corrections in the asymptotic form, i.e., the coefficients c_1 and c_2 , depend on the detailed definition of the gradient flow; in the Appendix, we compute the constants c_1 and c_2 and we have

$$c_1 = \ln \sqrt{\pi} + \frac{7}{22} \simeq 0.890547, \quad (4.30)$$

$$c_2 = \ln \sqrt{\pi} - \frac{7}{44} + \frac{b_1}{2b_0^2} \simeq 0.834762. \quad (4.31)$$

Finally, a possible method to determine the factor $\ln(\sqrt{8t}\Lambda)$ in Eq. (4.29), i.e., the flow time t in the unit of the one-loop Λ parameter (4.18), for small flow times is to use the expectation value of the density operator, Eq. (3.6). For this quantity, by applying Eqs. (4.13) and (4.17) to the result of the one-loop calculation, Eqs. (2.28) and (2.29) of Ref. [31] (specialized to the pure Yang–Mills theory), we have the asymptotic form,

$$t^2 \langle E(t, x) \rangle \stackrel{t \rightarrow 0}{\sim} \frac{3(N^2 - 1)}{128\pi^2} \frac{1}{-2b_0 [\ln(\sqrt{8t}\Lambda) + c]}, \quad (4.32)$$

where

$$c \equiv \ln(2\sqrt{\pi}) + \frac{26}{33} - \frac{9}{22} \ln 3 \simeq 1.60396. \quad (4.33)$$

One may use this asymptotic representation for $\ln(\sqrt{8t}\Lambda)$ in Eq. (4.29).⁷

5. Conclusion

In the present paper, we have derived a formula that relates the short flow-time behavior of some gauge-invariant local products generated by the Yang–Mills gradient flow and the correctly-normalized conserved energy–momentum tensor in the Yang–Mills theory. Our main result is Eq. (4.29). The right-hand side of Eq. (4.29) can be computed by the Wilson flow in lattice gauge theory with appropriate discretizations of operators, Eqs. (3.5) and (3.6) (see, e.g., Refs. [31, 34]). Here, the continuum limit $a \rightarrow 0$ must be taken first and then the $t \rightarrow 0$ limit is taken afterwards; otherwise our basic reasoning does not hold.

Although the formula (4.29) should be mathematically correct, the practical usefulness of Eq. (4.29) is a separate issue and has to be carefully examined numerically.⁸ Since the lattice spacing a must be sufficiently smaller than the square-root of the flow time $\sqrt{8t}$ for our reasoning to work, the reliable application of Eq. (4.29) will require rather small lattice spacings. One also worries about contamination by higher-dimensional operators (i.e., the $O(t)$ terms in Eqs. (3.7) and (3.8)) and the finite-size effect which we have not taken into account in the present paper. If our strategy turns to be practically feasible, it provides a completely new method to compute correlation functions containing a well-defined energy–momentum tensor. It is clear that the present approach to the energy–momentum tensor on the lattice is not limited to the pure Yang–Mills theory although the treatment might

⁷ In practice, one will use Eq. (4.29) to compute $t^2 \langle T_{\mu\nu} \rangle_R(x)$ from $t^2 U_{\mu\nu}(t, x)$ and $t^2 E(t, x)$. Then, from the value of $\sqrt{8t}\Lambda$, one can deduce $\langle T_{\mu\nu} \rangle_R(x)/\Lambda^4$.

⁸ We hope to return to this problem in the near future.

be slightly more complicated with the presence of other fields. The application will then include the determination of the shear and bulk viscosities (see, e.g., Refs. [43, 44]), the measurement of thermodynamical quantities (see Ref. [45] and references cited therein), the mass and the decay constant of the pseudo Nambu-Goldstone boson associated with the (approximate) dilatation invariance (see Ref. [46] and references cited therein), and so on.

It is also clear that our basic idea, that operators defined with lattice regularization and in the continuum theory can be related through the gradient flow is not limited to the energy-momentum tensor. For example, it might be possible to construct an ideal chiral current or an ideal supercurrent on the lattice, from the small flow-time limit of local products. It would be interesting to pursue this idea.

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A. One-loop calculation of coefficient functions

For calculational convenience, we define the coefficient functions $F(t)$ and $G(t)$ by

$$\begin{aligned} G_{\mu\rho}^a(t, x)G_{\nu\rho}^a(t, x) - \langle G_{\mu\rho}^a(t, x)G_{\nu\rho}^a(t, x) \rangle \\ = F(t) \{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) + G(t)\delta_{\mu\nu} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) + O(t). \end{aligned} \quad (A1)$$

Equation (3.5) then becomes (for $D = 4$),

$$\begin{aligned} U_{\mu\nu}(t, x) &= F(t) \left[\{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) - \frac{1}{4}\delta_{\mu\nu}\delta_{\rho\lambda} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) \right] + O(t) \\ &= F(t) \left[\{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) - \frac{1}{4}\delta_{\mu\nu} \left(1 - \frac{\beta}{2g}\right) \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) \right] + O(t), \end{aligned} \quad (A2)$$

where we have used Eq. (2.19). Rewriting this in favor of the energy-momentum tensor (2.12) with Eqs. (2.13) and (2.14), we have

$$\begin{aligned} U_{\mu\nu}(t, x) &= F(t) \left[\{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) - \frac{1}{4}\delta_{\mu\nu}\delta_{\rho\lambda} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) \right] + O(t) \\ &= F(t) \left[g^2 \{T_{\mu\nu}\}_R(x) + \frac{\beta}{8g}\delta_{\mu\nu} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) \right] + O(t). \end{aligned} \quad (A3)$$

Comparison with Eq. (3.7) then shows

$$c_T(t) = g^2 F(t), \quad c_S(t) = \frac{\beta}{2g} F(t). \quad (A4)$$

Similarly, for Eq. (3.8),

$$\begin{aligned} E(t, x) &= \langle E(t, x) \rangle + \frac{1}{4}F(t)\delta_{\rho\lambda} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) + G(t) \{F_{\mu\rho}^a F_{\mu\rho}^a\}_R(x) + O(t) \\ &= \left[\left(1 - \frac{\beta}{2g}\right) F(t) + 4G(t) \right] \left\{ \frac{1}{4}F_{\rho\sigma}^a F_{\rho\sigma}^a \right\}_R(x) + O(t), \end{aligned} \quad (A5)$$



Fig. A1

and therefore

$$c_E(t) = \left(1 - \frac{\beta}{2g}\right) F(t) + 4G(t). \quad (\text{A6})$$

This implies, for the ratio (4.23),

$$\begin{aligned} \frac{c_S(t)}{c_E(t)} &= \frac{\beta}{2g} \frac{1}{1 - \frac{\beta}{2g} + 4G(t)/F(t)} \\ &= -\frac{b_0}{2} g^2 \left\{ 1 + 2b_0 g^2 \left[-\frac{1}{4} + \frac{b_1}{2b_0^2} \right] - 4G(t) + O(g^4) \right\}. \end{aligned} \quad (\text{A7})$$

To find the coefficient functions $F(t)$ and $G(t)$ in Eq. (A1), we consider the correlation function

$$\langle G_{\rho\rho}^a(t, x) G_{\sigma\rho}^a(t, x) A_\mu^i(w) A_\nu^j(v) \rangle. \quad (\text{A8})$$

For $O(g_0^2)$, there are 17 flow-line Feynman diagrams (Figs. A1–A17) that contribute to this correlation function. In the figures, gauge potentials at the flow time t , $B_\mu(t, x)$, are represented by small filled squares; the open circle denotes the flow-time vertex and the full circle is the conventional vertex in the Yang-Mills theory. We refer the reader to Ref. [33] for the details of the Feynman rules for flow-line diagrams.

To read off the coefficient functions $F(t)$ and $G(t)$ in Eq. (A1) from the correlation function (A8), we consider the vertex functions, i.e., amputated diagrams in which the external propagators of the original Yang–Mills theory are truncated. Therefore, Figs. A10, A12 and A17, which provide only the conventional wave function renormalization, should be omitted in the computation of $F(t)$ and $G(t)$.⁹ On the other hand, the flow-line propagators [33], the arrowed straight lines in the diagrams, should not be truncated because these are not propagators in the quantum field theory but instead represent time evolution along the flow time.

The tree-level contribution to the vertex function is

$$\begin{aligned} \text{Fig. A1} &= \delta_{\rho\sigma} \int_{p,q} e^{i(p+q)x} \tilde{A}_\mu^a(p) \tilde{A}_\nu^a(q) e^{-ip^2} e^{-iq^2} i p_\mu i q_\nu \pm (\mu \leftrightarrow \rho, \nu \leftrightarrow \sigma) \\ &= F_{\rho\rho}^a(x) F_{\sigma\sigma}^a(x) + O(t), \end{aligned} \quad (\text{A9})$$

⁹ More precisely, these diagrams are different from conventional Feynman diagrams in that the propagators carry an additional factor e^{-ip^2} (in the Feynman gauge), where p is the external momentum. This factor is, however, irrelevant in the present computation of the coefficients of operators with the lowest number of derivatives.



Fig. A2



Fig. A3



Fig. A4



Fig. A5



Fig. A6



Fig. A7



Fig. A8



Fig. A9



Fig. A10



Fig. A11



Fig. A12



Fig. A13



Fig. A14



Fig. A15



Fig. A16



Fig. A17

where

$$A_\mu(x) = \int_p e^{ipx} \tilde{A}_\mu(p), \quad \int_p \equiv \int \frac{d^D p}{(2\pi)^D}, \quad (\text{A10})$$

and, here and in what follows, the alternating-sign symbol implies

$$\ell_{\mu\rho\sigma\nu} \pm (\mu \leftrightarrow \rho, \nu \leftrightarrow \sigma) \equiv \ell_{\mu\rho\sigma\nu} - \ell_{\rho\mu\sigma\nu} - \ell_{\mu\rho\nu\sigma} + \ell_{\rho\mu\nu\sigma}. \quad (\text{A11})$$

This tree-level result was used in obtaining Eq. (4.20).

The vacuum expectation value in the lowest order is

$$\begin{aligned} \text{Fig. A2} &= g_0^2 \delta^{an} \delta_{\rho\sigma} \left[\int_\ell \frac{1}{\ell^2} e^{-2i\ell^2} \ell_\mu \ell_\nu \delta_{\rho\sigma} \pm (\mu \leftrightarrow \rho, \nu \leftrightarrow \sigma) \right] \\ &= \frac{3}{128\pi^2} (N^2 - 1) g_0^2 \frac{1}{\ell^2} \delta_{\mu\nu}. \end{aligned} \quad (\text{A12})$$

Now, as an example of the computation of one-loop flow-line Feynman diagrams, we briefly illustrate the computation of Fig A13. A straightforward application of the Feynman rules in Ref. [33] in the ‘‘Feynman gauge’’ in which the gauge parameters are taken as $\lambda_0 = \alpha_0 = 1$, yields the expression,

$$\begin{aligned} \text{Fig. A13} &= N g_0^2 \delta_{\rho\sigma} \left(\int_{p,q} e^{i(p+q)x} \tilde{A}_n^b(p) \tilde{A}_\sigma^c(q) \right. \\ &\quad \times \int_\ell \frac{1}{(p+\ell)^2} \frac{1}{\ell^2} \frac{1}{(q-\ell)^2} e^{-i\ell(p+\ell)^2} e^{-i\ell(q-\ell)^2} \\ &\quad \times i(p+\ell)_\mu i(q-\ell)_\nu \\ &\quad \times [\delta_{\rho\lambda}(-p-2\ell)_\alpha + \delta_{\lambda\alpha}(\ell-p)_\rho + \delta_{\alpha\rho}(2p+\ell)_\lambda] \\ &\quad \times [\delta_{\lambda\sigma}(-2\ell+q)_\beta + \delta_{\sigma\beta}(-2q+\ell)_\lambda + \delta_{\beta\lambda}(q+\ell)_\sigma] \\ &\quad \left. \pm (\mu \leftrightarrow \rho, \nu \leftrightarrow \sigma) \right). \end{aligned} \quad (\text{A13})$$

To find the coefficients $F(t)$ and $G(t)$ in Eq. (A1), we write this vertex function as

$$\int_{p,q} e^{i(p+q)x} \tilde{A}_n^a(p) \tilde{A}_\beta^a(q) M_{\mu\nu,\alpha\beta}(p,q), \quad (\text{A14})$$

and find the coefficients of

$$-p_\mu q_\nu \delta_{\alpha\beta} \quad (\text{A15})$$

and

$$-4p \cdot q \delta_{\mu\nu} \delta_{\alpha\beta}, \quad (\text{A16})$$

respectively, in $M_{\mu\nu,\alpha\beta}(p, q)$. For this, we first exponentiate the denominators in Eq. (A13) by using

$$\frac{1}{(p+\ell)^2} \frac{1}{(q-\ell)^2} = \int_0^\infty d\xi \int_0^\infty d\eta e^{-\xi(p+\ell)^2} e^{-\eta(q-\ell)^2}. \quad (\text{A17})$$

We then simply expand the integrand with respect to the external momenta p and q to $O(p, q)$. The flow-time evolution factor $e^{-2t\epsilon^2}$ in the integrand makes the integral (A13) UV finite for any dimension D . On the other hand, there always exists a complex domain of D such that the integral is infrared finite; this provides the analytic continuation of the integral such that

$$\int_\ell \frac{1}{\ell^2} e^{-\alpha\ell^2} = \frac{1}{(4\pi)^{D/2}} \frac{1}{D/2-1} \alpha^{-D/2+1}, \quad (\text{A18})$$

$$\int_\ell \frac{1}{\ell^2} e^{-\alpha\ell^2} \ell_\mu \ell_\nu = \frac{1}{(4\pi)^{D/2}} \frac{1}{D} \alpha^{-D/2} \delta_{\mu\nu}, \quad (\text{A19})$$

$$\int_\ell \frac{1}{\ell^2} e^{-\alpha\ell^2} \ell_\mu \ell_\nu \ell_\rho \ell_\sigma = \frac{1}{(4\pi)^{D/2}} \frac{1}{2(D+2)} \alpha^{-D/2-1} (\delta_{\mu\nu} \delta_{\rho\sigma} + \delta_{\mu\rho} \delta_{\nu\sigma} + \delta_{\mu\sigma} \delta_{\nu\rho}), \quad (\text{A20})$$

$$\begin{aligned} \int_\ell \frac{1}{\ell^2} e^{-\alpha\ell^2} \ell_\mu \ell_\nu \ell_\rho \ell_\sigma \ell_\alpha \ell_\beta \\ = \frac{1}{(4\pi)^{D/2}} \frac{1}{4(D+4)} \alpha^{-D/2-2} (\delta_{\mu\nu} \delta_{\rho\sigma} \delta_{\alpha\beta} + 14 \text{ permutations}). \end{aligned} \quad (\text{A21})$$

Then it is straightforward to find the coefficients of Eqs. (A15) and (A16), which directly make a contribution to the functions $F(t)$ and $G(t)$.

In Table A1, we summarize the contribution of each diagram computed in the above method in the unit of

$$\frac{1}{16\pi^2} N g_0^2. \quad (\text{A22})$$

In the last line of the table, “Z factors” implies the contributions of the one-loop operator renormalization factors, Z_T (2.13) and Z_M (2.22), through the tree-level diagram, Eq. (A9) (recall Eq. (2.10)). We see that those operator renormalization factors precisely cancel the residues of $1/\epsilon$ and make the coefficients $F(t)$ and $G(t)$ finite; this is precisely what we expect from the general argument. From the results in the table, we then have

$$F(t) = 1 + 2\eta_0 g^2 \left[\ln(\sqrt{8t\mu}) + \ln\sqrt{\pi} + \frac{7}{22} \right], \quad (\text{A23})$$

$$G(t) = -\frac{1}{2} \eta_0 g^2 \left[\ln(\sqrt{8t\mu}) + \ln\sqrt{\pi} + \frac{1}{11} \right]. \quad (\text{A24})$$

Finally, comparison with the formulas (A4), (A7), (4.21) and (4.23) shows the results quoted in Sect. 4, Eqs. (4.30) and (4.31). Note that the coefficients of $\ln(\sqrt{8t\mu})$ in the explicit one-loop calculation (Eqs. (A23) and (A24)) are in agreement with those by the general argument on the basis of the renormalization group equations and the trace anomaly (Eqs. (4.21) and (4.23)). This agreement provides a consistency check for our one-loop calculation and supports the correctness of our reasoning.

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Table A1 The contributions of each flow-line Feynman diagram (in the Feynman gauge) to the coefficients of Eqs. (A15) and (A16), respectively, in the unit of Eq. (A22). These correspond to the coefficient functions $F(t)$ and $G(t)$ in Eq. (A1). The numbers with * are corrected from previous versions of the present paper.

	$F(t)$	$G(t)$
Fig. A3	0	0
Fig. A4	$-3\frac{1}{\epsilon} - 3\ln(8\pi t) - 1$	0
Fig. A5	$-\frac{7}{36}$	$-\frac{49}{144}$
Fig. A6	$2\frac{1}{\epsilon} + 2\ln(8\pi t) - \frac{1}{2}$	0
Fig. A7	$\frac{19}{288}$	$\frac{121}{384}$
Fig. A8	$\frac{35^*}{96}$	$\frac{143^*}{384}$
Fig. A9	$-\frac{25}{8}$	0
Fig. A11	$\frac{11}{3\epsilon} + \frac{1}{3}\ln(8\pi t) - \frac{17}{36}$	$\frac{7}{12\epsilon} + \frac{7}{12}\ln(8\pi t) + \frac{1}{144}$
Fig. A13	$-\frac{51}{3\epsilon} - \frac{5}{3}\ln(8\pi t) + \frac{25}{36}$	$-\frac{31}{2\epsilon} - \frac{3}{2}\ln(8\pi t) - \frac{29}{16}$
Fig. A14	$3\frac{1}{\epsilon} + 3\ln(8\pi t) + 3$	0
Fig. A15	$3\frac{1}{\epsilon} + 3\ln(8\pi t) + \frac{5}{2}$	0
Fig. A16	1*	$\frac{31}{24}$
Z factors	$-\frac{11}{3\epsilon}$	$\frac{11}{12\epsilon}$

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P28: 准分子分布与梯度流

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阅读方法

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Quasi parton distributions and the gradient flow

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ABSTRACT: We propose a new approach to determining quasi parton distribution functions (PDFs) from lattice quantum chromodynamics. By incorporating the gradient flow, this method guarantees that the lattice quasi PDFs are finite in the continuum limit and evades the thorny, and as yet unresolved, issue of the renormalization of quasi PDFs on the lattice. In the limit that the flow time is much smaller than the length scale set by the nucleon momentum, the moments of the smeared quasi PDF are proportional to those of the light-front PDF. We use this relation to derive evolution equations for the matching kernel that relates the smeared quasi PDF and the light-front PDF.

KEYWORDS: Lattice Quantum Field Theory, Lattice QCD

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1 Introduction

Quantum chromodynamics (QCD) is the theory of the strong nuclear force that connects hadronic bound states to their partonic constituents, quarks and gluons. Although quarks and gluons cannot be directly accessed in experiments, the connection between hadrons and partons can be characterized through parton distribution functions (PDFs).

PDFs capture aspects of hadron structure associated with the momentum, angular momentum and spin of the constituent quarks and gluons, and play a central role in our understanding of high energy hadronic scattering processes (see, for example, [1–3]). Through factorization, the scattering amplitudes of simple scattering processes, such as deep inelastic scattering and Drell-Yan production, can be expressed as the convolution of perturbative coefficients and PDFs, which encapsulate the nonperturbative dynamics of QCD at hadronic scales.

In principle, the direct calculation of PDFs from QCD will provide new insight into hadronic structure, more stringent tests of QCD, and reduced systematic uncertainties in high energy scattering experiments. At present, however, the only systematic method for ab initio, nonperturbative QCD calculations is lattice QCD, in which QCD is formulated on a discrete Euclidean hypercube. PDFs are defined in terms of matrix elements of light-front wave functions and cannot be directly determined from Euclidean lattice QCD. PDFs are currently determined from global analyses of a wide range of scattering data (see, for example, [4–10] for a selection of recent analyses).

Lattice QCD calculations have instead focused on the first few Mellin moments of PDFs, which can be related to matrix elements of local twist-two operators, where twist is the dimension minus the spin of the operator. The lattice regulator breaks rotational

symmetry, which induces mixing between lattice operators that would not mix in the continuum. The mixing between twist-two operators of different mass dimension introduces power-divergent mixing on the lattice, preventing the extraction of more than three moments of PDFs [11, 12].

Recently a new approach to determining PDFs on the lattice was proposed, via Euclidean counterparts of PDFs generally referred to as quasi PDFs [13–19]. A similar framework was proposed in [20]. The quasi PDFs are Euclidean matrix elements determined at finite nucleon momentum. At large Euclidean momentum, the quasi PDFs can be related to the true PDFs through an effective theory expansion, the Large Momentum Effective Theory (LaMET).

Preliminary lattice calculations have been encouraging [15, 18], although both calculations have incorporated only a single lattice spacing and a full understanding of systematic uncertainties is far from complete. In particular, there are three challenges for the approach as it stands: the restriction to low nucleon momentum with the computational resources currently available; a full understanding of the renormalisation of extended Euclidean operators; and the precise relation between light-front PDFs and Euclidean quasi PDFs.

These difficulties can be broadly classified as either chiefly practical, or chiefly theoretical. The first challenge, that associated with the systematic uncertainties corresponding to low values of nucleon momentum on current lattices, is largely a practical issue. Studies in the spectator di-quark model [16] suggest that moderate improvements in computational resources, and new algorithms tailored to nucleons with large momentum [21], will likely solve this difficulty, at least to a precision that can contribute to global analyses of the PDFs in regions of parameter space that are experimentally inaccessible. We will not consider these practical difficulties any further and focus instead on the theoretical aspects of quasi PDFs.

We address one of the theoretical challenges by proposing a new approach to calculating quasi PDFs on the lattice, in which the lattice degrees of freedom are smeared via the gradient flow [22–24]. Using ringed fermions, which do not require any multiplicative wavefunction renormalization [25, 26], the corresponding lattice matrix elements remain finite in the continuum limit. This approach evades the problem of the power-divergence associated with the Wilson line operator that defines the quasi PDF. The renormalization of quasi PDFs has been viewed through the lens of heavy quark effective theory [17] and, more recently, a counterterm procedure has been proposed to remove this power-divergence [27, 28], but neither approach has been established beyond two loops in perturbation theory.

Here we examine the relation between the smeared quasi PDF and the light-front PDF, and focus on the limit in which the flow time is small compared to the length scale set by the nucleon momentum and the nucleon momentum is sufficiently large that higher twist effects can be neglected. In this limit, we express the moments of the smeared quasi PDF in terms of moments of the light-front PDF via a small flow-time expansion. The primary advantage of our approach is the finite continuum limit for nonperturbative matrix elements determined using lattice QCD. The matching between the smeared quasi PDF, regulated by the flow time, and the light-front PDF in, say, the $\overline{MS}$ scheme can be carried out in

the continuum and is independent of the details of the nonperturbative lattice calculation.

We start by revisiting the definitions of the light-front and quasi PDFs in Section 2. We then analyze the relation between the light-front and quasi PDFs in Section 3 and derive an evolution equation for the matching kernel in Section 4. We present our summary and conclusions in Section 5.

2 Distribution functions

Throughout this work we focus on flavor non-singlet unpolarized quasi and light-front PDFs. The extension to polarized quasi PDFs is straightforward. The flavor singlet case introduces additional mixing with the gluon distribution, but the principles are similar. We also assume that the quarks are massless and ignore complications arising from the correct treatment of heavy flavors, a subject of continued study for light-front PDFs (for reviews, see, for example, [29–31]).

2.1 Bare PDFs

In the following section, when we use the term “bare” we mean finite matrix elements determined with some regulator at finite cutoff. We leave the regulator implicit in this discussion, although one can have in mind dimensional regularization if desired. These bare matrix elements require renormalization in some scheme before one can remove the regulator (or, on the lattice, take the continuum limit). This usage follows that of the extensive discussions of light-front PDFs in, for example, [3, 32].

We denote bare light-front PDFs by $f^{(0)}(\xi)$. Light-front PDFs are frequently represented by $f_{j/N}^{(0)}(\xi)$, where j denotes the quark flavor and N the nucleon species, but here we will be considering only non-singlet distributions, for which we can neglect mixing between parton species, and work with sufficient generality that the nucleon species is not relevant to our discussion. We use light-front coordinates, $(x^+, x^-, \mathbf{x}_T)$ such that $x^\pm = (t \pm z)/\sqrt{2}$, and define $\xi = k^+/P^+$. We use ξ to distinguish this variable from the Bjorken- x parameter that characterizes the kinematics of scattering experiments and is given in terms of the experimental momentum transfer $Q^2 = -q^2$ and hadron momentum P by $x = Q^2/(2P \cdot q)$. The bare PDF is defined as [3]

$$f^{(0)}(\xi) = \int_{-\infty}^{\infty} \frac{d\omega^-}{4\pi} e^{-i\xi P^+ \omega^-} \left\langle P \left| T \bar{\psi}(0, \omega^-, \mathbf{0}_T) W(\omega^-, 0) \gamma^+ \frac{\lambda^a}{2} \psi(0) \right| P \right\rangle_C. \quad (2.1)$$

Here T is the time-ordering operator, ψ is a quark field, and the subscript C indicates that the vacuum expectation value has been subtracted (in other words, only connected contributions are included). The operator $W(\omega^-, 0)$ is the Wilson line,

$$W(\omega^-, 0) = \mathcal{P} \exp \left[-ig_0 \int_0^{\omega^-} dy^- \mathcal{A}_+^a(0, y^-, \mathbf{0}_T) T_a \right], \quad (2.2)$$

with $\mathcal{P}$ the path-ordering operator, g_0 the QCD bare coupling, and $A^\mu = A_\alpha^\mu T_\alpha$ the $SU(3)$ gauge potential with generator T_α (summation over color index α is implicit). The target

state, $|P\rangle$, is a spin-averaged, exact momentum eigenstate with relativistic normalization

$$\langle P'|P\rangle = (2\pi)^3 2P^+ \delta(P^+ - P'^+) \delta^{(2)}(\mathbf{P}_T - \mathbf{P}'_T). \quad (2.3)$$

We define the moments of bare PDFs as

$$a_0^{(n)} = \int_0^1 d\xi \xi^{n-1} \left[f^{(0)}(\xi) + (-1)^n \bar{f}^{(0)}(\xi) \right] = \int_{-1}^1 d\xi \xi^{n-1} f(\xi), \quad (2.4)$$

where $\bar{f}^{(0)}(\xi)$ is the anti-quark PDF and the second equality follows from the relation of the quark to anti-quark PDFs

$$f^{(0)}(-\xi) = -\bar{f}^{(0)}(\xi), \quad (2.5)$$

which holds for the bare distributions if the quark and anti-quarks fields are classical, or quantized using light-front quantization [32].

We can relate these bare moments, $a_0^{(n)}$, to matrix elements of twist-two operators via

$$\langle P | \mathcal{O}_0^{\{\mu_1 \dots \mu_n\}} | P \rangle = 2a_0^{(n)} (P^{\mu_1} \dots P^{\mu_n} - \text{traces}). \quad (2.6)$$

Here the bare twist-two operators are

$$\mathcal{O}_0^{\{\mu_1 \dots \mu_n\}} = i^{n-1} \bar{\psi}(0) \gamma^{\{\mu_1} D^{\mu_2} \dots D^{\mu_n\}} \frac{\lambda^a}{2} \psi(0) - \text{traces}. \quad (2.7)$$

In these expressions the braces denote symmetrization, D^μ is the symmetric covariant derivative, λ^a are $SU(2)$ flavor matrices, and the subtraction of the trace terms ensures that the operator transforms irreducibly under $SU(2)_L \otimes SU(2)_R$.

2.2 Renormalized PDFs

To this point we have considered the bare light-front PDFs, with the understanding that such objects are evaluated with some regulator that renders the bare distributions finite. We now introduce renormalized light-front PDFs. We stress that in this section we consider a renormalization scheme that respects rotational symmetry and, for definiteness, one can have in mind the $\overline{MS}$ scheme. Complications will arise if a regulator that breaks rotational invariance, such as the lattice regulator, is used. We do not discuss such complications here, because we will avoid explicit computations of moments at finite lattice spacing. All correlation functions computed on the lattice can be renormalized and extrapolated to the continuum limit, provided that no power divergent mixing exists. In the next section, we propose a smeared correlation function that does not have power-divergent mixing.

In general, renormalized light-front PDFs are written in terms of a kernel, $Z(\zeta/\xi, \mu)$, as

$$f(\xi, \mu) = \int_t^1 \frac{d\zeta}{\zeta} \mathcal{Z}\left(\frac{\zeta}{\xi}, \mu\right) f^{(0)}(\zeta), \quad (2.8)$$

where μ is some renormalization scale. We do not need to consider mixing between parton species for non-singlet distributions. In terms of the renormalized light-front PDF, the renormalized Mellin moments are

$$a^{(n)}(\mu) = \int_0^1 d\xi \xi^{n-1} \left[f(\xi, \mu) + (-1)^n \bar{f}(\xi, \mu) \right] = \int_{-1}^1 d\xi \xi^{n-1} f(\xi, \mu), \quad (2.9)$$

which can be related to matrix elements of renormalized twist-two operators, $\mathcal{O}^{(\nu_1 \dots \nu_n)}(\mu) = Z_{\mathcal{O}}(\mu) \mathcal{O}_0^{(\nu_1 \dots \nu_n)}$, via

$$\left\langle P | \mathcal{O}^{(\nu_1 \dots \nu_n)}(\mu) | P \right\rangle = 2a^{(n)}(\mu) (P^{\nu_1} \dots P^{\nu_n} - \text{traces}). \quad (2.10)$$

This relation holds provided the light-front PDFs and twist-two operators are renormalized in the same scheme [32].

2.3 Smeared quasi PDFs

We construct a finite quasi PDF matrix element by smearing both the fermion and gauge fields via the gradient flow [22–24]. The gradient flow is a deterministic evolution of the original quark and gluon fields in a new dimension, generally referred to as the “flow time”, towards a classical minimum of the QCD action [23, 24]. The flow-time evolution is chosen to remove ultraviolet fluctuations, which corresponds to smearing out the quark and gluon fields in real space, with a smearing scale that is proportional to the square-root of the flow time. Here we will not describe in detail the gradient flow, but refer the reader to the recent reviews [33, 34] for more details and applications.

For our purposes, it is sufficient that the gradient flow has the following properties. First, the gradient flow serves as a gauge-invariant ultraviolet regulator. Second, given a renormalized theory at zero flow time, the matrix elements of smeared fields are automatically finite, up to a multiplicative wave-function renormalization for the fermion fields [24], which can be removed by introducing ringed fermion fields [25, 26]. Third, the lattice matrix elements of smeared fields remain finite in the continuum limit, provided the flow time is fixed in physical units [24, 35]. In essence, the gradient flow allows one to replace the lattice regulator with a new smearing-scale regulator. This last fact allows one to determine the continuum limit of lattice matrix elements of, for example, twist-two operators, without power-divergent mixing. In the continuum, because the gradient flow respects rotational symmetry, the mixing between twist-two operators is then reduced to ordinary mixing with coefficients that depend on the smearing scale and not powers of the inverse lattice spacing [35].

We denote the ringed fermion fields at flow time τ by $\overline{\chi}(x; \tau)$ and $\chi(x; \tau)$, and the corresponding Wilson line at the same flow time, constructed from the smeared gauge fields $B_\mu(x; \tau)$, by $\mathcal{W}(0, z; \tau)$. We start with the matrix element

$$h^{(s)} \left(\frac{z}{\sqrt{\tau}}, \sqrt{\tau} P_z, \sqrt{\tau} \Lambda_{\text{QCD}}, \sqrt{\tau} M_N \right) = \frac{1}{2P_z} \left\langle P_z \left| \overline{\chi}(z; \tau) \mathcal{W}(0, z; \tau) \gamma_z \frac{\lambda^a}{2} \chi(0; \tau) \right| P_z \right\rangle_C, \quad (2.11)$$

which, being dimensionless, depends only on dimensionless combinations of scales. We note that the flow time has units of length-squared. The subscript C indicates that disconnected contributions to this matrix element have been removed. The ringed fermion fields require no wave function renormalization and this smeared matrix element is finite provided the flow time, τ , is non-zero and fixed in physical units, because correlation functions constructed from smeared fields are finite [23, 24]. Note that divergences will appear in the limit of vanishing flow time and the matrix element will then require renormalization.

We then define the quasi PDF [13, 14] as

$$q^{(s)}(\xi, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N) = \int_{-\infty}^{\infty} \frac{dz}{2\pi} e^{i\xi z} P_z h^{(s)}(\sqrt{\tau}z, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N), \quad (2.12)$$

where ξ is a dimensionless parameter that can be naively interpreted as the longitudinal momentum fraction of the parton in the nucleon N . This interpretation is not correct in Euclidean space, however, and instead ξ should be viewed as a dimensionless momentum variable in a Fourier transformation.

In practice, the smeared matrix element h is determined from lattice computations at finite lattice spacing, a , as

$$h^{(s)}\left(\frac{z}{\sqrt{\tau}}, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N\right) = \lim_{a \rightarrow 0} h\left(\frac{z}{a}, \frac{\sqrt{\tau}}{a}, aP_z, a\Lambda_{\text{QCD}}, aM_N\right), \quad (2.13)$$

where $\sqrt{\tau}$ and P_z are held fixed and

$$h\left(\frac{z}{a}, \frac{\sqrt{\tau}}{a}, aP_z, a\Lambda_{\text{QCD}}, aM_N\right) = \frac{1}{2aP_z} \left\langle aP_z \left| \chi\left(\frac{z}{a}, \frac{\sqrt{\tau}}{a}\right) W\left(0, \frac{z}{a}, \frac{\sqrt{\tau}}{a}\right) \gamma_5 \frac{\lambda^a}{2} \chi\left(0, \frac{\sqrt{\tau}}{a}\right) \right| aP_z \right\rangle_C. \quad (2.14)$$

3 Relation to light-front distributions

We discuss the relation between quasi and light-front PDFs by examining the Mellin moments of these distributions, and using the connection between Mellin moments and matrix elements of local operators, which are twist-two in the case of light-front PDFs [36]. For the quasi PDFs, the local operators corresponding to the Mellin moments do not have a well-defined twist, but can be related to twist-two operators after subtracting higher twist effects and applying target-mass corrections [15, 18]. Although we consider smeared matrix elements in this work, the arguments regarding higher twist and target mass effects in [15, 18] still apply, because the flow time serves as an alternative gauge-invariant regulator to the lattice spacing.

We connect the Mellin moments of the quasi PDF to matrix elements of local operators in the following way. Working in axial gauge, $B_z(x; \tau) = 0$, the matrix element $h^{(s)}$ is

$$h^{(s)}\left(\frac{z}{\sqrt{\tau}}, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N\right) \Big|_{B_z=0} = \frac{1}{2P_z} \left\langle P_z \left| \chi(z; \tau) \gamma_5 \frac{\lambda^a}{2} \chi(0; \tau) \right| P_z \right\rangle_C. \quad (3.1)$$

We now substitute this expression into the definition of the quasi PDF, Equation (2.12), and integrate the resulting expression over the full range of ξ . In contrast to the light-front PDF, this range extends from negative to positive infinity, giving

$$\int_{-\infty}^{\infty} d\xi q^{(s)}(\xi, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N) \Big|_{B_z=0} = h^{(s)}(0, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N) \Big|_{B_z=0}. \quad (3.2)$$

Here we have used the relation $\delta(zP_z) = \delta(z)/P_z$, for $P_z > 0$. We see that the first Mellin moment of the quasi PDF can be expressed in terms of the Euclidean matrix element of a local (smeared) operator.

We extend this argument to arbitrary moments of quasi PDFs by considering derivatives of the quasi distribution with respect to the spatial separation z [3]. Inverting the Fourier transform in Equation (2.12), we have

$$h^{(s)}\left(\frac{z}{\sqrt{\tau}}, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N\right) = \int_{-\infty}^{\infty} d\xi e^{-i\xi z P_z} q^{(s)}(\xi, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N). \quad (3.3)$$

Applying derivatives with respect to the displacement z , we obtain

$$\left(\frac{i}{P_z} \frac{\partial}{\partial z}\right)^{n-1} h^{(s)}\left(\frac{z}{\sqrt{\tau}}, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N\right) = \int_{-\infty}^{\infty} d\xi \xi^{n-1} e^{-i\xi z P_z} q^{(s)}(\xi, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N). \quad (3.4)$$

Defining the moments of the smeared quasi PDF, $b_n^{(s)}$, as

$$b_n^{(s)}\left(\sqrt{\tau}P_z, \frac{\Lambda_{\text{QCD}}}{P_z}, \frac{M_N}{P_z}\right) = \int_{-\infty}^{\infty} d\xi \xi^{n-1} q^{(s)}(\xi, \sqrt{\tau}P_z, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}M_N), \quad (3.5)$$

and substituting the definition of the matrix element $h^{(s)}$, given in Equation (2.11), into Equation (3.4), in the limit that $z \rightarrow 0$ we obtain

$$b_n^{(s)}\left(\sqrt{\tau}P_z, \frac{\Lambda_{\text{QCD}}}{P_z}, \frac{M_N}{P_z}\right)_{B_z=0} = \frac{c_n^{(s)}(\sqrt{\tau}P_z)}{2P_z^n} \times \left\langle P_z \left| \left[\overline{\chi}(z; \tau) \gamma_z \left(i \overleftrightarrow{\partial}_z^{n-1} \right) \frac{\lambda^x}{2} \chi(0; \tau) \right]_{z=0} \right| P_z \right\rangle_C. \quad (3.6)$$

The perturbative coefficients, $c_n^{(s)}(\sqrt{\tau}P_z)$, capture potential singularities in the righthand side of Equation (3.4) in the limit of vanishing separation z and vanishing flow time τ , and follow from a smeared operator product expansion [35] approach to the nonlocal matrix element, as outlined in [15].

We restore gauge invariance to obtain our final expression for the moments of quasi PDFs in terms of Euclidean matrix elements of local operators:

$$b_n^{(s)}\left(\sqrt{\tau}P_z, \frac{\Lambda_{\text{QCD}}}{P_z}, \frac{M_N}{P_z}\right) = \frac{c_n^{(s)}(\sqrt{\tau}P_z)}{2P_z^n} \left\langle P_z \left| \left[\overline{\chi}(z; \tau) \gamma_z (i \overleftrightarrow{D}_z)^{n-1} \frac{\lambda^x}{2} \chi(0; \tau) \right]_{z=0} \right| P_z \right\rangle_C. \quad (3.7)$$

The local operators that appear in the matrix element on the right hand side of this expression are not twist-two operators: they are not symmetric and traceless. The discrepancy between these matrix elements and matrix elements of twist-two operators are given by corrections that appear in powers of $\Lambda_{\text{QCD}}^2/P_z^2$ and M_N^2/P_z^2 [15, 18]. The terms

that scale as $\mathcal{O}(M_N^2/P_z^2)$ correspond to target mass corrections [37, 38]. Although the appropriate interpretation of PDFs in the presence of these target mass corrections is subtle [39, 40], for our purposes it is sufficient that these non-leading corrections can be absorbed by writing [15, 18]

$$b_n^{(s)} \left(\sqrt{\tau} P_z, \frac{\Lambda_{\text{QCD}}^2}{P_z^2} \right) = \frac{c_n^{(s)}(\sqrt{\tau} P_z)}{2P_z^n} \left\langle P_z \left| \left[\overline{\chi}(z; \tau) \gamma_z (i \overleftrightarrow{D}_z)^{(n-1)} \frac{\lambda^s}{2} \chi(0; \tau) \right] \right| P_z \right\rangle_C \times K_n^{-1} \left(\frac{M_N^2}{4P_z^2} \right), \quad (3.8)$$

where

$$K_n \left(\frac{M_N^2}{4P_z^2} \right) = \sum_{j=0}^{n/2} \binom{n-j}{j} \left(\frac{M_N^2}{4P_z^2} \right)^j. \quad (3.9)$$

The corrected matrix elements on the right hand side of this equation can now be expanded in a Taylor series with respect to $\Lambda_{\text{QCD}}^2/P_z^2$. The coefficients in this expansion represent higher twist effects that arise because the original matrix element is not a matrix element of a twist-two operator. The leading coefficient in this expansion is a twist-two contribution that can depend only on the nucleon structure and the flow time:

$$b_n^{(s)} \left(\sqrt{\tau} P_z, \sqrt{\tau} \Lambda_{\text{QCD}} \right) = c_n^{(s)}(\sqrt{\tau} P_z) b_n^{(s, \text{twist}-2)} \left(\sqrt{\tau} \Lambda_{\text{QCD}} \right) + \mathcal{O} \left(\frac{\Lambda_{\text{QCD}}^2}{P_z^2} \right), \quad (3.10)$$

where, for $\Lambda_{\text{QCD}}^2/P_z^2 \ll 1$, the higher twist corrections can be ignored.

In summary, we assume that: first, we can correct exactly for target mass corrections; and second, we can take the momentum P_z sufficiently large that higher twist effects are negligible. Then, under these assumptions, the moments of the smeared quasi PDFs are dimensionless products of perturbative coefficients and pure twist-two matrix elements, which are only functions of the dimensionless quantity $\sqrt{\tau} \Lambda_{\text{QCD}}$, that contain information about the structure of the hadron.

3.1 Short-distance expansion

We can now relate the moments of the smeared quasi PDF $b_n^{(s, \text{twist}-2)}(\sqrt{\tau} \Lambda_{\text{QCD}})$, which are local matrix elements of smeared fields, to the renormalized moments of the light-front PDFs, by using the properties of the gradient flow that arise from a short distance expansion [23, 25, 26, 41, 42]. The exponentially local nature of the smearing procedure allows for a short distance expansion of the smeared local operators in terms of renormalized operators in some renormalization scheme, such as the $\overline{MS}$ scheme. It is straightforward to show that this short distance expansion leads to

$$b_n^{(s, \text{twist}-2)}(\sqrt{\tau} \Lambda_{\text{QCD}}) = \tilde{C}_n^{(0)}(\sqrt{\tau} \mu) a^{(n)}(\mu) + \mathcal{O}(\sqrt{\tau} \Lambda_{\text{QCD}}), \quad (3.11)$$

where μ is a renormalization scale. The leading order term in this expansion, $a^{(n)}(\mu)$, is the matrix element of a renormalized twist-two operator with the same gamma matrix and derivative structure as the smeared operator that appears in the matrix element on the left

hand side. The higher order terms arise from higher dimension operators that enter the short distance expansion of the smeared matrix element.

We now combine this short-distance expansion with Equation (3.10) to write

$$b_n^{(s)}(\sqrt{\tau}\Lambda_{\text{QCD}}) = C_n^{(0)}(\sqrt{\tau}\mu, \sqrt{\tau}P_z)a^{(s)}(\mu) + \mathcal{O}\left(\sqrt{\tau}\Lambda_{\text{QCD}}, \frac{\Lambda_{\text{QCD}}^2}{P_z^2}\right), \quad (3.12)$$

Both the leading short distance coefficient function, $C_n^{(0)}(\sqrt{\tau}\mu, \sqrt{\tau}P_z)$, and the higher order corrections can be computed in perturbation theory, so that this approximation can be systematically improved.

For the rest of this discussion, we will assume that we work in a regime in which there is a hierarchy of scales given by

$$\Lambda_{\text{QCD}}, M_N \ll P_z \ll \tau^{-1/2}, \quad (3.13)$$

so that power corrections and higher-twist effects can be ignored. We also assume that target mass corrections have been applied.

To relate the smeared quasi PDF with the light-front PDF, we introduce a kernel function, $Z(x, \sqrt{\tau}\mu, \sqrt{\tau}P_z)$, whose Mellin moments are given by

$$\left[C_n^{(0)}(\sqrt{\tau}\mu, \sqrt{\tau}P_z)\right]^{-1} = \int_{-\infty}^{\infty} dx x^{n-1} Z(x, \sqrt{\tau}\mu, \sqrt{\tau}P_z). \quad (3.14)$$

With this definition, and using the properties of multiplicative convolution, we find

$$f(x, \mu) = \int_{-\infty}^{\infty} \frac{d\xi}{\xi} Z\left(\frac{x}{\xi}, \sqrt{\tau}\mu, \sqrt{\tau}P_z\right) q^{(s)}\left(\xi, \sqrt{\tau}\Lambda_{\text{QCD}}\right) + \mathcal{O}(\sqrt{\tau}\Lambda_{\text{QCD}}). \quad (3.15)$$

We introduce the inverse kernel through

$$C_n^{(0)}(\sqrt{\tau}\mu, \sqrt{\tau}P_z) = \int_{-\infty}^{\infty} dx x^{n-1} \tilde{Z}(x, \sqrt{\tau}\mu, \sqrt{\tau}P_z), \quad (3.16)$$

which leads to

$$q^{(s)}\left(x, \sqrt{\tau}\Lambda_{\text{QCD}}, \sqrt{\tau}P_z\right) = \int_{-1}^1 \frac{d\xi}{\xi} \tilde{Z}\left(\frac{x}{\xi}, \sqrt{\tau}\mu, \sqrt{\tau}P_z\right) f(\xi, \mu) + \mathcal{O}(\sqrt{\tau}\Lambda_{\text{QCD}}) \quad (3.17)$$

Note that all of these relations are only valid if

$$\Lambda_{\text{QCD}}, M_N \ll P_z \ll \tau^{-1/2}. \quad (3.18)$$

The kernel function can be computed in continuum perturbation theory, following the methods introduced in [23] and the examples in [17, 20, 27, 28]. Given that those computations are performed in the continuum, they are independent of the lattice formulation used to extract the smeared quasi-PDFs introduced in this paper and lattice computations of smeared quasi-PDFs can be performed with a variety of lattice actions, each of which results in the same universal continuum quasi-PDF. Consequently, lattice computations

are disconnected from the matching procedure of these universal, continuum quasi-PDFs to the light-front PDFs.

In both our approach, as in Ji's approach, the large nucleon momentum serves only to suppress higher twist contributions. We have, however, introduced a new scale, the (inverse) flow time, τ^{-1} , that serves as a regulator of ultraviolet divergences that arise from external composite operators that define the matrix element. The scale τ^{-1} needs to be large relative to hadronic scales but remains finite. These requirements for the hierarchy of scales, expressed in Equation (3.18), are no different in nature than the requirements used to factor physical cross-sections into PDFs and Wilson coefficients and are similar in spirit to the factorization approach proposed in [20, 28]. In this approach, the renormalization scale, which plays a similar role as our flow time scale, and the factorization scale are distinct and separate from the large momentum, which suppresses higher twist effects.

4 DGLAP-like equation for the matching kernel

Ignoring mixing between quark flavors and gluons (i.e. looking at the non-singlet distributions) the renormalized PDFs satisfy a DGLAP equation [43–45] that describes their scale dependence

$$\mu \frac{d f(x, \mu)}{d \mu} = \frac{\alpha_s(\mu)}{\pi} \int_x^1 \frac{d y}{y} f(y, \mu) P\left(\frac{x}{y}\right). \quad (4.1)$$

Here $P(z)$ is a function whose moments are given by

$$\int_0^1 d x x^{n-1} P(x) = \gamma^{(n)}, \quad (4.2)$$

where

$$\left[\mu \frac{d}{d \mu} - \frac{\alpha_s(\mu)}{\pi} \gamma^{(n)} \right] a^{(n)}(\mu) = 0, \quad (4.3)$$

and $\alpha_s(\mu)$ is the (renormalized) strong coupling constant.

Similarly, we can derive a DGLAP-like equation for the matching kernel that relates smeared quasi PDFs and light-front PDFs. We start from the small distance expansion in Equation (3.12), apply the renormalization group operator $\mu d/(d\mu)$, and use Equation (4.3) to derive a renormalization group equation for the short distance coefficient

$$\left[\mu \frac{d}{d \mu} + \frac{\alpha_s(\mu)}{\pi} \gamma^{(n)} \right] C_n^{(0)}(\sqrt{\tau} \mu, \sqrt{\tau} P_z) = 0 + \mathcal{O}(\sqrt{\tau} \Lambda_{\text{QCD}}), \quad (4.4)$$

and its inverse

$$\left[\mu \frac{d}{d \mu} - \frac{\alpha_s(\mu)}{\pi} \gamma^{(n)} \right] \left[C_n^{(0)}(\sqrt{\tau} \mu, \sqrt{\tau} P_z) \right]^{-1} = 0 + \mathcal{O}(\sqrt{\tau} \Lambda_{\text{QCD}}). \quad (4.5)$$

We can obtain a DGLAP-like equation for the matching kernel by substituting Equations (3.14) and (4.2) into this renormalization group equation, to give

$$\mu \frac{d}{d \mu} Z(x, \sqrt{\tau} \mu, \sqrt{\tau} P_z) = \frac{\alpha_s(\mu)}{\pi} \int_x^\infty \frac{d y}{y} Z(y, \sqrt{\tau} \mu, \sqrt{\tau} P_z) P\left(\frac{x}{y}\right), \quad (4.6)$$

and

$$\mu \frac{d}{d\mu} \tilde{Z}(x, \sqrt{\tau}\mu, \sqrt{\tau}P_2) = -\frac{\alpha_s(\mu)}{\pi} \int_x^\infty \frac{dy}{y} \tilde{Z}(y, \sqrt{\tau}\mu, \sqrt{\tau}P_2) P\left(\frac{x}{y}\right), \quad (4.7)$$

up to corrections of $\mathcal{O}(\sqrt{\tau}\Lambda_{\text{QCD}})$.

5 Conclusion

Parton distribution functions (PDFs) characterize nucleon structure in terms of the nucleon's constituent quarks and gluons. These PDFs are defined as matrix elements of light-front wave functions and cannot be directly calculated in Euclidean lattice QCD. In principle, however, the Mellin moments of PDFs can be calculated in lattice QCD, through matrix elements of twist-two operators. Unfortunately, these calculations are limited to the first few moments, because the hypercubic symmetry of the lattice regulator induces power-divergent mixing between twist-two operators of different mass dimension, obscuring the continuum limit of matrix elements determined on the lattice. Current determinations of PDFs rely on global analyses of data in a wide range of experimental channels and a determination of PDFs from first principles is lacking.

A new approach to determining PDFs in lattice QCD was recently proposed by Ji and subsequently, through a related framework, by Qiu and Ma. In this approach, one calculates Euclidean quasi PDFs at large nucleon momentum. Two recent lattice calculations provided promising results, but several aspects of the approach are yet to be fully understood. First, there is the practical issue of the systematic uncertainties associated with finite nucleon momenta in lattice calculations. This issue is likely to be resolved, to the extent that lattice calculations will have sufficient precision that results will provide useful input into global analyses where experimental data are inadequate, with improvements in computational resources and the algorithmic advances already underway. Second, there are theoretical issues to be clarified: the renormalization of the extended operator that defines the quasi PDFs; and the relation between the Euclidean quasi PDF and the light-front PDF, which to date had been analyzed through a factorization formula at one loop in perturbation theory.

We have addressed the first of these theoretical considerations by introducing a quasi PDF constructed from fields smeared via the gradient flow. We explicitly demonstrated that there is a simple relation between the Mellin moments of the smeared Euclidean quasi PDF and the renormalized Mellin moments of the light-front PDF, once nucleon mass corrections are incorporated and provided the flow time is small relative to the inverse nucleon momentum. Corrections to this relation appear at $\mathcal{O}(\Lambda_{\text{QCD}}^2/P_2^2)$, where P_2 is the Euclidean momentum of the nucleon, and $\mathcal{O}(\sqrt{\tau}\Lambda_{\text{QCD}})$, where τ is the flow time. From this correspondence it follows that, provided $\Lambda_{\text{QCD}}, M_N \ll P_2 \ll \tau^{-1/2}$, the quasi PDF and light-front PDF can be matched through a convolution relation.

The chief advantage of our approach is that the gradient flow renders the quasi PDF finite in the continuum limit and evades the issues of the renormalization of the non-local operator that defines the quasi PDF on the lattice. The resulting continuum matrix elements are independent of the choices of discretized action used to undertake lattice QCD

calculations and can be matched directly to the corresponding light-front PDFs in the $\overline{MS}$ scheme using continuum perturbation theory. Combined with a nonperturbative step-scaling procedure, this matching can be carried out at an energy sufficiently high that perturbative truncation errors are no longer uncontrolled. The nonperturbative implementation of our proposal is work in progress.

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P35: 含费米子场的格点能动量张量

主题归属	梯度流、涂抹与流时算符
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页码范围：P35-p001 – P35-p032

Lattice energy–momentum tensor from the Yang–Mills gradient flow—inclusion of fermion fields

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 Local products of fields deformed by the so-called Yang–Mills gradient flow become renormalized composite operators. This fact has been utilized to construct a correctly normalized conserved energy–momentum tensor in the lattice formulation of the pure Yang–Mills theory. In the present paper, this construction is further generalized for vector-like gauge theories containing fermions.

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1. Introduction and the main result

Energy and momentum are fundamental notions in physics. In lattice field theory (the best-developed non-perturbative formulation of quantum field theory), however, the construction of the corresponding Noether current, the energy-momentum tensor [1–4], is not straightforward [5, 6], because the translational invariance is explicitly broken by the lattice structure. In a recent paper [7], a possible method to avoid this complication being inherent in the energy-momentum tensor on the lattice has been proposed on the basis of the Yang–Mills gradient flow (or the Wilson flow in the context of lattice gauge theory) [8–10].¹ See Ref. [18] for a recent review on the gradient flow and Refs. [19–25] for its applications in lattice gauge theory.

The basic idea of Ref. [7] is the following:² Consider the pure Yang–Mills theory. The gradient flow deforms the bare gauge field $A_\mu(x)$ according to a flow equation with a flow time t [Eq. (3.1) below] and this makes gauge field configurations “smooth” for positive flow times. It can then be shown to all orders in perturbation theory that any local products of the flowed gauge field $B_\mu(t, x)$ for any strictly positive flow time t is ultraviolet (UV) finite when expressed in terms of renormalized parameters [9]. In particular, no multiplicative renormalization factor is required to make those local products finite. In other words, they are renormalized composite operators. Such UV-finite quantities should be “universal” in the sense that they are independent of the UV regularization chosen, in the limit in which the regulator is removed. This suggests a possibility that by using the gradient flow as an intermediate tool one may bridge composite operators defined with the dimensional regularization, with which the translational invariance is manifest,³ and those in the lattice regularization with which one may carry out non-perturbative calculations.⁴ Following this idea, a formula that expresses the correctly normalized conserved energy-momentum tensor as a $t \rightarrow 0$ limit of a certain combination of the flowed gauge field was derived [7]. This formula provides a possible method to compute correlation functions of the energy-momentum tensor by using the lattice regularization because the universal combination in the formula should be independent of the regularization. An interesting point is that the small flow-time behavior of the (universal) coefficients in the formula can be determined by perturbation theory thanks to the asymptotic freedom. This implies that if the lattice spacing is fine enough a further non-perturbative determination of the coefficients is not necessary. (Practically, non-perturbative determination of those coefficients may be quite useful and how this determination can be carried out has been investigated in Ref. [26].) Although the validity of the formula in Ref. [7], especially the restoration of the conservation law in the continuum limit, still remains to be carefully investigated, the measurement of the interaction measure (the trace anomaly) and the entropy density of the Yang–Mills theory at finite temperature on

¹In Refs. [11–16], an interesting method to define a lattice energy-momentum tensor from shifted boundary conditions has been developed. In Ref. [17], a method on the basis of the $\mathcal{N} = 1$ supersymmetry has been proposed.

²The following reasoning was inspired by pioneering experimentation by E. Itou and M. Kitazawa (unpublished).

³The drawback of the dimensional regularization is, of course, that it is defined only in perturbation theory.

⁴We thus implicitly assume that the finiteness of the flowed fields that can rigorously be proven only in perturbation theory persists even in the non-perturbative level.

the basis of the formula [27] shows encouraging results; the method appears to be promising even practically.

In Ref. [7], the method was developed only for the pure Yang-Mills theory. It is then natural to ask for wider application if the method can be generalized to gauge theories containing matter fields, especially fermion fields. In the present paper, we work out this generalization. We thus suppose a vector-like gauge theory⁵ with a gauge group G that contains

$$N_f \text{ Dirac fermions in the gauge representation } R. \quad (1.1)$$

For simplicity, we assume that all N_f fermions possess a common mass; this restriction might be appropriately relaxed.

Our main result is Eq. (4.70), and a step-by-step derivation of this master formula is given in subsequent sections. For those who are interested mainly in the final result, here we give a brief explanation of how to read the master formula (4.70): The left-hand side is the correctly normalized conserved energy-momentum tensor (with the vacuum expectation value subtracted); our formula (4.70) holds only when the energy-momentum tensor is separated from other operators in correlation functions in the position space. The combinations in the right-hand side are defined by Eqs. (4.1)–(4.5). There, $G_{\mu\nu}^a(t, x)$ and D_μ are the field strength and the covariant derivative of the flowed gauge field, respectively (the definition of the flowed gauge field is identical to that of Refs. [8–10]); our *ringed* flowed fermion fields $\hat{\chi}(t, x)$ and $\hat{\bar{\chi}}(t, x)$ are, on the other hand, somewhat different from those of Ref. [10], $\chi(t, x)$ and $\bar{\chi}(t, x)$, and they are related by Eqs. (3.20) and (3.21). The coefficient functions $c_i(t)$ in Eq. (4.70) are given by Eqs. (4.72)–(4.76). There, $\bar{g}(q)$ and $\bar{m}(q)$ are the running coupling and the running mass parameter defined by Eqs. (4.20) and (4.21), respectively; throughout this paper, m denotes the (common) renormalized mass of the fermions. Equations (4.72)–(4.76) are for the minimal subtraction (MS) scheme and the result for the modified minimal subtraction ($\overline{\text{MS}}$) scheme can be obtained by the replacement (4.77). b_0 and d_0 are the first coefficients of renormalization group functions, Eq. (2.16) and Eq. (2.18), respectively. The energy-momentum tensor is given by the $t \rightarrow 0$ limit in the right-hand side of Eq. (4.70). As in the pure Yang-Mills case mentioned above, the combination in the right-hand side is UV finite and one may use the lattice regularization to compute the correlation functions of the combination in the right-hand side. In this way, correlation functions of the correctly normalized conserved energy-momentum tensor are obtained. To ensure the “universality,” however, the continuum limit has to be taken before the $t \rightarrow 0$ limit. Practically, with a finite lattice spacing a , the flow time t cannot be taken arbitrarily small to keep the contact with the continuum physics. Instead, we have a natural constraint,

$$a \ll \sqrt{8t} \ll R, \quad (1.2)$$

where R denotes a typical physical scale, such as the hadronic scale or the box size. The extrapolation for $t \rightarrow 0$ thus generally requires a sufficiently fine lattice.

Here is our definition of the quadratic Casimir operators: We set the normalization of anti-Hermitian generators T^a of the representation R as $\text{tr}_R(T^a T^b) = -T(R)\delta^{ab}$ and $T^a T^a = -C_2(R)1$. We also denote $\text{tr}_R(1) = \dim(R)$. From the structure constants in $[T^a, T^b] =$

⁵We assume that the theory is asymptotically free.

$f^{abc}T^c$, we define $f^{acd}f^{bcd} = C_2(G)\delta^{ab}$. For example, for the fundamental N representation of $SU(N)$ for which $\dim(N) = N$, the conventional normalization is

$$C_2(SU(N)) = N, \quad T(N) = \frac{1}{2}, \quad C_2(N) = \frac{N^2 - 1}{2N}. \quad (1.3)$$

2. Energy-momentum tensor with dimensional regularization

The description of the energy-momentum tensor in gauge theory [3, 4] is particularly simple with the dimensional regularization.⁶ This is because this regularization manifestly preserves the (vectorial) gauge symmetry and the translational invariance. Thus, in this section, we briefly recapitulate basic facts concerning the energy-momentum tensor on the basis of the dimensional regularization.

The action of the system under consideration in a D dimensional Euclidean space is given by

$$S = \frac{1}{4g_0^2} \int d^D x F_{\mu\nu}^a(x) F_{\mu\nu}^a(x) + \int d^D x \bar{\psi}(x)(\not{D} + m_0)\psi(x), \quad (2.1)$$

where g_0 and m_0 are bare gauge coupling and the mass parameter, respectively. The field strength is defined by

$$F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) + [A_\mu(x), A_\nu(x)], \quad (2.2)$$

for $A_\mu(x) = A_\mu^a(x)T^a$ and $F_{\mu\nu}(x) = F_{\mu\nu}^a(x)T^a$, and the covariant derivative on the fermion is

$$D_\mu = \partial_\mu + A_\mu. \quad (2.3)$$

Here, and in what follows, the summation over N_f fermion flavors is always suppressed. Our gamma matrices are Hermitian and for the trace over the spinor index we set $\text{tr } 1 = 4$ for any D . We also set

$$D = 4 - 2\epsilon. \quad (2.4)$$

Assuming the dimensional regularization, one can derive a Ward-Takahashi relation associated with the translational invariance straightforwardly. We consider the following infinitesimal variation of integration variables in the functional integral:

$$\delta A_\mu(x) = \xi_\nu(x) F_{\nu\mu}(x), \quad \delta \psi(x) = \xi_\mu(x) D_\mu \psi(x). \quad (2.5)$$

Then, since the action changes as⁷

$$\delta S = - \int d^D x \xi_\nu(x) \partial_\mu T_{\mu\nu}(x), \quad (2.6)$$

where the energy-momentum tensor $T_{\mu\nu}(x)$ is defined by

$$\begin{aligned} T_{\mu\nu}(x) &\equiv \frac{1}{g_0^2} \left[F_{\mu\rho}^a(x) F_{\nu\rho}^a(x) - \frac{1}{4} \delta_{\mu\nu} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x) \right] \\ &\quad + \frac{1}{4} \bar{\psi}(x) \left(\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right) \psi(x) - \delta_{\mu\nu} \bar{\psi}(x) \left(\frac{1}{2} \overleftrightarrow{D} + m_0 \right) \psi(x), \end{aligned} \quad (2.7)$$

⁶ Ref. [28] is a very nice exposition of the dimensional regularization.

⁷ Here, to make the perturbation theory well defined, we implicitly assume the existence of the gauge-fixing term and the Faddey-Popov ghost fields. However, since they do not explicitly appear in correlation functions of gauge-invariant operators, we neglect these elements in what follows.

with

$$\overrightarrow{D}_\mu \equiv D_\mu - \overleftarrow{D}_\mu, \quad \overleftarrow{D}_\mu \equiv \overleftarrow{\partial}_\mu - A_\mu, \quad (2.8)$$

we have

$$\left\langle \mathcal{O}_{out} \int_D d^D x \partial_\mu T_{\mu\nu}(x) \mathcal{O}_{in} \right\rangle = - \langle \mathcal{O}_{out} \partial_\mu \mathcal{O}_{in} \rangle, \quad (2.9)$$

where $\mathcal{O}_{in}$ ($\mathcal{O}_{out}$) is a collection of gauge-invariant operators localized inside (outside) the finite integration region D . This relation shows that the energy-momentum tensor generates the infinitesimal translation and, at the same time, the bare quantity (2.7) does not receive the multiplicative renormalization. Thus, we define a renormalized finite energy-momentum tensor by subtracting its (possibly divergent) vacuum expectation value:

$$\{T_{\mu\nu}\}_R(x) \equiv T_{\mu\nu}(x) - \langle T_{\mu\nu}(x) \rangle. \quad (2.10)$$

A fundamental property of the energy-momentum tensor is the trace anomaly [29–33]. One simple way to derive this [34, 35] is to set $\xi_\mu(x) \propto x_\mu$ in Eq. (2.5) and compare the resulting relation with the renormalization group equation. After some consideration, this yields

$$\delta_{\mu\nu} \{T_{\mu\nu}\}_R(x) = -\frac{\beta}{2g^3} \{F_{\mu\sigma}^a F_{\nu\sigma}^a\}_R(x) - (1 + \gamma_m) m \{\bar{\psi}\psi\}_R(x), \quad (2.11)$$

where we assume that the renormalized operators in the right-hand side are defined in the MS scheme [28].⁸ Throughout the present paper, we always assume that the vacuum expectation value is subtracted in renormalized operators. In Eq. (2.11), the renormalization group functions β and γ_m are defined by

$$\beta \equiv \left(\mu \frac{\partial}{\partial \mu} \right)_0 g = -\frac{1}{2} g \left(\mu \frac{\partial}{\partial \mu} \right)_0 \ln Z, \quad (2.12)$$

$$\gamma_m \equiv - \left(\mu \frac{\partial}{\partial \mu} \right)_0 \ln m = -\beta \frac{\partial}{\partial g} \ln Z_m, \quad (2.13)$$

where g and m are the renormalized gauge coupling and the renormalized mass, respectively, and the derivative with respect to the renormalization scale μ is taken with all bare quantities kept fixed. The renormalization constants are defined by

$$g_0^2 = \mu^{2\epsilon} g^2 Z, \quad m_0 = m Z_m^{-1}. \quad (2.14)$$

The first few terms of the perturbative expansion of those renormalization functions read

$$\beta = -b_0 g^3 - b_1 g^5 + O(g^7), \quad \gamma_m = d_0 g^2 + d_1 g^4 + O(g^6), \quad (2.15)$$

where [36, 37]

$$b_0 = \frac{1}{(4\pi)^2} \left[\frac{11}{3} C_2(G) - \frac{4}{3} T(R) N_f \right], \quad (2.16)$$

$$b_1 = \frac{1}{(4\pi)^4} \left\{ \frac{34}{3} C_2(G)^2 - \left[4C_2(R) + \frac{20}{3} C_2(G) \right] T(R) N_f \right\}, \quad (2.17)$$

⁸ The renormalization in the MS scheme to the one-loop order is summarized in Appendix A.

and [38, 39]

$$d_0 = \frac{1}{(4\pi)^2} 6C_2(R), \quad (2.18)$$

$$d_1 = \frac{1}{(4\pi)^4} \left\{ 3C_2(R)^2 + \left[\frac{97}{3}C_2(G) - \frac{20}{3}T(R)N_f \right] C_2(R) \right\}. \quad (2.19)$$

3. Yang–Mills gradient flow

3.1. Flow equations and the perturbative expansion

The Yang–Mills gradient flow is a deformation of a gauge field configuration generated by a gradient flow in which the Yang–Mills action integral is regarded as a potential height. To be explicit, for the gauge potential $A_\mu(x)$, the flow is defined by [8, 9]

$$\partial_t B_\mu(t, x) = D_\nu G_{\nu\mu}(t, x) + \alpha_0 D_\mu \partial_\nu B_\nu(t, x), \quad B_\mu(t=0, x) = A_\mu(x), \quad (3.1)$$

where t is the flow time and $G_{\mu\nu}(t, x)$ is the field strength of the flowed field,

$$G_{\mu\nu}(t, x) = \partial_\mu B_\nu(t, x) - \partial_\nu B_\mu(t, x) + [B_\mu(t, x), B_\nu(t, x)], \quad (3.2)$$

and the covariant derivative on the gauge field is

$$D_\mu = \partial_\mu + [B_\mu, \cdot]. \quad (3.3)$$

The first term in the right-hand side of Eq. (3.1) is the “gradient” in the functional space, $-g_0^2 \delta S / \delta B_\mu(t, x)$, where S is the Yang–Mills action integral for the flowed field. Note that since $D_\nu G_{\nu\mu}(t, x) = \Delta B_\mu(t, x) + O(B^2)$, Eq. (3.1) is a sort of diffusion equation and the flow for $t > 0$ effectively suppresses high-frequency modes in the configuration. The second term in the right-hand side of Eq. (3.1) with the parameter α_0 is a “gauge-fixing term” that makes the perturbative expansion well defined. It can be shown, however, that any gauge-invariant quantities are independent of α_0 . In actual perturbative calculation in the next section, we adopt the “Feynman gauge” $\alpha_0 = 1$ with which the expressions become simplest.

The formal solution of Eq. (3.1) is given by [8, 9]

$$B_\mu(t, x) = \int d^D y \left[K_t(x-y)_{\mu\nu} A_\nu(y) + \int_0^t ds K_{t-s}(x-y)_{\mu\nu} R_\nu(s, y) \right], \quad (3.4)$$

where⁹

$$K_t(x)_{\mu\nu} = \int_p \frac{e^{ipx}}{p^2} \left[(\delta_{\mu\nu} p^2 - p_\mu p_\nu) e^{-tp^2} + p_\mu p_\nu e^{-\alpha_0 t p^2} \right] \quad (3.6)$$

is the heat kernel and

$$R_\mu = 2[B_\nu, \partial_\nu B_\mu] - [B_\nu, \partial_\mu B_\nu] + (\alpha_0 - 1)[B_\mu, \partial_\nu B_\nu] + [B_\nu, [B_\nu, B_\mu]] \quad (3.7)$$

denotes non-linear interaction terms. By iteratively solving Eq. (3.4), we have a perturbative expansion for the flowed field $B_\mu(t, x)$ in terms of the initial value $A_\nu(y)$.

⁹ Throughout the present paper, we use the abbreviation,

$$\int_p = \int \frac{d^D p}{(2\pi)^D}. \quad (3.5)$$

A similar flow may also be considered for fermion fields [10]. For our purpose, it is not necessary that the flow of fermion fields is a gradient flow of the original fermion action. A possible choice introduced in Ref. [10] is

$$\partial_t \chi(t, x) = [\Delta - \alpha_0 \partial_\mu B_\mu(t, x)] \chi(t, x), \quad \chi(t=0, x) = \psi(x), \quad (3.8)$$

$$\partial_t \bar{\chi}(t, x) = \bar{\chi}(t, x) \left[\bar{\Delta} + \alpha_0 \partial_\mu B_\mu(t, x) \right], \quad \bar{\chi}(t=0, x) = \bar{\psi}(x), \quad (3.9)$$

where

$$\Delta = D_\mu D_\mu, \quad D_\mu = \partial_\mu + B_\mu, \quad (3.10)$$

$$\bar{\Delta} = \bar{D}_\mu \bar{D}_\mu, \quad \bar{D}_\mu \equiv \overleftarrow{\partial}_\mu - B_\mu. \quad (3.11)$$

The formal solutions of the above flow equations are

$$\chi(t, x) = \int d^D y \left[K_t(x-y) \psi(y) + \int_0^t ds K_{t-s}(x-y) \Delta' \chi(s, y) \right], \quad (3.12)$$

$$\bar{\chi}(t, x) = \int d^D y \left[\bar{\psi}(y) K_t(x-y) + \int_0^t ds \bar{\chi}(s, y) \bar{\Delta}' K_{t-s}(x-y) \right], \quad (3.13)$$

where

$$K_t(x) \equiv \int_p e^{ipx} e^{-tp^2} = \frac{e^{-x^2/4t}}{(4\pi t)^{D/2}}, \quad (3.14)$$

and

$$\Delta' \equiv (1 - \alpha_0) \partial_\mu B_\mu + 2B_\mu \partial_\mu + B_\mu B_\mu, \quad (3.15)$$

$$\bar{\Delta}' \equiv -(1 - \alpha_0) \partial_\mu B_\mu - 2\overleftarrow{\partial}_\mu B_\mu + B_\mu B_\mu. \quad (3.16)$$

The initial values for the above flow, $A_\mu(x)$, $\psi(x)$, and $\bar{\psi}(x)$, are quantum fields being subject to the functional integral. The quantum correlation functions of the flowed fields are thus obtained by expressing the flowed fields in terms of original un-flowed fields (the initial values) and taking the functional average of the latter. Equations (3.4), (3.12), and (3.13) provide an explicit method to carry this out. For example, in the lowest (tree-level) approximation, we have

$$\langle B_\mu^a(t, x) B_\nu^b(s, y) \rangle = g_0^2 \delta^{ab} \delta_{\mu\nu} \int_p e^{ip(x-y)} \frac{e^{-(t+s)p^2}}{p^2}, \quad (3.17)$$

in the ‘‘Feynman gauge’’ in which $\lambda_0 = \alpha_0 = 1$, where λ_0 is the conventional gauge-fixing parameter. Similarly, for the fermion field, in the tree-level approximation,

$$\langle \chi(t, x) \bar{\chi}(s, y) \rangle = \int_p e^{ip(x-y)} \frac{e^{-(t+s)p^2}}{i\not{p} + m_0}. \quad (3.18)$$

Besides these ‘‘quantum propagators,’’ we also have heat kernels, Eqs. (3.6) and (3.14), in the perturbative expansion of Eqs. (3.4), (3.12), and (3.13).

We now explain a diagrammatic representation of the perturbative expansion of flowed fields (the flow Feynman diagram). For quantum propagators (3.17) and (3.18), we use the standard convention that the free propagator of the gauge boson is denoted by a wavy line and the free propagator of the fermion is denoted by an arrowed straight line. We stick to these conventions because these are quite natural in a system containing fermions.



Fig. 1 C01



Fig. 2 C02



Fig. 3 C03

In Refs. [7] and [9], on the other hand, the arrowed straight line was adopted to represent the gauge boson heat kernel (3.6). Since we already used this in this paper for the fermion propagator, we instead use “doubled lines” to represent heat kernels (3.6) and (3.14). For instance, Figs. 1–5 are one-loop flow Feynman diagrams which contribute to the two-point function of the flowed fermion field. In these figures, the doubled straight line represents the fermion heat kernel (3.14); the arrow denotes the flow of the fermion number, not the direction of the flow time. Similarly, in Fig. 15, the doubled wavy line is the gauge boson heat kernel (3.6). In the present representation, we thus lose the information of the direction of the flow time, which is represented by an arrow in Refs. [7] and [9]. This information, however, can readily be traced back.

Another element of the flow Feynman diagram is the vertex. The vertices that come from the original action (2.1) are denoted by filled circles, while vertices that come through the flow equations, Eqs. (3.7), (3.15), and (3.16), are denoted by open circles as in Figs. 1–5; these conventions for the vertex are identical to those of Refs. [7] and [9].

3.2. Ringed fermion fields

A salient feature of the flowed fields is the UV finiteness: Any correlation functions of the flowed gauge field $B_\mu(t, x)$ with strictly positive t are, when expressed in terms of renormalized parameters, UV finite *without* the multiplicative (wave function) renormalization [9]. Moreover, this finiteness persists even for the equal-point limit. Thus, any correlation functions of any local products of $B_\mu(t, x)$ (with $t > 0$) are UV finite without further renormalization other than the parameter renormalization. In other words, although those local products are given by a certain combination of the bare gauge field $A_\mu(x)$ through the flow equation, they are renormalized finite quantities. A basic reason for this UV finiteness is that the propagator of the flowed gauge field (3.17) contains the Gaussian damping factor $\sim e^{-tp^2}$ which effectively provides a UV cutoff for $t > 0$. To prove the above finiteness, however, one has to also utilize a Becchi-Rouet-Stora (BRS) symmetry underlying the present system that is *inhomogeneous* with respect to the gauge potential [9].

Regrettably, the above finiteness in the first sense does not hold for the flowed fermion field. It *requires* the wave function renormalization. Although its propagator (3.18) also possesses the damping factor $\sim e^{-tp^2}$, the BRS transformation is *homogeneous* on the fermion field (and on general matter fields) and the finiteness proof in Ref. [9] does not apply. In fact, computation of the one-loop diagrams in Figs. 1–5 (and diagrams with opposite arrows) shows that the wave function renormalization

$$\chi_R(t, x) = Z_\chi^{1/2} \chi(t, x), \quad \bar{\chi}_R(t, x) = Z_\chi^{1/2} \bar{\chi}(t, x), \quad Z_\chi = 1 + \frac{g^2}{(4\pi)^2} C_2(R) 3 \frac{1}{\epsilon} + O(g^4) \quad (3.19)$$

makes correlation functions UV finite [10]. Finiteness in the above second sense still holds: Any correlation functions of any local products of $\chi_R(t, x)$ and $\bar{\chi}_R(t, x)$ remain UV finite [10].



Fig. 4 C04



Fig. 5 C05

Although the finiteness in the second sense is quite useful for our purpose, we still need to incorporate the renormalization factor Z_χ in Eq. (3.19). This introduces a complication to our problem, because we have to find a matching factor between Z_χ in the dimensional regularization and that in the lattice regularization.

One possible way to avoid this complication is to normalize the fermion fields by the vacuum expectation value of the fermion kinetic operator:¹⁰

$$\hat{\chi}(t, x) = \sqrt{\frac{-2 \dim(R) N_f}{(4\pi)^2 t^2 \langle \bar{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \rangle}} \chi(t, x), \quad (3.20)$$

$$\hat{\bar{\chi}}(t, x) = \sqrt{\frac{-2 \dim(R) N_f}{(4\pi)^2 t^2 \langle \bar{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \rangle}} \bar{\chi}(t, x), \quad (3.21)$$

where

$$\overleftrightarrow{D}_\mu \equiv D_\mu - \overleftarrow{D}_\mu, \quad (3.22)$$

so that the multiplicative renormalization factor Z_χ is cancelled out in the new ringed variables. Note that the mass dimension of $\hat{\chi}(t, x)$ and $\hat{\bar{\chi}}(t, x)$ is $3/2$ for any D , while that of $\chi(t, x)$ and $\bar{\chi}(t, x)$ is $(D-1)/2$.

The vacuum expectation value of the kinetic operator in the lowest (one-loop) order approximation is given by diagram D01 in Fig. 6. For $D = 4 - 2\epsilon$ dimensions, we have

$$\langle \bar{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \rangle = \frac{-2 \dim(R) N_f}{(4\pi)^2 t^2} (8\pi t)^\epsilon [1 + O(m_0^2 t)]. \quad (3.23)$$

Note that the mass scale, which is required for the vacuum expectation value, is supplied by the flow time t in the present setup. Having obtained this expression, the constant factors in Eqs. (3.20) and (3.21) have been chosen so that the difference between the ringed and the original variables becomes $O(g_0^2)$ for sufficiently small flow time ($m_0^2 t \ll 1$).

The next-to-leading-order (i.e., two-loop) expression for the expectation value (3.23) is given by the flow Feynman diagrams in Figs. 7–13 (and diagrams with arrows with the opposite direction). The computation of these diagrams is somewhat complicated but can be completed in a similar manner to the calculation in Appendix B of Ref. [8] [with the integration formulas in our Appendix B, Eqs. (B1) and (B2)]. The contribution of each

¹⁰ In the kinetic operator, the summation over N_f fermion flavors is understood. In the first version of the present paper, we used the scalar condensation to normalize the fermion fields. This choice causes another complication associated with the massless fermion and the use of the kinetic operator seems much more appropriate.



Fig. 6 D01



Fig. 7 D02



Fig. 8 D03



Fig. 9 D04



Fig. 10 D05



Fig. 11 D06



Fig. 12 D07



Fig. 13 D08

diagram is tabulated in Table 1. In total, we have

$$\begin{aligned}
 & \left\langle \bar{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \right\rangle \\
 &= \frac{-2 \dim(R) N_t}{(4\pi)^2 t^2} \left\{ (8\pi t)^e + \frac{g_0^2}{(4\pi)^2} C_2(R) \left[-3 \frac{1}{\epsilon} - 6 \ln(8\pi t) + \ln(432) \right] + O(m_0^2 t) + O(g_0^4) \right\}.
 \end{aligned}
 \tag{3.24}$$

Table 1 Contribution of each diagram to Eq. (3.24) in units of $\frac{-2\dim(R)N_t}{(4\pi)^2 t^2} \frac{g^2}{(4\pi)^2} C_2(R)$.

diagram	
D02	$-\frac{1}{\epsilon} - 2\ln(8\pi t) + O(m_0^2 t)$
D03	$2\frac{1}{\epsilon} + 4\ln(8\pi t) + 2 + 4\ln 2 - 2\ln 3 + O(m_0^2 t)$
D04	$-20\ln 2 + 16\ln 3 + O(m_0^2 t)$
D05	$12\ln 2 - 5\ln 3 + O(m_0^2 t)$
D06	$-4\frac{1}{\epsilon} - 8\ln(8\pi t) - 2 + O(m_0^2 t)$
D07	$8\ln 2 - 4\ln 3 + O(m_0^2 t)$
D08	$-2\ln 3 + O(m_0^2 t)$

Using Eq. (A1) for the normalization factor in Eqs. (3.20) and (3.21), we have

$$\frac{-2\dim(R)N_t}{(4\pi)^2 t^2 \langle \tilde{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \rangle} = Z(\epsilon) \left\{ 1 + \frac{g^2}{(4\pi)^2} C_2(R) \left[3\frac{1}{\epsilon} - \Phi(t) \right] + O(m^2 t) + O(g^4) \right\}, \quad (3.25)$$

where

$$Z(\epsilon) \equiv \frac{1}{(8\pi t)^\epsilon}, \quad (3.26)$$

and

$$\Phi(t) \equiv -3\ln(8\pi\mu^2 t) + \ln(432). \quad (3.27)$$

4. Energy-momentum tensor constructed from the flowed fields

4.1. Small flow-time expansion and the renormalization group argument

To express the energy-momentum tensor in Eqs. (2.7) and (2.10) in terms of the flowed fields, we consider following second-rank symmetric tensors (which are even under the CP transformation) constructed from the flowed fields:

$$\tilde{\mathcal{O}}_{1\mu\nu}(t, x) \equiv G_{\mu\rho}^a(t, x) G_{\nu\rho}^a(t, x), \quad (4.1)$$

$$\tilde{\mathcal{O}}_{2\mu\nu}(t, x) \equiv \delta_{\mu\nu} G_{\rho\sigma}^a(t, x) G_{\rho\sigma}^a(t, x), \quad (4.2)$$

$$\tilde{\mathcal{O}}_{3\mu\nu}(t, x) \equiv \tilde{\chi}(t, x) \left(\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right) \tilde{\chi}(t, x), \quad (4.3)$$

$$\tilde{\mathcal{O}}_{4\mu\nu}(t, x) \equiv \delta_{\mu\nu} \tilde{\chi}(t, x) \overleftrightarrow{D}^2 \tilde{\chi}(t, x), \quad (4.4)$$

$$\tilde{\mathcal{O}}_{5\mu\nu}(t, x) \equiv \delta_{\mu\nu} m \tilde{\chi}(t, x) \tilde{\chi}(t, x). \quad (4.5)$$

Note that all the above operators $\tilde{\mathcal{O}}_{\mu\nu}(t, x)$ are of dimension 4 for any D .

We also introduce corresponding bare operators in the D -dimensional x -space:

$$\mathcal{O}_{1\mu\nu}(x) \equiv F_{\mu\rho}^a(x) F_{\nu\rho}^a(x), \quad (4.6)$$

$$\mathcal{O}_{2\mu\nu}(x) \equiv \delta_{\mu\nu} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x), \quad (4.7)$$

$$\mathcal{O}_{3\mu\nu}(x) \equiv \bar{\psi}(x) \left(\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right) \psi(x), \quad (4.8)$$

$$\mathcal{O}_{4\mu\nu}(x) \equiv \delta_{\mu\nu} \bar{\psi}(x) \overleftrightarrow{\not{D}} \psi(x), \quad (4.9)$$

$$\mathcal{O}_{5\mu\nu}(x) \equiv \delta_{\mu\nu} m_0 \bar{\psi}(x) \psi(x). \quad (4.10)$$

The mass dimension of $\mathcal{O}_{1\mu\nu}(x)$ and $\mathcal{O}_{2\mu\nu}(x)$ is 4, while that of $\mathcal{O}_{3\mu\nu}(x)$, $\mathcal{O}_{4\mu\nu}(x)$, and $\mathcal{O}_{5\mu\nu}(x)$ is D .

We now consider the situation in which the flow time t in Eqs. (4.1)–(4.5) is very small. Since a flowed field at position x with a flow time t is a combination of un-flowed fields in the vicinity of x of radius $\sim \sqrt{8t}$,¹¹ the operators (4.1)–(4.5) can be regarded as local operators in x -space in the $t \rightarrow 0$ limit. Since the bare operators in Eqs. (4.6)–(4.10) span a complete set of symmetric second-rank gauge-invariant local operators of dimension 4 (for $D \rightarrow 4$) which are even under CP, we have the following (asymptotic) expansion for small t :

$$\tilde{\mathcal{O}}_{\mu\nu}(t, x) = \left\langle \tilde{\mathcal{O}}_{\mu\nu}(t, x) \right\rangle + \zeta_{ij}(t) [\mathcal{O}_{5\mu\nu}(x) - \langle \mathcal{O}_{5\mu\nu}(x) \rangle] + \mathcal{O}(t), \quad (4.11)$$

where the abbreviated terms are contributions of operators of mass dimension 6 (for $D \rightarrow 4$) or higher.

In writing down the small flow-time expansion (4.11), we have assumed that there is no other D -dimensional composite operator at the point x . That the expansion (4.11) cannot necessarily hold when there is another operator at the point x [say, $\mathcal{P}(x)$] can be seen by noting that the product of the left-hand side of Eq. (4.11) with $\mathcal{P}(x)$ does not possess any divergence for $t > 0$, while each term in the right-hand side can have an equal-point singularity with $\mathcal{P}(x)$ because the operators in the right-hand side of Eq. (4.11) are D -dimensional (i.e., non-flowed) composite operators. This is a contradiction if Eq. (4.11) holds. This implies that the formula we will derive for the energy-momentum tensor below holds *only when the energy-momentum tensor has no overlap with other operators*.¹² In particular, we cannot say anything about whether the Ward-Takahashi relation (2.9) is reproduced with our construction. Still, our construction is expected to have a correct normalization because it is determined from matching the energy-momentum tensor in the dimensional regularization which fulfills Eq. (2.9). Our lattice energy-momentum tensor is thus useful to compute correlation functions in which the energy-momentum tensor is separated from other operators. This is the case for correlation functions relevant to the viscosities [40–42], for example.

¹¹ This follows from the fact that the flow equations are basically the diffusion equation.

¹² This important point was not fully recognized in Ref. [7].

The expansion (4.11) may be inverted as

$$\mathcal{O}_{\mu\nu}(x) - \langle \mathcal{O}_{\mu\nu}(x) \rangle = (\zeta^{-1})_{ij}(t) \left[\tilde{\mathcal{O}}_{j\mu\nu}(t, x) - \langle \tilde{\mathcal{O}}_{j\mu\nu}(t, x) \rangle \right] + O(t), \quad (4.12)$$

where ζ^{-1} denotes the inverse matrix of ζ . Then, by substituting this relation into the energy-momentum tensor (2.7) and (2.10) in terms of the bare operators,

$$\begin{aligned} \{T_{\mu\nu}\}_R(x) &= \frac{1}{g_0^2} \left\{ \mathcal{O}_{1\mu\nu}(x) - \langle \mathcal{O}_{1\mu\nu}(x) \rangle - \frac{1}{4} [\mathcal{O}_{2\mu\nu}(x) - \langle \mathcal{O}_{2\mu\nu}(x) \rangle] \right\} \\ &\quad + \frac{1}{4} [\mathcal{O}_{3\mu\nu}(x) - \langle \mathcal{O}_{3\mu\nu}(x) \rangle] - \frac{1}{2} [\mathcal{O}_{4\mu\nu}(x) - \langle \mathcal{O}_{4\mu\nu}(x) \rangle] \\ &\quad - [\mathcal{O}_{5\mu\nu}(x) - \langle \mathcal{O}_{5\mu\nu}(x) \rangle], \end{aligned} \quad (4.13)$$

we have the expression (for $D = 4$)

$$\begin{aligned} \{T_{\mu\nu}\}_R(x) &= c_1(t) \left[\tilde{\mathcal{O}}_{1\mu\nu}(t, x) - \frac{1}{4} \tilde{\mathcal{O}}_{2\mu\nu}(t, x) \right] \\ &\quad + c_2(t) [\tilde{\mathcal{O}}_{2\mu\nu}(t, x) - \langle \tilde{\mathcal{O}}_{2\mu\nu}(t, x) \rangle] \\ &\quad + c_3(t) [\tilde{\mathcal{O}}_{3\mu\nu}(t, x) - 2\tilde{\mathcal{O}}_{4\mu\nu}(t, x) - \langle \tilde{\mathcal{O}}_{3\mu\nu}(t, x) - 2\tilde{\mathcal{O}}_{4\mu\nu}(t, x) \rangle] \\ &\quad + c_4(t) [\tilde{\mathcal{O}}_{4\mu\nu}(t, x) - \langle \tilde{\mathcal{O}}_{4\mu\nu}(t, x) \rangle] \\ &\quad + c_5(t) [\tilde{\mathcal{O}}_{5\mu\nu}(t, x) - \langle \tilde{\mathcal{O}}_{5\mu\nu}(t, x) \rangle] + O(t), \end{aligned} \quad (4.14)$$

where

$$\begin{aligned} c_1(t) &= \tilde{c}_1(t), & c_2(t) &= \tilde{c}_2(t) + \frac{1}{4}c_1(t), \\ c_3(t) &= \tilde{c}_3(t), & c_4(t) &= \tilde{c}_4(t) + 2c_3(t), & c_5(t) &= \tilde{c}_5(t), \end{aligned} \quad (4.15)$$

and

$$\tilde{c}_i(t) \equiv \frac{1}{g_0^2} \left\{ \left(\zeta^{-1} \right)_{1i}(t) - \frac{1}{4} \left(\zeta^{-1} \right)_{2i}(t) \right\} + \frac{1}{4} \left(\zeta^{-1} \right)_{3i}(t) - \frac{1}{2} \left(\zeta^{-1} \right)_{4i}(t) - \left(\zeta^{-1} \right)_{5i}(t). \quad (4.16)$$

In Eq. (4.14), we have used the fact that the finite operator $\tilde{\mathcal{O}}_{1\mu\nu}(t, x) - (1/4)\tilde{\mathcal{O}}_{2\mu\nu}(t, x)$ is traceless in $D = 4$ and thus has no vacuum expectation value. Equation (4.14) shows that if one knows the $t \rightarrow 0$ behavior of the coefficients $c_i(t)$, the energy-momentum tensor can be obtained as the $t \rightarrow 0$ limit of the combination in the right-hand side. As already noted, since the composite operators (4.1)–(4.5) constructed from (ringed) flowed fields should be independent of the regularization adopted, one may use the lattice regularization to compute correlation functions of the quantity in the right-hand side of Eq. (4.14). This provides a possible method to compute correlation functions of the correctly normalized conserved energy-momentum tensor with the lattice regularization.

Thus, we are interested in the $t \rightarrow 0$ behavior of the coefficients $c_i(t)$ in Eq. (4.14). Quite interestingly, one can argue that the coefficients $c_i(t)$ can be evaluated by the perturbation theory for $t \rightarrow 0$ thanks to the asymptotic freedom. To see this, we apply

$$\left(\mu \frac{\partial}{\partial \mu} \right)_0 \quad (4.17)$$

on both sides of Eq. (4.14), where μ is the renormalization scale and the subscript 0 implies that the derivative is taken while all bare quantities are

kept fixed. Since the energy-momentum tensor is not multiplicatively renormalized, $(\mu\partial/\partial\mu)_0(\text{left-hand side of Eq. (4.14)}) = 0$. On the right-hand side, since $\tilde{O}_{1,2,3,4\mu\nu}(t, x)$ and $(1/m)\tilde{O}_{5\mu\nu}(t, x)$ are entirely given by bare quantities through the flow equations, we have

$$\left(\mu\frac{\partial}{\partial\mu}\right)_0 \tilde{O}_{1,2,3,4\mu\nu}(t, x) = \left(\mu\frac{\partial}{\partial\mu}\right)_0 \frac{1}{m} \tilde{O}_{5\mu\nu}(t, x) = 0. \quad (4.18)$$

These observations imply

$$\left(\mu\frac{\partial}{\partial\mu}\right)_0 c_{1,2,3,4}(t) = \left(\mu\frac{\partial}{\partial\mu}\right)_0 mc_5(t) = 0. \quad (4.19)$$

Then the standard renormalization group argument says that $c_{1,2,3,4}(t)$ and $mc_5(t)$ are independent of the renormalization scale, if the renormalized parameters in these quantities are replaced by running parameters defined by

$$q\frac{d\bar{g}(q)}{dq} = \beta(\bar{g}(q)), \quad \bar{g}(q = \mu) = g, \quad (4.20)$$

$$q\frac{d\bar{m}(q)}{dq} = -\gamma_m(\bar{g}(q))\bar{m}(q), \quad \bar{m}(q = \mu) = m, \quad (4.21)$$

where μ is the original renormalization scale. Thus, since $c_{1,2,3,4}(t)$ and $mc_5(t)$ are independent of the renormalization scale, two possible choices, $q = \mu$ and $q = 1/\sqrt{8t}$, should give an identical result. In this way, we infer that

$$c_{1,2,3,4}(t)(g, m; \mu) = c_{1,2,3,4}(t)(\bar{g}(1/\sqrt{8t}), \bar{m}(1/\sqrt{8t}); 1/\sqrt{8t}), \quad (4.22)$$

$$c_5(t)(g, m; \mu) = \frac{\bar{m}(1/\sqrt{8t})}{m} c_5(t)(\bar{g}(1/\sqrt{8t}), \bar{m}(1/\sqrt{8t}); 1/\sqrt{8t}), \quad (4.23)$$

where we have explicitly written the dependence of $c_i(t)$ on renormalized parameters and on the renormalization scale. Finally, since the running gauge coupling $\bar{g}(1/\sqrt{8t}) \rightarrow 0$ for $t \rightarrow 0$ thanks to the asymptotic freedom, we expect that we can compute $c_i(t)$ for $t \rightarrow 0$ by using the perturbation theory. Although we are interested in low-energy physics for which the perturbation theory is ineffective, the coefficients $c_i(t)$ in Eq. (4.14) for $t \rightarrow 0$ can be evaluated by perturbation theory; this might be regarded as a sort of factorization.

4.2. $c_i(t)$ to the one-loop order

We thus evaluate the above coefficients $c_i(t)$ in Eq. (4.14) to the one-loop order approximation. For this, we compute the mixing coefficients $\zeta_{ij}(t)$ in Eq. (4.11) to the one-loop order. Then $c_i(t)$ are obtained by Eqs. (4.15) and (4.16). The loop expansion of $\zeta_{ij}(t)$ would yield

$$\zeta_{ij}(t) = \delta_{ij} + \zeta_{ij}^{(1)}(t) + \zeta_{ij}^{(2)}(t) + \cdots, \quad (4.24)$$

for $j = 1$ and $j = 2$, where the superscript denotes the loop order, and for $j = 3, 4$, and 5 , by taking the factor in Eq. (3.26) into account, we set

$$\zeta_{ij}(t) = Z(\epsilon) \left[\delta_{ij} + \zeta_{ij}^{(1)}(t) + \zeta_{ij}^{(2)}(t) + \cdots \right]. \quad (4.25)$$



Fig. 14 A03



Fig. 15 A04



Fig. 16 A05

As Ref. [7], it is straightforward to compute $\zeta_{ij}^{(1)}(t)$.¹³ For example, to obtain $\zeta_{ij}^{(1)}(t)$ with $j = 1$ and 2, we consider the correlation function

$$\left\langle \mathcal{O}_{\mu\nu}(t, x) A_{\beta}^k(y) A_{\gamma}^c(z) \right\rangle. \quad (4.26)$$

In the Feynman gauge which we use throughout the present paper, this has the structure

$$g_0^2 \int_k \frac{e^{ik(x-y)}}{k^2} g_0^2 \int_{\ell} \frac{e^{i\ell(x-z)}}{\ell^2} \delta^{bc} \mathcal{M}_{\mu\nu, \beta\gamma}(k, \ell). \quad (4.27)$$

After expanding $\mathcal{M}_{\mu\nu, \beta\gamma}(k, \ell)$ to $O(k, \ell)$, we can make use of the following correspondence to read off the operator mixing:

$$ik_{\mu} i\ell_{\nu} \delta_{\beta\gamma} \rightarrow F_{\mu\rho}^a(x) F_{\nu\rho}^a(x), \quad ik \cdot i\ell \delta_{\mu\nu} \delta_{\beta\gamma} \rightarrow \frac{1}{4} \delta_{\mu\nu} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x). \quad (4.28)$$

In this way, we obtain $\zeta_{ij}^{(1)}(t)$ with $j = 1$ and 2.¹⁴

Similarly, to obtain $\zeta_{ij}^{(1)}(t)$ with $j = 3, 4$, and 5, we consider

$$\left\langle \mathcal{O}_{\mu\nu}(t, x) \psi(y) \bar{\psi}(z) \right\rangle, \quad (4.29)$$

whose general structure reads

$$\int_k \frac{e^{ik(y-x)}}{i\vec{k} + m_0} \mathcal{M}_{\mu\nu}(k, \ell) \int_{\ell} \frac{e^{i\ell(x-z)}}{i\vec{\ell} + m_0}. \quad (4.30)$$

We then expand $\mathcal{M}_{\mu\nu}(k, \ell)$ to $O(k)$ and $O(\ell)$ and use the correspondence

$$\gamma_{\mu} i(k + \ell)_{\nu} + \gamma_{\nu} i(k + \ell)_{\mu} \rightarrow \bar{\psi}(x) \left[\gamma_{\mu} \overleftrightarrow{D}_{\nu} + \gamma_{\nu} \overleftrightarrow{D}_{\mu} \right] \psi(x), \quad \delta_{\mu\nu} \rightarrow \delta_{\mu\nu} \bar{\psi}(x) \psi(x), \quad (4.31)$$

to read off the operator mixing.

For $\zeta_{ij}^{(1)}(t)$, diagrams A03, A04, A05, A06, A07, A08, A09, A11, A13, A14, A15, A16, and A19 in Figs. 14–30 contribute.¹⁵ In these and following diagrams, the cross generically

¹³The justification for the following computational prescription was not well explained in previous versions of the present paper: One may wonder why the one-loop matching coefficients $\zeta_{ij}^{(1)}(t)$ can be read off from the correlation functions (4.26) and (4.29) alone, without computing corresponding correlation functions in which flowed composite operators $\mathcal{O}_{\mu\nu}(t, x)$ are replaced by bare ones $\mathcal{O}_{\mu\nu}(x)$. The justification is directly related to our way of treatment of infrared (IR) divergences and we supplement detailed explanation in Appendix D. We are quite grateful to a referee of the present paper for suggestions on this point.

¹⁴For the momentum integration, we use the integration formulas in Appendix B.

¹⁵Diagrams A10, A12, A17 (where the dotted line represents the ghost propagator), and A18 in these figures correspond to the conventional wave function renormalization and should be omitted in computing the operator mixing.



Fig. 17 A06



Fig. 18 A07



Fig. 19 A08



Fig. 20 A09



Fig. 21 A10



Fig. 22 A11



Fig. 23 A12



Fig. 24 A13



Fig. 25 A14



Fig. 26 A15



Fig. 27 A16



Fig. 28 A17

represents one of composite operators in Eqs. (4.1)–(4.5) [in the present case, $\hat{\mathcal{O}}_{1\mu\nu}(t, x)$]. Apart from A19, we can borrow the results from Ref. [7] for these diagrams. For completeness, these results are reproduced in the present convention in Table C1 of Appendix C. Combined



Fig. 29 A18



Fig. 30 A19



Fig. 31 B03



Fig. 32 B04



Fig. 33 B05

with the contribution of diagram A19, we have

$$\zeta_{11}^{(1)}(t) = \frac{g_0^2}{(4\pi)^2} C_2(G) \left[\frac{11}{3} \epsilon(t)^{-1} + \frac{7}{3} \right], \quad (4.32)$$

$$\zeta_{12}^{(1)}(t) = \frac{g_0^2}{(4\pi)^2} C_2(G) \left[-\frac{11}{12} \epsilon(t)^{-1} - \frac{1}{6} \right], \quad (4.33)$$

$$\zeta_{13}^{(1)}(t) = \frac{g_0^4}{(4\pi)^2} C_2(R) \left[-\frac{2}{3} \epsilon(t)^{-1} - \frac{7}{18} \right], \quad (4.34)$$

$$\zeta_{14}^{(1)}(t) = \frac{g_0^4}{(4\pi)^2} C_2(R) \left[\frac{1}{3} \epsilon(t)^{-1} - \frac{7}{18} \right], \quad (4.35)$$

$$\zeta_{15}^{(1)}(t) = \frac{g_0^4}{(4\pi)^2} C_2(R) \left[3\epsilon(t)^{-1} + \frac{5}{2} \right], \quad (4.36)$$

where

$$\epsilon(t)^{-1} \equiv \frac{1}{\epsilon} + \ln(8\pi t). \quad (4.37)$$

By considering the trace part of $\mathcal{O}_{1\mu\nu}$, from these we further have

$$\zeta_{21}^{(1)}(t) = 0, \quad (4.38)$$

$$\zeta_{22}^{(1)}(t) = \zeta_{11}^{(1)}(t) + \zeta_{12}^{(1)}(t)(4 - 2\epsilon) = \frac{g_0^2}{(4\pi)^2} C_2(G) \frac{7}{2}, \quad (4.39)$$

$$\zeta_{23}^{(1)}(t) = 0, \quad (4.40)$$

$$\zeta_{24}^{(1)}(t) = 2\zeta_{13}^{(1)}(t) + \zeta_{14}^{(1)}(t)(4 - 2\epsilon) = \frac{g_0^4}{(4\pi)^2} C_2(R)(-3), \quad (4.41)$$

$$\zeta_{25}^{(1)}(t) = \zeta_{15}^{(1)}(t)(4 - 2\epsilon) = \frac{g_0^4}{(4\pi)^2} C_2(R) [12\epsilon(t)^{-1} + 4]. \quad (4.42)$$

For $\zeta_{3j}^{(1)}(t)$ with $j = 1$ and 2 , diagrams B03, B04, B08, B09, B10, B11, and B12 in Figs. 31–40 contribute. The contribution of each diagram is tabulated in Table 2. For $\zeta_{3j}^{(1)}(t)$ with $j =$



Fig. 34 B06



Fig. 35 B07



Fig. 36 B08



Fig. 37 B09



Fig. 38 B10



Fig. 39 B11



Fig. 40 B12



Fig. 41 B13



Fig. 42 B14



Fig. 43 B15



Fig. 44 B16



Fig. 45 B17



Fig. 46 B18

3, 4, and 5, diagrams B06, B07, B13, B14, B15, B16, B17, and B18 in Figs. 34–46 contribute and their contributions are tabulated in Table 3. As the sum of these contributions, we have

Table 2 $\zeta_{3j}^{(1)}$ in units of $\frac{1}{(4\pi)^2}T(R)N_f$

diagram	$\zeta_{31}^{(1)}(t)$	$\zeta_{32}^{(1)}(t)$
B03	$-\frac{16}{3}\epsilon(t)^{-1} - \frac{64}{9}$	$\frac{4}{3}\epsilon(t)^{-1} + \frac{25}{9}$
B04	0	0
B08	0	$-\frac{1}{3}$
B09	$-\frac{5}{12}$	$-\frac{23}{144}$
B10	$\frac{5}{12}$	$-\frac{1}{16}$
B11	$\frac{10}{9}$	$\frac{7}{9}$
B12	0	0

$$\zeta_{31}^{(1)}(t) = \frac{1}{(4\pi)^2}T(R)N_f \left[-\frac{16}{3}\epsilon(t)^{-1} - 6 \right], \quad (4.43)$$

$$\zeta_{32}^{(1)}(t) = \frac{1}{(4\pi)^2}T(R)N_f \left[\frac{4}{3}\epsilon(t)^{-1} + 3 \right], \quad (4.44)$$

$$\zeta_{33}^{(1)}(t) = \frac{g^2}{(4\pi)^2}C_2(R) \left[3\frac{1}{\epsilon} - \Phi(t) \right] + \frac{g_0^2}{(4\pi)^2}C_2(R) \left[-\frac{1}{3}\epsilon(t)^{-1} + \frac{1}{18} \right], \quad (4.45)$$

$$\zeta_{34}^{(1)}(t) = \frac{g_0^2}{(4\pi)^2}C_2(R) \left[-\frac{4}{3}\epsilon(t)^{-1} - \frac{4}{9} \right], \quad (4.46)$$

$$\zeta_{35}^{(1)}(t) = 0, \quad (4.47)$$

where in $\zeta_{33}^{(1)}(t)$ (4.45) the first term in the right-hand side comes from the conversion from the un-ringed fields to the ringed fields in Eqs. (3.20) and (3.21)—recall Eq. (3.25); the

Table 3 $\zeta_{ij}^{(1)}$ in units of $\frac{g^2}{(4\pi)^2}C_2(R)$

diagram	$\zeta_{33}^{(1)}(t)$	$\zeta_{34}^{(1)}(t)$	$\zeta_{35}^{(1)}(t)$
B06	$-\frac{1}{3}\epsilon(t)^{-1} + \frac{1}{18}$	$\frac{2}{3}\epsilon(t)^{-1} + \frac{5}{9}$	$8\epsilon(t)^{-1} + 8$
B07	$2\epsilon(t)^{-1} + 2$	$-2\epsilon(t)^{-1} - 2$	$-8\epsilon(t)^{-1} - 8$
B13	0	1	0
B14	0	0	0
B15	$2\epsilon(t)^{-1} + 2$	0	0
B16	0	0	0
B17	-2	0	0
B18	$-4\epsilon(t)^{-1} - 2$	0	0

combination $\Phi(t)$ is given by Eq. (3.27). From these, we further have

$$\zeta_{41}^{(1)}(t) = 0, \quad (4.48)$$

$$\zeta_{42}^{(1)}(t) = \frac{1}{2}\zeta_{31}^{(1)}(t) + \frac{1}{2}\zeta_{32}^{(1)}(t)(4 - 2\epsilon) = \frac{1}{(4\pi)^2}T(R)N_1\frac{5}{3}, \quad (4.49)$$

$$\zeta_{43}^{(1)}(t) = 0, \quad (4.50)$$

$$\begin{aligned} \zeta_{44}^{(1)}(t) &= \zeta_{33}^{(1)}(t) + \frac{1}{2}\zeta_{34}^{(1)}(t)(4 - 2\epsilon) \\ &= \frac{g^2}{(4\pi)^2}C_2(R) \left[3\frac{1}{\epsilon} - \Phi(t) \right] + \frac{g_0^2}{(4\pi)^2}C_2(R) \left[-3\epsilon(t)^{-1} + \frac{1}{2} \right], \end{aligned} \quad (4.51)$$

$$\zeta_{45}^{(1)}(t) = 0. \quad (4.52)$$

Finally,

$$\zeta_{51}^{(1)}(t) = \zeta_{52}^{(1)}(t) = \zeta_{53}^{(1)}(t) = \zeta_{54}^{(1)}(t) = 0, \quad (4.53)$$

and $\zeta_{55}^{(1)}(t)$ is given by the sum of the contributions of one-loop diagrams in Table 4 and the conversion factor to the ringed fields:

$$\zeta_{55}^{(1)}(t) = \frac{g^2}{(4\pi)^2}C_2(R) \left[6\frac{1}{\epsilon} - \Phi(t) \right] + \frac{g_0^2}{(4\pi)^2}C_2(R) \left[-6\epsilon(t)^{-1} - 2 \right]. \quad (4.54)$$

We have now obtained all $\zeta_{ij}^{(1)}(t)$ in Eqs. (4.24) and (4.25). Then, since the matrix $\zeta_{ij}(t)$ in the tree-level approximation is a unit matrix, it is straightforward to invert the matrix $\zeta_{ij}(t)$

Table 4 $\zeta_{55}^{(1)}(t)$ in units of $\frac{g^2}{(4\pi)^2}C_2(R)$.

diagram	$\zeta_{55}^{(1)}(t)$
B06	$-4\epsilon(t)^{-1} - 2$
B13	0
B14	0
B15	$2\epsilon(t)^{-1} + 2$
B16	0
B18	$-4\epsilon(t)^{-1} - 2$

in the one-loop approximation; Eqs. (4.15) and (4.16) thus yield

$$c_1(t) = \frac{1}{g_0^2} \left[1 - \zeta_{11}^{(1)}(t) \right] - \frac{1}{4} \zeta_{31}^{(1)}(t), \quad (4.55)$$

$$c_2(t) = \frac{1}{g_0^2} \left(-\frac{1}{2}\epsilon \right) \zeta_{12}^{(1)}(t) + \left(\frac{3}{4} - \frac{1}{2}\epsilon \right) \zeta_{32}^{(1)}(t) + \frac{3}{16} \zeta_{31}^{(1)}(t), \quad (4.56)$$

$$c_3(t) = \left\{ \frac{1}{g_0^2} \left[-\zeta_{13}^{(1)}(t) \right] + \frac{1}{4} - \frac{1}{4} \zeta_{33}^{(1)}(t) \right\} Z(\epsilon)^{-1}, \quad (4.57)$$

$$c_4(t) = \left\{ \frac{1}{g_0^2} \left[-\frac{1}{2}\epsilon \zeta_{14}^{(1)}(t) - \frac{3}{2} \zeta_{13}^{(1)}(t) \right] + \left(\frac{3}{4} - \frac{1}{2}\epsilon \right) \zeta_{34}^{(1)}(t) \right\} Z(\epsilon)^{-1}, \quad (4.58)$$

$$c_5(t) = \left\{ \frac{1}{g_0^2} \left[-\frac{1}{2}\epsilon \zeta_{15}^{(1)}(t) \right] - 1 + \zeta_{55}^{(1)}(t) \right\} Z(\epsilon)^{-1}. \quad (4.59)$$

Then, by using the renormalized gauge coupling in the MS scheme (A1), to the one-loop order, we have (for $\epsilon \rightarrow 0$)

$$c_1(t) = \frac{1}{g^2} - b_0 \ln(8\pi\mu^2 t) - \frac{1}{(4\pi)^2} \left[\frac{7}{3} C_2(G) - \frac{3}{2} T(R) N_f \right], \quad (4.60)$$

$$c_2(t) = \frac{1}{8} \frac{1}{(4\pi)^2} \left[\frac{11}{3} C_2(G) + \frac{11}{3} T(R) N_f \right], \quad (4.61)$$

$$c_3(t) = \frac{1}{4} \left\{ 1 + \frac{g^2}{(4\pi)^2} C_2(R) \left[\frac{3}{2} + \ln(432) \right] \right\}, \quad (4.62)$$

$$c_4(t) = \frac{1}{8} d_0 g^2, \quad (4.63)$$

$$c_5(t) = - \left\{ 1 + \frac{g^2}{(4\pi)^2} C_2(R) \left[3 \ln(8\pi\mu^2 t) + \frac{7}{2} + \ln(432) \right] \right\}. \quad (4.64)$$

Since the $c_i(t)$ in Eq. (4.14) connect the finite energy-momentum tensor and UV-finite local products $\tilde{\mathcal{O}}_{\mu\nu}(t, x)$ constructed from (ringed) flowed fields, they should be UV finite. That our explicit one-loop calculation of $c_i(t)$ confirms this finiteness is quite reassuring.

If one prefers the $\overline{\text{MS}}$ scheme instead of the MS scheme assumed in above expressions, it suffices to make the replacement

$$\mu^2 \rightarrow \frac{e^{\gamma_E}}{4\pi} \mu^2, \quad (4.65)$$

where γ_E is Euler's constant.

4.3. A consistency check: The trace anomaly

It is interesting to see that Eq. (4.14) with $c_i(t)$ in Eqs. (4.60)–(4.64) in fact reproduces the trace anomaly (2.11) in the one-loop approximation. At first brief glance, $c_2(t)$ in Eq. (4.61) is incompatible with the correct trace anomaly, because $\delta_{\mu\nu}[\tilde{\mathcal{O}}_{\mu\nu} - (1/4)\tilde{\mathcal{O}}_{2\mu\nu}] = 0$ for $D = 4$ and $\delta_{\mu\nu}\tilde{\mathcal{O}}_{2\mu\nu} = 4 \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) + \mathcal{O}(g^2)$ [the last equality follows from Eq. (A17)]. On the other hand, $4c_2(t)$ from Eq. (4.61) is not identical to $b_0/2$, where b_0 is the first coefficient of the beta function (2.16), the correct one-loop coefficient of the trace anomaly.

This is a premature judgment, however. In fact, $\zeta_{42}^{(1)}(t)$ in Eq. (4.49) shows that there exists an operator mixing of the form

$$\hat{\chi}(t, x) \overleftrightarrow{D} \hat{\chi}(t, x) = \left\{ \bar{\psi} \overleftrightarrow{D} \psi \right\}_R(x) + \frac{1}{(4\pi)^2} T(R) N_f \frac{5}{3} \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) + \dots \quad (4.66)$$

Then the last term precisely fills the difference between $4c_2(t)$ and $b_0/2$.

In this way, to the one-loop order, we have

$$\begin{aligned} \delta_{\mu\nu} \{T_{\mu\nu}\}_R(x) &= \frac{1}{2} \frac{1}{(4\pi)^2} \left[\frac{11}{3} C_2(G) - \frac{4}{3} T(R) N_f \right] \{F_{\rho\sigma}^a F_{\rho\sigma}^a\}_R(x) \\ &\quad - \frac{3}{2} \left\{ \bar{\psi} \overleftrightarrow{D} \psi \right\}_R(x) - \left[4 + \frac{g^2}{(4\pi)^2} 6C_2(R) \right] m \{ \bar{\psi} \psi \}_R(x). \end{aligned} \quad (4.67)$$

This reproduces the trace anomaly (2.11) in the one-loop level if one uses the equation of motion of renormalized field,

$$\left\{ \bar{\psi} \overleftrightarrow{D} \psi \right\}_R(x) = -2m \{ \bar{\psi} \psi \}_R(x), \quad (4.68)$$

whose use is justified when there is no other operator in the point x as we are assuming. We observe that our one-loop result in Eqs. (4.60)–(4.64) is consistent with the trace anomaly.

In Ref. [7], for the pure Yang–Mills theory, the next-to-leading (two-loop order) term in $c_2(t)$ was determined as

$$c_2(t) = \frac{1}{8} b_0 - \frac{1}{8} b_0 \left[\frac{1}{(4\pi)^2} G_2(G) \frac{7}{2} - \frac{b_1}{b_0} \right] g^2, \quad (4.69)$$

where $b_0 = [1/(4\pi)^2](11/3)C_2(G)$ and $b_1 = [1/(4\pi)^4](34/3)C_2(G)^2$, by imposing that the expression (4.14) reproduces the trace anomaly (2.11) to the two-loop order. For the present system with fermions, however, it seems that this requirement alone cannot fix the next-to-leading terms in $c_2(t)$ and in $c_4(t)$; so we are content with the one-loop formulas, Eqs. (4.60)–(4.64), in the present paper treating a system containing fermions.

4.4. Master formula

From Eq. (4.14), the energy-momentum tensor is given by the $t \rightarrow 0$ limit,

$$\begin{aligned} \langle T_{\mu\nu} \rangle_R(x) = \lim_{t \rightarrow 0} & \left\{ c_1(t) \left[\langle \mathcal{O}_{1\mu\nu}(t, x) \rangle - \frac{1}{4} \langle \mathcal{O}_{2\mu\nu}(t, x) \rangle \right] \right. \\ & + c_2(t) \left[\langle \mathcal{O}_{2\mu\nu}(t, x) \rangle - \langle \mathcal{O}_{2\mu\nu}(t, x) \rangle \right] \\ & + c_3(t) \left[\langle \mathcal{O}_{3\mu\nu}(t, x) \rangle - 2 \langle \mathcal{O}_{4\mu\nu}(t, x) \rangle - \langle \mathcal{O}_{3\mu\nu}(t, x) - 2 \mathcal{O}_{4\mu\nu}(t, x) \rangle \right] \\ & + c_4(t) \left[\langle \mathcal{O}_{4\mu\nu}(t, x) \rangle - \langle \mathcal{O}_{4\mu\nu}(t, x) \rangle \right] \\ & \left. + c_5(t) \left[\langle \mathcal{O}_{5\mu\nu}(t, x) \rangle - \langle \mathcal{O}_{5\mu\nu}(t, x) \rangle \right] \right\}, \end{aligned} \quad (4.70)$$

where operators in the right-hand side are given by Eqs. (4.1)–(4.5). One may further use the identities

$$2 \langle \mathcal{O}_{3\mu\nu}(t, x) \rangle = \langle \mathcal{O}_{4\mu\nu}(t, x) \rangle = \frac{-2 \dim(R) N_f}{(4\pi)^2 t^2} \delta_{\mu\nu} \quad (4.71)$$

to make the expression a little simpler. Applying the consequence of the renormalization group argument, Eqs. (4.22)–(4.23), to Eqs. (4.60)–(4.64), we have

$$c_1(t) = \frac{1}{\bar{g}(1/\sqrt{8t})^2} - b_0 \ln \pi - \frac{1}{(4\pi)^2} \left[\frac{7}{3} C_2(G) - \frac{3}{2} T(R) N_f \right], \quad (4.72)$$

$$c_2(t) = \frac{1}{8} \frac{1}{(4\pi)^2} \left[\frac{11}{3} C_2(G) + \frac{11}{3} T(R) N_f \right], \quad (4.73)$$

$$c_3(t) = \frac{1}{4} \left\{ 1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} C_2(R) \left[\frac{3}{2} + \ln(432) \right] \right\}, \quad (4.74)$$

$$c_4(t) = \frac{1}{8} d_0 \bar{g}(1/\sqrt{8t})^2, \quad (4.75)$$

$$c_5(t) = -\frac{\bar{m}(1/\sqrt{8t})}{m} \left\{ 1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} C_2(R) \left[3 \ln \pi + \frac{7}{2} + \ln(432) \right] \right\}, \quad (4.76)$$

where $\bar{g}(q)$ is the running gauge coupling in the $\overline{\text{MS}}$ scheme. For going from the $\overline{\text{MS}}$ scheme to the $\overline{\text{MS}}$ scheme, it suffices to make the following replacement corresponding to Eq. (4.65),

$$\ln \pi \rightarrow \gamma_E - 2 \ln 2, \quad (4.77)$$

in the above expressions. Equation (4.70) with Eqs. (4.72)–(4.76) is our main result. Note that the renormalized mass parameter m in $c_5(t)$ (4.76) and that in $\mathcal{O}_{5\mu\nu}(t, x)$ (4.5) are redundant in Eq. (4.70) because they are cancelled out in the product. For the running mass parameter $\bar{m}(1/\sqrt{8t})$ in $c_5(t)$ for $t \rightarrow 0$, one may use the relation

$$\begin{aligned} \bar{m}(q) &= [2b_0 \bar{g}(q)^2 t^{d_0/2b_0} \exp \left\{ \int_0^{\bar{g}(q)} dg \left[-\frac{\gamma_m(g)}{\beta(g)} - \frac{d_0}{b_0 g} \right] \right\} m^\infty \\ &= \left(\frac{2}{t} \right)^{d_0/2b_0} \left[1 - \frac{d_0 b_1}{2b_0^2 t} (1 + \ln t) + \frac{d_1}{2b_0^2 t} + O(t^{-2}) \right] m^\infty, \quad \ell \equiv \ln(q^2/\Lambda^2), \end{aligned} \quad (4.78)$$

where m^∞ denotes the renormalization group invariant mass. For the massive fermion, m^∞ may be determined by using the method established in Ref. [43], for example. For the massless fermion, $m^\infty = 0$ and we can simply discard the last line of Eq. (4.70).

5. Equation of motion in the small flow-time limit

Let us consider the following representations for small flow-time:

$$\begin{aligned} \delta_{\mu\nu} \left[\bar{\psi}(x) \overleftrightarrow{D} \psi(x) + 2m_0 \bar{\psi}(x) \psi(x) \right] \\ = d_2(t) \bar{\mathcal{O}}_{2\mu\nu}(t, x) + d_4(t) \bar{\mathcal{O}}_{4\mu\nu}(t, x) + d_5(t) \bar{\mathcal{O}}_{5\mu\nu} + O(t), \end{aligned} \quad (5.1)$$

and

$$\delta_{\mu\nu} m_0 \bar{\psi}(x) \psi(x) = c_5(t) \bar{\mathcal{O}}_{5\mu\nu}(t, x) + O(t), \quad (5.2)$$

where it is understood that the vacuum expectation values are subtracted on both sides of the equations. By a renormalization group argument identical to that which led to Eqs. (4.22) and (4.23) and the one-loop calculation in Sect. 4, for $t \rightarrow 0$ we have

$$d_2(t) = -\frac{1}{(4\pi)^2} T(R) N_f \frac{5}{3}, \quad (5.3)$$

$$d_4(t) = 1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} C_2(R) \left[-\frac{1}{2} + \ln(432) \right], \quad (5.4)$$

$$d_5(t) = \frac{\bar{m}(1/\sqrt{8t})}{m} 2 \left\{ 1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} C_2(R) [3 \ln \pi + 2 + \ln(432)] \right\}, \quad (5.5)$$

and $c_5(t) = (1/2)d_5(t)$. We note that since the left-hand side of Eq. (5.1) is proportional to the equation of motion of the fermion field, when the position x does not coincide with positions of other operators in the position space, we may set the combination to zero (the Schwinger–Dyson equation). This implies that we can make the replacement (the subtraction of the vacuum expectation value is understood)

$$\begin{aligned} \bar{\mathcal{O}}_{4\mu\nu}(t, x) \rightarrow \frac{1}{(4\pi)^2} T(R) N_f \frac{5}{3} \bar{\mathcal{O}}_{2\mu\nu}(t, x) \\ - \frac{\bar{m}(1/\sqrt{8t})}{m} 2 \left[1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} C_2(R) \left(3 \ln \pi + \frac{5}{2} \right) \right] \bar{\mathcal{O}}_{5\mu\nu}(t, x) \end{aligned} \quad (5.6)$$

in the master formula (4.70) [for the $\overline{\text{MS}}$ scheme, one makes the substitution (4.77)], because throughout this paper we are assuming that the energy-momentum tensor $\{T_{\mu\nu}\}_R(x)$ is separated from other operators in correlation functions. This makes the expression of the energy-momentum tensor somewhat simpler.

If one is interested in the trace part of the energy-momentum tensor, that is, the total divergence of the dilatation current, the above procedure leads to

$$\begin{aligned} \delta_{\mu\nu} \{T_{\mu\nu}\}_R(x) = \lim_{t \rightarrow 0} \left(\frac{1}{2} b_0 \left[G_{\rho\sigma}^a(t, x) G_{\rho\sigma}^a(t, x) - \langle G_{\rho\sigma}^a(t, x) G_{\rho\sigma}^a(t, x) \rangle \right] \right. \\ \left. - \left\{ 1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} C_2(R) [3 \ln \pi + 8 + \ln(432)] \right\} \right. \\ \left. \times \bar{m}(1/\sqrt{8t}) [\hat{\chi}(t, x) \hat{\chi}(t, x) - \langle \hat{\chi}(t, x) \hat{\chi}(t, x) \rangle] \right), \end{aligned} \quad (5.7)$$

which is quite analogous to the trace anomaly (2.11). For the massless fermion, $\bar{m}(1/\sqrt{8t}) = 0$ and we end up with a quite simple expression for the trace part of the energy-momentum tensor.

6. Conclusion

In the present paper, on the basis of the Yang–Mills gradient flow, we constructed a formula (4.70) that provides a possible method to compute correlation functions containing the energy–momentum tensor in lattice gauge theory with fermions. This is a natural generalization of the construction in Ref. [7] for the pure Yang–Mills theory. Although the feasibility of the application in lattice Monte Carlo simulations remains to be carefully investigated, the experience in the thermodynamics of the quenched QCD [27] strongly indicates that, even with presently available lattice parameters, there exists a window (1.2) within which one can reliably carry out the extrapolation for $t \rightarrow 0$ in Eq. (4.70). We expect various applications of the present formulation. One is the application in many-flavor gauge theories with an infrared fixed point (which are subject to recent active investigations; see contributions in the last lattice conference [44, 45] for recent reviews).

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A. One-loop renormalization in the MS scheme

A.1. Parameters, elementary fields

The gauge coupling:

$$g_0^2 = \mu^{2\epsilon} g^2 \left\{ 1 + \frac{g^2}{(4\pi)^2} \left[-\frac{11}{3} C_2(G) + \frac{4}{3} T(R) N_f \right] \frac{1}{\epsilon} + O(g^4) \right\}. \quad (\text{A1})$$

The fermion mass:

$$m_0 = m \left[1 + \frac{g^2}{(4\pi)^2} C_2(R) (-3) \frac{1}{\epsilon} + O(g^4) \right]. \quad (\text{A2})$$

The gauge potential (in the Feynman gauge):

$$A_\mu^a(x) = \left[1 + \frac{g^2}{(4\pi)^2} C_2(G) (-1) \frac{1}{\epsilon} + O(g^4) \right] A_{\mu R}^a(x). \quad (\text{A3})$$

The fermion field:

$$\psi(x) = \left[1 + \frac{g^2}{(4\pi)^2} C_2(R) \left(-\frac{1}{2} \right) \frac{1}{\epsilon} + O(g^4) \right] \psi_R(x). \quad (\text{A4})$$

A.2. Composite operators

The bare operators (4.7)–(4.11) and renormalized counterparts

$$\{O_{1\mu\nu}\}_R(x) \equiv \{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x), \quad (A5)$$

$$\{O_{2\mu\nu}\}_R(x) \equiv \delta_{\mu\nu} \{F_{\mu\rho}^a F_{\rho\nu}^a\}_R(x), \quad (A6)$$

$$\{O_{3\mu\nu}\}_R(x) \equiv \left\{ \bar{\psi} \left(\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right) \psi \right\}_R(x), \quad (A7)$$

$$\{O_{4\mu\nu}\}_R(x) \equiv \delta_{\mu\nu} \left\{ \bar{\psi} \overleftrightarrow{D} \psi \right\}_R(x), \quad (A8)$$

$$\{O_{5\mu\nu}\}_R(x) \equiv \delta_{\mu\nu} m \left\{ \bar{\psi} \psi \right\}_R(x) \quad (A9)$$

are related as

$$O_{4\mu\nu}(x) = Z_{ij} \{O_{j\mu\nu}\}_R(x). \quad (A10)$$

The gluonic contribution to the operator renormalization of $O_{1\mu\nu}$ and $O_{2\mu\nu}$ was determined in Ref. [7]. By further computing fermionic contributions to the operator renormalization (corresponding to diagrams A18 and A19 in the main text) and taking the gauge coupling and wave function normalizations [Eqs. (A1) and (A3)] into account, we have

$$Z_{11} = 1 + \frac{g^2}{(4\pi)^2} C_2(G) \left(-\frac{11}{3} \right) \frac{1}{\epsilon} + O(g^4), \quad (A11)$$

$$Z_{12} = \frac{g^2}{(4\pi)^2} C_2(G) \frac{11}{12} \frac{1}{\epsilon} + O(g^4), \quad (A12)$$

$$Z_{13} = \frac{g^4}{(4\pi)^2} C_2(R) \frac{2}{3} \frac{1}{\epsilon} + O(g^6), \quad (A13)$$

$$Z_{14} = \frac{g^4}{(4\pi)^2} C_2(R) \left(-\frac{1}{3} \right) \frac{1}{\epsilon} + O(g^6), \quad (A14)$$

$$Z_{15} = \frac{g^4}{(4\pi)^2} C_2(R) (-3) \frac{1}{\epsilon} + O(g^6), \quad (A15)$$

and

$$Z_{21} = 0, \quad (A16)$$

$$Z_{22} = 1 + O(g^4), \quad (A17)$$

$$Z_{23} = 0, \quad (A18)$$

$$Z_{24} = O(g^6), \quad (A19)$$

$$Z_{25} = \frac{g^4}{(4\pi)^2} C_2(R) (-12) \frac{1}{\epsilon} + O(g^6). \quad (A20)$$

From these, to the one-loop order, we further have

$$\begin{aligned} & \delta_{\mu\nu} \{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) \\ &= \left[1 + \frac{g^2}{(4\pi)^2} C_2(G) \left(\frac{11}{6} \right) \right] \{F_{\mu\rho}^a F_{\nu\rho}^a\}_R(x) \\ &+ \frac{g^4}{(4\pi)^2} C_2(R) \left(-\frac{2}{3} \right) \left\{ \bar{\psi} \overleftrightarrow{D} \psi \right\}_R(x) + \frac{g^4}{(4\pi)^2} C_2(R) (-6) m \left\{ \bar{\psi} \psi \right\}_R(x). \end{aligned} \quad (A21)$$

On the other hand, the computation of diagrams B03, B04, B05, B06, and B07, combined with the wave function renormalization (A4), shows

$$Z_{31} = \frac{1}{(4\pi)^2} T(R) N_f \frac{16}{3} \frac{1}{\epsilon} + O(g^2), \quad (\text{A22})$$

$$Z_{32} = \frac{1}{(4\pi)^2} T(R) N_f \left(-\frac{4}{3} \right) \frac{1}{\epsilon} + O(g^2), \quad (\text{A23})$$

$$Z_{33} = 1 + \frac{g^2}{(4\pi)^2} C_2(R) \left(-\frac{8}{3} \right) \frac{1}{\epsilon} + O(g^4), \quad (\text{A24})$$

$$Z_{34} = \frac{g^2}{(4\pi)^2} C_2(R) \frac{4}{3} \frac{1}{\epsilon} + O(g^4), \quad (\text{A25})$$

$$Z_{35} = O(g^4), \quad (\text{A26})$$

and

$$Z_{41} = Z_{43} = 0, \quad Z_{42} = Z_{45} = O(g^4), \quad Z_{44} = 1 + O(g^4). \quad (\text{A27})$$

The consistency of these relations shows

$$\begin{aligned} & \delta_{\mu\nu} \left\{ \bar{\psi} \gamma_\mu \overleftrightarrow{D}_\nu \psi \right\}_R(x) \\ &= \left[1 + \frac{g^2}{(4\pi)^2} C_2(R) \frac{4}{3} \right] \left\{ \bar{\psi} \overleftrightarrow{D} \psi \right\}_R(x) + \frac{1}{(4\pi)^2} T(R) N_f \left(-\frac{4}{3} \right) \left\{ F_{\rho\sigma}^a F_{\rho\sigma}^a \right\}_R(x). \end{aligned} \quad (\text{A28})$$

Finally, a general theorem (see, for example, Ref. [28]) says that

$$Z_{51} = Z_{52} = Z_{53} = Z_{54} = 0, \quad Z_{55} = 1. \quad (\text{A29})$$

B. Integration formulas

$$\int_{\ell} \frac{1}{(\ell^2)^{\alpha}} e^{-s\ell^2} = \frac{\Gamma(D/2 - \alpha)}{(4\pi)^{D/2} \Gamma(D/2)} s^{\alpha - D/2}. \quad (\text{B1})$$

$$\int_k \int_{\ell} \frac{e^{-sk^2 - u\ell^2 - v(k+\ell)^2}}{k^2} = \frac{1}{(4\pi)^D (D/2 - 1)(u+v)} (su + uv + vs)^{1-D/2}. \quad (\text{B2})$$

$$\int_{\ell} e^{-s\ell^2} = \frac{1}{(4\pi)^{D/2}} s^{-D/2}. \quad (\text{B3})$$

$$\int_{\ell} e^{-s\ell^2} \ell_{\mu} \ell_{\nu} = \frac{1}{(4\pi)^{D/2}} s^{-D/2-1} \frac{1}{2} \delta_{\mu\nu}. \quad (\text{B4})$$

$$\int_{\ell} e^{-s\ell^2} \ell_{\mu} \ell_{\nu} \ell_{\rho} \ell_{\sigma} = \frac{1}{(4\pi)^{D/2}} s^{-D/2-2} \frac{1}{4} (\delta_{\mu\nu} \delta_{\rho\sigma} + \delta_{\mu\rho} \delta_{\nu\sigma} + \delta_{\mu\sigma} \delta_{\nu\rho}). \quad (\text{B5})$$

$$\int_{\ell} e^{-s\ell^2} \ell_{\mu} \ell_{\nu} \ell_{\rho} \ell_{\sigma} \ell_{\alpha} \ell_{\beta} = \frac{1}{(4\pi)^{D/2}} s^{-D/2-3} \frac{1}{8} (\delta_{\mu\nu} \delta_{\rho\sigma} \delta_{\alpha\beta} + 14 \text{ permutations}). \quad (\text{B6})$$

$$\int_{\ell} \frac{1}{\ell^2} e^{-s\ell^2} = \frac{1}{(4\pi)^{D/2}} \frac{1}{D/2 - 1} s^{-D/2+1}. \quad (\text{B7})$$

$$\int_{\ell} \frac{1}{\ell^2} e^{-s\ell^2} \ell_{\mu} \ell_{\nu} = \frac{1}{(4\pi)^{D/2}} \frac{1}{D} s^{-D/2} \delta_{\mu\nu}. \quad (\text{B8})$$

$$\int_{\ell} \frac{1}{\ell^2} e^{-s\ell^2} \ell_{\mu} \ell_{\nu} \ell_{\rho} \ell_{\sigma} = \frac{1}{(4\pi)^{D/2}} \frac{1}{2(D+2)} s^{-D/2-1} (\delta_{\mu\nu} \delta_{\rho\sigma} + \delta_{\rho\mu} \delta_{\nu\sigma} + \delta_{\mu\sigma} \delta_{\nu\rho}). \quad (\text{B9})$$

$$\begin{aligned} & \int_{\ell} \frac{1}{\ell^2} e^{-s\ell^2} \ell_{\mu} \ell_{\nu} \ell_{\rho} \ell_{\sigma} \ell_{\alpha} \ell_{\beta} \\ &= \frac{1}{(4\pi)^{D/2}} \frac{1}{4(D+4)} s^{-D/2-2} (\delta_{\mu\nu} \delta_{\rho\sigma} \delta_{\alpha\beta} + 14 \text{ permutations}). \end{aligned} \quad (\text{B10})$$

C. Gluonic contributions to $\zeta_{ij}^{(1)}(t)$

Gluonic contributions to $\zeta_{ij}^{(1)}(t)$ are tabulated in Table C1.

D. Justification of our computational prescription

In this appendix, we give detailed explanation why the one-loop matching coefficients $\zeta_{ij}^{(1)}(t)$ in Eq. (4.25) can be determined without computing correlation functions Eqs. (4.26) and (4.29) with operators $\tilde{\mathcal{O}}_{i\mu\nu}(t, x)$ are replaced by corresponding bare operators in Eqs. (4.6)–(4.10), that is,

$$\left\langle \mathcal{O}_{i\mu\nu}(x) A_{\beta}^b(y) A_{\gamma}^c(z) \right\rangle, \quad (\text{D1})$$

and

$$\left\langle \mathcal{O}_{i\mu\nu}(x) \psi(y) \bar{\psi}(z) \right\rangle. \quad (\text{D2})$$

We argue that one can neglect contributions of these correlation functions altogether, if one follows a regularization prescription for IR divergences we adopted in the main text.

Both Eqs. (D1) and (D2) can be treated in a similar manner, so we consider Eq. (D1). Let us take a particular one-loop 1PI Feynman diagram that contributes to Eq. (D1), for instance, diagram A11 in Fig. 22. This is a diagram in the D -dimensional gauge theory and thus consists only of ordinary (i.e., filled circle) vertices and ordinary (i.e., no Gaussian factor in Eq. (3.17)) propagators. Writing the contribution of this diagram to Eq. (D1) as Eq. (4.27), the vertex part $\mathcal{M}_{\mu\nu,\beta\gamma}(k, \ell)$ is given by a one-loop integral,

$$\mathcal{M}_{\mu\nu,\beta\gamma}(k, \ell) = \int_p \mathcal{I}_{\mu\nu,\beta\gamma}(p; k, \ell), \quad (\text{D3})$$

where dimensional counting says that the integrand $\mathcal{I}_{\mu\nu,\beta\gamma}(p; k, \ell)$ is of mass dimension -2 . In general, this integral is UV divergent and, if we further Taylor-expand the integrand with respect to the external momenta k and ℓ , exhibits also IR divergences.

Next, we note that for the above 1PI diagram in the D -dimensional gauge theory, there always exists a corresponding flow Feynman diagram with the same structure (i.e., diagram A11 in Fig. 22 in the present example). We write its contribution to the vertex

Table C1 $\zeta_{lj}^{(1)}$ in units of $\frac{g^2}{(4\pi)^2}C_2(G)$. The numbers with * are corrected from previous versions of the present paper.

diagram	$\zeta_{11}^{(1)}(t)$	$\zeta_{12}^{(1)}(t)$
A03	0	0
A04	$-3\epsilon(t)^{-1} - 1$	0
A05	$-\frac{7}{36}$	$-\frac{49}{144}$
A06	$2\epsilon(t)^{-1} - \frac{1}{2}$	0
A07	$\frac{19}{288}$	$\frac{121}{384}$
A08	$\frac{35^*}{96}$	$\frac{143^*}{384}$
A09	$-\frac{25}{8}$	0
A11	$\frac{1}{3}\epsilon(t)^{-1} - \frac{17}{36}$	$\frac{7}{12}\epsilon(t)^{-1} + \frac{1}{144}$
A13	$-\frac{5}{3}\epsilon(t)^{-1} + \frac{25}{36}$	$-\frac{3}{2}\epsilon(t)^{-1} - \frac{29}{16}$
A14	$3\epsilon(t)^{-1} + 3$	0
A15	$3\epsilon(t)^{-1} + \frac{5}{2}$	0
A16	1^*	$\frac{31}{24}$

part $\mathcal{M}_{\mu\nu,\beta\gamma}(k, l)$ in Eq. (4.27) as

$$\mathcal{M}_{\mu\nu,\beta\gamma}(k, \ell) = \int_p \mathcal{I}_{\mu\nu,\beta\gamma}(p; k, \ell; t), \quad (\text{D4})$$

where the Feynman rules for the vertices are completely identical to those for Eq. (D3), while propagators are identical except the Gaussian factor in Eq. (3.18) which depends on the flow time of the inserted composite operator $\mathcal{O}_{\mu\nu}(t, x)$; we have explicitly indicated the dependence of the integrand on t .

Now, in the *one-loop level*, it can be seen that what is relevant to $c_{ij}^{(1)}(\ell)$ is the difference between Eqs. (D4) and (D3):

$$\int_p [\mathcal{I}_{\mu\nu,\beta\gamma}(p; k, \ell; t) - \mathcal{I}_{\mu\nu,\beta\gamma}(p; k, \ell)]. \quad (\text{D5})$$

From the structure of two integrands explained above, (logarithmic) IR divergences are cancelled out in the difference and we may Taylor-expand the integrand with respect to the external momenta k and ℓ . The coefficient of the $O(k, \ell)$ -term is then given by

$$C \int_p \frac{e^{-2ip^2} - 1}{(p^2)^2} = C \frac{1}{(4\pi)^{D/2} (D-2)(D-4)} 4 (2\ell)^{2-D/2}, \quad (\text{D6})$$

where C is a constant and a complex dimension D is introduced *to regularize UV divergences*; the last expression has been obtained by the analytic continuation from the complex domain $2 < \text{Re}(D) < 4$. The combination (D6) (with $D \rightarrow 4$) is relevant for the contribution of the diagram under consideration to $c_{ij}^{(1)}$.

Now, although the above computation is a proper one, there exists a “facile method” that reproduces Eq. (D6) with much less effort; the argument proceeds as follows.

One considers only the contribution of the flow Feynman diagram (D4) and Taylor-expand the integrand with respect to the external momenta k and ℓ . In this way, one encounters IR divergences. One then introduces a complex dimension D *to regularize IR divergences*. Note that this is possible because the integral (D4) is UV finite thanks to the Gaussian factors; there always exists a complex domain of D with which the integral is well-defined. Then the coefficient of the $O(k, \ell)$ -term is given by

$$C \int_p \frac{e^{-2ip^2}}{(p^2)^2} = C \frac{1}{(4\pi)^{D/2} (D-2)(D-4)} 4 (2\ell)^{2-D/2}. \quad (\text{D7})$$

This is an expression obtained by the analytic continuation from the complex domain $4 < \text{Re}(D)$. The last expression is however identical to Eq. (D6).

Thus, we have arrived at the following facile method: We completely forget about the computation of correlation functions with bare operators, Eqs. (D1) and (D2). For correlation functions with flowed operators, Eqs. (4.26) and (4.29), we simply Taylor-expand the integrand of the Feynman integral with respect to the external momenta (and the fermion mass). Resulting IR divergences are regularized by “dimensional regularization”, i.e., the analytic continuation from $\text{Re}(D) > 4$. Then the result for the one-loop matching coefficients $c_{ij}^{(1)}$ is identical to the one obtained by a computation that properly takes the contribution of Eqs. (D1) and (D2) into account. This facile method is precisely the computational method we made use of in the main text.

E. Modified energy-momentum tensor

The argument in Sec. 2 shows that the energy-momentum tensor (2.7) fulfills the Ward-Takahashi relations,

$$\langle \partial_\mu T_{\mu\nu}(x) A_\rho(y) \cdots \rangle = -\delta(x-y) \langle [\partial_\nu A_\rho(y) - D_\rho A_\nu(y)] \cdots \rangle + \cdots, \quad (\text{E1})$$

and

$$\langle \partial_\mu T_{\mu\nu}(x) \psi(y) \cdots \rangle = -\delta(x-y) \langle D_\nu \psi(y) \cdots \rangle + \cdots. \quad (\text{E2})$$

In particular, there is no term being proportional to $\partial_\nu \delta(x-y)$ in the right-hand sides. These relations lead to

$$\begin{aligned} & \langle \partial_\mu [x_\nu T_{\mu\nu}(x)] A_\rho(y) \cdots \rangle \\ &= \langle T_{\mu\mu}(x) A_\rho(y) \cdots \rangle - x_\nu \delta(x-y) \langle [\partial_\nu A_\rho(y) - D_\rho A_\nu(y)] \cdots \rangle + \cdots, \end{aligned} \quad (\text{E3})$$

and

$$\langle \partial_\mu [x_\nu T_{\mu\nu}(x)] \psi(y) \cdots \rangle = \langle T_{\mu\mu}(x) \psi(y) \cdots \rangle - x_\nu \delta(x-y) \langle D_\nu \psi(y) \cdots \rangle + \cdots. \quad (\text{E4})$$

Integrating these over the whole space, we have

$$\begin{aligned} & \int d^D x \langle T_{\mu\mu}(x) A_\rho(y) \cdots \rangle = \langle [y_\nu \partial_\nu A_\rho(y) - y_\nu D_\rho A_\nu(y)] \cdots \rangle + \cdots \\ &= \langle \{ (y_\nu \partial_\nu + 1) A_\rho(y) - D_\rho [y_\nu A_\nu(y)] \} \cdots \rangle + \cdots, \end{aligned} \quad (\text{E5})$$

and

$$\begin{aligned} & \int d^D x \langle T_{\mu\mu}(x) \psi(y) \cdots \rangle = \langle y_\nu D_\nu \psi(y) \cdots \rangle + \cdots \\ &= \langle [y_\nu \partial_\nu \psi(y) + y_\nu A_\nu(y) \psi(y)] \cdots \rangle + \cdots. \end{aligned} \quad (\text{E6})$$

Now, we see that Eq. (E5) is accord with the naively-expected relation,

$$\int d^D x \langle T_{\mu\mu}(x) \mathcal{O}(y) \cdots \rangle = \langle (y_\nu \partial_\nu + d_{\mathcal{O}}) \mathcal{O}(y) \cdots \rangle + \cdots, \quad (\text{E7})$$

where $d_{\mathcal{O}}$ is the scale dimension of the field $\mathcal{O}$, up to the gauge transformation, but Eq. (E6) is not. One would expect instead

$$\int d^D x \langle \hat{T}_{\mu\mu}(x) \psi(y) \cdots \rangle = \left\langle \left(y_\nu \partial_\nu + \frac{D-1}{2} \right) \psi(y) \cdots \right\rangle + \cdots, \quad (\text{E8})$$

up to the gauge transformation. One can construct such a modified energy-momentum tensor by using a freedom to add the equation of motion to the energy-momentum tensor. We see that Eq. (E8) is realized, if

$$\left\langle \partial_\mu \hat{T}_{\mu\nu}(x) \psi(y) \cdots \right\rangle = -\delta(x-y) \langle D_\nu \psi(y) \cdots \rangle + \frac{D-1}{2D} \partial_\nu^2 \delta(x-y) \langle \psi(y) \cdots \rangle + \cdots. \quad (\text{E9})$$

It can be seen that the last term does not influence on Ward-Takahashi relations associated with the translation and the rotation (Poincaré transformations). This modification is accomplished by

$$\hat{T}_{\mu\nu}(x) = T_{\mu\nu}(x) + \frac{D-1}{2D} \delta_{\mu\nu} \bar{\psi}(x) \left(\vec{\not{\partial}} + 2m_0 \right) \psi(x). \quad (\text{E10})$$

This modified energy-momentum tensor, fulfilling the naively-expected relation (E7), leads to a simpler derivation of the trace anomaly. In terms of flowed operators, this modification amounts to adding the combination (5.1) times

$$\frac{D-1}{2D} \rightarrow \frac{3}{8}. \quad (\text{E11})$$

We note however that when the energy-momentum tensor is separated from other operators in position space, the modification has no effect because it is proportional to the equation of motion.

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P36：梯度流形式的双圈能动量张量

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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

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The two-loop energy-momentum tensor within the gradient-flow formalism

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The gradient-flow formulation of the energy-momentum tensor of QCD is extended to NNLO perturbation theory. This means that the Wilson coefficients which multiply the flowed operators in the corresponding expression for the regular energy-momentum tensor are calculated to this order. The result has been obtained by applying modern tools of regular perturbation theory, reducing the occurring two-loop integrals, which also include flow-time integrations, to a small set of master integrals which can be calculated analytically.

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1 Introduction

The gradient-flow formalism as introduced by Lüscher [1] and further formalized by Lüscher and Weisz [2] has proven useful in lattice QCD in many respects. One of its main virtues is that composite operators at finite flow time t do not require ultra-violet (UV) renormalization beyond the one of the involved parameters and fields. This means that the operators do not mix under renormalization-group running, which makes it particularly simple to combine results from different regularization schemes. This feature opens promising prospects for a cross-fertilization of lattice and perturbative calculations, such as a possible lattice determination of $\alpha_s(M_Z)$, for example [3].

A particularly powerful way to exhibit this possible interplay is obtained by considering the expansion of composite operators in the limit of small flow time, which expresses flowed operators in terms of QCD operators at $t = 0$, with t -dependent Wilson coefficients [2]. This method has been used by Makino and Suzuki [4,5] to derive a regularization-independent formula for the energy-momentum tensor (EMT) $T_{\mu\nu}$ which has already led to promising results (see, e.g., Refs. [6–10]).

The universal Wilson coefficients that occur in the formula of Ref. [5] for the EMT have been calculated through next-to-leading order (NLO) in perturbation theory [4,5]. This corresponds to a one-loop calculation in the sense that it involves integrals over a single D -dimensional momentum. In this paper, we will carry this calculation to the next perturbative order.¹ It is important to note at this point that the integrals which occur in

¹Note that we work in the limit of infinite volume. The inclusion of finite-volume effects requires different techniques, such as Numerical Stochastic Perturbation Theory, see Ref. [11].

the gradient-flow formalism are of a more general type than in regular QCD. They involve additional exponential factors which depend on loop and external momenta, as well as on flow-time variables, some of which are also integrated over. Nevertheless, the first two-loop result was already obtained in Ref. [1], even in analytic form. The extension to the three-loop level required significant aid from computer algebra and numerical tools [3]. From the quantum-field theoretical point of view, it closely followed the steps of Ref. [1] by directly expressing the Green's functions in terms of integrals with the help of Wick's theorem. The integrals themselves were evaluated using sector-decomposition [12,13] in order to isolate the poles in $D - 4$, whose coefficients were determined using high-precision numerical methods [14,15].

In the current calculation, we apply a completely independent setup. On the one hand, it applies the gradient-flow formalism described in terms of a five-dimensional quantum field theory [2], which leads to well-defined, albeit non-standard Feynman rules. On the other hand, rather than evaluating the resulting integrals numerically, we express them in terms of master integrals using the integration-by-parts method of Chetyrkin and Tkachov [16]. This reduces the NLO calculation of the Wilson coefficients of the EMT to a single one-loop integral without flow-time integration. The next-to-next-to-leading order (NNLO) calculation leads to four two-loop master integrals without flow-time integration, and two two-loop master integrals with a single flow-time integration. All master integrals can be calculated analytically by standard means for general values of D , the number of space-time dimensions.

By suitable renormalization, the Wilson coefficients of the EMT can be defined in such a way that they are formally renormalization-scale independent. For a fixed-order perturbative result, this means that the renormalization-scale dependence is formally of higher order. This allows one to estimate the perturbative uncertainty on the Wilson coefficients through their residual dependence on the renormalization scale μ around a particular "central" value. Based on the form of the analytical result, we argue for a specific choice of this central value. Our numerical study shows that the higher-order terms indeed lead to an appreciable reduction of the μ -variation. However, by comparison of the successive higher-order terms, it appears that the uncertainty estimate from a variation within $\mu \in [\mu_0/2, 2\mu_0]$, as it is common practice in regular perturbative QCD calculation, might be too optimistic.

While we consider the NNLO expressions for the Wilson coefficients of the EMT as our main result, our calculation allows us to obtain a number of additional results that might be useful in a broader context. Among these are the flowed quark-field renormalization constant Z_ψ , and the matrix of anomalous dimensions for the set of operators which form the energy-momentum tensor in regular (non-flowed) QCD through NNLO.

The remainder of the paper is structured as follows. After briefly introducing the perturbative gradient-flow formalism in order to define our notation in Sect. 2, we outline the approach of Refs. [4,5] for using this formalism to define the EMT in Sect. 3. Technical details of our calculation are described in Sect. 4. Section 5 contains our main result, the Wilson coefficients through NNLO QCD in the $\overline{\text{MS}}$ scheme. As pointed out in Ref. [5], the trace anomaly of the EMT allows for a welcome check of the calculation; we briefly describe the derivation of the resulting relations among the coefficient functions in Sect. 6. Finally,

in Sect. 7, we use the finiteness condition of the flowed operators in order to derive the anomalous-dimension matrix for the set of operators which occur in the EMT in regular QCD. Section 8 presents our conclusions.

2 QCD gradient flow in perturbation theory

In the following, we will work in D -dimensional Euclidean space-time with $D = 4 - 2\epsilon$. The gradient-flow formalism continues the gluon and quark fields $A_\mu^a(x)$ and $\psi(x)$ of regular² QCD to $(D + 1)$ -dimensional fields $B_\mu^a(t, x)$ and $\chi(t, x)$ through the boundary conditions

$$B_\mu^a(t = 0, x) = A_\mu^a(x), \quad \chi(t = 0, x) = \psi(x), \quad (1)$$

and the flow equations

$$\begin{aligned} \partial_t B_\mu^a &= D_\nu^{ab} G_{\nu\mu}^b + \kappa D_\mu^{ab} \partial_\nu B_\nu^b, \\ \partial_t \chi &= \Delta \chi - \kappa \partial_\mu B_\mu^a T^a \chi, \\ \partial_t \bar{\chi} &= \bar{\chi} \overleftarrow{\Delta} + \kappa \bar{\chi} \partial_\mu B_\mu^a T^a, \end{aligned} \quad (2)$$

where the “flow time” t is a parameter of mass dimension minus two, and κ is an additional gauge parameter which drops out of physical observables.

The $(D + 1)$ -dimensional field-strength tensor is defined as

$$G_{\mu\nu}^a = \partial_\mu B_\nu^a - \partial_\nu B_\mu^a + f^{abc} B_\mu^b B_\nu^c, \quad (3)$$

the covariant derivative in the adjoint representation is given by

$$D_\mu^{ab} = \delta^{ab} \partial_\mu - f^{abc} B_\mu^c, \quad (4)$$

and

$$\Delta = (\partial_\mu + B_\mu)(\partial_\mu + B_\mu), \quad \overleftarrow{\Delta} = (\overleftarrow{\partial}_\mu - B_\mu)(\overleftarrow{\partial}_\mu - B_\mu). \quad (5)$$

As usual, the color indices of the adjoint representation are denoted by $a, b, c, \dots$, while $\mu, \nu, \rho, \dots$ are D -dimensional Lorentz indices. Color indices of the fundamental representation are suppressed throughout this paper, unless required by clarity. The symmetry generators T^a obey the commutation relation

$$[T^a, T^b] = f^{abc} T^c, \quad (6)$$

with the structure constants f^{abc} .

The flow-field equation leads to a smearing of gauge-field configurations at finite flow time $t > 0$. As a consequence, composite operators at $t > 0$ do not require renormalization beyond the renormalization of the parameters and fields of the Lagrangian. For the strong coupling and the quark mass, the renormalization constants are identical to those at $t = 0$: the flowed gluon fields do not require renormalization at finite flow time as was pointed out in Ref. [2]. The renormalization constant for the flowed quark field through NNLO is a by-product of this paper and will be given below.

²We will use the terms “flowed” and “regular” QCD to distinguish quantities defined at $t > 0$ from those defined at $t = 0$.

3 Energy-momentum tensor

In a continuous D -dimensional space-time, the gauge invariant part of the EMT reads

$$T_{\mu\nu}(x) = \frac{1}{g_0^2} \left[\mathcal{O}_{1,\mu\nu}(x) - \frac{1}{4} \mathcal{O}_{2,\mu\nu}(x) \right] + \frac{1}{4} \mathcal{O}_{3,\mu\nu}(x) - \frac{1}{2} \mathcal{O}_{4,\mu\nu}(x) - \mathcal{O}_{5,\mu\nu}(x), \quad (7)$$

where g_0 is the bare coupling constant of QCD. The operators are defined as

$$\begin{aligned} \mathcal{O}_{1,\mu\nu}(x) &\equiv F_{\mu\rho}^a(x) F_{\nu\rho}^a(x), \\ \mathcal{O}_{2,\mu\nu}(x) &\equiv \delta_{\mu\nu} F_{\rho\sigma}^a(x) F_{\rho\sigma}^a(x), \\ \mathcal{O}_{3f,\mu\nu}(x) &\equiv \bar{\psi}_f(x) \left(\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right) \psi_f(x), \\ \mathcal{O}_{4f,\mu\nu}(x) &\equiv \delta_{\mu\nu} \bar{\psi}_f(x) \overleftrightarrow{D} \psi_f(x), \\ \mathcal{O}_{5f,\mu\nu}(x) &\equiv \delta_{\mu\nu} m_{f,\beta} \bar{\psi}_f(x) \psi_f(x), \\ \mathcal{O}_{i,\mu\nu}(x) &= \sum_{f=1}^{n_F} \mathcal{O}_{if,\mu\nu}(x), \quad i \in \{3, 4, 5\}, \end{aligned} \quad (8)$$

where f labels the n_F different quark flavors, $m_{f,\beta}$ is the bare quark mass, and

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c, \quad \overleftrightarrow{D}_\mu = \partial_\mu - \overleftarrow{\partial}_\mu + 2A_\mu. \quad (9)$$

The notation $i_f \in \{i_1, \dots, i_{n_F}\}$ for the indices which label different flavors will be useful later on in this paper. In general, $T_{\mu\nu}$ may contain gauge-dependent operators which vanish when evaluating physical matrix elements [17]. Here and in what follows, we implicitly assume that the vacuum expectations values of all composite operators have been subtracted³ so that $\langle \mathcal{O}_{i,\mu\nu}(x) \rangle = 0 \forall i$.

In this paper, we will focus on the case where physical matrix elements of the EMT itself are considered, i.e. no other operator multiplies the EMT at the same space-time point. In this case, the equations-of-motion (EOM) render the set of operators in Eq. (8) redundant. In particular, the EOM for the quark fields in regular QCD implies

$$0 = \mathcal{O}_{4,\mu\nu}(x) + 2 \mathcal{O}_{5,\mu\nu}(x), \quad (10)$$

which allows us to eliminate $\mathcal{O}_{5,\mu\nu}$ from the set of operators in Eq. (8). Note that due to this relation, the last two terms in Eq. (7) cancel.

We will further assume all quark masses to be equal to each other

$$m_{f,\beta} = m_0, \quad f = 1, \dots, n_F. \quad (11)$$

Therefore, the different quarks are indistinguishable and the mixing between two different quark flavors cannot depend on the flavors.

³In other words, the precise definition of $\mathcal{O}_{i,\mu\nu}$, for example, would be given by $F_{\mu\nu}^a F_{\mu\nu}^a - \langle F_{\mu\nu}^a F_{\mu\nu}^a \rangle$.

Defining the analogous operators of Eq. (8) for flowed fields, we write

$$\begin{aligned}
\tilde{\mathcal{O}}_{1,\mu\nu}(t, x) &= G_{\mu\rho}^a(t, x) G_{\nu\rho}^a(t, x), \\
\tilde{\mathcal{O}}_{2,\mu\nu}(t, x) &= \delta_{\mu\nu} G_{\rho\sigma}^a(t, x) G_{\rho\sigma}^a(t, x), \\
\tilde{\mathcal{O}}_{3f,\mu\nu}(t, x) &= Z_{\chi f} \bar{\chi}_f(t, x) \left(\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right) \chi_f(t, x), \\
\tilde{\mathcal{O}}_{4f,\mu\nu}(t, x) &= Z_{\chi f} \delta_{\mu\nu} \bar{\chi}_f(t, x) \overleftrightarrow{\not{D}} \chi_f(t, x), \\
\tilde{\mathcal{O}}_{i,\mu\nu} &= \sum_{f=1}^{n_f} \tilde{\mathcal{O}}_{if,\mu\nu}, \quad i \in \{3, 4\},
\end{aligned} \tag{12}$$

where $Z_{\chi f} = Z_\chi$ is the renormalization constant for the flowed quark fields, and

$$\overleftrightarrow{D}_\mu = \partial_\mu - \overleftarrow{\partial}_\mu + 2B_\mu. \tag{13}$$

Since we have eliminated $\mathcal{O}_{5,\mu\nu}$ from the set of operators by using Eq. (10), we do not need to include a flowed version of this operator in Eq. (12). Similar to the composite operators of regular QCD, we assume that the vacuum expectation values of the flowed composite operators have been subtracted, i.e. $\langle \tilde{\mathcal{O}}_{i,\mu\nu}(t, x) \rangle = 0 \ \forall i$.

We can now use the expansion in small flow time [2]

$$\tilde{\mathcal{O}}_{i,\mu\nu}(t, x) = \zeta_{ij}(t) \mathcal{O}_{j,\mu\nu}(x) + \dots, \tag{14}$$

to get a relation between flowed and regular QCD operators. In Eq. (14), and similarly in what follows, a sum $\sum_{j=1}^4$ is understood. The ellipsis denotes terms that vanish as $t \rightarrow 0$ which will be neglected throughout this paper. As discussed above, matrix elements of the l.h.s. of this equation are finite after renormalization of the QCD parameters, while those of the regular QCD operators on the r.h.s. are in general divergent. The mixing matrix $\zeta_{ij}(t)$ will therefore be divergent as well.

Inverting Eq. (14) and using it to re-express the regular QCD operators in the energy-momentum tensor in terms of flowed fields, one arrives at

$$T_{\mu\nu}(x) = c_i(t) \tilde{\mathcal{O}}_{i,\mu\nu}(t, x), \tag{15}$$

where

$$c_i(t) = \frac{1}{g_0^2} \left(\zeta_{1i}^{-1}(t) - \frac{1}{4} \zeta_{2i}^{-1}(t) \right) + \frac{1}{4} \zeta_{3i}^{-1}(t), \quad i = 1, \dots, 4. \tag{16}$$

Since matrix elements of the $\tilde{\mathcal{O}}_i$ as well as the energy-momentum tensor itself are finite (after mass and charge renormalization), the universal coefficients $c_i(t)$ of Eq. (16) are finite as well. In Ref. [5], they have been calculated in perturbation theory through NLO QCD. The goal of the current paper is to evaluate them through NNLO QCD.

4 Calculation of the Wilson coefficients

4.1 Method of Projectors

To compute the coefficients $\zeta_{ij}(t)$ we use the so-called "method of projectors" [18,19], which consists of constructing external states $|k\rangle$ and differential operators D_k for which

$$P_k[\mathcal{O}_i(x)] \equiv D_k(0|\mathcal{O}_i(x)|k) = \delta_{ik}, \quad (17)$$

where we have dropped the Lorentz indices for convenience, and we define the matrix element to include only diagrams which are one-particle irreducible (1PI) with respect to (w.r.t.) QCD particles. Applying P_k on both sides of Eq. (14), one obtains

$$P_k[\tilde{\mathcal{O}}_i(t, x)] = \zeta_{ij}(t) P_k[\mathcal{O}_j(x)]. \quad (18)$$

Since the $\zeta_{ij}(t)$ only depend on the flow time t and the renormalization scale μ , we can choose arbitrary values for all other dimensional parameters in this equation. Setting them to zero turns all higher-order corrections on the r.h.s. into massless tadpoles, so that Eq. (17) is only required to hold at tree-level. One thus obtains

$$\zeta_{ij}(t) = P_j[\tilde{\mathcal{O}}_i(t, x)] \Big|_{p=m=0}, \quad (19)$$

where m and p collectively denote all masses and external momenta. The right-hand side thus results in vacuum diagrams whose only dimensional scale is ℓ .

In order to find suitable projectors, we first derive the Feynman rules for the relevant terms of the operators. For example,⁴

$$\mathcal{O}_{1,\mu\nu} = \partial_\mu A^c_\rho \partial_\nu A^c_\rho + \dots \Rightarrow$$
$$= -p_{1,\mu} p_{2,\nu} \delta_a \delta_b \delta^{ab}, \quad (20)$$

where the momenta are defined to be outgoing. This suggests to use

$$P_1[X] = -\frac{\delta^{ab}}{N_A} P_{\alpha\beta[\mu\nu]\sigma\nu} \frac{\partial}{\partial p_{1,\alpha}} \frac{\partial}{\partial p_{2,\sigma}} \langle 0 | A_\alpha^a(p_1) A_\beta^b(p_2) X_{\mu\nu} | 0 \rangle, \quad (21)$$

where N_A is the dimension of the adjoint representation of the gauge group; for $SU(N_c)$, it is $N_A = N_c^2 - 1$. The projector onto the Lorentz structure is defined by

$$P_{\alpha_1\beta_1|\cdots|\alpha_n\beta_n}T_{\alpha_1\beta_1\cdots\alpha_n\beta_n} = \begin{cases} 1 & \text{for } T_{\alpha_1\beta_1\cdots\alpha_n\beta_n} = \delta_{\alpha_1\beta_1}\cdots\delta_{\alpha_n\beta_n}, \\ 0 & \text{for any other linearly} \\ & \text{independent Lorentz tensor.} \end{cases} \quad (22)$$

^aAll Feynman diagrams in this paper were drawn using TikZ-Feynman [20].

In the appendix, one can find the relevant parts of the Feynman rules for the other operators, which in a similar way lead to the projectors

$$\begin{aligned}
P_2[X] &= -\frac{\delta^{ab}}{4N_A} P_{\alpha\beta|\mu\nu\rho\sigma} \frac{\partial}{\partial p_{1,\rho}} \frac{\partial}{\partial p_{2,\sigma}} \langle 0 | A_\alpha^a(p_1) A_\beta^b(p_2) X_{\mu\nu} | 0 \rangle, \\
P_3[X] &= -i \frac{\delta^{ij}}{4N_c} P_{\mu\nu|\rho\sigma} \frac{\partial}{\partial p_{2,\sigma}} \text{Tr} \left[\gamma_\rho \langle 0 | \psi_f^j(p_2) \bar{\psi}_f^i(p_1) X_{\mu\nu} | 0 \rangle \right], \\
P_4[X] &= -i \frac{\delta^{ij}}{4N_c} P_{\mu\nu|\rho\sigma} \frac{\partial}{\partial p_{2,\sigma}} \text{Tr} \left[\gamma_\rho \langle 0 | \psi_f^j(p_2) \bar{\psi}_f^i(p_1) X_{\mu\nu} | 0 \rangle \right] \\
&\quad - \frac{1}{2} \frac{\delta^{ij}}{4N_c} P_{\mu\nu} \frac{\partial}{\partial m_0} \text{Tr} \left[\langle 0 | \psi_f^j(p_2) \bar{\psi}_f^i(p_1) X_{\mu\nu} | 0 \rangle \right],
\end{aligned} \tag{23}$$

where N_c is the dimension of the fundamental representation of the gauge group, i.e. the number of colors, and i and j are the corresponding indices. The trace appearing in the projectors P_3 and P_4 is taken w.r.t. the spinor indices of the Green's function, and f denotes the associated quark flavor. Note that P_{4_f} is constructed such that

$$P_{4_f}[\mathcal{O}_4 + 2\mathcal{O}_5] = 0 \tag{24}$$

in order to ensure that only those fermionic operators are taken into account which do not vanish according to the EOM, see Eq. (10).

With this procedure, we get a mixing matrix which distinguishes between different quark flavors. To avoid confusion with $\zeta_{ij}(t)$, which is the mixing matrix between operators summed over all flavors, we will call it $\Omega_{ij}(t)$. This matrix is then defined by

$$\tilde{\mathcal{O}}_{i,\mu\nu}(t, x) = \Omega_{ij}(t) \mathcal{O}_{j,\mu\nu}(x), \tag{25}$$

where double indices in this expression are summed over $\{1, 2, 3_1, \dots, 3_{n_F}, 4_1, \dots, 4_{n_F}\}$ (see Eq. (8)), in contrast to Eq. (15), where double indices are summed over $\{1, 2, 3, 4\}$. Its general structure is given by

$$\Omega = \begin{pmatrix} \Omega_{11} & \Omega_{12} & \underline{\Omega_{13}}^T & \underline{\Omega_{14}}^T \\ \Omega_{21} & \Omega_{22} & \underline{\Omega_{23}}^T & \underline{\Omega_{24}}^T \\ \underline{\Omega_{31}} & \underline{\Omega_{32}} & \underline{\underline{\Omega_{33}}} & \underline{\underline{\Omega_{34}}} \\ \underline{\Omega_{41}} & \underline{\Omega_{42}} & \underline{\underline{\Omega_{43}}} & \underline{\underline{\Omega_{44}}} \end{pmatrix}, \tag{26}$$

where an element Ω_{ij} represents the mixing between $\mathcal{O}_i$ and $\tilde{\mathcal{O}}_j$, taking into account individual flavors. Therefore, an underlined and a double underlined element denotes an n_F -dimensional vector and an $n_F \times n_F$ dimensional matrix, respectively. Their elements describe the mixing between different flavors. As the quarks are indistinguishable, the n_F and n_F^2 dimensional objects appearing in Eq. (26) can each be described by two independent parameters, named ω_{ij} and $\bar{\omega}_{ij}$:

$$\Omega_{ij} = \omega_{ij} \quad \text{for } i, j < 3, \quad \underline{\Omega_{ij}} = \omega_{ij} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \quad \text{for } i < 3, j > 2 \quad \text{or } i > 2, j < 3,$$

$$\underline{\underline{\Omega}}_{ij} = \begin{pmatrix} \omega_{ij} & \bar{\omega}_{ij} & \bar{\omega}_{ij} & \dots & \bar{\omega}_{ij} \\ \bar{\omega}_{ij} & \omega_{ij} & \bar{\omega}_{ij} & \dots & \bar{\omega}_{ij} \\ \bar{\omega}_{ij} & \bar{\omega}_{ij} & \omega_{ij} & \dots & \bar{\omega}_{ij} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \bar{\omega}_{ij} & \bar{\omega}_{ij} & \bar{\omega}_{ij} & \dots & \omega_{ij} \end{pmatrix} \quad \text{for } i, j > 2. \quad (27)$$

Summing over the different flavors occurring in Eq. (26), the relation between $\Omega(t)$ and $\zeta(t)$ can be easily established by

$$\zeta_{ij} = \omega_{ij} \quad \text{for } i < 3, \quad (28)$$

$$\zeta_{ij} = n_F \omega_{ij} \quad \text{for } i > 2, j < 3, \quad (29)$$

$$\zeta_{ij} = \omega_{ij} + (n_F - 1) \bar{\omega}_{ij} \quad \text{for } i > 2, j > 2. \quad (30)$$

4.2 Computational Methods

The gradient-flow formalism in perturbation theory can be formulated in terms of a Lagrangian field theory, where the flow equations (2) are implemented with the help of Lagrange-multiplier fields [2]. The crucial difference between the regular QCD Feynman rules and those in the gradient-flow formalism is the occurrence of exponential factors $\exp(-sp^2)$, where s is a “flow-time variable”, and p the linear combination of D -dimensional external and/or loop momenta. Vertices involving flowed fields induce an integration over all positive values of the corresponding flow-time variable, which is, however, bounded from above by “propagators” of the Lagrange-multiplier fields, since they introduce step functions of the flow-time variables.

We have implemented the Feynman rules into the program **qgraf** [21,22], which generates the Feynman diagrams for the desired matrix elements. Its output is then transformed to **FORM** [23,24] notation by **q2e/exp** [25,26]. An in-house set of **FORM** routines inserts the Feynman rules, performs the projections onto the relevant color and Lorentz structures according to the P_j of Eq. (21) and (23), and evaluates the Dirac and color traces using the **color** package [27]. The result is then expressed in terms of a linear combination of integrals whose general form is as follows:

$$I_l((d_1, \dots, d_f), (b_1, \dots, b_n), (a_1, \dots, a_n)) \\ = \frac{1}{\pi^{D/2}} t^{D/2 - \sum_{j=1}^n a_j} \left[\prod_{i=0}^f \int_0^1 du_i u_i^{d_i} \right] \left[\prod_{r=1}^l \int d^D k_r \right] \frac{\exp(-t \sum_{j=1}^n b_j q_j^2)}{(q_1^2)^{a_1} \dots (q_n^2)^{a_n}}, \quad (31)$$

where the a_i and d_i are integers ($d_i \geq 0$), f and l is the number of flow-time and loop integrations, respectively, the b_j are polynomials in (rescaled) flow-time parameters u_i and the q_i are linear combinations of the loop momenta k_j . For the problem and the perturbative order under consideration, it is $0 \leq f \leq 4$, $1 \leq l \leq 2$, and $0 \leq n \leq 3$, respectively. Note that the projectors defined in Eqs. (21) and (23) eliminate all dependence on external momenta and masses, so that, after making a suitable ansatz for the index structure of the integrals, we only have to evaluate scalar vacuum integrals. Using the identities [16]

$$\int d^D k \left(\frac{\partial}{\partial k} \cdot q \right) f(k, q, \dots) = 0, \quad (32)$$

and similar ones for the flow-time integrations,

$$\int_0^1 ds \frac{\partial}{\partial s} f(s, \dots) = f(1, \dots) - f(0, \dots), \quad (33)$$

one can derive relations among these integrals by explicitly performing the derivatives in the integrand on the l.h.s. These so-called “integration-by-parts (IBP) relations” were fed to Kira [28] which allowed us to reduce all occurring integrals to a single master integral at one-loop level, and six master integrals at two-loop level using the Laporta algorithm [29].⁵ Their analytical evaluation is possible along the lines of Ref. [1]:

$$I_1((), (2), (0)) = 2^{-D/2},$$

$$I_2((), (0, 2, 2), (0, 0, 0)) = 2^{-D},$$

$$I_2((), (1, 1, 1), (0, 0, 0)) = 3^{-D/2},$$

$$I_2((), (1, 1, 1), (1, 1, 0)) = \frac{1}{D-2} \left[-2\pi \csc\left(\frac{D\pi}{2}\right) + \frac{3^{2-D/2}}{D-4} {}_2F_1\left(1, 1; 3 - \frac{D}{2}; \frac{3}{4}\right) \right], \quad (34)$$

$$I_2((), (0, 0, 2), (1, 1, 0)) = -\frac{2^{3-D}\pi}{D-2} \csc\left(\frac{D\pi}{2}\right),$$

$$I_2((0), (2 - u_1, u_1, u_1), (0, 0, 0)) = 2^{2-2D} B_{1/4} \left(1 - \frac{D}{2}, 1 - \frac{D}{2}\right),$$

$$I_2((0), (1 + u_1, 1 + u_1, 1 - u_1), (0, 0, 0)) = 2^{2-2D} \left[B_{3/4} \left(1 - \frac{D}{2}, 1 - \frac{D}{2}\right) - B_{1/2} \left(1 - \frac{D}{2}, 1 - \frac{D}{2}\right) \right].$$

In these expressions, we used $\csc(z) = 1/\sin(z)$, and the hypergeometric function defined as

$${}_2F_1(a, b; c; z) \equiv \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \quad (35)$$

⁵The reduction with Kira 1.0 takes about 20 minutes on 8 CPU threads and requires less than 13 GB of RAM.

with the Pochhammer symbol

$$(x)_n = \frac{\Gamma(x+n)}{\Gamma(x)}. \quad (36)$$

Furthermore, the incomplete beta function is defined by

$$B_z(a, b) = \int_0^z dt t^{a-1} (1-t)^{b-1} \quad (37)$$

and can be expressed as

$$B_z(a, b) = z^a \sum_{n=0}^{\infty} \frac{(1-b)_n}{n!(a+n)} z^n = \frac{z^a}{a} {}_2F_1(a, 1-b; a+1; z). \quad (38)$$

The expansions of the hypergeometric function in the limit $\epsilon \rightarrow 0$ can be obtained with the help of the `Mathematica` [30] package `HypExp` [31,32].

A more detailed description of parts of our setup will be described in a forthcoming publication [33]. As a check, we evaluated the correlators $\langle G_{\mu\nu}^a G_{\mu\nu}^a \rangle$, $\langle \chi \chi \rangle$ and $\langle \chi \not{D} \chi \rangle$ through NLO. They lead to the same set of master integrals as given in Eq. (34). Comparing our results to Ref. [1] and Ref. [5]⁶, we find full agreement.

5 Coefficient functions through NNLO QCD

The strong coupling and the quark mass require the regular QCD renormalization according to

$$g_0 = \left(\frac{\mu e^{\gamma_E/2}}{\sqrt{4\pi}} \right)^\epsilon Z_g g, \quad m_0 = Z_m m, \quad (39)$$

where we write the renormalization constants Z_g and Z_m as

$$\begin{aligned} Z_g &= 1 - \frac{g^2}{(4\pi)^2} \frac{\beta_0}{2\epsilon} + \frac{g^4}{(4\pi)^4} \left(\frac{3\beta_0^2}{8\epsilon^2} - \frac{\beta_1}{4\epsilon} \right) + \mathcal{O}(g^6), \\ Z_m &= 1 - \frac{g^2}{(4\pi)^2} \frac{\gamma_{m,0}}{2\epsilon} + \frac{g^4}{(4\pi)^4} \left[\frac{1}{\epsilon^2} \left(\frac{\gamma_{m,0}^2}{8} + \frac{\beta_0 \gamma_{m,0}}{4} \right) - \frac{\gamma_{m,1}}{4\epsilon} \right] + \mathcal{O}(g^6), \end{aligned} \quad (40)$$

with

$$\begin{aligned} \beta_0 &= \frac{11}{3} C_A - \frac{4}{3} T_F, & \beta_1 &= \frac{34}{3} C_A^2 - \left(4C_F + \frac{20}{3} C_A \right) T_F, \\ \gamma_{m,0} &= 6C_F, & \gamma_{m,1} &= \frac{97}{3} C_A C_F + 3C_F^2 - \frac{20}{3} C_F T_F. \end{aligned} \quad (41)$$

C_F and C_A are the quadratic Casimir eigenvalues of the fundamental and the adjoint representation of the gauge group, respectively. Furthermore, $T_F = T_{n_F}$, with n_F the

⁶We compare to `arXiv` versions 2 and 5 of that paper.

number of quark flavors, and T the trace normalization in the fundamental representation. For $SU(N_c)$, it is $C_F = (N_c^2 - 1)/(2N_c)$, $C_A = N_c$, and $T = 1/2$.

In addition, the flowed quark fields also require renormalization according to

$$\chi_{f,R} = \sqrt{Z_\chi} \chi_f, \quad (42)$$

leading to the factor Z_χ in the definition of the operators $\tilde{O}_{3,4}$ in Eq. (12). Z_χ differs from the quark-field renormalization of regular QCD. Through NLO, the $\overline{\text{MS}}$ result has been evaluated in Ref. [34]. To determine the renormalization constant at NNLO as required by our calculation, we may use the fact that the coefficient $c_3(t)$ must be finite after renormalization. Writing the $\overline{\text{MS}}$ expression as

$$Z_\chi^{-1} = 1 - \frac{g^2}{(4\pi)^2} \frac{\gamma_{\chi,0}}{2\epsilon} + \frac{g^4}{(4\pi)^4} \left[\frac{1}{\epsilon^2} \left(\frac{\gamma_{\chi,0}^2}{8} + \frac{\beta_0 \gamma_{\chi,0}}{4} \right) - \frac{\gamma_{\chi,1}}{4\epsilon} \right] + \mathcal{O}(g^6), \quad (43)$$

we find⁷

$$\begin{aligned} \gamma_{\chi,0} &= 6C_F, \\ \gamma_{\chi,1} &= C_A C_F \left(\frac{223}{3} - 16 \ln 2 \right) - C_F^2 (3 + 16 \ln 2) - \frac{44}{3} C_F T_F. \end{aligned} \quad (44)$$

This allows us to evaluate the coefficients of the energy-momentum tensor in the $\overline{\text{MS}}$ scheme through NNLO QCD:⁸

$$\begin{aligned} c_1(t) &= \frac{1}{g^2} \left\{ 1 + \frac{g^2}{(4\pi)^2} \left[-\frac{7}{3} C_A + \frac{3}{2} T_F - \beta_0 L(\mu, t) \right] \right. \\ &\quad + \frac{g^4}{(4\pi)^4} \left[-\beta_1 L(\mu, t) + C_A^2 \left(-\frac{14482}{405} - \frac{16546}{135} \ln 2 + \frac{1187}{10} \ln 3 \right) \right. \\ &\quad + C_A T_F \left(\frac{59}{9} \text{Li}_2 \left(\frac{1}{4} \right) + \frac{10873}{810} + \frac{73}{54} \pi^2 - \frac{2773}{135} \ln 2 + \frac{302}{45} \ln 3 \right) \\ &\quad \left. \left. + C_F T_F \left(-\frac{256}{9} \text{Li}_2 \left(\frac{1}{4} \right) + \frac{2587}{108} - \frac{7}{9} \pi^2 - \frac{106}{9} \ln 2 - \frac{161}{18} \ln 3 \right) \right] \right. \\ &\quad \left. + \mathcal{O}(g^6) \right\}, \end{aligned} \quad (45)$$

⁷Note that since $c_2 = 0$ at leading order, this coefficient only requires NLO renormalization.

⁸For convenience of the reader, we provide the expressions for $c_1, \dots, c_4$ also in electronic form in an ancillary file with this paper.

$$\begin{aligned}
c_2(t) = & \frac{1}{4g^2} \left\{ -1 + \frac{g^2}{(4\pi)^2} \left[\frac{25}{6} C_A - 3T_F + \beta_0 L(\mu, t) \right] \right. \\
& + \frac{g^4}{(4\pi)^4} \left[\beta_1 L(\mu, t) + C_A^2 \left(\frac{56713}{1620} - \frac{1187}{10} \ln 3 + \frac{16546}{135} \ln 2 \right) \right. \\
& + C_A T_F \left(-\frac{59}{9} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{6071}{405} - \frac{73}{54} \pi^2 + \frac{2287}{135} \ln 2 - \frac{361}{90} \ln 3 \right) \\
& + C_F T_F \left(\frac{220}{9} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{1757}{54} + \frac{10}{9} \pi^2 - \frac{164}{9} \ln 2 + \frac{247}{9} \ln 3 \right) \left. \right] \\
& \left. + \mathcal{O}(g^6) \right\},
\end{aligned} \tag{46}$$

$$\begin{aligned}
c_3(t) = & \frac{1}{4} \left\{ 1 + \frac{g^2}{(4\pi)^2} \left(\frac{3}{2} C_F + \frac{\gamma_{\bar{\psi}\psi}}{2} L(\mu, t) \right) \right. \\
& + \frac{g^4}{(4\pi)^4} \left[\frac{\gamma_{\bar{\psi}\psi}}{4} \left(\beta_0 + \frac{\gamma_{\bar{\psi}\psi}}{2} \right) \left(L^2(\mu, t) + L(\mu, t) \right) + \frac{\gamma_{\bar{\psi}\psi}}{2} L(\mu, t) \right. \\
& + C_F^2 \left(-\frac{137}{9} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{559}{216} + \frac{103}{108} \pi^2 - \frac{1736}{27} \ln 2 + \frac{122}{3} \ln 3 - 4 \ln^2 2 \right) \\
& + C_F T_F \left(-\frac{136}{9} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{3377}{810} - \frac{7}{9} \pi^2 + \frac{1232}{135} \ln 2 - \frac{136}{15} \ln 3 \right) \\
& + C_A C_F \left(-\frac{365}{9} \text{Li}_2 \left(\frac{1}{4} \right) + \frac{261829}{3240} + \frac{77}{108} \pi^2 + \frac{5788}{45} \ln 2 \right. \\
& \left. \left. - \frac{2102}{15} \ln 3 - 4 \ln^2 2 \right) \right] \\
& \left. + \mathcal{O}(g^6) \right\},
\end{aligned} \tag{47}$$

$$\begin{aligned}
c_4(t) = & \frac{C_F}{2} \left\{ \frac{g^2}{(4\pi)^2} + \frac{g^4}{(4\pi)^4} \left[\left(\beta_0 + \frac{\gamma_\chi \bar{a}}{2} \right) L(\mu, t) \right. \right. \\
& + C_F \left(-\frac{161}{18} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{41}{54} - \frac{55}{108} \pi^2 - \frac{1105}{27} \ln 2 + \frac{101}{6} \ln 3 \right) \\
& + T_F \left(\frac{25}{9} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{20573}{1620} + \frac{5}{18} \pi^2 + \frac{6559}{135} \ln 2 - \frac{679}{30} \ln 3 \right) \\
& + C_A \left(\frac{257}{36} \text{Li}_2 \left(\frac{1}{4} \right) - \frac{137}{405} + \frac{11}{216} \pi^2 - \frac{419}{90} \ln 2 + \frac{1157}{60} \ln 3 \right) \Bigg] \\
& \left. + \mathcal{O}(g^6) \right\},
\end{aligned} \tag{48}$$

where we introduced the parameter⁹

$$L(\mu, t) \equiv \ln(2\mu^2 t) + \gamma_E, \tag{49}$$

with the Euler-Mascheroni constant $\gamma_E = 0.57721 \dots$. Through NLO, these results are in full agreement with those of Ref. [4,5]. We have carried out the calculation in the general R_ξ gauge of regular QCD; the fact that the gauge-parameter dependence cancels in the final result serves as another welcome check. The gauge parameter κ of Eq. (2) has been set to 1.

While the energy-momentum tensor $T_{\mu\nu}(x)$ is renormalization-scheme independent, this is not necessarily the case for the operators $\mathcal{O}_{i,\mu\nu}(t, x)$ and the coefficient functions $c_i(t)$. Since $\bar{\mathcal{O}}_{1,\mu\nu}$ and $\bar{\mathcal{O}}_{2,\mu\nu}$ do not require operator renormalization, their matrix elements as well as the coefficient function are indeed renormalization-scheme independent. On the other hand, using the quark-field renormalization Z_χ of Eq. (43) in the $\overline{\text{MS}}$ scheme, matrix elements of $\bar{\mathcal{O}}_{i,\mu\nu}$ and coefficient functions $c_i(t)$ become explicitly dependent on the renormalization scale μ for $i \in \{3, 4\}$.

However, this renormalization-scheme dependence can be avoided by introducing so-called “ringed” quark fields as suggested in Ref. [5]. This corresponds to replacing Z_χ in Eq. (12) by

$$\tilde{Z}_\chi(t) = \frac{-2N_c n_F}{(4\pi t)^2 \langle \bar{\chi}_f(t, x) \not{D} \chi_f(t, x) \rangle}. \tag{50}$$

Currently, $\tilde{Z}_\chi(t)$ is available only through NLO QCD. Its explicitly μ -dependent terms can be reconstructed from the requirement that $c_i(t)$ must be finite and μ -independent for $i \in \{3, 4\}$ though. In this way we find for the ratio to the $\overline{\text{MS}}$ quark-field renormalization

⁹This parameter is motivated by the product of the typical factor $(8\pi t)^*$ occurring in flow-time integrals [1], and the usual definition of the renormalization scale in the $\overline{\text{MS}}$ scheme, see Eq. (39): $(8\pi t)^* (\mu^2 e^{\gamma_E}/(4\pi))^* = 1 + \epsilon L(\mu, t) + \mathcal{O}(\epsilon^2)$.

constant of Eq. (43):

$$\begin{aligned} \zeta_\chi = \frac{\tilde{Z}_\chi}{Z_\chi} = 1 + \frac{g^2}{(4\pi)^2} & \left(\frac{\gamma_{\chi,0}}{2} L(\mu, t) - 3C_F \ln 3 - 4C_F \ln 2 \right) \\ & + \frac{g^4}{(4\pi)^4} \left\{ \frac{\gamma_{\chi,0}}{4} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) L^2(\mu, t) + \left[\frac{\gamma_{\chi,1}}{2} - \frac{\gamma_{\chi,0}}{2} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) \ln 3 \right. \right. \\ & \left. \left. - \frac{2}{3} \gamma_{\chi,0} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) \ln 2 \right] L(\mu, t) + C_2 \right\} \\ & + \mathcal{O}(g^6). \end{aligned} \quad (51)$$

The constant C_2 cannot be determined in this way, but requires a dedicated three-loop calculation of the two-point function occurring in the denominator of Eq. (50). A detailed outline of this calculation is beyond the scope of this paper; it will be presented together with a more complete description of our setup in a forthcoming publication [33]. At this point, we simply quote the numerical value of this result up to three significant digits, which is more than sufficient in the light of the theoretical uncertainties to be discussed below:¹⁰

$$C_2 = -23.8 C_A C_F + 30.4 C_F^2 - 3.92 C_F T_F. \quad (52)$$

Multiplication of $c_3(t)$ and $c_4(t)$ in Eqs. (47) and (48) by this ratio makes also these coefficients formally μ -independent, i.e.,

$$\mu \frac{d}{d\mu} \{c_1, c_2, \hat{c}_3, \hat{c}_4\} = 0, \quad \text{where} \quad \hat{c}_i \equiv \zeta_\chi^{-1} c_i. \quad (53)$$

As in any perturbative calculation, the μ -independence only holds up to higher orders in g . The decrease of the residual μ -dependence is thus commonly used as a qualitative check of the perturbation expansion for the specific observable under consideration. We thus study the μ -dependence of the four coefficients after dividing c_3 and c_4 by the ratio ζ_χ defined in Eq. (51). We fix a characteristic value for the flow time t and vary the renormalization scale μ around the central value μ_0 , which we define such that $L(\mu_0, t) = 0$, cf. Eq. (49), i.e.

$$\mu_0 = \frac{e^{-\gamma_E/2}}{\sqrt{2t}}. \quad (54)$$

Figures 1 and 2 show the leading order (LO), NLO, and the NNLO approximation of c_1 , c_2 , $\hat{c}_3$, and $\hat{c}_4$ as functions of the renormalization scale for two different values of the flow time t , corresponding to $\mu_0 = 3$ GeV and $\mu_0 = 130$ GeV, respectively. In the former case, we set $n_F = 3$, in the latter $n_F = 5$. We use $\alpha_s^{(n_F=5)}(M_Z) = 0.118$ in order to evaluate the input values for the couplings, $g^{(n_F=3)}(3 \text{ GeV}) = 1.77$ and $g^{(n_F=5)}(130 \text{ GeV}) = 1.19$. The μ -variation of the strong coupling constant $g(\mu)$ is determined by numerically solving the corresponding renormalization-group equation with the help of `RunDec` [35, 36] at one-, two-, and three-loop level for the LO, NLO, and the NNLO curve, respectively. In Fig. 1, the value of t is chosen such that the central scale of Eq. (54) is $\mu_0 = 3$ GeV. At this

¹⁰Note that the calculation of $\tilde{Z}_\chi$ also provided an independent check for Z_χ .

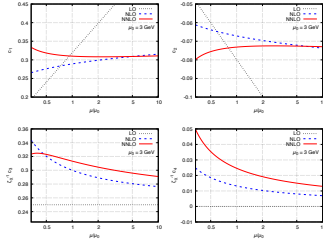


Figure 1: Renormalization-scale dependence of the coefficients c_1 , c_2 , $\hat{c}_3 = \zeta_h^{-1}c_3$, $\hat{c}_4 = \zeta_h^{-1}c_4$, defined in Eqs. (45)–(48), with ζ_h from Eq. (51). The dotted black, dashed blue, and solid red curve correspond to keeping terms up to order $(g^2)^{n-1}$ in c_1 and c_2 , and $(g^2)^n$ in $\hat{c}_3$ and $\hat{c}_4$, with $n = 0, 1, 2$, respectively. The central scale is set to $\mu_0 = 3 \text{ GeV}$, corresponding to $t = 3.12 \cdot 10^{-2}/\text{GeV}^2$, see Eq. (54). The number of flavors is set to $n_f = 3$.

central scale, the NNLO corrections increase the modulus of the coefficients c_1 and c_2 by 10% and 13% relative to NLO, respectively. This is within twice the NLO uncertainty due to missing higher-order effects as estimated by varying μ/μ_0 between 1/2 and 2, where one finds 7.3% for c_1 , and 8.0% for c_2 . We are therefore confident that the NNLO uncertainty estimated in the same way is rather reliable: it is given by 5.7% for c_1 and 7.2% for c_2 . Note that the dominant contribution to these numbers comes from the downward variation of μ , where $g(\mu)$ starts to become sensitive to the non-perturbative region. The behavior of c_1 and c_2 towards larger values of μ seems to suggest that this uncertainty estimate may actually be too conservative.

As opposed to the gluonic coefficients c_1 and c_2 , the coefficients of the fermionic operators c_3 and c_4 exhibit a residual scale dependence only from the NLO term onwards. One therefore expects a stronger μ -dependence at NNLO for these terms. Nevertheless, for $\hat{c}_3$, the estimate of the theory uncertainty due to scale variation still decreases from 9.8% to

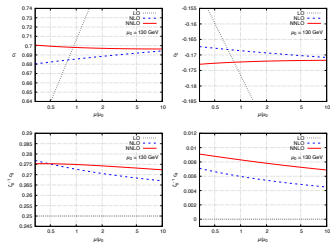


Figure 2: Same as Fig. 1, but for $\mu_0 = 130$ GeV (or $t = 1.66 \cdot 10^{-5}/\text{GeV}^2$), and $n_F = 5$.

8.1%. The increase of the result due to the NNLO effects is 8.3% relative to the NLO result at $\mu = \mu_0$.

The behavior of $\hat{c}_4$, on the other hand, is less satisfactory at $\mu_0 = 3$ GeV. The NNLO effects more than double the NLO result in this case, and the uncertainty estimate due to scale variation actually *increases* from 47% to 71% when going from NLO to NNLO. Note, however, that $c_4 = 0$ at LO, which means that this coefficient is numerically sub-dominant.

As one would expect, for $\mu_0 = 130$ GeV, the perturbative behavior of all coefficients is significantly improved, cf. Fig. 2. For c_1 , c_2 , and $\hat{c}_3$, the scale uncertainty is at the sub-percent level already at NLO; at NNLO, it amounts to less than 0.2% in all three cases. The effect of the NNLO corrections relative to the NLO result is about 2% for c_1 and c_2 , and 0.8% for $\hat{c}_3$. Also for $\hat{c}_4$, the situation improves significantly: the NNLO terms add 38% to the NLO result, and the uncertainty goes down from 10% at NLO to 5.8% at NNLO.

It is also worth pointing out that the choice of the central scale μ_0 as defined in Eq. (54) seems justified by the behavior of the successively higher orders. In almost all cases, the NLO and the NNLO corrections are *both* relatively small at $\mu = \mu_0$. At the same time, the NNLO corrections relative to the NLO result are always smaller than the NLO corrections compared to the LO result. The only exception to this is $\hat{c}_4$ at $\mu_0 = 3$ GeV, where, however, no choice of μ seems to stand out over any other.

In summary, we conclude that the NNLO terms lead to a significant improvement of the perturbative accuracy of the Wilson coefficients.

6 Trace anomaly

As a test of our result, we use the trace anomaly of the EMT. As suggested in Ref. [17], a simple derivation consists of taking the trace of the EMT in $D = 4 - 2\epsilon$ dimensions. By use of the equations of motion, this gives for the gauge invariant part

$$T_{\mu\mu} = \frac{\epsilon}{2g_0^2} F_{\mu\alpha}^a F^{\alpha}_{\mu} - \sum_{f=1}^{n_f} m_f \bar{\psi}_f \psi_f = \frac{1}{2D} \left(\frac{\epsilon}{g_0^2} \mathcal{O}_{2,\mu\mu} + \mathcal{O}_{4,\mu\mu} \right), \quad (55)$$

where we have used Eq. (10) in the last step. Using the mixing matrix $\zeta_{ij}(t)$, we can rewrite this in terms of flowed operators:

$$T_{\mu\mu} = \bar{c}_1(t) \bar{\mathcal{O}}_{4,\mu\mu}(t, x), \quad \bar{c}_1(t) = \frac{1}{2D} \left(\frac{\epsilon}{g_0^2} \zeta_{21}^{-1}(t) + \zeta_{44}^{-1}(t) \right). \quad (56)$$

Note that $\bar{c}_1(t) = \bar{c}_3(t) = 0$, as $\bar{\mathcal{O}}_{1,\mu\nu}(t, x)$ and $\bar{\mathcal{O}}_{3,\mu\nu}(t, x)$ have a non-trivial index structure and therefore $\mathcal{O}_{2,\mu\nu}(x)$ and $\mathcal{O}_{4,\mu\nu}(x)$ cannot mix with them. Since $\bar{\mathcal{O}}_{2,\mu\mu} = D\bar{\mathcal{O}}_{1,\mu\mu}$ and $2\bar{\mathcal{O}}_{4,\mu\mu} = D\bar{\mathcal{O}}_{3,\mu\mu}$, we cannot equate coefficients with Eq. (15) for all i individually. Instead, only the weaker conditions

$$c_1(t) + Dc_2(t) = D\bar{c}_2(t), \quad 2c_3(t) + Dc_4(t) = D\bar{c}_4(t). \quad (57)$$

can be derived. We checked that these equations are indeed fulfilled by our result.

7 Operator renormalization

Using the fact that flowed operators are finite after mass and field renormalization, we can also compute the renormalization matrix for the regular QCD operators $\{\mathcal{O}_{1,\mu\nu}(x), \mathcal{O}_{2,\mu\nu}(x), \mathcal{O}_{3,\mu\nu}(x), \mathcal{O}_{4,\mu\nu}(x)\}$ defined in Eq. (8). It is convenient to define an equivalent set of operators as

$$\hat{\mathcal{O}}_{i,\mu\nu}(x) = H_{ij} \mathcal{O}_{j,\mu\nu}(x), \quad \text{where} \quad H_{ij} = \begin{cases} 1/g_0^2 & \text{for } i = j \in \{1, 2\}, \\ 1 & \text{for } i = j \in \{3, 4\}, \\ 0 & \text{for } i \neq j \end{cases}. \quad (58)$$

This multiplication of $\mathcal{O}_{1,\mu\nu}$ and $\mathcal{O}_{2,\mu\nu}$ by $1/g_0^2$ ensures that the mass dimension of all operators $\hat{\mathcal{O}}_{i,\mu\nu}$ is equal to D . The renormalization matrix is then defined as

$$\{\hat{\mathcal{O}}_i\}_R(x) = Z_{ij} \hat{\mathcal{O}}_j(x). \quad (59)$$

Expressing the $\hat{\mathcal{O}}_{i,\mu\nu}(x)$ in terms of flowed operators, one can determine its entries in the $\overline{\text{MS}}$ scheme by demanding that

$$\{\hat{\mathcal{O}}_i\}_R(x) = Z_{ij} H_{jk} \zeta_{\text{kl}}^{-1}(t) \hat{\mathcal{O}}_l(t, x) \quad (60)$$

be finite. In analogy to Eqs. (40) and (43), we write

$$Z_{ij} = \delta_{ij} - \frac{g^2}{(4\pi)^2} \frac{\gamma_{0,ij}}{2\epsilon} + \frac{g^4}{(4\pi)^4} \left[\frac{1}{\epsilon^2} \left(\frac{\gamma_{0,ik} \gamma_{0,kj}}{8} + \frac{\beta_0}{4} \gamma_{0,ij} \right) - \frac{\gamma_{1,ij}}{4\epsilon} \right] + \mathcal{O}(g^6). \quad (61)$$

Our result for the coefficients of the anomalous dimension at NLO is in agreement with Ref. [5]:

$$\gamma_0 = \begin{pmatrix} -\frac{8}{3} T_F & \frac{11}{6} C_A & \frac{4}{3} C_F & \frac{7}{3} C_F \\ 0 & \frac{22}{3} C_A - \frac{8}{3} T_F & 0 & 12 C_F \\ \frac{32}{9} T_F & -\frac{8}{3} T_F & -\frac{16}{9} C_F & \frac{4}{3} C_F \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (62)$$

At NNLO, we find

$$\begin{aligned} \gamma_{1,11} &= -\frac{140}{27} C_A T_F - \frac{296}{27} C_F T_F, & \gamma_{1,12} &= \frac{34}{3} C_A^2 - \frac{112}{27} C_A T_F - \frac{10}{27} C_F T_F, \\ \gamma_{1,13} &= \frac{188}{27} C_A C_F - \frac{56}{27} C_F^2 - \frac{104}{27} C_F T_F, & \gamma_{1,14} &= \frac{812}{27} C_A C_F + \frac{85}{27} C_F^2 - \frac{44}{27} C_F T_F, \\ \gamma_{1,21} &= 0, & \gamma_{1,22} &= \frac{136}{3} C_A^2 - \frac{80}{3} C_A T_F - 16 C_F T_F, \\ \gamma_{1,23} &= 0, & \gamma_{1,24} &= \frac{388}{3} C_A C_F + 12 C_F^2 - \frac{80}{3} C_F T_F, \\ \gamma_{1,31} &= \frac{560}{27} C_A T_F + \frac{1184}{27} C_F T_F, & \gamma_{1,32} &= -\frac{272}{27} C_A T_F - \frac{392}{27} C_F T_F, \\ \gamma_{1,33} &= -\frac{752}{27} C_A C_F + \frac{224}{27} C_F^2 + \frac{416}{27} C_F T_F, & \gamma_{1,34} &= \frac{244}{27} C_A C_F - \frac{16}{27} C_F^2 - \frac{544}{27} C_F T_F, \\ \gamma_{1,41} &= \gamma_{1,42} = \gamma_{1,43} = \gamma_{1,44} = 0. \end{aligned} \quad (63)$$

The renormalization matrix Z_{ij} and the mixing matrix ζ_{ij} are provided in ancillary files with this paper. In this way, one obtains the following expression for the energy-density operator in terms of flowed operators, for example:

$$\begin{aligned}
\left\{ \frac{1}{g_0^2} F_{\mu\nu}(x) F_{\mu\nu}(x) \right\}_R &= \frac{Z_{2j} H_{jk}}{D} \left[\zeta_{k2}^{-1}(t) \tilde{\mathcal{O}}_{2,\mu\nu}(t, x) + \zeta_{k4}^{-1}(t) \tilde{\mathcal{O}}_{4,\mu\nu}(t, x) \right] \\
&= \frac{1}{g^2} G_{\mu\nu}(t, x) G_{\mu\nu}(t, x) \left\{ 1 - \frac{7}{2} C_A \frac{g^2}{(4\pi)^2} \right. \\
&\quad + \frac{g^4}{(4\pi)^4} \left[\left(-\frac{3}{2} C_A^2 - 2 C_A T_F - 14 C_F T_F \right) L(\mu, t) \right. \\
&\quad \left. + C_A^2 \left(-\frac{1427}{180} + \frac{87}{5} \ln 2 - \frac{54}{5} \ln 3 \right) + \frac{8}{9} C_A T_F - \frac{34}{3} C_F T_F \right] \Big\} \\
&+ Z_\chi \sum_{f=1}^{n_f} \bar{\chi}_f(t, x) \overleftrightarrow{\mathcal{D}} \chi_f(t, x) \left\{ \frac{g^2}{(4\pi)^2} C_F (5 + 6L(\mu, t)) \right. \\
&\quad + \frac{g^4}{(4\pi)^4} \left[\gamma_{\chi,0} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) L^2(\mu, t) \right. \\
&\quad + \left(\frac{304}{3} C_A C_F + 18 C_F^2 - \frac{80}{3} C_F T_F \right) L(\mu, t) \\
&\quad + C_A C_F \left(-2\text{Li}_2\left(\frac{1}{4}\right) + \frac{2923}{30} - \frac{4}{3}\pi^2 + \frac{8546}{15} \ln 2 - \frac{2139}{5} \ln 3 \right) \\
&\quad \left. \left. + 6 C_F^2 - C_F T_F \left(30 + \frac{4}{3}\pi^2 \right) \right] \right\}. \tag{64}
\end{aligned}$$

Through NLO, this result agrees with Refs. [5,37]. Similar relations can be derived for all other operators of Eq. (8), of course.

8 Conclusions

We have presented the universal Wilson coefficients for the gradient-flow definition of the energy-momentum tensor through NNLO QCD. The NNLO corrections modify the three numerically dominant coefficients c_1 , c_2 , c_3 at the level of 10% (1-2%) for a central scale of $\mu_0 = 3 \text{ GeV}$ ($\mu_0 = 130 \text{ GeV}$), where μ_0 is related to the flow time t according to Eq. (54). We observe a reduction of the theoretical uncertainty relative to the NLO result as derived from varying the renormalization scale by a factor of two around its central value. The behavior of the fourth coefficient c_4 is less satisfactory, but its impact is expected to be numerically suppressed.

Aside from this main outcome, new results presented in this paper include the flowed quark-field renormalization constant to NNLO in the $\overline{\text{MS}}$ scheme, and the anomalous dimension matrix for the regular QCD operators which make up the EMT.

In conclusion, we hope that our results will help to improve the studies of the EMT on

the lattice. They are the first outcome of a systematic setup for higher-order perturbative calculations within the gradient-flow formalism [33], which should prove useful also in other applications of this theoretical framework.

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A Feynman rules

In this section, we present the Feynman rules for the operators defined in Eq. (8) in regular QCD. Only the terms which are relevant for the construction of the projectors in Eqs. (21) and (23) are listed explicitly. All momenta are understood to be outgoing.

$$\mathcal{O}_{1,\mu\nu} = \partial_\mu A_\rho^c \partial_\nu A_\rho^c + \dots \quad \Rightarrow \quad \begin{array}{c} a, \alpha \quad b, \beta \\ \swarrow \quad \searrow \\ p_1 \quad p_2 \\ \text{---} \mu \quad \nu \end{array} = -p_{1,\rho} p_{2,\sigma} \delta_{\alpha\beta} \delta^{\sigma\mu} \quad (65)$$

$$\mathcal{O}_{2,\mu\nu} = 2\delta_{\mu\nu} \partial_\rho A_\rho^c \partial_\sigma A_\sigma^c + \dots \quad \Rightarrow \quad \begin{array}{c} a, \alpha \quad b, \beta \\ \swarrow \quad \searrow \\ p_1 \quad p_2 \\ \text{---} \mu \quad \nu \end{array} = -4\delta_{\mu\sigma} p_1 \cdot p_2 \delta_{\alpha\beta} \delta^{\sigma\mu} \quad (66)$$

$$\mathcal{O}_{3f,\mu\nu} = \bar{\psi}_f \gamma_\mu \partial_\nu \psi_f + \dots \quad \Rightarrow \quad \begin{array}{c} i \quad j \\ \swarrow \quad \searrow \\ p_1 \quad p_2 \\ \text{---} \mu \quad \nu \end{array} = i\gamma_\mu p_{2\nu} \delta^{ij} \quad (67)$$

$$\mathcal{O}_{4f,\mu\nu} = \delta_{\mu\nu} \bar{\psi}_f \not{\partial} \psi_f + \dots \quad \Rightarrow \quad \begin{array}{c} i \quad j \\ \swarrow \quad \searrow \\ p_1 \quad p_2 \\ \text{---} \mu \quad \nu \end{array} = i\delta_{\mu\nu} \not{p}_2 \delta^{ij} \quad (68)$$

$$\mathcal{O}_{5f,\mu\nu} = \delta_{\mu\nu} m_0 \bar{\psi}_f \psi_f \quad \Rightarrow \quad \begin{array}{c} i \quad j \\ \swarrow \quad \searrow \\ p_1 \quad p_2 \\ \text{---} \mu \quad \nu \end{array} = \delta_{\mu\nu} m_0 \delta^{ij} \quad (69)$$

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P37: 梯度流高阶计算的技术与结果

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Results and techniques for higher order calculations within the gradient-flow formalism

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Abstract

We describe in detail the implementation of a systematic perturbative approach to observables in the QCD gradient-flow formalism. This includes a collection of all relevant Feynman rules of the five-dimensional field theory and the composite operators considered in this paper. Tools from standard perturbative calculations are used to obtain Green's functions at finite flow time t at higher orders in perturbation theory. The three-loop results for the quark condensate at finite t and the conversion factor for the “ringed” quark fields to the $\overline{\text{MS}}$ scheme are presented as applications. We also re-evaluate an earlier result for the three-loop gluon condensate, improving on its accuracy.

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1 Introduction

The gradient flow in field theory has proven to be a very useful concept. Originally introduced as a method to regularize divergences and ultraviolet fluctuations in lattice calculations [1, 2], it has mainly gained traction through its use for scale-setting in lattice QCD [3–5] and has meanwhile become an indispensable tool in many practical calculations in this field (see, e.g. Refs. [6–9]). While first focused on QCD, generalizations of the gradient-flow formalism (GFF) yield a much wider range of applications, for example the study of dualities in field theory [10–15].

In QCD, the formulation as a five-dimensional field theory has been presented in Ref. [16], where the additional dimension is associated with the so-called flow time. The form of

the fundamental theory (i.e., actual QCD) serves as the boundary condition at vanishing flow time. It has been proven that, when expressed in terms of renormalized parameters and fields, composite operators at positive flow time are finite [16]. This property and the realization that the gradient flow can be used as a matching scheme between lattice and perturbative calculations shed light on a whole new set of applications that are currently being explored.

One of the first possible cross-fertilizations among perturbative and lattice QCD is given by the definition of a new scheme for the strong coupling, defined by the gluon condensate (QCD action density) at finite flow time [3]. Its perturbative relation to the $\overline{\text{MS}}$ coupling is known through three loops [17].¹ Another cross-boundary application is the gradient-flow definition of the energy-momentum tensor, which uses the small-flow-time expansion in order to express it in terms of well-defined composite operators at positive flow time and perturbatively accessible coefficient functions [19–22]. Yet another example is the proposal to relate Euclidean quasi PDFs² on the lattice to perturbative light-front PDFs by using the gradient flow [23, 24].

All these applications rely on input from the lattice as well as from perturbative calculations, both at finite flow time. It was found that higher order corrections in perturbation theory are crucial for reducing the perturbative truncation error as estimated by the variation of the renormalization scale [17, 21]. Such corrections were also found to significantly stabilize the required extrapolations of the corresponding lattice results [22].

Despite the fact that loop integrals in the GFF involve additional exponential factors as well as integrations over flow-time variables, many important techniques for regular perturbative calculations retain their usefulness, albeit in slightly generalized form. It is the goal of this paper to provide the basis for further perturbative calculations within the GFF, and thus to contribute to the further exploration of the capabilities of this approach. Sect. 2 reviews the QCD gradient flow in perturbation theory, recapitulating results of Refs. [3, 6, 16]. The various stages of automation in a perturbative multi-loop calculation of correlation functions are described in Sect. 3: the generation of Feynman diagrams and insertion and evaluation of Feynman rules, followed by a reduction of the loop integrals to a set of master integrals, and finally the numerical evaluation of the latter. As an application, Sect. 4 presents the three-loop results for the gluon condensate, obtained before in Ref. [17], the quark condensate, as well as the “ringed” quark field renormalization constant, which are new results. These quantities allow one to define a precision gradient-flow coupling and gradient-flow mass scheme and their matching to $\overline{\text{MS}}$

¹Unless stated otherwise, we refer to perturbative calculations as performed at infinite volume in this paper. The inclusion of finite-volume effects requires different techniques, such as Numerical Stochastic Perturbation Theory, see Ref. [18].

²PDF=parton distribution function.

as shown in Sect. 5. Our conclusions are presented in Sect. 6.

2 The QCD gradient flow in perturbation theory

For completeness, we collect the main steps of the original derivations of Refs. [3, 16] in this section.

2.1 Definition of the gradient flow

In the following, we work in D -dimensional Euclidean space-time with $D = 4 - 2\epsilon$. The GFF continues the gluon and quark fields $A_\mu^a(x)$ and $\psi_\alpha^i(x)$ of regular³ QCD to $(D+1)$ -dimensional fields $B_\mu^a(t, x)$ and $\chi_\alpha^i(t, x)$ through the boundary conditions

$$B_\mu^a(t=0, x) = A_\mu^a(x), \quad \chi_\alpha^i(t=0, x) = \psi_\alpha^i(x) \quad (1)$$

and the flow equations [3, 6]

$$\begin{aligned} \partial_t B_\mu^a &= \mathcal{D}_\nu^{ab} G_{\nu\mu}^b + \kappa \mathcal{D}_\mu^{ab} \partial_\nu B_\nu^b, \\ \partial_t \chi &= \Delta \chi - \kappa \overleftarrow{\partial}_\mu B_\mu^a T^a \chi, \\ \partial_t \bar{\chi} &= \bar{\chi} \overleftarrow{\Delta} + \kappa \bar{\chi} \overleftarrow{\partial}_\mu B_\mu^a T^a, \end{aligned} \quad (2)$$

where the “flow time” t is a parameter of mass dimension minus two, and κ is a gauge parameter which drops out of physical observables (see below).

The $(D+1)$ -dimensional field-strength tensor is defined as

$$G_{\mu\nu}^a = \partial_\mu B_\nu^a - \partial_\nu B_\mu^a + f^{abc} B_\mu^b B_\nu^c, \quad (3)$$

the covariant derivative in the adjoint representation is given by

$$\mathcal{D}_\mu^{ab} = \delta^{ab} \partial_\mu - f^{abc} B_\mu^c, \quad (4)$$

and

$$\Delta = \mathcal{D}_\mu^F \mathcal{D}_\mu^F, \quad \overleftarrow{\Delta} = \overleftarrow{\mathcal{D}}_\mu^F \overleftarrow{\mathcal{D}}_\mu^F, \quad (5)$$

with the covariant derivative in the fundamental representation,

$$\mathcal{D}_\mu^F = \partial_\mu + B_\mu^a T^a, \quad \overleftarrow{\mathcal{D}}_\mu^F = \overleftarrow{\partial}_\mu - B_\mu^a T^a. \quad (6)$$

³We use the terms “flowed” and “regular” QCD to distinguish quantities defined at $t > 0$ from those defined at $t = 0$.

As usual, the color indices of the adjoint representation are denoted by $a, b, c, \dots$, while $\mu, \nu, \rho, \dots$ are D -dimensional Lorentz indices. Color indices of the fundamental representation are denoted by $i, j, k, \dots$, but they are suppressed throughout this paper, unless required by clarity; similarly for spinor indices $\alpha, \beta, \gamma, \dots$. The symmetry generators T^a are understood in the fundamental representation. They obey the commutation relation

$$[T^a, T^b] = f^{abc} T^c, \quad (7)$$

with the structure constants f^{abc} and the trace is normalized to

$$\text{Tr}(T^a T^b) = -T_R \delta^{ab}, \quad (8)$$

with $T_R > 0$.

Different choices of κ in Eq. (2) correspond to gauge transformations of the form

$$\chi \rightarrow \chi' = \Lambda \chi \quad \text{and} \quad B_\mu^a T^a \rightarrow B_\mu'^a T^a = \Lambda B_\mu^a T^a \Lambda^{-1} + \Lambda \partial_\mu \Lambda^{-1}, \quad (9)$$

where

$$\Lambda(t, x) = e^{-\int_0^t ds \kappa \partial_\mu B_\mu^a(s, x) T^a}. \quad (10)$$

Of course, all observables are independent of the gauge parameter κ [3]. In perturbative calculations, it is usually most convenient to set $\kappa = 1$.

The flow equations (2) can be incorporated in a Lagrangian formalism by defining

$$\mathcal{L} = \mathcal{L}_{\text{QCD}} + \mathcal{L}_{\text{gauge-fixing}} + \mathcal{L}_{\text{ghost}} + \mathcal{L}_B + \mathcal{L}_\chi. \quad (11)$$

The first three terms constitute the regular Yang-Mills Lagrangian, with fermions added in the fundamental representation (quarks). Introducing an index f in order to distinguish different quark flavors of mass m_f , the classical, gauge-fixing, and Faddeev-Popov ghost part are given by

$$\begin{aligned} \mathcal{L}_{\text{QCD}} &= \frac{1}{4g^2} F_{\mu\nu}^a F_{\mu\nu}^a + \sum_{f=1}^{n_f} \bar{\psi}_f (\not{D} \not{=} + m_f) \psi_f, \\ \mathcal{L}_{\text{gauge-fixing}} &= \frac{1}{2g^2 \xi} (\partial_\mu A_\mu^a)^2, \\ \mathcal{L}_{\text{ghost}} &= \frac{1}{g^2} \partial_\mu \bar{c}^a D_\mu^{ab} c^b, \end{aligned} \quad (12)$$

respectively, where

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c \quad (13)$$

is the regular field strength tensor and

$$D_\mu^F = \partial_\mu + A_\mu^a T^a, \quad D_\mu^{ab} = \delta^{ab} \partial_\mu - f^{abc} A_\mu^c \quad (14)$$

are the regular covariant derivatives in the fundamental and adjoint representation, respectively, g is the gauge coupling, ξ the QCD gauge parameter, and n_F the number of different quark flavors. The flow equations are incorporated by introducing Lagrange multiplier fields

$$L_\mu^a(t, x) \quad \text{and} \quad \lambda_f(t, x), \quad \bar{\lambda}_f(t, x), \quad (15)$$

of mass dimensions 3 and 5/2 that otherwise carry the same quantum numbers as the flowed gluon and quark/antiquark fields B_μ^a and $\chi, \bar{\chi}$, respectively. Their Euler-Lagrange equations derived from

$$\begin{aligned} \mathcal{L}_B = & -2 \int_0^\infty dt \text{Tr} \left[L_\mu^a T^a \left(\partial_t B_\mu^b T^b - D_\mu^{bc} G_{\nu\mu}^c T^b - \kappa D_\mu^{bc} \partial_b B_\nu^c T^b \right) \right], \\ \mathcal{L}_\chi = & \sum_{f=1}^{n_F} \int_0^\infty dt \left(\bar{\lambda}_f \left(\partial_t - \Delta + \kappa \left(\partial_\mu B_\mu^a \right) T^a \right) \chi_f \right. \\ & \left. + \bar{\chi}_f \left(\bar{\partial}_t - \bar{\Delta} - \kappa \left(\partial_\mu B_\mu^a \right) T^a \right) \lambda_f \right), \end{aligned} \quad (16)$$

indeed lead to Eq. (2) [16].

Note that the Lagrangian (11) does not include flowed ghost fields $d^a(t, x)$ and $\bar{d}^a(t, x)$. They arise in the same way as the usual Faddeev-Popov ghosts $c^a(x)$ and $\bar{c}^a(x)$ due to gauge-fixing, and obey the initial condition

$$d^a(t, x)|_{t=0} = c^a(x). \quad (17)$$

Similar to the ghosts of the regular gauge fields, they always form closed loops as long as there are no external d^a or $\bar{d}^a$ fields (which can be avoided by considering only physical degrees of freedom in amplitudes with external gluons). As becomes clear later, closed loops of only flowed fields vanish, so that one can omit d^a and $\bar{d}^a$ already at the level of the Lagrangian.

2.2 Perturbative solution of the flow equations

Let us introduce the short-hand notation

$$\int_p \equiv \int \frac{d^D p}{(2\pi)^D}, \quad \int_x \equiv \int d^D x, \quad (18)$$

where it should be clear from the context whether an integration variable is in position $(x, y, z, \dots)$ or momentum space $(p, k, q, \dots)$.

It is helpful to separate the flow equation (2) of the flowed gauge field $B_\mu^a(t, x)$ into a linear and a non-linear part:

$$\begin{aligned} \partial_t B_\mu^a &= \partial_\nu \partial_\nu B_\mu^a + (\kappa - 1) \partial_\mu \partial_\nu B_\nu^a + R_\mu^a, \\ R_\mu^a &= 2f^{abc} B_\nu^b \partial_\nu B_\mu^c - f^{abc} B_\mu^b \partial_\nu B_\nu^c + (\kappa - 1) f^{abc} B_\nu^b \partial_\nu B_\mu^c + f^{abc} f^{ade} B_\nu^b B_\nu^c B_\mu^d. \end{aligned} \quad (19)$$

The linear equation can be solved by introducing the integration kernel

$$K_{\mu\nu}(t, x) = \int_p \frac{e^{ipx}}{p^2} \left((\delta_{\mu\nu} p^2 - p_\mu p_\nu) e^{-tp^2} + p_\mu p_\nu e^{-\kappa tp^2} \right) \equiv \int_p e^{ipx} \tilde{K}_{\mu\nu}(t, p), \quad (20)$$

which fulfills

$$\lim_{t \rightarrow 0} K_{\mu\nu}(t, x) = \delta_{\mu\nu} \delta^{(D)}(x). \quad (21)$$

Taking into account the initial condition (1), the full solution of the flow equation is then given by

$$B_\mu^a(t, x) = \int_y K_{\mu\nu}(t, x - y) A_\nu^a(y) + \int_y \int_0^t ds K_{\mu\nu}(t - s, x - y) R_\nu^a(s, y), \quad (22)$$

or, in momentum space,

$$\tilde{B}_\mu^a(t, p) = \int_x e^{-ipx} B_\mu^a(t, x) = \tilde{K}_{\mu\nu}(t, p) \tilde{A}_\nu^a(p) + \int_0^t ds \tilde{K}_{\mu\nu}(t - s, p) \tilde{R}_\nu^a(s, p). \quad (23)$$

By inserting the solution iteratively into itself, one can express the Fourier transform of the non-linear part of Eq. (19) as

$$\begin{aligned} \tilde{R}_\mu^a(t, p) &= \int_{q, l, k} (2\pi)^D \delta^{(D)}(p - q - l - k) \left[\delta^{(D)}(k) \cdot X_{2, \mu\nu\rho}^{abc}(q, l) \tilde{B}_\nu^b(t, q) \tilde{B}_\rho^c(t, l) \right. \\ &\quad \left. + X_{3, \mu\nu\rho\sigma}^{abcd} \tilde{B}_\nu^b(t, q) \tilde{B}_\rho^c(t, l) \tilde{B}_\sigma^d(t, k) \right], \end{aligned} \quad (24)$$

where

$$\begin{aligned} X_{2, \mu\nu\rho}^{abc}(q, l) &= -i f^{abc} [(l - q)_\mu \delta_{\nu\rho} + 2q_\nu \delta_{\mu\rho} - 2l_\nu \delta_{\mu\rho} + (\kappa - 1)(q_\mu \delta_{\nu\rho} - l_\mu \delta_{\nu\rho})], \\ X_{3, \mu\nu\rho\sigma}^{abcd} &= f^{abc} f^{cd\epsilon} (\delta_{\mu\sigma} \delta_{\nu\rho} - \delta_{\mu\rho} \delta_{\nu\sigma}) + f^{ade} f^{bec} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) \\ &\quad + f^{ace} f^{dbe} (\delta_{\mu\sigma} \delta_{\nu\rho} - \delta_{\mu\rho} \delta_{\nu\sigma}). \end{aligned} \quad (25)$$

The structure of X_3 is identical to the four-gluon vertex of regular QCD. When formulating the Feynman rules later, X_2 and X_3 describe the three- and four-point vertices of the flowed gluon fields.

The flow equation (2) for the flowed quark fields can be solved by again splitting it into a linear and a non-linear part,

$$\partial_t \chi = \partial_\mu \partial_\mu \chi + \Delta' \chi \quad \text{with} \quad \Delta' = (1 - \kappa) \partial_\mu B_\mu^a T^a + 2 B_\mu^a T^a \partial_\mu + B_\mu^a B_\mu^b T^a T^b. \quad (26)$$

The linear equation is solved by the integration kernel

$$K(t, x) = \int_p e^{ipx} e^{-tp^2} \equiv \int_p e^{ipx} \tilde{K}(t, p), \quad (27)$$

with the help of which we can write the full solution as

$$\chi(t, x) = \int_y K(t, x - y) \psi(y) + \int_y \int_0^t ds K(t - s, x - y) \Delta' \chi(s, y). \quad (28)$$

Here and in what follows, we suppress the flavor index f unless required for clarity. The non-linear part of the Fourier-transformed field

$$\tilde{\chi}(t, p) = \tilde{K}(t, p) \tilde{\psi}(p) + \int_0^t ds \tilde{K}(t - s, p) \tilde{\Delta' \chi}(s, p) \quad (29)$$

can be expressed as

$$\begin{aligned} \tilde{\Delta' \chi}(t, p) = & \int_{q, r} (2\pi)^D \delta^{(D)}(p - q - l - r) \left[\delta^{(D)}(l) \cdot Y_{1, \nu}^b(p, q, r) \tilde{B}_\nu^b(t, q) \right. \\ & \left. + \frac{1}{2} Y_{2, \nu \rho}^{bc}(p, q, l, r) \tilde{B}_\nu^b(t, q) \tilde{B}_\rho^c(t, l) \right] \tilde{\chi}(t, r), \end{aligned} \quad (30)$$

where

$$Y_{1, \nu}^b(q, r) = i(2r_\nu + (1 - \kappa)q_\nu)T^b, \quad Y_{2, \nu \rho}^{bc} = \delta_{\nu \rho} \{T^b, T^c\}. \quad (31)$$

These expressions lead to the three- and four-point vertices of the flowed quark fields. For $\tilde{\chi}$ one proceeds analogously.

2.3 Feynman rules

(Flowed) Propagators

Plugging in the solution of the flowed gluon field in momentum space Eq. (23) into the two-point function, one finds [16]

$$\begin{aligned} \left\langle \tilde{B}_\mu^a(t, p) \tilde{B}_\nu^b(s, q) \right\rangle \Big|_{\text{LO}} &= \tilde{K}_{\mu\rho}(t, p) \tilde{K}_{\nu\sigma}(s, q) \left\langle \tilde{A}_\rho^a(p) \tilde{A}_\sigma^b(q) \right\rangle \\ &= (2\pi)^D \delta^{(D)}(p + q) g^2 D_{\mu\nu}^{ab}(p, t + s, \xi, \kappa), \end{aligned} \quad (32)$$

where

$$D_{\mu\nu}^{ab}(p, t, \xi, \kappa) = \delta^{ab} \frac{1}{p^2} \left(\left(\delta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2} \right) e^{-tp^2} + \xi \frac{p_\mu p_\nu}{p^2} e^{-\kappa tp^2} \right), \quad (33)$$

and we have used the result for the fundamental gluon propagator,

$$\left\langle \tilde{A}_\mu^a(p) \tilde{A}_\nu^b(q) \right\rangle \Big|_{\text{LO}} = (2\pi)^D \delta^{(D)}(p+q) g^2 D_{\mu\nu}^{ab}(p, 0, \xi, 0). \quad (34)$$

The factor g^2 is taken into account in the corresponding vertices further below. Since the flowed gluon propagator coincides with the fundamental gluon propagator at $t = s = 0$, we can express both of them by the same Feynman rule:

$$s, \nu, b \xrightarrow{p} t, \mu, a = D_{\mu\nu}^{ab}(p, t+s, \xi, \kappa). \quad (35)$$

We refer to this as the (flowed) gluon propagator. Eq. (35) also applies to the mixed propagator $\langle \tilde{A}\tilde{B} \rangle$, which is obtained by setting one of the two flow-time variables to zero.

The same can be done for flowed quark fields. Inserting their momentum space solution in Eq. (29) into the two-point function results in

$$\begin{aligned} \left\langle \tilde{\chi}_\alpha^i(t, p) \tilde{\chi}_\beta^j(s, q) \right\rangle \Big|_{\text{LO}} &= \tilde{K}(t, p) \tilde{K}(s, q) \left\langle \tilde{\psi}_\alpha^i(p) \tilde{\psi}_\beta^j(q) \right\rangle \\ &= (2\pi)^D \delta^{(D)}(p+q) S_{F, \alpha\beta}^{ij}(p, m, t+s), \end{aligned} \quad (36)$$

where

$$S_F^{ij}(p, m, t) = \delta^{ij} \frac{-i\not{p} + m}{p^2 + m^2} e^{-tp^2}, \quad (37)$$

and we have used the result for the fundamental quark propagator

$$\left\langle \tilde{\psi}_\alpha^i(p) \tilde{\psi}_\beta^j(q) \right\rangle = (2\pi)^D \delta^{(D)}(p+q) S_{F, \alpha\beta}^{ij}(p, m, 0). \quad (38)$$

Since one can express the fundamental propagator through the flowed propagator at vanishing flow times, both can be represented by the same Feynman rule:

$$s, \beta, j \xrightarrow{p} t, \alpha, i = S_{F, \alpha\beta}^{ij}(p, m, t+s). \quad (39)$$

We refer to this as the (flowed) quark propagator, which again includes the mixed propagators $\langle \psi\chi \rangle$ and $\langle \chi\bar{\psi} \rangle$.

Flow lines

Since there are no quadratic terms of the Lagrange multiplier fields L_μ^a , λ , and $\bar{\lambda}$ in Eq. (16), there are no propagators for these fields. However, the Lagrangian contains bilinear terms of a flowed field and a Lagrange multiplier field. Using standard methods, one derives the two-point function at leading order as

$$\left\langle B_\mu^a(t, x) L_\nu^b(s, y) \right\rangle = \delta^{ab} H_{\mu\nu}(t - s, x - y), \quad (40)$$

where $H_{\mu\nu}(t, x)$ obeys the equation

$$[(\partial_t - \partial_\sigma \partial_\sigma) \delta_{\mu\nu} + (1 - \kappa) \partial_\mu \partial_\nu] H_{\nu\rho}(t, x) = \frac{1}{2T_R} \delta_{\mu\rho} \delta(t) \delta^{(D)}(x), \quad (41)$$

and T_R is the usual color trace normalization. Since the fundamental gluon field $A_\mu^a(x) = B_\mu^a(0, x)$ does not couple to the Lagrange multiplier field L_μ^a , and all flow-time variables are positive, one can impose the initial condition

$$\left\langle B_\mu^a(t, x) L_\nu^b(s, y) \right\rangle \Big|_{t=0} = 0 \quad \Rightarrow \quad H_{\mu\nu}(-s, x) = 0, \quad (42)$$

so that the unique solution becomes

$$H_{\mu\nu}(t, x) = \frac{1}{2T_R} \theta(t) K_{\mu\nu}(t, x). \quad (43)$$

We refer to the $\langle BL \rangle$ bilinear as “gluon flow line”. The inverse of the factor $1/(2T_R)$ appears in the corresponding “flow vertices” further below. We can therefore discard it altogether and write

$$s, \nu, b \xrightarrow{p} t, \mu, a = \delta^{ab} \theta(t - s) \tilde{K}_{\mu\nu}(t - s, p), \quad (44)$$

where $\tilde{K}_{\mu\nu}(t, p)$ has been defined in Eq. (20), and the adjacent arrow indicates the direction towards increasing flow time as implied by the θ -distribution. As opposed to an actual propagator, the gluon flow line is a regular function for all p .

Similarly, one determines the mixed fermionic two-point function at leading order as

$$\left\langle \chi_\alpha^i(t, x) \bar{\lambda}_\beta^j(s, y) \right\rangle = \delta_{\alpha\beta} \delta^{ij} G(t - s, x - y), \quad (45)$$

where $G(t, x)$ obeys

$$(\partial_t - \partial_\mu \partial_\mu) G(t, x) = \delta(t) \delta^{(D)}(x), \quad (46)$$

with the condition

$$\left\langle \chi_{\alpha}^i(t, x) \lambda_{\beta}^j(s, y) \right\rangle \Big|_{t=0} = 0 \quad \Rightarrow \quad G(-s, x) = 0. \quad (47)$$

The unique solution reads

$$G(t, x) = \theta(t) K(t, x). \quad (48)$$

The Fourier transformed expression defines the “fermion flow line” Feynman rule, where the θ -distribution again imposes a direction as indicated by the adjacent arrow pointing towards increasing flow time:

$$s, \beta, j \xrightarrow[p]{\quad} t, \alpha, i = \delta_{\alpha\beta} \delta_{ij} \theta(t-s) \tilde{K}(t-s, p), \quad (49)$$

where $\tilde{K}(t, p)$ has been defined in Eq. (27). As usual, the arrow *on* the fermion line denotes the “charge flow” of the fermion. In this way, we have a unified Feynman rule for both the $\langle \chi \bar{\chi} \rangle$ and the $\langle \lambda \bar{\lambda} \rangle$ bilinear, where the latter can be obtained analogously as above and simply corresponds to reversing the direction of the charge flow:

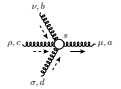
$$s, \beta, j \xleftarrow[p]{\quad} t, \alpha, i = \delta_{\alpha\beta} \delta_{ij} \theta(t-s) \tilde{K}(t-s, p). \quad (50)$$

Since $\tilde{K}(t, p)$ only depends on p only quadratically, the momentum direction for the fermion flow lines is irrelevant.

Flow vertices

Since $\mathcal{L}_B$ and $\mathcal{L}_{\chi}$ of Eq. (16) are proportional to a Lagrange multiplier field, the resulting vertices always involve at least one flow line. Such vertices are always associated with a flow-time parameter which is integrated over. We denote them by “flow vertices” and represent them by empty circles in Feynman diagrams. The corresponding Feynman rules can be derived straightforward. In this paper, we define Feynman rules by assuming all momenta to be outgoing.

Similarly, the four-point gluon flow vertex is governed by N_3 :



$$= -g^2 \int_0^\infty ds \left(f^{abc} f^{cde} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) + f^{ace} f^{bde} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) + f^{ade} f^{bce} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\rho} \delta_{\nu\sigma}) \right). \quad (52)$$

Again, there is exactly one outgoing flow line, while the other three lines are either flowed gluons or incoming flow lines. This means that the Feynman rule (52) actually represents four different vertices.

Note that the factor $2T_R$, arising from the trace in Eq. (16), has been discarded in both vertices in accordance with the normalization of Eq. (44).

Similar considerations applied to $\mathcal{L}_\chi$ lead to vertices involving flowed quark fields. For example, the quark flow vertex with one gluon is described by the following Feynman rule:



$$= ig \delta_{\alpha\beta} (T^a)_{ij} \int_0^\infty ds (2p_\rho + (1 - \kappa)q_\rho). \quad (53)$$

In this case, the fermion line with outgoing charge flow is also outgoing in the gradient flow, while the other two lines can be either flowed propagators, or ingoing flow lines. Thus, this Feynman rule actually represents four different vertices.

A complete list of the Feynman rules can be found in Appendix A.

Let us summarize the Feynman rules for the GFF and point out a few more features:

1. Propagators and flow lines always carry an exponential factor $e^{\pm tp^2}$ for each vertex, where t is the flow time of the vertex.
2. Flow lines always start at a flow vertex. They end either at another flow vertex or an external operator. The θ -distribution $\theta(t - s)$ implies a flow-time direction, which we denote by an adjacent arrow in Feynman diagrams.

3. A flow vertex always has an outgoing flow line connected to it. The other lines are either incoming flow lines, or flowed propagators. Each flow vertex is defined at a flow time s and implies an integration $\int_0^\infty ds$. The θ -distribution of the outgoing flow line restricts the integration to a finite upper limit t , i.e. $\int_0^t ds$.
4. Diagrams with closed flow-line loops vanish, because the integration interval shrinks to a point. Hence, only diagrams whose flow lines form trees contribute to any observable.

3 Automated implementation

3.1 Generation of Feynman diagram expressions

We generate the Feynman diagrams including symmetry factors and signs (from closed fermion loops) with the help of the program **qgraf** [25,26]. In the notation of this program, the propagators for the gluon, the ghost, and a single quark flavor can be defined as

$$[\mathbf{g}, \mathbf{g}, +], \quad [c, c, -], \quad [f\mathbf{q}, f\mathbf{q}, -].$$

The sign in the third entry of each square bracket denotes whether the particle is a boson or a fermion.

The flow lines for the gluon and the quark are implemented by introducing separate fields $\mathbf{b}$, $\mathbf{fr}$, and $\mathbf{fs}$ for L , λ , and $\bar{\lambda}$, respectively. They are implemented as

$$[\mathbf{b}, \mathbf{b}, +], \quad [f\mathbf{r}, f\mathbf{r}, -], \quad [f\mathbf{s}, f\mathbf{s}, -],$$

where the latter two represent the $\langle \chi \bar{\chi} \rangle$ and $\langle \lambda \bar{\lambda} \rangle$ bilinears (see Eqs. (49) and (50)). With these fields, we can then define the regular as well as the flow vertices. For example, for the trilinear flow vertex of the pure gauge theory defined in Eq. (51), we define

$$[\mathbf{B}, \mathbf{b}, \mathbf{b}], \quad [\mathbf{B}, \mathbf{b}, \mathbf{g}], \quad [\mathbf{B}, \mathbf{g}, \mathbf{g}],$$

which corresponds to the three combinations of Fig. 1.

Already in regular QCD it is convenient to separate the color structure from the rest of the calculation. This becomes non-trivial in cases which involve the four-gluon vertex. It is thus convenient to introduce an auxiliary "particle" $\Sigma_{\mu\nu}^a$ whose "propagator" is given by [27]

$$\rho, \sigma, b \circ \circ \circ \circ \circ \circ \mu, \nu, a \quad = \quad \delta^{ab} \delta_{\rho\sigma} \delta_{\nu\sigma}. \quad (54)$$

The four-gluon vertex can then be replaced by a trilinear $gg\Sigma$ vertex whose Feynman rule reads



$$= \frac{ig}{\sqrt{2}} f^{abc} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\nu\rho} \delta_{\mu\sigma}) . \quad (55)$$

This allows to factorize the color factor off all Feynman diagrams, albeit at the cost of increasing their number. We proceed correspondingly for the quartic gluon flow vertices by introducing an additional Σ particle which is directed in flow time. It forms trilinear vertices with gluons and gluon flow lines, as shown explicitly in Appendix A. Again, they contain exactly one outgoing (Σ or gluon) flow line.

By default, **qgraf** also generates diagrams with closed flow-line loops. We use a **Perl** script to parse the **qgraf** output in order to eliminate such diagrams before the actual calculation. They vanish trivially algebraically though, as discussed earlier. In the same way we multiply diagrams by a factor n_f for each closed fermion loop. We then process the diagrams with the help of **q2e/exp** [28, 29] in order to insert the Feynman rules and convert the diagrams to **FORM** code.

Within **FORM** [30, 31], we contract the Lorentz indices, take the fermion traces, and simplify the resulting expressions to a standard form of scalar integrals, as defined below. The color factor is evaluated separately with the help of the **color** package [32].

For example at the three-loop level (as needed for the results in this paper) the flow-time integrals can be described in the following form

$$I(t, \mathbf{n}, \mathbf{a}, \mathbf{c}, D) = \left(\prod_{r=1}^N \int_0^{t_r^{\text{up}}} dt_r t_r^{c_r} \right) \int_{p_1, p_2, p_3} \frac{\exp[i \sum_{k,j,j} a_{kij} t_k p_i \cdot p_j]}{p_1^{2n_1} p_2^{2n_2} p_3^{2n_3} p_4^{2n_4} p_5^{2n_5} p_6^{2n_6}}, \quad (56)$$

where

$$\begin{aligned} \mathbf{n} &= \{n_1, \dots, n_6\}, & \mathbf{c} &= \{c_1, \dots, c_N\}, \\ \mathbf{a} &= \{a_{kij} : k = 0, \dots, N; i = 1, 2, 3; j = 1, 2, 3\}, \end{aligned} \quad (57)$$

are sets of integers, $N \leq 4$, $t_0 \equiv t$, and the upper limits for the flow-time integrations are linear combinations of the other flow-time variables, $t_r^{\text{up}} = t_r^{\text{up}}(t_0, \dots, t_{r-1})$. Initially, all the c_i are zero. It is helpful to introduce these parameters for the subsequent discussion

though. The momenta p_4, p_5, p_6 are linear combinations of the integration momenta p_1, p_2, p_3 . For the three-loop quantities considered in this paper, the flow time t is the only dimensionful external scale. A generalization of this notation to a different number of loops and to additional external scales is straightforward.

3.2 IBP reduction of flow-time loop integrals

In the first step, we reduce all occurring integrals to so-called master integrals by employing integration-by-parts (IBP) identities [33, 34]. They are based on the observation that integrals over total derivatives vanish in dimensional regularization. Applying the operator

$$\mathcal{D}_{ij} \equiv \frac{\partial}{\partial p_i} \cdot p_j = \delta_{ij} D + p_j \cdot \frac{\partial}{\partial p_i} \quad (58)$$

to the integrand of $I(t, \mathbf{n}, \mathbf{a}, \mathbf{c}, D)$ in Eq. (56) thus results in a vanishing integral for all $i, j \in \{1, 2, 3\}$. On the other hand, explicitly acting with the derivative on the integrand leads to a sum of integrals where some of the indices n_i and c_r are shifted by ± 1 .⁵ Repeated application of the $\mathcal{D}_{ij}$ thus leads to linear relations among integrals with different $\mathbf{n}$ and $\mathbf{c}$. However, the number of flow-time integrations and the exponential function in the integrand still remain unaltered by this procedure.

We may apply an analogous strategy for the flow-time parameters though. Inserting a derivative with respect to one of the flow-time integration variables, one arrives at the sum of two integrals with fewer flow-time integrations and an altered exponential,

$$\int_0^{t_r^{\text{up}}} dt_r \frac{\partial}{\partial t_r} f(t_r, \dots) = f(t_r^{\text{up}}, \dots) - f(0, \dots). \quad (59)$$

On the other hand, explicit evaluation of the derivative at the integrand level either reduces one of the indices c_k or one of the indices n_i by one. One therefore arrives at linear relations among integrals with different $\mathbf{n}$, $\mathbf{c}$, $\mathbf{a}$, and different number of flow-time integrations N .

Applying the above operations to “seed integrals”, i.e., integrals with fixed numerical values for the $\mathbf{n}$, $\mathbf{c}$, and $\mathbf{a}$, allows one to build a system of linear relations among the $I(t, \mathbf{n}, \mathbf{a}, \mathbf{c}, D)$. Defining an ordering (complexity) criterion allows one to solve the system using a Gaussian elimination type reduction for a minimal set of integrals (master integrals) [35]. Through this procedure all integrals of the type defined in Eq. (56) are reduced to the minimal set of master integrals, which are simplest by the ordering criterion introduced. The system of linear relations is process dependent and for the observables in this

⁵To be precise: At most one n_i is shifted by -1 and at most one of the n_i and at most one of the c_r are shifted by $+1$.

study we construct it in `Mathematica` [36]. We use the specialized software `Kira` [37, 38] for the reduction of such linear equation systems to solve it.

While `Kira` alone is sufficient for the reduction of all our integrals at the two-loop level (see e.g. Ref. [21]), we find that the algebraic solution of the system at the three-loop level would require more than 750 GiB of RAM and thus exceeds our available computing resources.⁶ Using finite field and reconstruction techniques, which over the last decade have gained an increased use for higher-order perturbative calculations (see, e.g. Refs. [39–41]), one can improve on the required computational resources as follows.

Since t is the only dimensionful scale and can thus be factored out, the coefficients of the integrals in the linear system are simply polynomials in D . The individual steps for the reduction of the linear system only involve elementary arithmetic operations, which means that the coefficients of the master integrals are rational functions in D . One can then solve the linear system numerically over a finite field using a sufficiently large number of different integer values for D (“probes”) and reconstruct the rational functions in D exactly. With this approach the size of the coefficients remain simple numbers and the requirements on computational resources can be improved.

While `Kira` already uses `pyRed` as a first step to remove linearly dependent equations over a finite field, in our approach `pyRed` is used to reduce the system itself multiple times over a finite field. The resulting numbers are processed with the library `FireFly` [42], which provides an efficient implementation of interpolation [43, 44] and rational-reconstruction algorithms [45, 46].⁷ In our case, we require 201 probes, chosen from three different finite fields, defined as prime fields $\mathbb{Z}_p$ with p a 63-bit prime number, to reconstruct all coefficients, plus one additional probe in a fourth prime field to verify the reconstruction. For each probe, the solution of the system with `Kira` now just requires about 70 GiB of RAM and takes about three CPU hours. In total, the reduction took less than three days, using ten threads on two Intel Xeon Gold 6138 processors.

For the observables considered further below in this paper at the three-loop level we start with a total of 3195 integrals and can reduce them to a minimal set of 188 master integrals.⁸ Their numerical evaluation is described in the next section.

⁶This calculation was performed before the release of `Kira` 1.2 [38]. It is well possible that these statements could change with the newly implemented features.

⁷We remark that `FireFly` also works for multi-variate rational functions.

⁸When our observables are expressed in terms of the master integrals, the dependence of six of them drops out.

3.3 Numerical computation of flow-time loop integrals

As a first step of the numerical evaluation of the gradient-flow integrals given by Eq. (56), we express the propagators through Schwinger-parameter integrals and map them from $x \in (0, \infty)$ to $y \in (0, 1)$ using the simple transformation $x = y/(1 - y)$. Momentum integrations are performed as D -dimensional Gaussian integrals after a diagonalization. Similarly, all flow-time integrations are mapped to the unit interval with simple linear transformations. The overall result is an integral over a unit hypercube.

The integrand typically involves a number of (overlapping) singularities, which we factorize using **FIESTA** [47], an implementation of the sector decomposition algorithm [48]. As opposed to regular Feynman integrals where sector decomposition is typically employed in combination with Feynman parameterization, it appears that we cannot restrict the singularities to the lower integration bound only.⁹ If an integral involves singularities both at the lower and the upper bound, we split all integration intervals in the middle, and map the singularities to the lower bound by an appropriate change of variable.

The sector decomposed integrals obtained from **FIESTA** are then integrated with our implementation of fully symmetric integration rules of order 13 [17, 50], which can handle integrable logarithmic-like singularities at the integration boundaries very well. We perform all arithmetic with 256-bit precision using the **MPFR** library [51], and used a local adaptive bisection in the direction of the largest fourth difference. A high precision arithmetic turns out to be necessary to achieve a relative numerical accuracy of 10^{-10} or better. The integration uncertainty is estimated by the difference between the integration results of rules of order 13 and 11.

As a check of our results further below, we apply the numerical integration method to our unreduced set of 3195 integrals as well as to the set of 188 master integrals. Finding full agreement for the results obtained with both sets within numerical uncertainties serves as a check of the numerical integration as well as for the reduction procedure.

Analytical computation. For some master integrals we can find analytical results either through elementary methods with the help of **Mathematica** [36] or by using **HyperInt** [52]. The integration with **Mathematica** sometimes yields hypergeometric functions which can be expanded with the package **HypExp** [53, 54]. For example as shown further below, we find analytical results for all contributions which contain at least one factor of T_R in the color structure. A fully analytical calculation of integrals with other color factors requires a more detailed investigation. However, for all practical purposes, the numerical results provided in this paper are (more than) sufficient.

⁹To identify cases where singularities appear at the upper bound, we determine the degree of divergence for each subset of integration variables at the level of the Schwinger parameterization [49].

4 Observables

Among the simplest quantities one can consider within the GFF are vacuum expectation values of gauge-invariant operators at finite flow time. It is one of the remarkable properties of the GFF that these operators do not require any renormalization beyond that of regular QCD, and that of the involved flowed fields. For the gauge coupling and the quark masses, the $\overline{\text{MS}}$ renormalization is given by the replacement

$$g \rightarrow g_0 \equiv \left(\frac{\mu e^{\gamma_E/2}}{\sqrt{4\pi}} \right)^\epsilon Z_g(\alpha_s(\mu)) g(\mu), \quad m_f \rightarrow m_{f,0} \equiv Z_m(\alpha_s(\mu)) m_f(\mu) \quad (60)$$

in Eq. (12), where μ is the renormalization scale, $\alpha_s = g^2/(4\pi)$, and $\gamma_E = 0.5772\dots$ the Euler-Mascheroni constant. Through the perturbative order required in this paper, the renormalization constants are given by

$$\begin{aligned} Z_g(\alpha_s) &= 1 - \frac{\alpha_s}{4\pi} \frac{\beta_0}{2\epsilon} + \left(\frac{\alpha_s}{4\pi} \right)^2 \left(\frac{3\beta_0^2}{8\epsilon^2} - \frac{\beta_1}{4\epsilon} \right) + \mathcal{O}(\alpha_s^3), \\ Z_m(\alpha_s) &= 1 - \frac{\alpha_s}{4\pi} \frac{\gamma_{m,0}}{2\epsilon} + \left(\frac{\alpha_s}{4\pi} \right)^2 \left[\frac{1}{\epsilon^2} \left(\frac{\gamma_{m,0}^2}{8} + \frac{\beta_0 \gamma_{m,0}}{4} \right) - \frac{\gamma_{m,1}}{4\epsilon} \right] + \mathcal{O}(\alpha_s^3), \end{aligned} \quad (61)$$

with

$$\begin{aligned} \beta_0 &= \frac{11}{3} C_A - \frac{4}{3} T_F, \quad \beta_1 = \frac{34}{3} C_A^2 - \left(4C_F + \frac{20}{3} C_A \right) T_F, \\ \gamma_{m,0} &= 6C_F, \quad \gamma_{m,1} = \frac{97}{3} C_A C_F + 3C_F^2 - \frac{20}{3} C_F T_F. \end{aligned} \quad (62)$$

C_F and C_A are the quadratic Casimir eigenvalues of the fundamental and the adjoint representation of the gauge group, respectively. Furthermore, $T_F = T_R n_F$, with n_F the number of quark flavors, and T_R the trace normalization in the fundamental representation. For $\text{SU}(N_c)$, it is $C_F = (N_c^2 - 1)/(2N_c)$, $C_A = N_c$, and $T_R = 1/2$.

The flowed gauge field $B_\mu^a(t, x)$ does not require renormalization, so that for example matrix elements of the gluon action density,

$$E(t, x) \equiv \frac{1}{4} G_{\mu\nu}^a(t, x) G_{\mu\nu}^a(t, x), \quad (63)$$

are finite after just the renormalization of g and m_f . This allows for a direct comparison of results obtained in different regularization schemes (lattice and perturbation theory, for example).

On the contrary, flowed quark fields require a renormalization factor $Z_\chi^{1/2}(\alpha_s)$ in order to render Green's functions finite. In the $\overline{\text{MS}}$ scheme, it is [3, 21]

$$Z_\chi^{-1}(\alpha_s) = 1 - \frac{\alpha_s}{4\pi} \frac{\gamma_{\chi,0}}{2\epsilon} + \left(\frac{\alpha_s}{4\pi}\right)^2 \left[\frac{1}{\epsilon^2} \left(\frac{\gamma_{\chi,0}^2}{8} + \frac{\beta_0 \gamma_{\chi,0}}{4} \right) - \frac{\gamma_{\chi,1}}{4\epsilon} \right] + \mathcal{O}(\alpha_s^3), \quad (64)$$

with

$$\begin{aligned} \gamma_{\chi,0} &= 6 C_F, \\ \gamma_{\chi,1} &= \left(\frac{223}{3} - 16 \ln 2 \right) C_A C_F - (3 + 16 \ln 2) C_F^2 - \frac{44}{3} C_F T_F. \end{aligned} \quad (65)$$

The quantity

$$S(t, x) \equiv Z_\chi \sum_{f=1}^{n_F} \bar{\chi}_f(t, x) \chi_f(t, x) \quad (66)$$

thus acquires an anomalous dimension, which prevents a direct comparison of results from different regularization schemes. Alternatively, one may work with “ringed quark fields” [20], which amounts to using

$$\hat{Z}_\chi(t, \mu) = -\frac{2N_C n_F}{(4\pi t)^2} \langle R(t) \rangle^{-1}, \quad \text{with} \quad R(t, x) = \sum_{f=1}^{n_F} \bar{\chi}_f(t, x) \overleftrightarrow{\mathcal{D}}^F \chi_f(t, x) \quad (67)$$

instead of Z_χ in order to renormalize the quark fields, where

$$\overleftrightarrow{\mathcal{D}}^F = \mathcal{D}^F - \overleftarrow{\mathcal{D}}^F. \quad (68)$$

This corresponds to a “physical” renormalization scheme, which means that the anomalous dimension of the operator

$$\hat{S}(t, x) = \zeta_\chi(t, \mu) S(t, x), \quad \text{with} \quad \zeta_\chi(t, \mu) \equiv Z_\chi^{-1} \hat{Z}_\chi(t, \mu), \quad (69)$$

vanishes. In Eq. (67), we used the notation

$$\langle \mathcal{O}(t) \rangle \equiv \int d^4x \langle \mathcal{O}(t, x) \rangle \quad (70)$$

for the zero-momentum Fourier transform of the vacuum expectation value of an operator $\mathcal{O}(t, x)$, which we adopt in the following.

The Feynman rules for the operators $E(t, x)$, $S(t, x)$, and $R(t, x)$ introduced above are obtained in the same manner as those described in Sect. 2.3. Expressing them in terms of

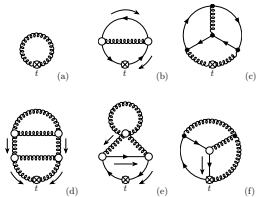


Figure 2: The only diagram contributing to $\langle E(t) \rangle$ at one loop (a); a two-loop diagram for $\langle S(t) \rangle$ and $\langle R(t) \rangle$ (b); and four three-loop diagrams: two diagrams for $\langle E(t) \rangle$ (c,d); one diagram for $\langle S(t) \rangle$ and $\langle R(t) \rangle$ (e); and one diagram for $\langle R(t) \rangle$ (f).

the flowed fields $B(t, x)$ and $\chi_f(t, x)$, the operator $E(t, x)$ results in a bilinear, trilinear, and a quartic gluon vertex as displayed in Eqs. (114)–(116). $S(t, x)$ corresponds to the single bilinear quark vertex given in Eq. (117), and $R(t, x)$ to the bilinear quark vertex of Eq. (118) and the quark-gluon vertex of (119). Similar to the flow vertices, the lines attached to these vertices are depicted with dashed arrows, indicating that they can be both flow lines and propagators. In contrast to the flow vertices, the flow time of all lines in these vertices is directed towards the corresponding operator. Therefore, these vertices set the largest flow time in the Feynman diagrams.

Using the methods described in Sects. 2 and 3, we can calculate the vacuum expectation values of $E(t, x)$, $S(t, x)$, and $R(t, x)$ through three loops. Sample diagrams for these quantities are shown in Fig. 2. The number of diagrams in each case is given in Table 1; recall, however, that this takes into account the re-writing of the four-gluon vertex into trilinear vertices as described in Sect. 3. Not included in this number are diagrams with closed flow-line loops, but integrals with scale-less sub-loops are counted in.

Table 1: The number of Feynman diagrams contributing to the quantities computed in this section. At three-loop level, the number for “quenched QCD” (marked as $T_R = 0$) is shown separately.

#loops	1	2	3	3 ($T_R = 0$)
$\langle E(t) \rangle$	1	11	232	211
$\langle S(t) \rangle$	1	8	210	202
$\langle \bar{R}(t) \rangle$	1	11	311	300

4.1 Results for the gluon condensate at three loops

We write the perturbative result for the gluon condensate as

$$\langle E(t) \rangle = \frac{3\alpha_s N_A}{4\pi t^2} \frac{1}{8} \left[e_0 + \frac{\alpha_s}{4\pi} e_1 + \left(\frac{\alpha_s}{4\pi} \right)^2 e_2 + \mathcal{O}(\alpha_s^3) \right] + \mathcal{O}(m), \quad (71)$$

where $\alpha_s = \alpha_s(\mu)$, with μ the renormalization scale. As indicated, quark mass terms are neglected here. Their effect at NLO has been studied in Ref. [17] and it was found to be negligible. In this case, the e_i depend only on the product $z \equiv \mu^2 t$. For reasons that will become clear shortly, it is convenient to parameterize this dependence in terms of the variable

$$L(z) \equiv \ln(2z) + \gamma_E. \quad (72)$$

A natural choice for the renormalization scale is defined by the inverse “smearing radius” $q_8 \equiv 1/\sqrt{8t}$ of the gradient flow. However, from explicit higher order calculations, it was found that the slightly larger value

$$\mu_0 = \frac{e^{-\gamma_E/2}}{\sqrt{2t}} \approx 1.5 \, q_8, \quad (73)$$

corresponding to $L(\mu_0^2 t) = 0$, improves the stability of the perturbative corrections [21] (see also Ref. [17]). This was corroborated by the behavior of the $t \rightarrow 0$ extrapolation for thermodynamical quantities evaluated in Ref. [22] using the two-loop result of Ref. [21].

We thus write the coefficients of Eq. (71) as

$$\begin{aligned} e_0(z) &= e_{0,0}, & e_1(z) &= e_{1,0} + \beta_0 L(z), \\ e_2(z) &= e_{2,0} + (2\beta_0 e_{1,0} + \beta_1) L(z) + \beta_0^2 L^2(z), \end{aligned} \quad (74)$$

with β_0, β_1 from Eq. (62). For the coefficients $e_{i,j}$, we find

$$\begin{aligned} e_{0,0} &= 1, & e_{1,0} &= \left(\frac{52}{9} + \frac{22}{3} \ln 2 - 3 \ln 3 \right) C_A - \frac{8}{9} T_F \\ e_{2,0} &= 27.9786 C_A^2 - (31.5652 \dots) T_F C_A + \left(16\zeta(3) - \frac{43}{3} \right) T_F C_F + \left(\frac{8\pi^2}{27} - \frac{80}{81} \right) T_F^2, \end{aligned} \quad (75)$$

where $\zeta(z)$ is Riemann's ζ function with $\zeta(3) = 1.20206 \dots$. The three dots in the coefficient of $T_F C_A$ indicate that we were able to obtain an expression in analytical form. It can be found in Appendix B, Eq. (128). Only four decimal places of the numerical result for the C_A^2 coefficient are displayed here, while our estimate of the numerical accuracy, obtained by propagating the uncertainty of the individual integrals to the final result, is about six digits beyond that. This estimate matches the observed differences between the numerical and analytical results in the cases where the latter have been computed. The same statements hold for the other observables listed below. The NLO coefficient e_1 was first evaluated in Ref. [3]. Setting $\mu = q_s$, one finds that the NNLO result agrees with Ref. [17] at the sub-percent level.¹⁰

The logarithmic terms $e_{n,k} L^k(z)$ obey the all-order recursion formula ($1 \leq k \leq n$)

$$k e_{n,k} = \sum_{l=0}^{n-1} (n-l) e_{n-l-1, k-1} \beta_l, \quad (76)$$

which follows from renormalization-group invariance, i.e.

$$\mu \frac{d}{d\mu} \langle E(t) \rangle = 0, \quad \mu^2 \frac{d}{d\mu^2} \alpha_s = \alpha_s \beta(\alpha_s) = -\frac{\alpha_s^2}{4\pi} \left(\beta_0 + \frac{\alpha_s}{4\pi} \beta_1 + \dots \right). \quad (77)$$

This provides a welcome check of our calculation. The residual dependence on μ can be used as probe of the perturbative behavior. We refer to Ref. [17] for such a study.

Since $\langle E(t) \rangle$ is also gauge independent, we are free to choose Feynman gauge for the QCD gauge parameter ($\xi = 1$) and $\kappa = 1$ for the gradient-flow gauge parameter for the evaluation of the results presented above, which facilitates the actual calculation significantly. However, we also check gauge invariance explicitly by evaluating the terms with the highest power in the gauge parameter ξ , i.e. ξ^4 , and show that its coefficient vanishes. Indeed, this already happens algebraically after the reduction to master integrals, even before their numerical values are inserted. Keeping the gauge parameter fully general unfortunately leads to intermediate expressions which exceed our available computing resources.

¹⁰To be precise, all color coefficients of Ref. [17] are compatible with our new calculation, except for the C_A^2 term, for which “only” the first three digits agree, corresponding to an overly optimistic uncertainty estimate of Ref. [17] for that term. The effect for any practical application should be irrelevant.

4.2 Results for the quark condensate at three loops

To obtain the quark condensate with ringed fields we first consider the conversion factor $\zeta_\chi(t, \mu)$ from $\overline{\text{MS}}$ renormalized to ringed quark fields as defined in Eq. (69). Applying the methods described in Sect. 2, we can calculate the diagrams contributing to the Green's function $\langle R(t) \rangle$ for massless quarks, which eventually leads to the result

$$\begin{aligned} \zeta_\chi(t, \mu) = 1 + \frac{\alpha_s}{4\pi} & \left(\frac{\gamma_{\chi,0}}{2} L(\mu^2 t) - 3C_F \ln 3 - 4C_F \ln 2 \right) \\ & + \left(\frac{\alpha_s}{4\pi} \right)^2 \left\{ \frac{\gamma_{\chi,0}}{4} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) L^2(\mu^2 t) + \left[\frac{\gamma_{\chi,1}}{2} - \frac{\gamma_{\chi,0}}{2} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) \ln 3 \right. \right. \\ & \left. \left. - \frac{2}{3} \gamma_{\chi,0} \left(\beta_0 + \frac{\gamma_{\chi,0}}{2} \right) \ln 2 \right] L(\mu^2 t) + C_2 \right\} + \mathcal{O}(\alpha_s^3, m). \end{aligned} \quad (78)$$

The NLO result has been obtained in Ref. [20], and the μ -dependent NNLO terms were derived in Ref. [21]. That paper also used a preliminary result for the coefficient C_2 , derived for the first time in the current paper. Keeping four decimal places for the individual color coefficients, we find

$$C_2 = -23.7947 C_A C_F + 30.3914 C_F^2 - (3.9226 \dots) C_F T_F, \quad (79)$$

where again, as indicated by the three dots, the coefficient of $C_F T_F$ was obtained in analytical form and is given in Appendix B, Eq. (130).

Let us now turn to the quark condensate $\langle S(t) \rangle$. Since it vanishes in the chiral limit $m_f = 0$, we compute the leading coefficient of the expansion in the quark masses m_f ,

$$\langle \zeta_\chi S(t) \rangle \equiv \langle \dot{S}(t) \rangle = \sum_{f=1}^{np} s_f(t) + \mathcal{O}(m^2), \quad s_f(t) \equiv m_f \frac{d\langle \dot{S}(t) \rangle}{dm_f} \Big|_{m=0}, \quad (80)$$

where it is understood that *all* quark masses are set to zero after taking the derivative. This expansion can be applied before the loop and flow-time integrations are carried out, which means that, in the quark propagators, we write

$$\frac{1}{i\rlap{/}{q} + m_f} = \frac{1}{i\rlap{/}{q}} + \frac{m_f}{q^2} + \mathcal{O}(m_f^2), \quad (81)$$

and keep only the leading linear term in m of the integrand. The resulting integrals are then again of the form shown in Eq. (56). We thus write

$$s_f(t) = -\frac{N_c m_f}{8\pi^2 t} \left[s_0 + \frac{\alpha_s}{4\pi} s_1 + \left(\frac{\alpha_s}{4\pi} \right)^2 s_2 + \mathcal{O}(\alpha_s^3) \right] + \mathcal{O}(m^2). \quad (82)$$

Renormalizing the quark masses in the $\overline{\text{MS}}$ scheme according to Eq. (60), we find

$$\begin{aligned}s_0(z) &= s_{0,0}, & s_1(z) &= s_{1,0} + \frac{\gamma_{m,0}}{2} L(z), \\ s_2(z) &= s_{2,0} + \frac{1}{2} [(\gamma_{m,0} + 2\beta_0) s_{1,0} + \gamma_{m,1}] L(z) + \frac{1}{8} (\gamma_{m,0} + 2\beta_0) \gamma_{m,0} L^2(z),\end{aligned}\quad (83)$$

with $z = \mu^2 t$ and the $\gamma_{m,i}$ and β_i from Eq. (62). The non-logarithmic terms are given by

$$\begin{aligned}s_{0,0} &= 1, & s_{1,0} &= (4 + 4 \ln 2 - 3 \ln 3) C_F \\ s_{2,0} &= 19.6422 C_F^2 + 41.3897 C_F C_A - (15.7975 \dots) T_F C_F,\end{aligned}\quad (84)$$

where again the $T_F C_F$ term is known analytically, as indicated by the three dots. It can be found in Appendix B, Eq. (131). The NLO term has been obtained in Ref. [55], the NNLO term is new. Again, only four decimal places are displayed and our uncertainty estimate for the numerical integration is several digits beyond that.

The logarithmic terms $s_{n,k} L^k(z)$ obey the all-order recursion formula ($1 \leq k \leq n$)

$$k s_{n,k} = \sum_{l=0}^{n-1} \left[\frac{\gamma_{m,l}}{2} + (n-l-1) \beta_l \right] s_{n-l-1,k-1}, \quad (85)$$

which follows from renormalization group invariance, i.e.

$$\mu \frac{d}{d\mu} s_f(t) = 0, \quad \mu \frac{d}{d\mu} m_f = m_f \gamma_m(\alpha_s) = -m_f \frac{\alpha_s}{4\pi} \left(\gamma_{m,0} + \frac{\alpha_s}{4\pi} \gamma_{m,1} + \dots \right), \quad (86)$$

and the renormalization group equation for α_s given in Eq. (77). This again provides a welcome check of our calculation. The residual dependence on μ can be used as probe of the perturbative behavior. It is displayed in Fig. 3 for three values of the central energy scale $\mu_0 = (2te^{\gamma_E})^{-1/2}$, see Eq. (73). We observe a qualitatively similar behavior as was found for $\langle t^2 E(t) \rangle$ in Ref. [17]. While for large μ_0 , the reduction of the renormalization scale with increasing perturbative order is quite significant, this improvement reduces as μ_0 gets closer to the non-perturbative regime. Taking the variation of μ by a factor of two around the central scale as an estimate of the uncertainty due to missing higher orders, one finds that they overlap between successive perturbative orders for all three values of μ_0 . However, the trend of the curves does not explicitly support μ_0 as the central scale in this case, but seems to favor a slightly smaller scale.

Another important consistency check is that $\langle S(t) \rangle$ and $\langle R(t) \rangle$ can be renormalized with the same flowed-quark field renormalization constant Z_ψ as obtained in Ref. [21]. $\langle S(t) \rangle$ and $\langle R(t) \rangle$ are also gauge independent. For the calculation of the full NNLO result, we again chose Feynman gauge ($\xi = 1$) and $\kappa = 1$. We check gauge invariance explicitly by evaluating the terms with the highest power in the gauge parameter ξ , which is ξ^3 for $\langle S(t) \rangle$ and $\langle R(t) \rangle$, and show that their coefficients vanish. This happens here also immediately after the reduction to master integrals.

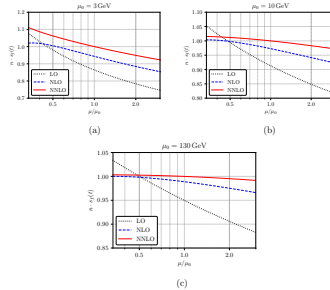


Figure 3: Renormalization scale dependence of $s_f(t)$, defined in Eq. (80), for three different values of the central scale $\mu_0 = e^{-\gamma_E/2}/\sqrt{2L}$. (a) $\mu_0 = 3 \text{ GeV}$, corresponding to $t = (0.03 \text{ fm})^2$, where we use $n_F = 3$ and $\alpha_s^{(3)}(3 \text{ GeV}) = 0.248$; (b) $\mu_0 = 10 \text{ GeV}$ ($t = (0.01 \text{ fm})^2$), where $n_F = 5$ and $\alpha_s^{(5)}(10 \text{ GeV}) = 0.178$; (c) $\mu_0 = 130 \text{ GeV}$ ($t = (8 \cdot 10^{-4} \text{ fm})^2$), where $n_F = 5$ and $\alpha_s^{(5)}(130 \text{ GeV}) = 0.112$. The running of $\alpha_s(\mu)$ and $m_f(\mu)$ is evaluated with the help of `RunDec` [56, 57] at the corresponding loop order. All curves are normalized to the NNLO result at $\mu = \mu_0$.

5 Gradient-flow coupling and mass

5.1 Gradient-flow coupling

The renormalization group invariance of $\langle E(t) \rangle$ allows us to fix a scale $\mu = \sqrt{\rho/t}$ with constant $\rho \in \mathbb{R}$, and its proportionality to α_s suggests to define a “gradient-flow coupling”

$$\hat{\alpha}_\rho(\mu) \equiv \frac{32\pi\rho^2}{3N_A\mu^4} \langle E(\rho/\mu^2) \rangle \equiv \alpha_s(\mu) \sum_{n=0}^{\infty} \left(\frac{\alpha_s(\mu)}{4\pi} \right)^n e_n(\rho). \quad (87)$$

For the usual definition of the gradient-flow coupling [3], ρ is set to $1/8 = 0.125$, while $\rho = 1/(2e^{\pi}) \approx 0.281\dots$ appears to be more suitable from a perturbative point of view (cf. Eq. (73)).

Perturbatively solving Eq. (71) for α_s results in

$$\alpha_s = \hat{\alpha}_\rho \left[1 - c_1(\rho) \frac{\hat{\alpha}_\rho}{4\pi} + (2c_1^2(\rho) - c_2(\rho)) \left(\frac{\hat{\alpha}_\rho}{4\pi} \right)^2 + \dots \right], \quad (88)$$

with the coefficients $e_n(\rho)$ given in Eq. (74). Such a transformation between mass-independent renormalization schemes leaves the first two coefficients of the β function invariant, i.e.

$$\mu^2 \frac{d}{d\mu^2} \hat{\alpha}_\rho(\mu) = \hat{\alpha}_\rho(\mu) \hat{\beta}_\rho(\hat{\alpha}_\rho), \quad (89)$$

$$\hat{\beta}_\rho(\hat{\alpha}_\rho) = - \sum_{n=0}^{\infty} \hat{\beta}_{\rho,n} \left(\frac{\hat{\alpha}_\rho}{4\pi} \right)^n = - \frac{\hat{\alpha}_\rho}{4\pi} \hat{\beta}_0 - \left(\frac{\hat{\alpha}_\rho}{4\pi} \right)^2 \hat{\beta}_1 - \sum_{n=2}^{\infty} \hat{\beta}_{\rho,n} \left(\frac{\hat{\alpha}_\rho}{4\pi} \right)^n,$$

with $\hat{\beta}_0$ and $\hat{\beta}_1$ from Eq. (62). The third coefficient is given by

$$\hat{\beta}_{\rho,2} = \hat{\beta}_2 - c_1(\rho) \hat{\beta}_1 + (c_2(\rho) - c_1^2(\rho)) \hat{\beta}_0, \quad (90)$$

with the $\overline{\text{MS}}$ coefficient $\hat{\beta}_2$ which we quote here for the $\text{SU}(3)$ gauge group:

$$\hat{\beta}_2 = \frac{2857}{2} - \frac{5033}{18} n_F + \frac{325}{54} n_F^2. \quad (91)$$

5.2 Gradient-flow mass

Similarly, we may define a gradient-flow quark mass as

$$\hat{m}_f^{(\rho)}(\mu) = - \frac{8\pi^2\rho}{N_c\mu^2} m_f \frac{d\langle \bar{\psi}\psi(\rho/\mu^2) \rangle}{dm_f} \Big|_{m=0} = m_f(\mu) \sum_{n=0}^{\infty} \hat{s}_n(\rho) \left(\frac{\hat{\alpha}_\rho(\mu)}{4\pi} \right)^n, \quad (92)$$

where

$$\hat{s}_0(\rho) = s_0(\rho), \quad \hat{s}_1(\rho) = s_1(\rho), \quad \hat{s}_2(\rho) = s_2(\rho) - s_1(\rho) e_1(\rho), \quad (93)$$

with the s_n and e_1 given in Eqs. (83) and (74). The inverse relation is

$$m_f = \hat{m}_f^{(\rho)} \left[1 - \hat{s}_1(\rho) \frac{\hat{\alpha}_\rho}{4\pi} + (\hat{s}_1^2(\rho) - \hat{s}_2(\rho)) \left(\frac{\hat{\alpha}_\rho}{4\pi} \right)^2 + \dots \right]. \quad (94)$$

The new mass obeys the renormalization group equation

$$\mu \frac{d}{d\mu} \hat{m}_f^{(\rho)}(\mu) = \hat{\gamma}_m^{(\rho)}(\hat{\alpha}_\rho) \hat{m}_f^{(\rho)}(\mu) \equiv -\hat{m}_f^{(\rho)}(\mu) \sum_{n=0}^{\infty} \left(\frac{\hat{\alpha}_\rho(\mu)}{4\pi} \right)^{n+1} \hat{\gamma}_{m,n}^{(\rho)} \quad (95)$$

where

$$\begin{aligned} \hat{\gamma}_{m,0}^{(\rho)} &= \gamma_{m,0}, & \hat{\gamma}_{m,1}^{(\rho)} &= \gamma_{m,1} - \gamma_{m,0} e_1(\rho) + 2 \beta_0 \hat{s}_1(\rho), \\ \hat{\gamma}_{m,2}^{(\rho)} &= \gamma_{m,2} - 2 \gamma_{m,1} e_1(\rho) + \gamma_{m,0} [2 e_1^2(\rho) - e_2(\rho)] \\ &\quad + 2 \beta_1 \hat{s}_1(\rho) + 2 \beta_0 (2 \hat{s}_2(\rho) - \hat{s}_1^2(\rho)), \end{aligned} \quad (96)$$

with β_0 , β_1 , $\gamma_{m,0}$, $\gamma_{m,1}$ given in Eq. (62), and $\gamma_{m,2}$ the three-loop $\overline{\text{MS}}$ coefficient which we quote for the $\text{SU}(3)$ gauge group:

$$\gamma_{m,2} = 2498 - \left(\frac{4432}{27} + \frac{320}{3} \zeta(3) \right) n_F - \frac{280}{81} n_F^2. \quad (97)$$

6 Conclusions

A framework for the calculation of correlation functions in the perturbative gradient-flow formalism using tools and techniques from standard perturbative QCD calculations has been presented. It employs a Feynman-diagrammatic approach to the gradient-flow formalism, based on the five-dimensional field-theoretical formulation of Ref. [16]. We implemented the corresponding QCD Feynman rules within the gradient-flow formalism as well as those of the composite operators relevant for the quantities considered in this paper, and automated the reduction and calculation of algebraic expressions and loop integrals to a large extent. An important ingredient of our setup is the reduction to master integrals with the help of integration-by-parts identities that also involve the flow-time variables. Except for a few cases where an analytic result was derived, the master integrals were evaluated numerically.

To demonstrate the viability of this setup, we reproduced and improved the three-loop result for the gluon condensate of an earlier calculation, which was based on explicit field-theoretic calculations at the level of Wick contractions [17].

Using this setup, we then evaluated the three-loop approximations to the quark condensate and the conversion factor from $\overline{\text{MS}}$ renormalized to ringed quark fields, both of which are new results. Checks on gauge parameter independence, renormalization group invariance, as well as alternative calculational approaches (with and without reduction to master integrals) convinced us of the correctness of these results.

The gluon and the quark condensate lend themselves for straightforward definitions of a gradient-flow gauge coupling and a gradient-flow quark mass, for which we derive the three-loop matching relations to the $\overline{\text{MS}}$ scheme. Provided that sufficiently accurate lattice results for these quantities become available, the conversion factors evaluated in this paper should allow for precise first-principle determinations of the gauge coupling,¹¹ and possibly also quark masses.

While focusing on the calculation of vacuum matrix elements in this paper, our setup should be applicable in principle to more general problems which may involve a larger number of scales, for example.

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¹¹First steps in this direction have been taken in Ref. [58].

A Feynman rules

In this section we present all the Feynman rules required for our calculation. All momenta are considered to be outgoing. ξ is the usual QCD gauge parameter and κ the additional gradient-flow gauge parameter introduced in the flow equations (2). Lines with a solid arrow are flow lines and lines with a dashed arrow can be both flow lines or flowed propagators. The symbols p, q, r denote Euclidean four-momenta, while s, t are real-valued flow-time variables.

Propagators and flow lines

- gluon propagator:

$$s, \nu, b \overset{p}{\text{~~~~~}} t, \mu, a = \delta^{ab} \frac{1}{p^2} \left(\left(\delta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2} \right) e^{-(t+s)p^2} + \xi \frac{p_\mu p_\nu}{p^2} e^{-\kappa(t+s)p^2} \right) \quad (98)$$

- (anti)quark propagator:

$$s, \beta, j \overset{p}{\longrightarrow} t, \alpha, i = \delta_{ij} \frac{(-i\not{p} + m)_{\alpha\beta}}{p^2 + m^2} e^{-(t+s)p^2} \quad (99)$$

- ghost propagator:

$$b \overset{p}{\cdots\cdots\cdots} a = \delta^{ab} \frac{1}{p^2} \quad (100)$$

- gluon flow line:

$$s, \nu, b \overset{p}{\text{~~~~~}} t, \mu, a = \delta^{ab} \theta(t-s) \left(\left(\delta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2} \right) e^{-(t-s)p^2} + \frac{p_\mu p_\nu}{p^2} e^{-\kappa(t-s)p^2} \right) \quad (101)$$

- quark flow line:

$$s, \beta, j \overset{p}{\longrightarrow} t, \alpha, i = \delta^{ij} \delta_{\alpha\beta} \theta(t-s) e^{-(t-s)p^2} \quad (102)$$

- antiquark flow line:

$$s, \beta, j \xrightarrow[p]{\leftarrow} t, \alpha, i = \delta^{ij} \delta_{\alpha\beta} \theta(t-s) e^{-(t-s)p^2} \quad (103)$$

Vertices

- three-gluon vertex:

$$\begin{array}{c} \nu, b \\ \swarrow \\ \text{---} q \\ \searrow \\ r \\ \swarrow \\ \rho, c \end{array} \begin{array}{c} \mu, a \\ \leftarrow p \\ \rightarrow \end{array} = ig f^{abc} (\delta_{\rho\nu} (p-q)_\rho + \delta_{\nu\rho} (q-r)_\mu + \delta_{\rho\mu} (r-p)_\nu) \quad (104)$$

- four-gluon vertex:

$$\begin{array}{c} \mu, a \\ \swarrow \\ \text{---} q \\ \searrow \\ \nu, b \end{array} \begin{array}{c} \sigma, d \\ \swarrow \\ \text{---} r \\ \searrow \\ \rho, c \end{array} = -g^2 (f^{abc} f^{ade} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) + f^{ace} f^{bde} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) + f^{ade} f^{bce} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\rho} \delta_{\nu\sigma})) \quad (105)$$

- ghost-gluon vertex:

$$\begin{array}{c} b \\ \swarrow \\ \text{---} p \\ \searrow \\ c \end{array} \begin{array}{c} \mu, a \\ \leftarrow p \\ \rightarrow \end{array} = -ig f^{abc} p_\mu \quad (106)$$

- quark-gluon vertex:



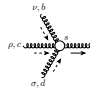
$$= g (T^a)_{ij} (\gamma_\mu)_{\alpha\beta} \quad (107)$$

- two-plus-one gluon flow vertex:



$$= -igf^{abc} \int_0^\infty ds (\delta_{\nu\rho}(r-q)_\mu + 2\delta_{\mu\nu}q_\rho - 2\delta_{\mu\rho}r_\nu + (\kappa-1)(\delta_{\mu\rho}q_\nu - \delta_{\mu\nu}r_\rho)) \quad (108)$$

- three-plus-one gluon flow vertex:



$$= -g^2 \int_0^\infty ds (f^{abc}f^{ade}(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + f^{ace}f^{bde}(\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + f^{ade}f^{bce}(\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\rho}\delta_{\nu\sigma})) \quad (109)$$

- quark-one-gluon flow vertex:



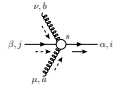
$$= ig \delta_{\alpha\beta} (T^a)_{ij} \int_0^\infty ds (2p_\mu + (1-\kappa)q_\mu) \quad (110)$$

- antiquark-one-gluon flow vertex:



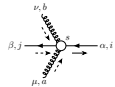
$$= -ig \delta_{\alpha\beta} (T^a)_{ji} \int_0^\infty ds (2p_\mu + (1-\kappa)q_\mu) \quad (111)$$

- quark-two-gluons flow vertex:



$$= g^2 \delta_{\alpha\beta} \delta_{\mu\nu} \{T^a, T^b\}_{ij} \int_0^\infty ds \quad (112)$$

- antiquark-two-gluons flow vertex:



$$= g^2 \delta_{\alpha\beta} \delta_{\mu\nu} \{T^a, T^b\}_{ji} \int_0^\infty ds \quad (113)$$

Feynman rules for the composite flow-time operators:

- $G_{\mu\nu}^a(t, x)G_{\mu\nu}^a(t, x)$:

- two-gluon vertex:

$$\mu, a \begin{array}{c} p \\ \text{---} \otimes \text{---} \\ \text{---} \end{array} \begin{array}{c} t \\ \text{---} \otimes \text{---} \\ \text{---} \end{array} \begin{array}{c} q \\ \text{---} \otimes \text{---} \\ \text{---} \end{array} \nu, b = -g^2 \delta^{ab} (\delta_{\mu\nu} p \cdot q - p_\mu q_\nu) \quad (114)$$

- three-gluon vertex:

$$\begin{array}{c}
\mu, a \\
\swarrow \\
p \\
\searrow \\
t \\
\swarrow \quad \searrow \\
q \quad k \\
\swarrow \quad \searrow \\
\nu, b \\
\end{array}
= ig^3 f^{abc} (\delta_{\mu\nu} (q-p)_\rho + \delta_{\nu\rho} (k-q)_\mu + \delta_{\rho\mu} (p-k)_\nu) \quad (115)$$

- four-gluon vertex:

$$\begin{aligned}
& \mu, a \quad \sigma, d \\
& \nu, b \quad \rho, c
\end{aligned}
=
\begin{aligned}
& g^4 (f^{abc} f^{cde} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) \\
& + f^{ace} f^{bde} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}) \\
& + f^{ade} f^{bce} (\delta_{\mu\nu} \delta_{\rho\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho}))
\end{aligned} \quad (116)$$

- $\bar{\chi}(t, x)\chi(t, x)$:

$$\beta, j \longrightarrow \begin{array}{c} t \\ \otimes \\ \longrightarrow \end{array} \longrightarrow \alpha, i \quad = \quad \delta_{ij} \delta_{\alpha\beta} \quad (117)$$

- $\bar{\chi}(t, x) \overset{\leftrightarrow}{D}^F \chi(t, x)$:

- quark-antiquark vertex:

$$\beta, j \quad \begin{array}{c} \xrightarrow{q} \\ \text{---} \end{array} \quad \begin{array}{c} t \\ \otimes \end{array} \quad \begin{array}{c} \xrightarrow{p} \\ \text{---} \end{array} \quad \alpha, i \quad = \quad i\delta^{ij}(p-q)_{\alpha\beta} \quad (118)$$

- quark-antiquark-gluon vertex:



$$= -2g(T^a)^{ij}(\gamma_\mu)_{\alpha\beta} \quad (119)$$

Auxiliary Feynman rules

The following Feynman rules are auxiliary Feynman rules for our implementation as described in Sect. 3. They represent the vertices with four gluons or gluon-flow lines and allow us to factorise the color structure of all Feynman diagrams.

- four-gluon vertex:

- Σ propagator:



$$= \delta^{ab} \delta_{\rho\mu} \delta_{\nu\sigma} \quad (120)$$

- Σ -gluon-gluon vertex:



$$= \frac{ig}{\sqrt{2}} f^{abc} (\delta_{\rho\mu} \delta_{\nu\sigma} - \delta_{\sigma\rho} \delta_{\mu\nu}) \quad (121)$$

- three-plus-one gluon flow vertex:

- Σ flow line:



$$= \delta^{ab} \delta_{\rho\mu} \delta_{\nu\sigma} \delta(t-s) \quad (122)$$

– outgoing Σ -gluon-gluon flow vertex:



$$= \frac{ig}{\sqrt{2}} f^{abc} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\nu\rho} \delta_{\mu\sigma}) \int_0^\infty ds \quad (123)$$

– incoming Σ -gluon-gluon flow vertex:



$$= \frac{ig}{\sqrt{2}} f^{abc} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\nu\rho} \delta_{\mu\sigma}) \int_0^\infty ds \quad (124)$$

• four-gluon vertex from $G_{\mu\nu}^a(t, x) G_{\rho\sigma}^a(t, x)$:

– Σ_E flow line:



$$= \delta^{ab} \delta_{\rho\mu} \delta_{\nu\sigma} \delta(t - s) \quad (125)$$

– outgoing Σ_E -gluon-gluon vertex:



$$= \frac{g^2}{\sqrt{2}} f^{abc} (\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\nu\rho} \delta_{\mu\sigma}) \int_0^\infty ds \quad (126)$$

– incoming Σ_E -gluon-gluon vertex:



$$= \frac{g^2}{\sqrt{2}} f^{abc} (\delta_{\rho\sigma} \delta_{\mu\nu} - \delta_{\nu\rho} \delta_{\mu\sigma}) \quad (127)$$

B Analytical results

Here, we provide the analytical results for the color coefficients as far as they are available. For the $T_F C_A$ term of $e_{2,0}$ in Eq. (75), we find

$$\begin{aligned} -(31.5652\dots) = & \frac{2}{81} \left(-108 \text{Li}_3 \left(-\frac{1}{3} \right) - 792 \text{Li}_2 \left(\frac{3}{4} \right) + 216 \text{Li}_2 \left(-\frac{1}{3} \right) \right. \\ & + 108 \text{Li}_{1,2} \left(-3, -\frac{1}{3} \right) - 216 \text{Li}_{1,2} \left(1, -\frac{1}{3} \right) + 216 \text{Li}_{2,1} \left(-3, -\frac{1}{3} \right) \\ & - 216 \text{Li}_{2,1} \left(1, -\frac{1}{3} \right) + 216 \text{Li}_{1,1,1} \left(-1, -3, -\frac{1}{3} \right) \\ & - 864 \text{Li}_{1,1,1} \left(-1, 3, -\frac{1}{3} \right) + 216 \text{Li}_{1,1,1} \left(1, 3, -\frac{1}{3} \right) \\ & + 432 \text{Li}_{1,1,1} \left(3, 1, -\frac{1}{3} \right) - 1134 \zeta(3) + 90\pi^2 - 641 - 54 \ln^3 3 \\ & + 324 \ln^3 2 + 324 \ln 2 \ln^2 3 + 270 \ln^2 3 - 648 \ln^2 2 \ln 3 - 2232 \ln^2 2 \\ & \left. + 504 \ln 2 \ln 3 - 18\pi^2 \ln 3 + 1890 \ln 3 + 108\pi^2 \ln 2 - 2148 \ln 2 \right), \end{aligned} \quad (128)$$

where

$$\text{Li}_{n_1, \dots, n_r}(z_1, \dots, z_r) = \sum_{0 < k_1 < \dots < k_r} \frac{z_1^{k_1} \dots z_r^{k_r}}{k_1^{n_1} \dots k_r^{n_r}} \quad (129)$$

are multiple polylogarithms [60, 61].

The $C_F T_F$ coefficient of C_2 in Eq. (79) reads

$$-(3.9226\dots) = -\frac{1}{18} \left(48 \operatorname{Li}_2 \left(\frac{1}{9} \right) - 2400 \operatorname{Li}_2 \left(\frac{1}{3} \right) + 672 \operatorname{Li}_2 \left(\frac{3}{4} \right) - 131 + 46\pi^2 \right. \\ \left. + 960 \ln^2 2 - 1068 \ln^2 3 + 1888 \ln 2 - 1032 \ln 3 + 624 \ln 2 \ln 3 \right). \quad (130)$$

And finally, the $C_F T_F$ coefficient of $s_{2,0}$ in Eq. (84) reads

$$-(15.7975\dots) = \frac{16}{3} \operatorname{Li}_2 \left(\frac{1}{9} \right) - \frac{800}{3} \operatorname{Li}_2 \left(\frac{1}{3} \right) + 104 \operatorname{Li}_2 \left(\frac{3}{4} \right) - \frac{71}{6} + \frac{23\pi^2}{9} \\ + 192 \ln^2 2 - \frac{362}{3} \ln^2 3 + \frac{544}{9} \ln 2 - \frac{100}{3} \ln 3 + \frac{56}{3} \ln 2 \ln 3. \quad (131)$$

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P38：局域涂抹算符乘积展开

主题归属	梯度流、涂抹与流时算符
底稿	arXiv:1501.05348
arXiv / DOI	1501.05348
完整原始 PDF	12 页；来源 PyQCD；SHA-256 f16003f5503a...
转排索引	EN 19 页；ZH 19 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

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Locally smeared operator product expansions in scalar field theory

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We propose a new locally smeared operator product expansion to decompose nonlocal operators in terms of a basis of smeared operators. The smeared operator product expansion formally connects nonperturbative matrix elements determined numerically using lattice field theory to matrix elements of nonlocal operators in the continuum. These nonperturbative matrix elements do not suffer from power-divergent mixing on the lattice, which significantly complicates calculations of quantities such as the moments of parton distribution functions, provided the smearing scale is kept fixed in the continuum limit. The presence of this smearing scale complicates the connection to the Wilson coefficients of the standard operator product expansion and requires the construction of a suitable formalism. We demonstrate the feasibility of our approach with examples in real scalar field theory.

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I. INTRODUCTION

Quantum chromodynamics (QCD) connects hadronic matter to its fundamental constituents, quarks, and gluons. Many aspects of QCD are poorly understood, in spite of four decades of intense experimental and theoretical effort. As part of this effort, deep inelastic scattering (DIS) of leptons from nucleons has played a central role in establishing QCD as the reigning theory of the strong interaction and continues to serve as a mainstay for attempts to unravel QCD's mysteries (for complete reviews see, for example, Refs. [1] and [2]).

Theoretically, inclusive DIS cross sections are separated into leptonic and hadronic tensors, which capture the electroweak and strong dynamics of the scattering process, respectively. The hadronic tensor factorizes into a convolution of infrared-safe perturbative coefficients and parton distribution functions (PDFs), which incorporate the low-energy QCD physics and therefore must be determined nonperturbatively. PDFs are independent of the scattering process but depend on the target nucleon, while the perturbative coefficients are independent of the external scattering states.

PDFs are important for two reasons. First, they furnish direct knowledge of the constituents of hadronic states—the dominant form of visible matter in the universe—in terms of the fundamental theory of the strong force. Second, they provide constraints on hadronic backgrounds at collider experiments, such as the Large Hadron Collider. These backgrounds affect the sensitivity of a variety of high energy experiments, including studies of the properties of the Higgs boson [3–

5], and searches for heavy W' and Z' boson production [6].

PDFs cannot currently be calculated directly from QCD using *ab initio* lattice QCD, because they are defined in terms of light-cone matrix elements that are not directly accessible on Euclidean lattices. Hence, PDFs are usually extracted from global analyses of different experiments [7–15].

Naturally a direct nonperturbative computation of PDFs is desirable: with sufficient precision, such a calculation would further constrain global fits in regions that may be experimentally inaccessible.

Until recently, lattice calculations have focused on determining the Mellin moments of PDFs, which are related to matrix elements of twist-2 operators that can be determined on the lattice [16–18]. The lattice regulator breaks Lorentz symmetry, which induces radiative mixing between operators of different mass dimensions. The resulting power divergences in the lattice spacing completely obscure the continuum limit. Moments up to the fourth moment can be extracted by carefully choosing the external momenta and operators, but this significantly limits the precision with which one can extract meaningful results for PDFs [19, 20]. Beyond the fourth moment, power divergent mixing is inevitable and this method cannot provide reliable calculations.

Ji recently proposed a new approach to directly extract PDFs from lattice calculations [21, 22], based on a large-momentum effective theory [23]. Preliminary results first appeared in Ref. [24], but a number of issues remain, including a complete understanding of the renormalization of the relevant lattice matrix elements and the practical ability to resolve sufficiently large momenta on the lattice.

Here we propose a new formalism that removes mixing in the continuum limit and, in principle, enables the extraction of higher moments of PDFs from lattice QCD.

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We call this formalism the “smeared operator product expansion” (sOPE). We expand continuum matrix elements in a basis of locally-smeared lattice degrees of freedom. The resulting matrix elements are functions of two scales, the smearing scale and the renormalization scale. The sOPE provides the framework necessary to relate these matrix elements, via Wilson coefficients, to phenomenologically useful quantities, such as the hadronic tensor of DIS.

Smearing has been widely used in lattice QCD to reduce ultraviolet fluctuations, partially restore rotational symmetry and thereby systematically improve the precision of lattice calculations [25–28]. For a pedagogical overview of smearing in lattice calculations, see, for example, Ref. [29]. In the sOPE, we implement smearing via the gradient flow, a classical evolution of the fields in a new dimension, the flow time, toward the stationary points of the action [30–34]. Matrix elements determined nonperturbatively on the lattice require no further renormalization, up to a fermionic wave function renormalization [30], and remain finite, provided the smearing scale is kept fixed in physical units in the continuum limit [31, 35]. There are two further advantages: the gradient flow allows one to use smearing lengths of only one or two lattice spacings, much smaller than hadronic length scales on typical lattices, and therefore does not distort the low energy physics [36], and also the gradient flow is computationally very cheap.

The gradient flow is now well-established as a tool to study lattice gauge theories, with applications from scale-setting, *i.e.* determining the lattice spacing in physical units [37–41], to studying renormalization in lattice gauge theories. For example, the gradient flow has been used to define finite-volume renormalization schemes for the strong coupling constant [42–46], for operator renormalization [47], and to understand the nonperturbative scale dependence of renormalized matrix elements [48]. Related work has used Ward identities to study chiral fermions on the lattice [49] and the energy-momentum tensor [50–52].

In this paper we introduce the gradient flow for a single real scalar field with quartic interactions and outline some of the properties of the sOPE applied to real scalar field theory. Scalar field theories describe some important critical phenomena in nature, such as the antiferromagnetic phase transition, and have a long history as a testing ground for fundamental ideas in quantum field theory in four dimensions. For our purposes, the chief advantage is the simplicity of Euclidean scalar field theory, which lays bare the structure of the sOPE and highlights the points of similarity and contrast with the local operator product expansion (OPE).

In the next section, we review Wilson’s OPE and introduce the sOPE. We then apply the sOPE to scalar field theory in Sec. III, illustrate how the gradient flow removes power-divergent mixing in Sec. IV, and calculate the perturbative Wilson coefficients to two loops in Sec. V. Finally, in Sec. VI, we study the scale dependence

of the sOPE through renormalization group equations for the Wilson coefficients.

II. THE OPERATOR PRODUCT EXPANSION

Wilson’s approach to the OPE for a nonlocal operator is widely known—see, for example, Ref. [53]—and here we review some notation necessary for our discussion.

We write the OPE for a nonlocal operator, $\mathcal{Q}(x)$, as

$$\mathcal{Q}(x) \stackrel{x \rightarrow 0}{\sim} \sum_k c_k(x, \mu) [\mathcal{O}^{(k)}(0)]_{\text{R}}. \quad (1)$$

The Wilson coefficients $c_k(x, \mu)$ are complex functions of the spacetime separation, x , and renormalization scale, μ , that capture the short-distance physics associated with the renormalized local operator $[\mathcal{O}^{(k)}(0)]_{\text{R}}$. We represent renormalized operators by $[\dots]_{\text{R}}$ and suppress their dependence on the renormalization scale, μ , for notational simplicity. The free-field mass dimension of the local operator determines the leading spacetime dependence of the corresponding Wilson coefficient.

As an example, let us consider the time-ordered two-point function of two scalar fields with spacetime separation x : $\mathcal{T}\{\phi(x)\phi(0)\}$. In free scalar field theory, the OPE is a Laurent expansion. Interactions generate sub-leading dependence on the spacetime separation in the Wilson coefficients, which become functions of the spacetime separation, the (renormalized) mass m_{R} , and the renormalization scale, μ :

$$\mathcal{T}\{\phi(x)\phi(0)\} = \frac{c_1(\mu x, m_{\text{R}} x)}{4\pi^2 x^2} \mathbb{1} + c_{\phi^2}(\mu x, m_{\text{R}} x) [\phi^2(0)]_{\text{R}} + \mathcal{O}(x). \quad (2)$$

Here we have factored out the leading spacetime dependence from the Wilson coefficients. The $\mathcal{O}(x)$ indicates terms of order x , up to logarithmic corrections generated by interactions.

We propose replacing the set of local operators in the OPE by their locally smeared counterparts

$$\mathcal{Q}(x) \stackrel{x \rightarrow 0}{\sim} \sum_k d_k(\tau, x, \mu) \mathcal{S}^{(k)}(\tau, 0). \quad (3)$$

The Wilson coefficients, $d_k(\tau, x, \mu)$, and the smeared operators, $\mathcal{S}^{(k)}(\tau, 0)$, are now functions of an extra scale, the flow time, τ . Nevertheless, the leading spacetime dependence of the Wilson coefficients is dictated by the canonical mass dimension of the corresponding smeared operator and is therefore just that of the standard OPE.

Just as the OPE is only valid for small spacetime separations, the sOPE requires small flow times. We will see that we require that $\tau \propto x^2$, where x is assumed to be small, to ensure that the Wilson coefficients are independent of the external states.

For example, returning to the time-ordered two-point function, the sOPE is

$$\begin{aligned} \mathcal{T}\{\phi(x)\phi(0)\} = & \frac{d_0(\mu x, \mu^2\tau, m_{\text{R}}x)}{4\pi^2 x^2} \\ & + d_2(\mu x, \mu^2\tau, m_{\text{R}}x)\rho^2(\tau, 0) + \mathcal{O}(x, \tau), \end{aligned} \quad (4)$$

where we denote smeared fields at flow time τ by $\rho(\tau, x)$.

In the following sections, we will observe four features of the smeared Wilson coefficients in the sOPE.

1. The logarithmic spacetime dependence of the original OPE is preserved in the sOPE.
2. The flow time serves as an ultraviolet regulator for the smeared Wilson coefficients, to leading-order in perturbation theory.
3. Beyond leading-order the flow time cannot regularize the Wilson coefficients, because the flow evolution is classical. We can, however, absorb renormalization scale dependence into the renormalization parameters of the original theory.
4. For any OPE to be meaningful, the Wilson coefficients must be independent of the external states. We can ensure this for the sOPE by choosing $s_{\text{max}} < x$; i.e. the mean smearing radius must be smaller than the spacetime extent of the nonlocal operator. This ensures that the sOPE remains an expansion in approximately local operators. In other words, if the gradient flow probes length scales on the order of the nonlocal operator, then the sOPE becomes a poor expansion for the original operator.

Although we refer to this expansion as the *smeared* OPE, we really have in mind that the smearing is implemented via the gradient flow. The gradient flow acts as a smoothing operation that drives the degrees of freedom of the theory to the stationary points of the action. In QCD, the gradient flow corresponds to a continuous stout-smearing procedure, an analytic method for constructing lattice gauge fields with damped ultraviolet fluctuations [25]. The direct comparison of the gradient flow to other smearing schemes was first undertaken in Ref. [54].

Many studies of the gradient flow have incorporated a small flow-time expansion of fields at nonvanishing flow time in terms of local fields at zero flow time [48–52, 55]. We can view such an expansion as an OPE in the flow time and thereby relate renormalized quantities calculated at nonzero flow time to the corresponding quantities in the original theory at vanishing flow time, which would otherwise be difficult to compute.

In this work, we take a different approach. We do not expand the flowed fields in terms of original fields at nonzero flow time, but rather take as the fundamental objects of study the (matrix elements of) fields at positive flow time. Both approaches reflect the physically-motivated expectation that smearing scales much smaller

than the physical scales of the system should not distort the physics in question. The small flow-time expansion quantifies any deviations from this expectation and, furthermore, decouples analytic calculations of Wilson coefficients in the continuum from lattice calculations with smeared degrees of freedom. The sOPE incorporates the smearing scale as an inherent scale of the system, which requires new Wilson coefficients to be determined. Thus, both the small flow-time expansion and the sOPE are shaped by the role of the smearing scale as ultraviolet regulator and are related, but distinct, conceptual approaches to the same physics: partially restoring rotational symmetry on the lattice.

Smearing, in general, is a tool that partially restores rotational symmetry [36] and thereby improves the continuum limit of lattice calculations. Smearing via the gradient flow has a number of advantages. In particular, matrix elements determined nonperturbatively on the lattice using smeared degrees of freedom require no further renormalization [32], up to fermionic wave function renormalization [30], and remain finite, provided the smearing scale is kept fixed in physical units as the continuum limit is taken.

We now turn to a more complete study of the sOPE applied to ϕ^4 scalar field theory, a particularly straightforward theory in which to examine the sOPE. We can solve the flow-time equations exactly, because the flow-time evolution is linear in the scalar field. We do not need to consider the complications associated with gauge fixing [31, 32]—nor the extra renormalization of fermions [30]. Furthermore, for scalar fields the sOPE can be understood to be simply a resummation of the original OPE. Although it is not necessary for our work, it is also interesting to note that the OPE is known to converge for Euclidean ϕ^4 theory in four dimensions, at a fixed, but arbitrary order in the perturbative expansion [56]. This result holds at arbitrary spacetime separations, provided the external states are of compact support.

Looking toward future calculations in QCD, we anticipate that the technical issues associated with a flow equation that incorporates gauge field interactions will slightly complicate the perturbative calculations by increasing the number of diagrams at a given order in perturbation theory, but will not alter our conclusions. Within the gauge sector, there are no loops of flowed fields, because renormalized correlation functions remain finite at nonzero flow time [31]. Therefore, at leading-order in perturbation theory, the flow time regulates ultraviolet divergences; beyond leading-order an appropriate renormalization procedure must be incorporated. With fermionic fields there is an extra fermion renormalization parameter, calculated in Ref. [30], but this can be removed by considering renormalization group invariant quantities [30]. We also note that all perturbative calculations for the sOPE can be carried out in the continuum and do not require lattice perturbation theory, which is generally more involved [57, 58].

III. GRADIENT FLOW FOR SCALAR FIELD THEORY

We work in four-dimensional Euclidean scalar field theory with quartic self-interactions and bare mass m , defined by the action

$$S_\phi[\phi] = \frac{1}{2} \int d^4x \left[(\partial_\mu \phi)^2 + m^2 \phi^2 + \frac{\lambda}{12} \phi^4 \right]. \quad (5)$$

To study the sOPE, we introduce the scalar gradient flow, which we define through the flow-time evolution equation

$$\frac{\partial \rho(\tau, x)}{\partial \tau} = \partial^2 \rho, \quad (6)$$

where ρ is a scalar field at nonzero flow time, τ , and ∂^2 is the Euclidean, four-dimensional Laplacian operator. Note that the flow time has units $[\tau] = [x]^2$.

Imposing the Dirichlet boundary condition $\rho(0, x) = \phi(x)$, we may write the exact solution of the flow time equation as

$$\rho(\tau, x) = e^{-\partial^2 \tau} \phi(x), \quad (7)$$

or, in the momentum representation, as

$$\tilde{\rho}(\tau, p) = e^{-\tau p^2} \tilde{\phi}(p). \quad (8)$$

The full solution is

$$\begin{aligned} \rho(\tau, x) &= \int d^4y \int \frac{d^4p}{(2\pi)^4} e^{ip \cdot (x-y)} e^{-\tau p^2} \phi(y) \\ &= \frac{1}{16\pi^2 \tau^2} \int d^4y e^{-(x-y)^2/4\tau} \phi(y), \end{aligned} \quad (9)$$

which demonstrates explicitly the “smearing” effect of the gradient flow: the flow time exponentially suppresses ultraviolet modes. We parametrize the smearing radius via the root-mean-square smearing length, $s_{\text{rms}} = \sqrt{8\tau}$.

The (Euclidean) smeared scalar propagator, for two fields at flow times τ_1 and τ_2 , is given by

$$\tilde{G}_\rho(\tau_1, \tau_2, k) = \frac{e^{-(\tau_1 + \tau_2)k^2}}{k^2 + m^2}. \quad (10)$$

The flow-time evolution is classical, so any interactions must occur at zero flow time. The corresponding Feynman rule for the four-point vertex is just that of the standard (Euclidean) four-point vertex, $V^{(4)} = -\lambda/24$.

IV. MIXING ON THE LATTICE

Before we examine the sOPE in detail, we demonstrate how the gradient flow removes power-divergent mixing on the lattice. We consider the example of twist-2 operators in scalar field theory, which are symmetric and traceless and given by

$$T_{\mu_1 \dots \mu_n}(x) = \phi(x) \partial_{\mu_1} \dots \partial_{\mu_n} \phi(x) - \text{traces}. \quad (11)$$

On the lattice, we replace the partial derivatives with discrete difference operators

$$T_{\mu_1 \dots \mu_n}^{\text{latt}}(x) = \phi(x) \Delta_{\mu_1} \dots \Delta_{\mu_n} \phi(x) - \text{traces}. \quad (12)$$

The spacetime point x is now a node in the lattice, $x_\mu = a n_\mu$, where a is the lattice spacing and n_μ is a four-component lattice vector. The discrete difference operators can be improved to remove discretization effects, but the simplest such symmetric operator acts on a scalar field as

$$\Delta_\mu \phi(x) = \frac{1}{2a} (\phi(x + \hat{\mu}) - \phi(x - \hat{\mu})), \quad (13)$$

where $\hat{\mu}$ is the unit vector in the μ^{th} direction.

The lattice regulator breaks rotational symmetry, which induces mixing between twist-2 operators of different mass dimension. On dimensional grounds, the mixing coefficients scale with inverse powers of the lattice spacing and these coefficients diverge in the continuum limit. This is the problem of power-divergent mixing on the lattice. For example, the simple bilinear $T^{\text{latt}}(x) = \phi(x) \phi(x)$ mixes with the operator $T_{\mu_1 \mu_2}^{\text{latt}}(x)$ with a coefficient that scales as $1/a^2$. More generally, the mixing between $T^{\text{latt}}(x)$ and any twist-2 operator with an even number of derivatives, $T_{\mu_1 \dots \mu_n}^{\text{latt}}(x)$, scales as $1/a^{2n}$.

The smeared counterparts of these operators, which are given by

$$S_{\mu_1 \dots \mu_n}^{\text{latt}}(x) = \rho(\tau, x) \Delta_{\mu_1} \dots \Delta_{\mu_n} \rho(\tau, x) - \text{traces}. \quad (14)$$

do not suffer from this problem. The mixing coefficient between $S^{\text{latt}}(x) = \rho(\tau, x) \rho(\tau, x)$ and $S_{\mu_1 \dots \mu_n}^{\text{latt}}(x)$ scales as $1/\tau^n$, where τ is the flow time in *physical units*. Provided we keep the dimensionful scale $\tau = \bar{\tau} a^2$ fixed, where $\bar{\tau}$ is dimensionless, then as the lattice spacing a decreases, the mixing coefficient remains finite, because matrix elements at nonzero flow time cannot contain any additional divergences [31, 35].

As a simple illustration of this behavior, let us consider the matrix element of two twist-2 operators of different mass dimension:

$$M_{\text{cont}} = \langle \Omega | \phi(0) \phi(0) \phi(0) \partial_\mu \partial_\nu \phi(0) | \Omega \rangle. \quad (15)$$

This matrix element vanishes for massless scalar fields. On the lattice, however, the corresponding matrix element,

$$M_{\text{latt}} = \langle \Omega | \phi(0) \phi(0) \phi(0) \Delta_\mu \Delta_\nu \phi(0) | \Omega \rangle, \quad (16)$$

is nonzero.

We show the Feynman diagram for the leading contribution to this matrix element in the left-hand diagram of Fig. 1, which is given by

$$M_{\text{latt}}^{(0)} = \frac{1}{a^2} \int_{-\pi/a}^{\pi/a} \frac{d^4(ak)}{(2\pi)^4} \frac{\hat{k}_\mu \hat{k}_\nu}{(k^2 + m_0^2)^2}, \quad (17)$$



FIG. 1. Diagrams representing the leading contributions to the mixing matrix elements with unsmeared fields, $M_{\text{lat}}^{(0)}$ and smeared fields, $M_{\text{lat}}^{(0)}(\tau)$. The black solid lines in the left-hand diagram represent unsmeared propagators and the black solid square and diamond are the operators $\phi^2(0)$ and $\phi(0)\Delta_\mu\Delta_\nu\phi(0)$. The blue dashed lines in the right-hand diagram are smeared propagators and the two different blue blobs, filled and unfilled, represent the smeared operators $\rho^2(\tau, 0)$ and $\rho(\tau, 0)\Delta_\mu\Delta_\nu\rho(\tau, 0)$.

where $\tilde{k}_\mu = (2/a)\sin(ak_\mu/2)$. We have included a bare mass m_0 to remove spurious infrared divergences, but the integral is finite and we can take the massless limit, $m_0 \rightarrow 0$, to match to the continuum massless theory. We expand this in powers of the lattice spacing to obtain

$$\begin{aligned} M_{\text{lat}}^{(0)} &= -\frac{\delta_{\mu\nu}}{4} \int_{-\pi/a}^{\pi/a} \frac{d^4k}{(2\pi)^4} \frac{k^2}{(k^2 + m_0^2)^2} + \mathcal{O}(a^2) \\ &= -\frac{\delta_{\mu\nu}}{48a^2} + \mathcal{O}(a^0) \end{aligned} \quad (18)$$

in the massless limit. This result signals the appearance of power-divergent mixing: the $1/a^2$ term diverges in the continuum limit. Although this calculation is only leading-order in perturbation theory, higher-order contributions do not modify this power-divergent dependence on the lattice spacing [36].

If we calculate this matrix element with smeared degrees of freedom (which we depict in the right-hand diagram of Fig. 1), however, we find

$$\begin{aligned} M_{\text{lat}}^{(0)}(\tau) &= \langle \Omega | \rho(\tau, 0) \rho(\tau, 0) \Delta_\mu \Delta_\nu \rho(\tau, 0) | \Omega \rangle \\ &= -\frac{\delta_{\mu\nu}}{4} \int_{-\pi/a}^{\pi/a} \frac{d^4k}{(2\pi)^4} \frac{k^2 e^{-k^2\tau}}{(k^2 + m_0^2)^2} + \mathcal{O}(a^2) \\ &= \delta_{\mu\nu} \frac{e^{-4\pi^2\tau/a^2} - 1}{256a^2\tau} + \mathcal{O}(a^0). \end{aligned} \quad (19)$$

Once again we have taken the massless limit. In the continuum limit, keeping the flow time τ fixed in physical units, this matrix element tends to a constant, signaling the suppression of power-divergent mixing for smeared degrees of freedom.

V. WILSON COEFFICIENTS FOR THE SOPE

The procedure for calculating smeared Wilson coefficients parallels that for the OPE, discussed in, for example, Ref. [53]. With the Feynman rules of Sec. III in hand, the calculation of smeared Wilson coefficients up to next-to-leading-order in perturbation theory is straightforward. We determine the smeared Wilson coefficients

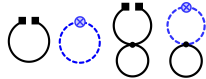


FIG. 2. Diagrams representing the contributions to the Wilson coefficient d_1 at leading-order (left two diagrams) and next-to-leading-order (right two diagrams). Solid black and dashed blue lines are propagators at vanishing and nonvanishing flow times, respectively. The black squares are unsmeared fields $\phi(0)$, black dots are interaction vertices at vanishing flow time and the blue blob represents the smeared operator $\rho^2(\tau, 0)$.

for the leading connected and disconnected operators, starting with the disconnected contribution.

A. Disconnected contributions

1. Leading order

We illustrate the leading-order (tree-level) and next-to-leading-order (one-loop) contributions to the smeared Wilson coefficient for the disconnected operator, $\mathbb{I}$, in Fig. 2. As we will see, the leading-order contribution is independent of the renormalization scale, μ , because at this order the flow time serves as the ultraviolet regulator. Beyond leading order, however, the smeared Wilson coefficient will have a renormalization scale dependence.

We extract the disconnected smeared Wilson coefficient by considering matrix elements of each of the operators in the sOPE in the vacuum, which removes any connected contributions. To $\mathcal{O}(x)$ we have

$$\begin{aligned} \langle \Omega | \mathcal{T} \{ \phi(x) \phi(0) \} | \Omega \rangle &= \frac{d_1(\mu x, \mu^2 \tau, m x)}{4\pi^2 x^2} \langle \Omega | \mathbb{I} | \Omega \rangle \\ &+ d_{\rho^2}(\mu x, \mu^2 \tau, m x) \langle \Omega | \rho^2(\tau, 0) | \Omega \rangle + \mathcal{O}(x), \end{aligned} \quad (20)$$

Here we have chosen the normalization of d_1 so that at leading order in the free theory d_1 is unity. We expand each quantity in this expression to one loop according to

$$f = f^{(0)} - \lambda f^{(1)} + \mathcal{O}(\lambda^2), \quad (21)$$

where f stands for either a matrix element or Wilson coefficient. We have chosen the sign of the one-loop contribution to factor out the sign of the coupling constant arising from the Feynman rule for the four-point vertex $V^{(4)} = -\lambda/24$.

The tree-level Wilson coefficient is then given by the

small spacetime behavior of

$$\frac{d_1^{(0)}}{4\pi^2 x^2} \stackrel{x \rightarrow 0}{=} \left\{ \langle \Omega | \mathcal{T} \{ \phi(x) \phi(0) \} | \Omega \rangle - \langle \Omega | \rho^2(\tau, 0) | \Omega \rangle \right\}_{\mathcal{O}(m^2)}^{(0)}, \quad (22)$$

where we have neglected the arguments of the coefficient for clarity and used the fact that the tree-level connected coefficient is $d_{\mu}^{(0)} = 1$. The subscript indicates that we must expand the result to $\mathcal{O}(m^2)$. For more details about the calculation of Wilson coefficients see, for example, Ref. [53]. The corresponding Feynman integral representation is

$$\begin{aligned} d_1^{(0)} \stackrel{x \rightarrow 0}{=} 4\pi^2 x^2 \left\{ \int_k \frac{e^{ik \cdot x} - e^{-2k^2 \tau}}{k^2 + m^2} \right\}_{\mathcal{O}(m^2)} \\ = 4\pi^2 x^2 \int_k \left(e^{ik \cdot x} - e^{-2k^2 \tau} \right) \left(\frac{1}{k^2} - \frac{m^2}{(k^2)^2} \right), \quad (23) \end{aligned}$$

where

$$\int_k = \int \frac{d^4 k}{(2\pi)^4}. \quad (24)$$

The smeared Wilson coefficient is

$$d_1^{(0)} = 1 - \frac{x^2}{8\tau} + \frac{m^2 x^2}{4} \left[\gamma_E - 1 + \log \left(\frac{x^2}{8\tau} \right) \right], \quad (25)$$

with $\gamma_E \simeq 0.577216$ the Euler-Mascheroni constant.

We can compare this result with the Wilson coefficient for the OPE for $d = 4 - 2\epsilon$ dimensions [53]:

$$\tilde{c}_1^{(0)} = 1 + \frac{m^2 x^2}{4} \left[\frac{1}{\epsilon} + 1 + \gamma_E + \log \left(\frac{\pi \mu^2 x^2}{4} \right) \right]. \quad (26)$$

Then in the $\overline{MS}$ scheme, this becomes

$$\tilde{c}_1^{(0)} = 1 + \frac{m^2 x^2}{4} \left[1 + 2\gamma_E + \log \left(\frac{\mu^2 x^2}{16} \right) \right]. \quad (27)$$

Although the finite contribution to these expressions cannot be directly compared, because we have expressed the Wilson coefficients in two different renormalization schemes, we note three important features. First, the logarithmic dependence on the spacetime separation is identical. Second, the flow time τ plays the role of the renormalization scale at leading order. Third, we see that for small spacetime separations, we require a small flow-time parameter. If we do not choose the flow-time parameter appropriately, we generate large logarithmic contributions to the smeared Wilson coefficients and the sOPE exhibits poor convergence properties, even at small spacetime separations.

2. Next-to-leading-order

At one loop, the smeared Wilson coefficient is given by

$$\begin{aligned} -\lambda \frac{d_1^{(1)}}{4\pi^2 x^2} = \left\{ \langle \Omega | \mathcal{T} \{ \phi(x) \phi(0) \} | \Omega \rangle - \langle \Omega | \rho^2(\tau, 0) | \Omega \rangle \right\}_{\mathcal{O}(m^2)}^{(1)}. \quad (28) \end{aligned}$$

The four-point interaction in this diagram, which we show in Fig. 2, appears at zero flow time. Therefore the flow time cannot act as a regulator for the momentum integral over k_2 and we must introduce a renormalization procedure. We use dimensional regularization and the $\overline{MS}$ scheme. The double integral is straightforward, however, because the two integrals can be carried out separately:

$$\frac{d_1^{(1)}}{4\pi^2 x^2} = \left\{ \int_{k_1} \frac{e^{ik_1 \cdot x} - e^{-2k_1^2 \tau}}{(k_1^2 + m^2)^2} \frac{1}{2} \int_{k_2} \frac{1}{k_2^2 + m^2} \right\}_{\mathcal{O}(m^2)}. \quad (29)$$

We find, for $m > 0$,

$$d_1^{(1)} = -\frac{m^2 x^2}{128\pi^2} \left[\gamma_E - 1 + \log \left(\frac{x^2}{8\tau} \right) \right] \left[1 + \log \left(\frac{\mu^2}{m^2} \right) \right]. \quad (30)$$

Here we see that the second term, which is a function of the new renormalization scale, μ , and the bare mass, m , is nothing other than the one-loop contribution to the mass renormalization of the original theory, in the $\overline{MS}$ scheme [59]. Moreover, the factor containing the flow time is identical to the $\mathcal{O}(m^2)$ term from the leading-order contribution, $d_1^{(0)}$, in Eq. (25).

Therefore, we can simply combine the leading-order and next-to-leading-order terms to give

$$d_1 = 1 - \frac{x^2}{8\tau} + \frac{m_R^2 x^2}{4} \left[\gamma_E - 1 + \log \left(\frac{x^2}{8\tau} \right) \right] + \mathcal{O}(\lambda^2), \quad (31)$$

where m_R is the renormalized mass in the $\overline{MS}$ scheme, given by $m_R^2 = Z_m^{-1} m^2$ and Z_m is the mass renormalization [59]:

$$Z_m = 1 + \frac{\lambda}{16\pi^2} \left[1 + \log \left(\frac{\mu^2}{m^2} \right) \right] + \mathcal{O}(\lambda^2). \quad (32)$$

This is a clear, next-to-leading-order example of how the divergences of the theory at nonzero flow time are absorbed by the renormalization parameters of the original theory at zero flow time. In other words, the renormalized theory at zero flow time remains ultraviolet finite at nonzero flow time.

B. Connected contribution

We illustrate the leading- and next-to-leading-order contributions to the connected Wilson coefficient



FIG. 3. Diagrams representing the leading-order (one-loop) contributions to the Wilson coefficient $d_{\mu^2, \tau}$. For details of the Feynman diagrams provided in the caption of Fig. 2.



FIG. 4. Diagrams representing the contributions to the next-to-leading-order (two-loop) contributions to the Wilson coefficient $d_{\mu^2, \tau}$. For details of the Feynman diagrams, see the caption of Fig. 2.

$d_{\mu^2}(\mu x, \mu^2 \tau, mx)$ of Eq. (4) in Figs. 3 and 4, respectively. Throughout this section, we neglect diagrams that are trivially incorporated as part of the wave function renormalization of the external fields, such as the one-loop examples illustrated in Fig. 5. Provided the original theory at zero flow time is renormalized, counterterms cancel these contributions completely.

1. Leading order

We extract the leading-order connected Wilson coefficient by considering matrix elements of each of the operators in the sOPE coupled to two external fields, which removes any disconnected contributions. We can then read off the one-loop contribution to the Wilson coefficient, $d_{\mu^2}^{(1)}$, by matching terms at $\mathcal{O}(\lambda)$. This contribution



FIG. 5. Example diagrams that are naturally incorporated in the renormalized external propagators. We do not include these contributions in our explicit calculations of the Wilson coefficients. For details of the Feynman diagrams, see the caption of Fig. 2.

is given by the small spacetime behavior of

$$-\lambda d_{\mu^2}^{(1)} \stackrel{x \rightarrow 0}{\sim} \left\{ \langle \Omega | \mathcal{T} \{ \phi(x) \phi(0) \} \tilde{\phi}(p_1) \tilde{\phi}(p_2) | \Omega \rangle^{(1)} - \langle \Omega | p^2(\tau, 0) \tilde{\phi}(p_1) \tilde{\phi}(p_2) | \Omega \rangle^{(1)} \right\}_{\mathcal{O}(m^0)}, \quad (33)$$

The corresponding Feynman integral is

$$\frac{1}{(p_1^2 + m^2)(p_2^2 + m^2)} \left\{ \frac{1}{2} \int_k \frac{e^{ikx} - e^{-(k^2 + q^2)\tau}}{(k^2 + m^2)(q^2 + m^2)} \right\}, \quad (34)$$

where $q = k - p_1 - p_2$. We extract the smeared Wilson coefficient by examining the small spacetime behavior of the integral in curly braces, expanded to $\mathcal{O}(m^0)$:

$$d_{\mu^2}^{(1)} \stackrel{x \rightarrow 0}{\sim} \left\{ \frac{1}{2} \int_k \frac{e^{ikx} - e^{-(k^2 + q^2)\tau}}{(k^2 + m^2)(q^2 + m^2)} \right\}_{\mathcal{O}(m^0)}. \quad (35)$$

The smeared Wilson coefficients must be independent of the external states to ensure that the sOPE is truly an operator expansion. In this particular case, we require that $d_{\mu^2}^{(1)}$ is independent of the external momenta p_1 and p_2 and the mass. By taking a derivative with respect to one of the external momenta,

$$\frac{d}{dp_1} d_{\mu^2}^{(1)} = \int_k \frac{q_1 [e^{ikx} - e^{-(k^2 + q^2)\tau}] (1 + (q^2 + m^2)\tau)}{(k^2 + m^2)(q^2 + m^2)^2}, \quad (36)$$

we obtain a convergent integral. The $x \rightarrow 0$ limit of this integral is now well defined and only vanishes if the flow time is related to the spacetime separation. An analogous result holds if we take a derivative with respect to the external mass, m . On dimensional grounds, then, we guarantee that the smeared Wilson coefficient is independent of the external states by choosing $\tau \propto x^2$.

On physical grounds, however, we require that the smearing radius is smaller than the spacetime extent of the nonlocal operator: $s_{\text{min}} < x$. This choice ensures that the sOPE remains an expansion in local operators. Physically speaking, if the gradient flow probes length scales on the order of the nonlocal operator, which we represent in Fig. 6, then smeared operators would cease to be (approximately) local. In other words, the sOPE would be a poor expansion for the original operator. This is the physical origin of the third feature that we saw in the leading-order disconnected contribution, Eq. (25): small spacetime separations require small flow times to ensure convergence. The OPE requires small spacetime separations for good convergence, so it follows that the sOPE requires small flow times as well.

There is a complementary viewpoint that elucidates the small flow-time requirement more quantitatively: the small flow-time expansion [31, 50, 52]. In this case we see that the derivative of Eq. (36) is independent of the external momenta in the $x \rightarrow 0$ limit, up to terms linear in the flow time:

$$\frac{d}{dp_1} d_{\mu^2}^{(1)} = \int_k \frac{q_1 (e^{ikx} - 1)}{(k^2 + m^2)(q^2 + m^2)^2} + \mathcal{O}(\tau). \quad (37)$$

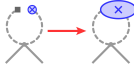


FIG. 6. At small flow times (left-hand diagram), the smeared operators are localized relative to the spacetime separation of the nonlocal operator. At large flow-time values (right-hand diagram), however, the smearing radius probes the scale of the nonlocal and the sOPE is a poor approximation to the original nonlocal operator. For details of the Feynman diagrams, see the caption of Fig. 2.

From this, it is clear that the Wilson coefficients will be independent of the external states, up to terms linear in the flow time. Therefore, provided the flow time is small relative to the spacetime separation, the sOPE is an expansion in approximately local operators, in a quantifiable sense. Moreover, the flow-time dependence will cancel in the product of the Wilson coefficient and its associated matrix element, to the desired order in the flow time.

We are free to set $p_1 = p_2 = 0$ in Eq. (35), because the smeared Wilson coefficient is independent of the external momenta and the mass to the order at which we are working [53]. Expanding in the mass, we obtain

$$d_{\rho^2}^{(1)} = \frac{1}{2} \int_k \frac{e^{ikx} - e^{-2k^2\tau}}{(k^2)^2} = -\frac{1}{32\pi^2} \left[\gamma_E - 1 + \log\left(\frac{x^2}{8\tau}\right) \right]. \quad (38)$$

Combining this with the leading-order contribution, which is just unity, we have

$$d_{\rho^2} = 1 + \frac{\lambda}{32\pi^2} \left[\gamma_E - 1 + \log\left(\frac{x^2}{8\tau}\right) \right] + \mathcal{O}(\lambda^2). \quad (39)$$

In contrast, the Wilson coefficient for the OPE in the $\overline{MS}$ scheme, denoted by $\bar{\pi}$, is

$$\bar{\pi}_{\rho^2} = 1 + \frac{\lambda}{32\pi^2} \left[1 + 2\gamma_E + \log\left(\frac{\mu^2 x^2}{16}\right) \right] + \mathcal{O}(\lambda^2). \quad (40)$$

We note the occurrence of the three features we observed for the leading-order disconnected contribution: the same spacetime dependence for both smeared and unsmeared coefficients; the appearance of the flow time as a leading-order regulator; and the need to choose a small flow time for small spacetime separations to avoid large logarithmic contributions. From the small flow-time expansion viewpoint, we can confirm that the flow-time dependence ultimately cancels to the desired order. For example, the matrix element of $\rho^2(\tau, 0)$ coupled to two external fields



FIG. 7. Two-loop diagrams that appear at $\mathcal{O}(m^2)$ and therefore do not contribute to d_{ρ^2} . For details of the Feynman diagrams, see the caption of Fig. 2.

is

$$\langle \Omega | \rho^2(\tau, 0) \tilde{\phi}(p_1) \tilde{\phi}(p_2) | \Omega \rangle = 1 + \frac{\lambda}{32\pi^2} \left[1 + \gamma_E + \log(2m^2\tau) \right] + \mathcal{O}(\tau, \lambda^2) \quad (41)$$

for sufficiently small flow times. Here we have dropped the external fields, which we take to be on shell, for simplicity. The product of this matrix element with the Wilson coefficient is independent of the flow time to one loop and $\mathcal{O}(\tau)$, as we would expect:

$$d_{\rho^2} \langle \Omega | \rho^2(\tau, 0) \tilde{\phi}(p_1) \tilde{\phi}(p_2) | \Omega \rangle = 1 + \frac{\lambda}{32\pi^2} \left[2\gamma_E + \log\left(\frac{m^2 x^2}{4}\right) \right] + \mathcal{O}(\tau, \lambda^2). \quad (42)$$

2. Next-to-leading-order

To determine the next-to-leading-order contribution to the connected Wilson coefficient, we must incorporate the Feynman diagrams of Fig. 4. The two-loop diagrams of Fig. 7 do not contribute to the Wilson coefficient d_{ρ^2} , because they appear at $\mathcal{O}(m^2)$.

Looking at the Feynman diagrams of Fig. 4, and bearing in mind our experience of the next-to-leading-order contribution to d_i , we immediately observe that these diagrams are simply the product of the one-loop Wilson coefficient, $d_i^{(1)}$, and the next-to-leading-order renormalized four-point vertex. This contribution to the vertex is, of course, nothing other than the next-to-leading-order contribution to the renormalized coupling constant. Thus, we write the Wilson coefficient quite simply as

$$d_{\rho^2} = 1 + \frac{\lambda_R}{32\pi^2} \left[\gamma_E - 1 + \log\left(\frac{x^2}{8\tau}\right) \right] + \mathcal{O}(\lambda^3), \quad (43)$$

where λ_R is the renormalized coupling. In the $\overline{MS}$ scheme, the renormalized coupling is given by

$$\lambda_R = \lambda - \frac{3\lambda^2}{2(16\pi^2)} \log\left(\frac{\mu^2}{m^2}\right) + \mathcal{O}(\lambda^3). \quad (44)$$

For a calculation of this to five loops, see [59].

As we move beyond the leading-order Wilson coefficients, i.e. $d_i^{(0)}$ and $d_{\rho^2}^{(1)}$, divergent radiative corrections appear in our calculations, because all field interaction

vertices appear at zero flow time. These divergences can be removed by the renormalization parameters of the original theory and the renormalization scale dependence of the smeared operators is completely contained in the renormalization parameters of the original theory. This is to be expected: it follows from the fact that renormalized matrix elements at zero flow time remain finite at nonzero flow time and require no further renormalization [31, 35].

Ultimately, for realistic calculations in lattice QCD, the perturbative calculation of smeared Wilson coefficients must be combined with nonperturbative determinations of matrix elements at hadronic energy scales. Scalar ϕ^4 theory is not asymptotically free in four dimensions, but we can understand the mathematical features of the sOPE in more detail by studying the renormalization group equations for the simple example of scalar fields.

VI. RENORMALIZATION GROUP EQUATIONS

We consider the matrix elements of scalar operators coupled to N external, unsmeared scalar fields. We can derive renormalization group equations for the Wilson coefficients of the OPE by considering the scale dependence of suitably chosen Green functions [53]. The Green function for $N + 2$ external scalar fields obeys the renormalization group equation

$$\left[\mu \frac{d}{d\mu} + (N + 2) \gamma \right] \langle \Omega | \tilde{\phi}(p_1) \dots \tilde{\phi}(p_{N+2}) | \Omega \rangle = 0, \quad (45)$$

while the Green function of the renormalized operator $\phi_R^2(0)$ coupled to N external scalar fields satisfies

$$\left[\mu \frac{d}{d\mu} - \gamma_m + N \gamma \right] \langle \Omega | \phi_R^2(0) \tilde{\phi}(p_1) \dots \tilde{\phi}(p_N) | \Omega \rangle = 0. \quad (46)$$

Here the renormalization group operator for scalar field theory is

$$\mu \frac{d}{d\mu} = \mu \frac{\partial}{\partial \mu} \Big|_{\lambda, m_{\text{B}}} + \beta \frac{\partial}{\partial \lambda} \Big|_{\mu, m_{\text{B}}} - \gamma_m m_{\text{B}} \frac{\partial}{\partial m_{\text{B}}} \Big|_{\mu, \lambda} \quad (47)$$

and the coefficients are [59]

$$\beta^{\overline{\text{MS}}}(\lambda) = \mu \frac{d\lambda}{d\mu} = \frac{3\lambda^2}{(16\pi^2)^2} + \mathcal{O}(\lambda^3), \quad (48)$$

$$\begin{aligned} \gamma_m^{\overline{\text{MS}}}(\lambda) &= -\frac{\mu}{2} \frac{d \log(m_{\text{B}})}{d\mu} \\ &= -\frac{\lambda}{2(16\pi^2)} + \frac{5\lambda^2}{12(16\pi^2)^2} + \mathcal{O}(\lambda^3), \end{aligned} \quad (49)$$

$$\gamma^{\overline{\text{MS}}}(\lambda) = \frac{\mu}{2} \frac{d \log(Z_\phi)}{d\mu} = \frac{\lambda^2}{12(16\pi^2)^2} + \mathcal{O}(\lambda^3). \quad (50)$$

In general these coefficients depend on the mass, the renormalization scale, and the renormalized coupling

constant, but in a mass independent renormalization scheme, such as the $\overline{\text{MS}}$ scheme, they depend on the renormalization scale only through the renormalized coupling constant. These functions are known to five loops for the $O(N)$ -symmetric theory, given by the N -multiplet $\Phi = \{\phi_1, \dots, \phi_N\}$ [59, 60].

Returning again to our example of the OPE for the two-point function, Eq. (2), we can derive a renormalization group equation for the Wilson coefficient c_ϕ by coupling these operators to two external scalar fields and using Eqs. (45) and (46) (for further details, see, for example, Ref. [53]):

$$\left[\mu \frac{d}{d\mu} + 2(\gamma + \gamma_m) \right] c_\phi = 0. \quad (51)$$

Just as we might expect, the anomalous dimension of the Wilson coefficient c_ϕ is equal to the difference between the anomalous dimension of $\phi(x)\phi(0)$ and that of $[\phi^2(0)]_{\text{R}}$.

A. Renormalization group equations for the sOPE Wilson coefficients

The flow time, τ , introduces a new scale into the problem. In principle one can view the flow time as just another external scale, unrelated to the renormalization scale μ . In this case, it is natural to modify the renormalization group equations to account for the change in the Green functions as we change both μ and τ :

$$\mu \frac{d}{d\mu} \rightarrow \mu \frac{d}{d\mu} - 2\tau \frac{d}{d\tau} \quad (52)$$

Our choice of differential operator is constrained by the mass dimension of each scale, μ and τ , but is not unique. In particular, one could choose the operator $\mu d/d\mu + \tau d/d\tau$. This freedom does not affect the logic of our discussion nor our conclusion, because alternative conventions can be absorbed into the definition of the renormalization parameters and anomalous dimensions in, for example, Eq. (58).

At this stage it is worth commenting on the two scales in the problem, μ and τ . In DIS, the spacetime separation of the corresponding OPE, x , and renormalization scale μ are, in principle, two distinct scales. The spacetime separation is provided by the inverse momentum transfer of a particular DIS experiment or set of experiments. The renormalization scale, however, is a theoretical choice, and ultimately physical quantities should not depend on the renormalization scale. It is generally convenient to choose $\mu = 1/x$, but it is not strictly necessary.

The relationship between the flow time and the renormalization scale is analogous and these two scales are distinct. For the sOPE, the flow time can be considered as simply an external scale, imposed by some particular lattice “experiment”, and the renormalization scale is a convenient theoretical choice. We will see that it is helpful to tie these scales together, to reduce the two-scale

problem to a single scale, but this is not formally necessary. Therefore, in the following analysis, the flow and renormalization scale should be understood as completely independent scales.

Considering again the sOPE for the two-point function, Eq. (4), we determine the renormalization group equation for the smeared Wilson coefficient, d_{ρ^2} , by following a procedure analogous to that outlined above.

We assume that we are working in the small flow-time regime, which allows us to relate operators at vanishing and nonvanishing flow time [31, 50, 52]:

$$[\phi^2(0)]_R = \mathcal{Z}_{\rho^2}(\tau, \mu) \rho^2(\tau, 0) + \mathcal{O}(\tau). \quad (53)$$

This coefficient satisfies

$$\mu \frac{d}{d\mu} \log(\mathcal{Z}_{\rho^2}(\tau, \mu^2)) = 2\gamma_m, \quad (54)$$

and to one loop is given by

$$\mathcal{Z}_{\rho^2}(\tau, \mu^2) = 1 - \frac{\lambda}{32\pi^2} [1 + \gamma_E + \log(2\tau\mu^2)] + \mathcal{O}(\lambda^2, \tau). \quad (55)$$

We apply the renormalization group operator

$$\mu \frac{d}{d\mu} - 2\tau \frac{d}{d\tau} + (N+2)\gamma \quad (56)$$

to matrix elements of the operators in Eq. (53) coupled to N external scalar fields to obtain

$$\left[\mu \frac{d}{d\mu} - 2\tau \frac{d}{d\tau} + 2(\zeta_{\rho^2} - \gamma) \right] d_{\rho^2} = \mathcal{O}(\tau). \quad (57)$$

Here

$$\zeta_{\rho^2} = \tau \frac{d}{d\tau} \log(\mathcal{Z}_{\rho^2}(\tau, \mu^2)) \quad (58)$$

is an anomalous dimension associated with the flow-time dependence of the operator $\rho^2(\tau, 0)$. The renormalization group equation for the corresponding matrix element of $\rho^2(\tau, 0)$ coupled to N external fields is given by

$$\left[\mu \frac{d}{d\mu} - 2\tau \frac{d}{d\tau} + 2\zeta_{\rho^2} + N\gamma \right] \times \langle \Omega | \rho^2(\tau, 0) \tilde{\phi}(p_1) \dots \tilde{\phi}(p_N) | \Omega \rangle = \mathcal{O}(\tau). \quad (59)$$

We note that this equation only holds provided the flow time is small compared with the momenta of the external particles, which for DIS would be of the order of hadronic scales.

If we now demand that the smearing scale, τ , and the inverse of the renormalization scale, μ , are proportional to each other, i.e., $\tau = b/\mu^2$ with b real, then the renormalization group equation becomes

$$\left[2\mu \frac{d}{d\mu} + 2\zeta_{\rho^2} + N\gamma \right] \times \langle \Omega | \rho^2(b/\mu^2, 0) \tilde{\phi}(p_1) \dots \tilde{\phi}(p_N) | \Omega \rangle = \mathcal{O}(b). \quad (60)$$

This renormalization group equation provides the starting point for a nonperturbative step-scaling method [47, 61] that evolves nonperturbative matrix elements to a high scale, where they can be combined with perturbative smeared Wilson coefficients. Here the renormalization group equation holds in the small flow time limit, or in other words, provided $b \ll 1$. This constraint automatically ensures that the flow time is also smaller than any hadronic length scales, $\Lambda_{\text{QCD}} \ll \mu^2 \ll 1/\tau$, and is generally true for practical step-scaling methods [47, 61]. We are currently investigating this approach in QCD.

VII. CONCLUSION

We have proposed a new method, the smeared operator product expansion (sOPE), to extract matrix elements from numerical nonperturbative calculations without power divergent mixing. The smeared operator product expansion is a general framework relevant to any asymptotically free theory with nonperturbative matrix elements that suffer from power divergent mixing. Within QCD, the most obvious application is to deep inelastic scattering, but other applications include nonperturbative determinations of $K \rightarrow \pi\pi$ decays [62, 63] and B -meson mixing [64]. Beyond QCD, applications include nonperturbative studies of critical phenomena in the Heisenberg model, spin systems and other condensed matter systems.

In the sOPE, we expand nonlocal operators in a basis of smeared operators, the matrix elements of which can be determined on the lattice. We implement the smearing via the gradient flow, a classical evolution of the theory in a new dimension that smooths ultraviolet fluctuations. The continuum limit of these matrix elements is free of power divergent mixing, provided the localization scale, the smearing length, is kept fixed in the continuum limit. The resulting matrix elements are functions of two scales, the renormalization scale and the smearing length. The sOPE systematically relates these matrix elements to smeared Wilson coefficients, which can be calculated in perturbation theory, thereby providing a complete determination of the nonlocal operators.

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P39: 微扰论中的涂抹准分布

主题归属	梯度流、涂抹与流时算符
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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

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Smeared quasidistributions in perturbation theory

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Quasi- and pseudodistributions provide a new approach to determining parton distribution functions from first principles^a calculations of QCD. Here I calculate the flavor nonsinglet unpolarized quasidistribution at one loop in perturbation theory, using the gradient flow to remove ultraviolet divergences. I demonstrate that, as expected, the gradient flow does not change the infrared structure of the quasidistribution at one loop and use the results to match the smeared matrix elements to those in the $\overline{\text{MS}}$ scheme. This matching calculation is required to relate numerical results obtained from nonperturbative lattice QCD computations to light-front parton distribution functions extracted from global analyses of experimental data.

Keywords: Lattice Quantum Field Theory, Lattice QCD

I. INTRODUCTION

A central challenge for QCD, the gauge theory of the strong nuclear force, is the direct prediction of hadron structure. In particular, parton distribution functions (PDFs) and generalized parton distributions (GPDs) have, until recently, posed intractable difficulties for first principles^a calculations. Although a complete, *ab initio* determination of PDFs and GPDs has yet to appear, a promising new method was recently proposed in Ref. [1]. In this approach, PDFs and GPDs are extracted from matrix elements of spatially extended operators between nucleon states at finite momentum. These matrix elements are generally referred to as quasi-PDFs or quasidistributions. Related frameworks have also been proposed; in Refs. [2–4] quasidistributions were treated as “lattice cross sections” from which light-front PDFs can be factorized, and Refs. [5–7] introduced and studied the closely related pseudodistributions.

Quasidistributions are matrix elements of time-local operators and therefore can be directly calculated using lattice QCD [8, 9], in which QCD is discretized on a Euclidean hypercubic lattice. Preliminary nonperturbative results for both quasi- and pseudodistributions are encouraging [7, 10–14]. Moreover, a number of theoretical issues [8, 15, 16] have been clarified or solved, including a proof of the multiplicative renormalization of the spatial Wilson-line operator [17, 18], a proof of the factorization of light-front PDFs from quasidistributions [2, 3], and a proof that the matrix element extracted from Euclidean correlation functions is identical to that defined through a Lehmann-Symanzik-Zimmermann reduction procedure in Minkowski space [9, 19, 20].

One of the computational challenges that must be addressed is the presence of a power divergence induced in the quasi- and pseudodistributions by the finite lattice regulator. The Wilson-line operator, which defines

these distributions, has a divergence that scales exponentially with the length of the Wilson line divided by the lattice spacing, and this divergence must be removed nonperturbatively. Several approaches have been suggested: the authors of Refs. [21] and [22] proposed removing the power divergence through an exponentiated mass renormalization, and, more recently, regularization invariant momentum subtraction (RI/MOM [23] and RI [24]) schemes have been suggested as nonperturbative renormalization procedures. In Ref. [25], we introduced an alternative approach, the smeared quasidistribution, which circumvents some of the challenges of the power divergence by taking advantage of the properties of the gradient flow [26–28].

The gradient flow is a classical evolution, or one-parameter mapping, of the original quark and gluon degrees of freedom in a new parameter, the flow time. The flow time exponentially suppresses UV field fluctuations, which corresponds to smearing out the original degrees of freedom in real space. The critical property of the gradient flow is that, up to a multiplicative fermion wave function renormalization [28], finite correlation functions of the original theory remain finite at nonzero flow time [27]. By fixing the flow time in physical units, one can ensure that matrix elements determined nonperturbatively at finite lattice spacing remain finite in the continuum limit. The gradient flow therefore provides a nonperturbative, gauge-invariant method to render the quasidistribution finite, even in the continuum [25]. The resulting continuum matrix element can then be related directly to the light-front PDF, or to the quasi- or pseudodistributions renormalized in, for example, the $\overline{\text{MS}}$ scheme. In practice, it is simpler to carry out this matching procedure in several steps: first relate the smeared distribution to the distribution without smearing, and then use the known relation between the unsmeared distribution and the light-front PDF [29–32]. The perturbative calculation of the relevant matrix elements in the $\overline{\text{MS}}$ scheme appears in Ref. [33], and here my focus is the determination of the matrix elements at finite flow time.

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I study the matrix element that defines the smeared quasidistribution at one loop in perturbation theory. Computing the matrix element of the smeared Wilson-line operator between external massless quark states at rest, enables all integrals to be evaluated analytically. At one loop, the Wilson-line power divergence manifests as a contribution linear in $\bar{\tau} = z/r_r$, where z is the length of the Wilson line and r_r is the gradient flow smearing radius, for $\bar{\tau} \gg 1$. Subtracting this contribution, the remaining matrix element is finite, and has a well-defined $\bar{\tau} \rightarrow 0$ limit. In the small flow-time regime, for which $\bar{\tau} \gg 1$, the matrix element depends only logarithmically on $\bar{\tau}$ and satisfies a relation analogous to the usual renormalization-group equation.

I start by briefly reviewing light-front PDFs, smeared quasidistributions and their relation in Sec. II. I discuss the perturbative calculation of the relevant matrix element in Sec. III, provide some numerical results in Sec. IV, and summarize in Sec. V.

II. LIGHT-FRONT AND QUASIDISTRIBUTIONS

Throughout this work, I focus on flavor nonsinglet unpolarized quasi and light-front PDFs. The extension to polarized quasi-PDFs is straightforward and much of the discussion applies equally to pseudodistributions. I do not consider the flavor singlet case, which introduces additional mixing with the gluon distribution; this mixing is studied for the case of the quasidistribution at one loop in perturbation theory in Ref. [31].

A. Light-front PDFs

I denote renormalized light-front PDFs by $f(\xi, \mu)$, where the renormalization scale is μ , and define $\xi = k^+/P^+$. Here the light-front coordinates are $(x^+, x^-, \mathbf{x}_\perp)$ such that $x^\pm = (t \pm z)/\sqrt{2}$. In general, one can write the renormalized light-front PDFs in terms of the bare light-front PDFs, $f^{(0)}(\xi)$, as

$$f(\xi, \mu) = \int_{\xi}^1 \frac{d\zeta}{\zeta} Z\left(\frac{\xi}{\zeta}, \mu\right) f^{(0)}(\zeta). \quad (1)$$

The bare PDF is defined as [34]

$$f^{(0)}(\xi) = \int_{-\infty}^{\infty} \frac{d\omega^-}{4\pi} e^{-i\xi P^+ \omega^-} \times \left\langle P \left| T \bar{\psi}(0, \omega^-, \mathbf{0}_\perp) W(\omega^-, 0) \gamma^+ \frac{\lambda^a}{2} \psi(0) \right| P \right\rangle_C, \quad (2)$$

where T is the time-ordering operator, ψ is a quark field, λ^a is an $SU(2)$ -flavor Pauli matrix, and the subscript C indicates that the vacuum expectation value has been subtracted (in other words, only connected contributions

are included). The operator $W(\omega^-, 0)$ is the Wilson line,

$$W(\omega^-, 0) = \mathcal{P} \exp \left[-ig_0 \int_0^{\omega^-} dy^- A_\perp^a(0, y^-, \mathbf{0}_\perp) T_c \right], \quad (3)$$

with $\mathcal{P}$ the path-ordering operator, g_0 the QCD bare coupling, and $A^a = A_\perp^a T_c$ the $SU(3)$ gauge potential with generator T_c (summation over color index c is implicit). The target state, $|P\rangle$, is a spin-averaged, exact momentum eigenstate with relativistic normalization

$$\langle P' | P \rangle = (2\pi)^3 2P^+ \delta(P^+ - P'^+) \delta^{(2)}(\mathbf{P}_\perp - \mathbf{P}'_\perp). \quad (4)$$

B. Smeared quasidistributions

One of the central challenges for nonperturbative calculations of quasidistributions is a power divergence associated with the length of the Wilson line in the presence of the lattice regulator. A number of approaches have been proposed to remove this power divergence [7, 21–24], including circumventing the issue completely by using alternative operators [35–39]. In Ref. [25], we constructed a finite matrix element to extract the continuum quasidistribution from lattice calculations by smearing both the fermion and gauge fields via the gradient flow [26–28]. For a more complete discussion of the gradient flow, I refer the reader to the recent reviews [40, 41].

At finite flow time, the gradient flow guarantees that the matrix elements constructed from flowed fields is UV finite. Provided one holds the flow time fixed in physical units, the lattice regulator can be removed and one can extract a continuum quasidistribution that is a function of the flow time. This smeared quasidistribution can then be related to the light-front distributions through a perturbative matching relation [25], analogous to the relation that holds for the ordinary (unsmeared) quasidistribution. Here, I study, at one loop in perturbation theory, the matrix element that defines the smeared quasidistribution. This result is necessary to relate the smeared matrix element to the corresponding matrix element in the $\overline{MS}$ scheme.

The continuum smeared matrix element is finite but, because of the presence of the flow-time scale, is expected to contain a power divergence as the flow time is taken to zero [25]. This contribution must be removed nonperturbatively. The gradient flow provides a continuous parameter, in contrast to the lattice regulator, which should provide a better lever through which to study the power-divergent contribution. At one loop in perturbation theory, this divergence manifests as a term linear in the length of the Wilson line, an expectation confirmed by the results presented in Sec. III.

I denote the ringed fermion fields [42, 43] at flow time τ by $\tilde{\chi}(x; \tau)$ and $\chi(x; \tau)$, and the corresponding Wilson line at the same flow time, constructed from the smeared gauge fields $B_a(x; \tau)$, by $\mathcal{W}(x_1, x_2; \tau)$. I start with the

connected matrix element

$$h^{(\alpha)} \left(\frac{n^2}{\tau}, n \cdot P, \sqrt{\tau} \Lambda_{\text{QCD}}, \sqrt{\tau} M_N \right) = \frac{1}{2P} \left\langle P \left| \bar{\chi}(n; \tau) \mathcal{W}(n, 0; \tau) \gamma_n \frac{\lambda^\alpha}{2} \chi(0; \tau) \right| P \right\rangle_C, \quad (5)$$

which, being dimensionless, depends only on dimensionless combinations of scales. Here, n is a 4-vector that is usually taken to be in the z -direction, $n = (0, z, 0)$; the dependence of the matrix element on 4-vectors is constrained by Euclidean $\text{SO}(4)$ invariance [5]. I note that the flow time has units of length squared. The ringed fermion fields require no wave function renormalization and this smeared matrix element is finite provided the flow time, τ , is nonzero and fixed in physical units, because correlation functions constructed from smeared fields are finite [27, 28]. Divergences will appear in the limit of vanishing flow time and the matrix element will then require renormalization. The Lorentz index α can take any value from 1 to 4, but the choice $\alpha = 4$ removes some higher-twist contamination [5] and avoids mixing, at least for the unpolarized flavor-nonsinglet distribution, at finite lattice spacing [33]. In lattice calculations, it is common to choose the spatial momentum of the nucleon state to be $\mathbf{P} = (0, 0, P_z)$, in which case the index α is restricted to be $\alpha = \{3, 4\}$.

The smeared quasisdistribution [25] is then defined, assuming $n = (0, z, 0)$, as

$$q^{(\alpha)}(\xi, \sqrt{\tau} P_z, \sqrt{\tau} \Lambda_{\text{QCD}}, \sqrt{\tau} M_N) = \int_{-\infty}^{\infty} \frac{dz}{2\pi} e^{i\xi z} P_z \times h^{(\alpha)} \left(\frac{n^2}{\tau}, n \cdot P, \sqrt{\tau} \Lambda_{\text{QCD}}, \sqrt{\tau} M_N \right), \quad (6)$$

where ξ is a dimensionless parameter that in Euclidean space should be viewed as a (dimensionless) Fourier-conjugate variable.

C. Relating smeared quasisdistributions and PDFs

The smeared quasisdistribution can be directly related to the light-front distribution by [25]

$$q^{(\alpha)}(x, \sqrt{\tau} \Lambda_{\text{QCD}}, \sqrt{\tau} P_z) = \int_{-1}^1 \frac{d\xi}{\xi} \bar{Z} \left(\frac{x}{\xi}, \sqrt{\tau} \mu, \sqrt{\tau} P_z \right) f(\xi, \mu) + \mathcal{O}(\sqrt{\tau} \Lambda_{\text{QCD}}), \quad (7)$$

provided

$$\Lambda_{\text{QCD}}, M_N \ll P_z \ll \tau^{-1/2}. \quad (8)$$

The quasisdistribution and the light-front PDF have the same IR structure [2, 3, 8, 9], so that the matching kernel can be determined in perturbation theory. In practice, it is simplest to relate the smeared quasisdistribution to the quasisdistribution at zero flow time first and then match

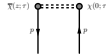


FIG. 1. Tree-level contribution to the smeared nonsinglet quasisdistribution. The gray circles indicate fermion fields at finite flow time τ and solid lines represent fermion propagators.

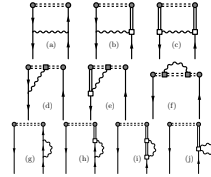


FIG. 2. Diagrams representing the one-loop contributions to the smeared nonsinglet quasisdistribution. The gray circles indicate fermion fields at finite flow time τ ; solid lines represent fermion propagators; double lines are fermion flow kernels; and open squares are flow vertices at arbitrary flow time τ_1 . The diagrams labeled (a), (d), (f), and (g) are in one-to-one correspondence with the diagrams that contribute to the unsmeared quasisdistribution in the $\overline{\text{MS}}$ scheme, for which the fermion fields that appear in the extended operator are replaced by fermion fields at vanishing flow time.

this (unsmeared) quasisdistribution to the light-front PDF [29, 30].

I show the diagrams representing the tree-level and one-loop contributions to the smeared quasisdistribution in generalized Feynman gauge (with $\alpha = \lambda = 1$, where α is the usual gauge-fixing parameter and λ is a parameter introduced to fix the gauge of the smeared gauge fields [27, 44]) in Figs. 1 and 2, respectively. These diagrams correspond to the “quark-in-quark” distributions at one loop in perturbation theory. Although the quasisdistribution can be more easily evaluated in axial gauge, this gauge cannot be consistently generalized to arbitrary flow time. In principle, it is possible to evaluate these nine diagrams and determine the matching kernel directly at one loop in perturbation theory. In practice, however, the corresponding integrals cannot be evaluated analytically, and it is simpler to use a procedure similar to that

employed in Refs. [22, 33] and match at a fixed value of the Wilson-line length, z . This procedure, which corresponds to matching at the level of the matrix element itself, rather than dealing with the quasidistributions, is preferable largely because the matrix elements do not have any dependence on ξ . In the rest of this work, I focus on the matrix element, $h^{(s)}$, itself.

III. MATRIX ELEMENTS IN PERTURBATION THEORY

I illustrate the leading-order, or tree-level, contribution to the smeared nonsinglet matrix element in Fig. 1. In both Figs. 1 and 2, I use the following conventions: solid lines represent fermion propagators; double lines are fermion flow kernels; open squares are flow vertices at arbitrary flow time τ ; gray circles indicate fermion fields at fixed flow time τ ; gray squares indicate vertices at fixed flow time τ ; and the dashed line is the spatially extended Wilson-line operator. Note that gauge kernels do not contribute to the smeared nonsinglet quasidistribution at one loop in perturbation theory. The tree-level contribution is given by

$$\begin{aligned} h^{(0)} &= \frac{1}{2p} \langle p, s | \bar{\chi}(n; \tau) \gamma_n \frac{\lambda^a}{2} \chi(0; \tau) | p, s \rangle \\ &= \frac{1}{2p} e^{i p \cdot z} e^{-2ip^2 \tau} \bar{u}^s(p) \frac{\lambda^a}{2} \gamma_n u^s(p), \end{aligned} \quad (9)$$

where I have dropped the arguments of the matrix element for clarity, and I have used

$$\chi(x; \tau) | p, s \rangle = e^{-i p^2 \tau} \psi(x) | p, s \rangle = e^{-i p^2 \tau} e^{i p \cdot x} u^s(p), \quad (10)$$

and an analogous relation for the antiquark.

Beyond tree-level, the operator receives radiative corrections that can be written as

$$\begin{aligned} h^{(s)} &= \mathcal{Z}_h h^{(0)} \\ &= \mathcal{Z}_h^{(1)}(\bar{\tau}^2, \mu^2 \tau) \mathcal{Z}_\chi^{(1)}(\mu^2 \tau) (\mathcal{Z}_\psi^{(1)})^{-1} h^{(0)} + \mathcal{O}(\alpha_s^2). \end{aligned} \quad (11)$$

Here, $\alpha_s = g^2/(4\pi)$ is the renormalized coupling constant, which is equal to the bare coupling constant to this order and is most naturally evaluated at the scale $\mu^2 = 1/\tau^2$. I show the individual one-loop diagrams that contribute to the correction parameter $\mathcal{Z}_h$ in Fig. 2. Diagrams (a) to (f) contribute to the operator renormalization parameter, $\mathcal{Z}_h$. The wave function renormalization $\mathcal{Z}_\psi$ receives contributions from diagram (g). The extra fermion wave function renormalization $\mathcal{Z}_\chi$, which receives contributions from diagrams (h), (i) and (j), is automatically removed via the ringed fermions.

The determination of the operator renormalization parameter, $\mathcal{Z}_h$, is the central result of this paper. Both the wave function renormalization parameters appear elsewhere (in particular, $\mathcal{Z}_\chi$ is calculated in Refs. [28, 42]);

I calculate these contributions for completeness and as a check of my methods and combine these results with $\mathcal{Z}_h$ to obtain the complete one-loop correction $\mathcal{Z}_h$. I show that, in the small flow-time and local vector current limits, the parameter $\mathcal{Z}_h$ reduces to the corresponding results in the literature [42, 43], a nontrivial check of my results. The existence of a well-defined local vector-current limit for the extended operator is in contrast to, for example, the extended operator in the $\overline{MS}$ scheme in the absence of the flow time [33] and is one attractive feature of the smeared quasidistribution.

I split the following discussion into two parts, starting with an analytic study of the UV structure of the matrix elements. This is sufficient to extract $\mathcal{Z}_h^{(1)}$ and is possible because I assume the renormalization scale is much larger than the external momentum and set the momentum to zero [22]. Diagrams (b), (c), (e), and (i) are proportional to the external momentum and therefore vanish in this case. Moreover, the remaining integrals themselves are considerably simplified and become analytically tractable. I regulate the resulting IR divergences with dimensional regularization and demonstrate that these divergences cancel in the matching procedure between the matrix elements in the $\overline{MS}$ scheme and at nonzero flow time. In the second part of my discussion, Sec. IV, I numerically evaluate all nine flow-time-dependent diagrams as a function of the external momentum, quark mass, and flow time to illustrate the behavior of the full matrix elements at one loop in perturbation theory.

A. Vertex-type diagrams

At one loop in perturbation theory, there are three non-vanishing “vertex-type” diagrams associated with the extended operator, which are diagrams (a), (d), and (f) in Fig. 2, and four “wave-function-type” diagrams, which are diagrams (g) to (j) in Fig. 2. At finite Wilson-line length, z , the diagrams exhibit the following UV and collinear behavior: diagram (a) is UV finite and logarithmically IR divergent, while diagrams (d) and (f) would be logarithmically UV divergent without the presence of the flow time, which renders them UV finite and IR finite. Thus, I evaluate diagrams (d) and (f) directly in four dimensions, but diagram (a) requires dimensional regularization to regularize the IR behavior of the integral; I choose $d = (4 + 2\epsilon_m)$. At zero external momentum, the symmetry and dimension of the bare matrix element ensures that the final result is only a function of $\bar{\tau}^2 = z^2/r_\tau^2$ [5], although individual diagrams may also depend on dimensionless combinations of μ^2 , z^2 and τ . I do not impose any constraints on the functional dependence of any intermediate results, so establishing that the final result depends only on $\bar{\tau}^2$ is a useful cross-check of the calculation.

To illustrate the calculation, I will consider diagram (a). The calculation of diagrams (d) and (f) proceeds in

a very similar manner. At zero external momentum, the contribution to Z_h from this diagram is given by

$$Z^{(a)} = C_F (-ig_0 \gamma_\rho) \int_k \frac{e^{-\tau k^2}}{i\tilde{k}^2} \gamma_\alpha e^{-ik \cdot n} \frac{e^{-\tau k^2}}{i\tilde{k}^2} (-ig_0 \gamma_\alpha) \frac{\delta_{\rho\sigma}}{k^2}. \quad (12)$$

Here $C_F = 4/3$ is the color factor; g_0 is the bare coupling; all repeated Lorentz indices are summed over; and I define

$$\int_k = \mu^{4-d} \int \frac{d^d k}{(2\pi)^d}. \quad (13)$$

The Lorentz structure of the numerator is straightforward and, using the Schwinger representation for the denominator, this integral becomes

$$Z_h^{(a)} = g_0^2 C_F \frac{\mu^{4-d}}{(2\pi)^d} \int_0^\infty d\alpha \frac{\alpha}{2} \times \int d^d k N_\alpha(k, d) e^{-(2\tau+\alpha)k^2} e^{-ik \cdot n}. \quad (14)$$

The numerator $N(k, d)$ depends, at finite flow time and spatial separation, on the choice of Lorentz structure in the original matrix element. Introducing momentum components parallel and perpendicular to the spacelike separation, k_z and $k_\perp = (k_x, k_y, k_t)$, respectively, this numerator is given by

$$N(k, d) = (2-d)(k_z^2 - k_\perp^2) \quad (15)$$

for the choice $\gamma_\alpha = \gamma_z$ in $h^{(a)}$ in Eq. (5) and

$$N(k, d) = (2-d) \left(\frac{3-d}{d-1} k_\perp^2 - k_z^2 \right) \quad (16)$$

otherwise.

The momentum integral in Eq. (14) can be carried out by integrating over k_z and $k_\perp$ individually, leaving a Schwinger parameter integral over α that can be done analytically, giving

$$Z_h^{(a)} = \left(\frac{g_0}{4\pi} \right)^2 C_F \left[\frac{1}{\epsilon_{\text{in}}} - \gamma_E - \log(\pi \mu^2 z^2) + 3 - \frac{3}{2\tau} (e^{-\tau z^2} - 1) - \frac{1}{2\tau} (4 - e^{-\tau z^2}) + \text{Ei}(-\tau z^2) \right], \quad (17)$$

and

$$Z_h^{(a)} = \left(\frac{g_0}{4\pi} \right)^2 C_F \left[\frac{1}{\epsilon_{\text{in}}} - \gamma_E - \log(\pi \mu^2 z^2) + 1 - \frac{1}{2\tau} (1 - e^{-\tau z^2} (1 + \tau z^2)) + \text{Ei}(-\tau z^2) \right], \quad (18)$$

respectively. Here, $\text{Ei}(z)$ is the exponential integral

$$\text{Ei}(z) = - \int_{-z}^\infty \frac{e^{-t}}{t} dt. \quad (19)$$

As one would expect, the gradient flow does not affect the IR structure of this diagram (see the Appendix for

the equivalent result without the presence of the gradient flow).

I treat the other diagrams, (d) and (f), in a similar fashion and find

$$Z_h^{(d)} = 2g_0^2 C_F \frac{\pi^2}{(2\pi)^4} \int_0^\infty d\alpha \frac{\alpha}{(\alpha + 2\tau)^2} \times \left(1 - e^{-z^2/(4(\alpha+2\tau))} \right), \quad (20)$$

$$Z_h^{(f)} = g_0^2 C_F \frac{\pi^{d/2}}{(2\pi)^4} \int_0^\infty d\alpha \frac{1}{(\alpha + 2\tau)^{3/2}} \times \int_0^\infty d\beta \frac{1}{\sqrt{\alpha + \beta + 2\tau}} \left(e^{-z^2/(4(\alpha+\beta+2\tau))} - 1 \right). \quad (21)$$

The results are

$$Z_h^{(d)} = - \left(\frac{g_0}{4\pi} \right)^2 C_F \cdot 2 \left[1 - \gamma_E - \log(\tau z^2) + \text{Ei}(-\tau z^2) + \frac{1}{2\tau} (e^{-\tau z^2} - 1) \right], \quad (22)$$

$$Z_h^{(f)} = \left(\frac{g_0}{4\pi} \right)^2 C_F \cdot 2 \left[\gamma_E + \log(\tau z^2) - \text{Ei}(-\tau z^2) - 2(e^{-\tau z^2} - 1) - 2\sqrt{\pi\tau z^2} \text{erf}(\sqrt{\tau z^2}) \right], \quad (23)$$

where $\text{erf}(z)$ is the error function

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt. \quad (24)$$

This term is the one-loop contribution to the expected power divergence and scales linearly with the length of the Wilson line for $\tau \gg 1$.

B. Wave-function-type diagrams

The “wave-function-type” diagrams are diagrams (g), (h), and (j) in Fig. 2 [diagram (i) is proportional to the external momentum]. The contribution from diagram (g) is just the standard quark wave function renormalization, which will cancel exactly in the one-loop matching relation between the smeared and $\overline{\text{MS}}$ distributions, because it is independent of the flow time. In general, this diagram vanishes in dimensional regularization for vanishing external momentum and in the absence of any other scales, but it is helpful to expose the UV and IR divergences by introducing a separation scale, and I discuss this in more detail in the Appendix.

The divergent pieces of the flow-time-dependent diagrams (h) and (j) are removed through the use of ringed fermions, which incorporates the extra fermion renormalization Z_N associated with the fermions at finite flow time. Diagrams (h) and (j) include flow vertices that occur at arbitrary flow times τ_i and must be integrated

from zero to τ . The contributions from these diagrams are represented by

$$Z_h^{(h)} = 2g_0^2 C_F \cdot 2 \int_0^\tau d\tau_1 \int_k \frac{e^{-2k^2\tau_1}}{k^2} \\ = \left(\frac{g_0}{4\pi}\right)^2 C_F \cdot 2 \left[\frac{1}{\epsilon_{uv}} + 1 + \log(8\pi\mu^2\tau) \right], \quad (25)$$

$$Z_h^{(l)}(\mu^2\tau) = -2g_0^2 C_F d \int_0^\tau d\tau_1 \int_k \frac{e^{-2k^2\tau_1}}{k^2} \\ = -\left(\frac{g_0}{4\pi}\right)^2 C_F \cdot 4 \left[\frac{1}{\epsilon_{uv}} + \frac{1}{2} + \log(8\pi\mu^2\tau) \right], \quad (26)$$

where I have included a factor of 2 to account for the corresponding antiquark corrections. These contributions, which are independent of the nonlocal operator structure and are canceled by terms in Z_γ , were first calculated for local quark bilinear operators in Ref. [45].

C. One-loop renormalization parameter

Drawing together these results, I obtain the one-loop expression for the parameter Z_Δ ,

$$Z_\Delta^{(1)} = 1 + \frac{\alpha_s}{3\pi} \left[Z_h^{(h)} + Z_h^{(d)} + Z_h^{(f)} + Z_h^{(g)} + Z_h^{(h)} + Z_h^{(l)} \right], \quad (27)$$

which is

$$Z_\Delta^{(1)} = 1 + \frac{\alpha_s}{3\pi} \left[-\frac{3}{\epsilon_{uv}} + 3\gamma_E + 3\log(\tau^2) - 3\text{Ei}(-\tau^2) \right. \\ \left. - 4\sqrt{\tau^2} \text{erf}(\sqrt{\tau^2}) - 3\log(8\pi\mu^2\tau) + a(\tau^2) \right], \quad (28)$$

where

$$a(\tau^2) = 5 + \frac{3}{\tau^2} (1 - e^{-\tau^2}) - \frac{1}{\tau^2} (2 + e^{-\tau^2}) - 4e^{-\tau^2}, \quad (29)$$

for $\gamma_\alpha = \gamma_z$ and

$$a(\tau^2) = 3 - \frac{1}{\tau^2} (1 - e^{-\tau^2}) + \frac{1}{\tau^2} (2 - e^{-\tau^2}) - 4e^{-\tau^2} \quad (30)$$

otherwise. These expressions are the central results of this work.

1. Asymptotic behavior

These one-loop expressions satisfy a number of cross-checks, based on the structure expected in the asymptotic regimes $\tau \ll 1$ and $\tau \gg 1$, which correspond to the local vector-current and small flow-time limits, respectively.

a. Vector-current limit In principle, in contrast to the dimensionally regularized case (see the Appendix), there are no contact terms in the limit of vanishing quark-field separation $z \rightarrow 0$ [33], provided one has regulated the IR divergences through an appropriate regulator,

such as an IR cutoff or external momentum. In this case, one could directly take the limit $\tau \rightarrow 0$. Here, however, I have used dimensional regularization for the IR divergences, which now manifest in a logarithmic dependence on $\mu^2 z^2$, and this complicates the procedure. It is simplest to examine the $\tau \ll 1$ regime at the level of each integrand, in which case one finds $Z_h^{(d)} = Z_h^{(f)} = 0$ and

$$\lim_{\tau \rightarrow 0} Z_h^{(h)} = \left(\frac{g_0}{4\pi}\right)^2 C_F \left[\frac{1}{\epsilon_{in}} + \frac{1}{2} - \log(8\pi\mu^2\tau) \right], \quad (31)$$

independent of the choice of the matrix γ_α in the original matrix element, as expected.

Combining this result with the expressions for diagrams (h) and (j), which are independent of the spatial separation z (but not, of course, the flow time τ), I find

$$\lim_{\tau \rightarrow 0} Z_h^{(1)} = \left(\frac{g_0}{4\pi}\right)^2 C_F \left\{ \frac{1}{2} + \frac{1}{\epsilon_{in}} - \frac{3}{\epsilon_{uv}} + \log(8\pi\mu^2\tau) \right\}, \quad (32)$$

As I show in the next section, when combined with Z_ψ and Z_N , the parameter $Z_h^{(1)}$ reduces to the local vector-current result of Ref. [43], an important check of the calculation.

b. Small flow-time limit In the small flow-time limit, one anticipates that the diagrams reduce to the expressions obtained in the $\overline{MS}$ scheme. Diagram (a) reduces to

$$Z_h^{(a)} \stackrel{\tau \gg 1}{\simeq} \left(\frac{g_0}{4\pi}\right)^2 C_F \left[\frac{1}{\epsilon_{in}} + C_\alpha^{(\infty)} - \gamma_E - \log(\pi\mu^2 z^2) \right], \quad (33)$$

where $C_z^{(\infty)} = 3$ and $C_\alpha^{(\infty)} = 1$. These results are in exact agreement with the $\overline{MS}$ -scheme expressions in Eq. A6. Taking $\tau \gg 1$, diagrams (d) and (f) become

$$Z_h^{(d)} \stackrel{\tau \gg 1}{\simeq} -\left(\frac{g_0}{4\pi}\right)^2 C_F 2 \left[1 - \gamma_E - \log(\tau^2) \right], \quad (34)$$

$$Z_h^{(f)} \stackrel{\tau \gg 1}{\simeq} \left(\frac{g_0}{4\pi}\right)^2 C_F 2 \left[1 + \gamma_E + \log(\tau^2) + (1 - 2\sqrt{\tau^2}) \right]. \quad (35)$$

The contribution to diagram (f) in parentheses is the linear divergence that must be subtracted before comparison can be made to the calculation carried out with dimensional regularization. Removing this contribution, the sum of these two diagrams is

$$\left(Z_h^{(d)} + Z_h^{(f)} \right) \stackrel{\tau \gg 1}{\simeq} 4 \left(\frac{g_0}{4\pi}\right)^2 C_F, \quad (36)$$

in agreement with the sum of Eqs. (A7) and (A8). In short, the small flow-time limit of Z_Δ exactly matches the corresponding result directly calculated in the $\overline{MS}$ scheme, as one would expect.

D. One-loop smeared matrix element

We can now write down the complete smeared matrix element, $h^{(s)}$ at one loop in perturbation theory, by

combining the wave function renormalization Z_ψ and Z_χ [28, 42] with the results for Z_h from the previous section. Recalling Eq. (11), the one-loop matrix element is $h^{(s)} = Z_h h^{(0)}$, where

$$Z_h^{(1)} = 1 + \frac{\alpha_s}{3\pi} \left[3 \log(\tau^2) + 6\gamma_E - 3 \log(4\pi) - \log(432) \right. \\ \left. - 3 \text{Ei}(-\tau^2) - 4\sqrt{\pi}\tau^2 \text{erf}(\sqrt{\tau^2}) + a(\tau^2) \right] \quad (37)$$

and here

$$c(\tau^2) = 3 \left[3 + \frac{1}{2\tau^2} (e^{-\tau^2} - 2) + \frac{1}{\tau^2} (1 - e^{-\tau^2}) \right], \quad (38)$$

$$c(\tau^2) = 7 + \frac{1}{\tau^2} (3e^{-\tau^2} - 2) + \frac{1}{\tau^2} (e^{-\tau^2} - 1), \quad (39)$$

for $\gamma_\psi = \gamma_\chi$ and $\gamma_h = \gamma_\perp$, respectively. I note that the IR divergences in the vertex correction of diagram (a) and the wave function renormalization, diagram (g), have canceled. This matches the IR behavior in the $\overline{MS}$ scheme, Eq. (A10), and demonstrates that the gradient flow does not modify the IR behavior of the quasisdistribution. Moreover, the total one-loop result is UV finite (that is, there are no poles in ϵ_{UV}), as is guaranteed for nonzero flow time, provided one uses ringed fermions. The logarithmic terms combine in such a way that all dependence on μ has been eliminated, leaving only $\log(\tau)$. There is a linear divergence generated by the Wilson-line self-energy, diagram (f), that scales as $z/\sqrt{\tau}$, as one would expect on dimensional grounds. At higher orders in perturbation theory, one anticipates that this divergence exponentiates in the form $e^{c_2 z/\sqrt{\tau}}$.

I plot both $Z_{\text{sub}}^{(s)}(\tau^2)$ and $Z_{\text{sub}}^{(\perp)}(\tau^2)$ as functions of τ in Fig. 3, represented by the purple and blue curves, respectively. I have removed the linear divergence, which would otherwise dominate for values of $\tau \lesssim 2$, as illustrated by the dot-dashed gray line, which shows $Z^{(s)}(\tau^2)$ including the linear divergence, denoted by $Z^{(s)}(\tau^2)$ in the legend. I also plot the asymptotic behavior in the $\tau \ll 1$ and $\tau \gg 1$ regimes as orange dashed lines.

a. Vector-current limit In contrast to the matrix element in dimensional regularization, the limit of vanishing spatial separation of the quark fields, $\tau \rightarrow 0$, is well defined and is given by

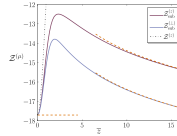
$$\lim_{\tau \rightarrow 0} Z_h = \frac{1}{\tau} - \log(432), \quad (40)$$

independent of the choice one makes for γ_ψ and in exact agreement with the vector-current result of [43]. This result exhibits a feature characteristic of composite operators constructed from ringed fermions: perturbative corrections are finite, as they must be for a conserved vector current, but not necessarily zero (compare, for example, with the nonsinglet axial-vector current defined in Ref. [45]).

b. Small flow-time limit Conversely, the small flow-time expansion gives

$$Z_h^{(\text{sub})} \simeq b - \gamma_E - \log(432) - \log(\tau^2), \quad (41)$$

FIG. 3. Renormalization parameter Z_h^{sub} for $\gamma_\psi = \gamma_\chi$ and $\gamma_h = \gamma_\perp$ as a function of τ , where the superscript “sub” indicates that the linear divergence has been subtracted. The dot-dashed gray line is Z_h including the linear divergence in τ , denoted by Z_h^{div} in the legend. I plot the asymptotic behavior for $\tau \ll 1$ and $\tau \gg 1$, from Eqs. (40) and (41) as orange dashed lines.



with $b = 9$ and $b = 7$, for $\gamma_\psi = \gamma_\chi$ and $\gamma_h = \gamma_\perp$, respectively. In this regime, the parameter $Z_h^{\text{sub}}(\tau^2)$ depends only logarithmically on τ , and therefore satisfies a relation akin to the usual renormalization-group equation [46]:

$$\left[\frac{d}{d \log \tau^2} + \gamma \right] Z_h^{\text{sub}}(\tau^2) = 0. \quad (42)$$

Here, $\gamma\tau = \gamma\tau^{(1)}\alpha_s + \mathcal{O}(\alpha_s^2)$, with $\gamma\tau^{(1)} = 1$, is analogous to the usual anomalous dimension. Beyond one loop in perturbation theory, a complete renormalization-group analysis would necessarily take into account the implicit scale dependence of the coupling constant $\alpha_s(\mu)$. A renormalization-group trajectory is then a path in the two-dimensional space (τ, μ) , which can be simplified by tying the two scales together by choosing, for example, $\mu = 1/\tau$ [46]. In principle, this could provide an opportunity to construct a nonperturbative step-scaling method for the matrix element $h^{(s)}(\tau^2)$, which would connect lattice calculations at hadronic scales to a high scale at which the matching to the $\overline{MS}$ scheme can be carried out without significant perturbative truncation uncertainties. Equation (42) only holds in the small flow-time regime, however, which, from Fig. 3, applies for $\tau \simeq 10$, at least at one loop in perturbation theory. Imposing this constraint throughout the entire step-scaling procedure may not be feasible for current lattices, but systematic uncertainties associated with deviations from the small flow-time expectation can be studied nonperturbatively.

E. Matching in coordinate space

With the complete one-loop expression for the smeared quasidistribution in hand, I can match the results to the renormalized matrix element in the $\overline{MS}$ scheme, $h^{\overline{MS}}(\mu^2 z^2)$ (see the Appendix). Writing

$$h^{\overline{MS}}(\mu^2 z^2) = C_h^{ub}(\mu^2 z^2, \bar{z}^2) h^{(u)}(\bar{z}^2), \quad (43)$$

the matching factor is given by

$$C_h^{ub}(\mu^2 z^2, \bar{z}^2) = 1 + \frac{\alpha_s}{3\pi} \left[a(\bar{z}^2) - 3\bar{z} - \log(432) - 3\text{Ei}(-\bar{z}^2) - 3\log(8\pi^2 t) - 4\sqrt{\pi\bar{z}^2} \text{erf}\left(\sqrt{\bar{z}^2}\right) \right]. \quad (44)$$

Here $c(\bar{z}^2)$ is given in Eqs. (38) and (39), $\bar{z} = 5/3$, and $\bar{z} = 7/3$. In general, the linear divergence will be subtracted nonperturbatively and I have removed this contribution from the matching relation, indicated by the subscript “sub.” At zero external momentum, this matching parameter is real, at finite external momentum, discussed in the next section, the matching parameter becomes imaginary [33].

IV. NUMERICAL RESULTS

The central aim of this work is to study the matrix element $h^{(u)}$ at one loop in perturbation theory, and thereby relate the smeared matrix element to the unsmeared matrix element in the $\overline{MS}$ scheme.

In general, relating two renormalized operators can be done with (massless) external quark states with vanishing spatial momentum, provided that momentum does not provide the renormalization scale (such as in the RI/MOM scheme). For completeness, however, I would like to explore the momentum structure of the smeared matrix element in perturbation theory. In this section, I therefore briefly study the momentum-dependent finite pieces of $h^{(u)}$, which were not necessary for the matching relation of the previous section.

a. Analytic approaches Before I present some sample results, I briefly comment on the calculation at finite external momentum. Introducing a nonzero external momentum considerably complicates the calculation. There are four new diagrams, diagrams (b), (c), (e), and (i), but this increase in the number of diagrams is not too onerous at one loop. More challenging are the momentum integrals themselves. The key difficulty is the exponential damping factor provided by the gradient flow, which prevents the easy application of the residue theorem to the integrals. Moreover, these damping factors also complicate the use of Feynman parameters, because the momentum combinations that appear in the denominator of the integrand are generally not those that appear as arguments of the exponentials, so that one cannot

reduce the integrands to radially symmetric integrands. The exponential factors naturally lead one to introduce Schwinger representations of the propagators in the diagram, but these generically lead to integrals over positive Schwinger parameters that are of the form

$$\sum_i \int_0^\infty \prod_j d\alpha^{(j)} f^{(i)}(\alpha^{(j)}) e^{-A^{(i)}\alpha^{(i)} + B^{(i)}\alpha^{(i)}}. \quad (45)$$

Here, the $f^{(i)}$ are polynomials, which may include negative and fractional powers, of the (multiple) Schwinger parameters $\alpha^{(j)}$ (and the physical scales in the problem), and $A^{(i)}$ and $B^{(i)}$ are independent of $\alpha^{(j)}$. These integrals can be carried out numerically but are generally very cumbersome expressions. These integrals seem to be ubiquitous in this approach and also arise from integration over intermediate flow times.

b. Numeric approach I determine each momentum integral using the adaptive Monte Carlo VEGAS algorithm [47]. I reduce the two-dimensional integral over $\mathbf{k}_\perp$ to a single integral over the positive radial direction in the (k_x, k_y) plane, because the diagrams depend only on $\mathbf{k}_\perp^2$ and not the individual components k_x and k_y . I evaluate the Lorentz structure of each integral by hand, and confirm the results using FeynCalc [48, 49]. These manipulations are straightforward at one loop. For simplicity, I take the gamma matrix insertion to be $\gamma_\alpha = \gamma_4$ and express all results in units of $\alpha_s/(3\pi)$. Thus, for the case of diagram (a), I evaluate

$$h_4^{(a)} = \frac{1}{2p} g_0^2 C_F 2 \int_k \frac{e^{-2\tau k^2} e^{-ik_\perp z}}{(k^2 + m^2)^2 (p - k)^2} \times \text{Tr} \left\{ \left(\frac{-i\not{p} + m}{2} \right) (4imk_4 - 2\not{k}k_4 + (k^2 + m^2)\gamma_4) \right\} \\ = \frac{\alpha_s}{3\pi} \mathcal{I}_4^{(a)}, \quad (46)$$

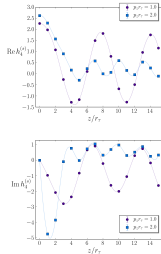
where the numerical integral is

$$\mathcal{I}_4^{(a)} = \frac{1}{2\bar{p}} \frac{8i}{\pi} \int_0^{\bar{\Lambda}} d\ell \ell_\perp \int_{-\bar{\Lambda}}^{\bar{\Lambda}} d\ell_4 d\ell_z e^{-\ell^2} e^{-i\ell_z \bar{z}} \\ \times \frac{2(\bar{p} \cdot \ell + 2\bar{m}^2)\ell_\alpha - (\ell^2 + \bar{m}^2)\bar{p}_\alpha}{(\ell^2 + \bar{m}^2)^2 (\bar{p} - \ell)^2}. \quad (47)$$

The dimensionless integration variable is $\ell = k r_\tau = k\sqrt{8\tau}$, and the bar indicates the dimensionless combinations $\bar{m} = m r_\tau$, $\bar{p} = p r_\tau$, and $\bar{z} = z/r_\tau$.

In Fig. 4, I provide numerical results over an appropriate range of parameters, based on typical scales in units of the smearing radius. Current lattice calculations use typical lattice spacings of approximately $a \sim 0.1 \text{ fm} \simeq 0.5 \text{ GeV}^{-1}$ and momenta around $p_z \sim 2 - 3 \text{ GeV}$. Typical flow times are of the order $r_\tau/a \sim 1$ [50]—one advantage of the gradient flow is that, in practice, it seems relatively little smearing is required—and so it is reasonable to take $z/r_\tau \sim 0 - 15$ and $p_z r_\tau \sim 1 - 2$ as illustrative values.

FIG. 4. The real (upper panel) and imaginary (lower panel) parts of the smeared matrix element at one loop, $h_1^{(s)}$, as a function of the spatial Wilson-line length, z/r_T , for three different values of the external quark momentum, $P_z/r_T = \{1, 2\}$. I take $m_q r_T = 0.001$ and $P_z/\Lambda = 2$. For these results, I use 20 iterations of 1.5×10^6 measurements following 10 thermalization iterations of 10^6 measurements each. The smooth curves are interpolations, not fits, to ease comparison between data at different momenta, and the statistical uncertainties are too small to be visible at this scale.



I plot the real and imaginary parts of the smeared matrix element, as a function of z/r_T and for two different values of the quark momentum, in the upper and lower panels of Figs. 4, respectively. The lines are smooth interpolations to the data, not fits, to ease comparison between data points at the two different momenta. The statistical uncertainties from the numerical integration are approximately $\sigma_{\text{VEGAS}} \sim 10^{-4}$, which are too small to be visible at this scale.

V. SUMMARY

Quasi and pseudodistributions have opened a new window into nucleon structure and provided an opportunity to extract, for the first time, parton distribution functions directly from lattice QCD. Preliminary results of nonperturbative computations have been encouraging, but there are still challenges to be navigated before the

lattice community can claim a robust determination of light-front PDFs. One particular difficulty for lattice QCD is the presence of a power divergence associated with the length of the Wilson line, which must be removed nonperturbatively before one can take a continuum limit. In Ref. [25], we introduced the smeared quasisdistribution as a tool to circumvent the issue of the power divergence associated with the Wilson line. By taking advantage of the renormalization properties of the gradient flow, a classical evolution of the original degrees of freedom in a new parameter, the flow time, the smeared quasisdistribution remains finite in the continuum limit.

Here, I study the smeared quasisdistribution at one loop in perturbation theory. Although the gradient flow complicates the perturbative calculation, by introducing new diagrams not present at vanishing flow time and complicating the corresponding integrals, the advantage is significant: provided one fixes the flow time in physical units, the smeared quasisdistribution is finite in the continuum limit. The resulting continuum smeared quasisdistributions can then be related, directly or through the quasisdistribution at zero flow time, to the light-front PDFs in the $\overline{MS}$ scheme. The IR behavior of the quasisdistribution is unaffected by the flow time, which serves only to regulate the UV behavior, so the matching procedure can be carried out perturbatively, by evaluating the Wilson-line operator between external, gauge-fixed quark states.

I undertake the matching at fixed Wilson-line length, in which case the external quark momentum can be set to zero. This reduces the number of diagrams that contribute at a given order in perturbation theory and simplifies the corresponding one-loop integrals, which can then be carried out analytically. I use dimensional regularization to regularize the logarithmic IR divergences and demonstrate that the resulting poles in the space-time dimension cancel in the matching relation between the smeared and unsmeared quasisdistributions. I show that the limit of vanishing Wilson-line length is well defined and reduces to the known vector current result at finite flow time. In the small flow-time limit, the matrix element depends only logarithmically on the ratio of the Wilson-line length and the gradient flow smearing radius. The smeared matrix element therefore satisfies an equation similar to the typical renormalization-group equation, which provides the potential for defining a nonperturbative step-scaling procedure. Finally, I study the smeared matrix element at finite external momentum by evaluating the corresponding one-loop diagrams numerically. Extending this study to two loops would likely require an automated approach, because the extra vertices induced by the gradient flow cause the number of diagrams to increase rapidly with each order in perturbation theory. A nonperturbative study of the systematic uncertainties associated with the smeared quasi- and pseudodistributions is underway.

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Appendix A: Matrix elements in dimensional regularization

In this Appendix, I determine the matrix element at vanishing flow time using dimensional regularization to regularize both IR and UV singularities. These results were first calculated in Refs. [33, 51, 52]. I repeat the calculations here because: the intermediate results demonstrate the usefulness of the Schwinger representation of the diagrams; provide a useful cross-check of my new results at nonzero flow time, and demonstrate that both the smeared and unsmeared quasidistributions have the same IR behavior. There are four diagrams that contribute, corresponding to diagrams (a), (d), (f), and (g) in Fig. 2 with the flow time set to zero. The resulting integrals can be reduced to

$$Z_h^{(a)} = g_0^2 C_F \frac{\pi^{d/2}}{(2\pi)^d} \frac{(d-2)^2}{d} \mu^{4-d} \int_0^\infty \frac{d\alpha}{\alpha^{d/2-1}} e^{-z^2/(4\alpha)}, \quad (A1)$$

$$Z_h^{(d)} = 2g_0^2 C_F \frac{\pi^{d/2}}{(2\pi)^d} \mu^{4-d} \int_0^\infty \frac{d\alpha}{\alpha^{d/2-1}} \left(1 - e^{-z^2/(4\alpha)}\right), \quad (A2)$$

$$Z_h^{(f)} = g_0^2 C_F \frac{\pi^{d/2}}{(2\pi)^d} \mu^{4-d} \int_0^\infty \frac{d\alpha}{\alpha^{d/2-1}} \int_0^\infty \frac{d\beta}{\sqrt{\alpha+\beta}} \times \left(e^{-z^2/(4(\alpha+\beta))} - 1\right), \quad (A3)$$

$$Z_h^{(g)} = g_0^2 C_F \frac{\pi^{d/2}}{(2\pi)^d} \frac{2(2-d)}{\Gamma(d/2)} \mu^{4-d} \int_0^\infty \frac{d\alpha}{\alpha^{d/2-1}}. \quad (A4)$$

I carry out these integrals analytically [with the Schwinger parametrization, this is a straightforward modification to the analogous smeared integrals, with the replacement $(\alpha+2r) \rightarrow \alpha$]. I regulate the UV divergences in diagrams (d) and (f) by setting the spacetime dimension to $d = (4 - 2\epsilon_{\text{UV}})$ and the IR divergence in diagram (a) by setting $d = (4 + 2\epsilon_{\text{IR}})$. By the rules of dimensional regularization, diagram (g) should vanish, but I wish to expose the UV and IR divergences. I therefore split the Schwinger parameter integral into two regions

$$\int_0^\infty d\alpha \alpha^{-1-d/2} = \int_0^\lambda d\alpha \alpha^{-1-d/2} + \int_\lambda^\infty d\alpha \alpha^{-1-d/2}, \quad (A5)$$

and set $d = (4 - 2\epsilon_{\text{UV}})$ in the UV region, $[0, \lambda]$, and $d = (4 + 2\epsilon_{\text{IR}})$ in the IR region. The result is independent of the choice of λ .

Then, expanding the results around $\epsilon_{\text{UV}} = \epsilon_{\text{IR}} = 0$, I

have

$$Z_h^{(a)} = \left(\frac{g_0}{4\pi}\right)^2 C_F \left[\frac{1}{\epsilon_{\text{IR}}} - \gamma_E + \bar{C}_\alpha - \log(\pi\mu^2 z^2)\right], \quad (A6)$$

$$Z_h^{(d)} = \left(\frac{g_0}{4\pi}\right)^2 C_F 2 \left[\frac{1}{\epsilon_{\text{UV}}} + \gamma_E + \log(\pi\mu^2 z^2)\right], \quad (A7)$$

$$Z_h^{(f)} = \left(\frac{g_0}{4\pi}\right)^2 C_F 2 \left[\frac{1}{\epsilon_{\text{UV}}} + 2 + \gamma_E + \log(\pi\mu^2 z^2)\right]. \quad (A8)$$

$$Z_h^{(g)} = -\left(\frac{g_0}{4\pi}\right)^2 C_F \left[\frac{1}{\epsilon_{\text{UV}}} + \frac{1}{\epsilon_{\text{IR}}}\right]. \quad (A9)$$

Here $\bar{C}_\alpha = 3$ and $\bar{C}_\beta = 1$. As one would anticipate, there is no term proportional to z , because power-divergent contributions are explicitly removed by dimensional regularization. Diagram (a) has a pole in $1/\epsilon_{\text{IR}}$, which is not removed by any renormalization procedure, but is canceled by the IR pole in the wave function renormalization, diagram (g). From these expressions, one sees that the $\overline{MS}$ -scheme renormalization parameter is

$$Z_h^{\overline{MS}} = 1 + \frac{\alpha_s}{3\pi} \frac{3}{\epsilon_{\text{UV}}}, \quad (A10)$$

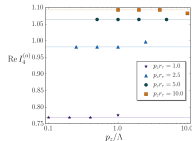
in agreement with [33, 51, 52].

Appendix B: Numerical integration

In this Appendix, I highlight some details of the numerical integration of Sec. IV, including some tests of the numerical convergence of the results. Once the wave function renormalization contributions are properly incorporated, all diagrams in Fig. 2 are UV and IR finite, so I integrate up to an arbitrary cutoff, $\bar{\Lambda}$, chosen to ensure convergence at a given level of numerical precision. I illustrate this convergence in Fig. 5 for the real part of diagram (a) and observe that, at the current level of precision, convergence occurs for $p_z/\Lambda \simeq 0.5$ for $P_z r_\tau = 1$, and this ratio increases to $p_z/\Lambda = 4$ for $p_z r_\tau = 10.0$. The value of the cutoff at which convergence occurs is diagram dependent, but generally between $p_z/\Lambda \simeq 0.2 - 0.5$, which ensures that cutoff effects can be neglected, without unnecessarily increasing the statistical uncertainty.

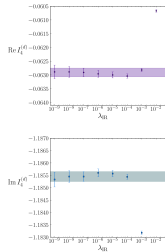
Diagrams (d), (e) and (f) suffer from spurious IR instabilities introduced by the numerical evaluation of $(p_z - k_z)$ terms in the denominator. I remove these numerical effects by introducing an IR cutoff λ_{IR} , such that momentum points with $|p_z - k_z| \leq \lambda_{\text{IR}}$ are excluded, and demonstrate the numerical convergence of these results, within statistical uncertainties, for the real (upper plot) and imaginary (lower plot) parts of diagram (d) in Fig. 6. For these diagrams, I choose $\lambda_{\text{IR}} = 10^{-7}$, which is illustrated by the shaded band in each plot and is sufficiently small to ensure convergence without unduly increasing statistical uncertainties.

FIG. 5. Numerical convergence for the evaluation of the real part of diagram (a), for different values of the ratio p_c/Λ and for four different values of the (dimensionless) external momentum $P_c r_c$. The horizontal bands indicate the corresponding results at the smallest value of P_c/Λ . I take $m_{\pi} r_c = 0.005$ and $z/r_c = 10$. For these results, I use 20 iterations of 5×10^6 measurements following 10 thermalization iterations of 10^4 measurements each. The statistical uncertainties are generally too small to be visible at this scale.



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FIG. 6. Numerical convergence for the evaluation of the real (upper plot) and imaginary (lower plot) part of diagram (d). The horizontal bands indicate the corresponding result at $\Lambda_{\text{RI}} = 10^{-2}$. I take $m_\pi r_\pi = 0.001$, $z/r_\pi = 1$, and $P_z/\Lambda = 2$. For these results, I use 20 iterations of 5×10^5 measurements following 10 thermalization iterations of 10^6 measurements each, taking approximately 70 sec on three CPU cores.



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Off-lightcone Wilson-line operators in gradient flow

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ABSTRACT: Off-lightcone Wilson-line operators are constructed using local operators connected by time-like or space-like Wilson lines, which ensure gauge invariance. Off-lightcone Wilson-line operators have broad applications in various contexts. For instance, space-like Wilson-line operators play a crucial role in determining quasi-distribution functions (quasi-PDFs), while time-like Wilson-line operators are essential for understanding quarkonium decay and production within the potential non-relativistic QCD (pNRQCD) framework. In this work, we establish a systematic approach for calculating the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme in the limit of small flow time for off-lightcone Wilson-line operators. By employing the one-dimensional auxiliary-field formalism, we simplify the matching procedure, reducing it to the matching of local current operators. We provide one-loop level matching coefficients for these local current operators. For the case of hadronic matrix element related to the quark quasi-PDFs, we show at one-loop level that the finite flow time effect is very small as long as the flow radius is smaller than the physical distance z , which is usually satisfied in lattice gradient flow computations. Applications include lattice gradient flow computations of quark/gluon quasi-PDFs, gluonic correlators related to quarkonium decay and production in pNRQCD, and spin-dependent potentials in terms of chromoelectric and chromomagnetic field insertions into a Wilson loop.

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1 Introduction

The path-ordered phase factor $\text{Pexp}\left(ig \int A_\mu dx^\mu\right)$, referred to as the Wilson line¹, is a fundamental object in gauge theories, playing a crucial role in connecting local fields to construct gauge-invariant operators. The study of the renormalization of Wilson lines and Wilson loops was initiated by Polyakov [1] and Gervais, Neveu [2]. The all-order proof of the multiplicative renormalizability of Wilson lines was made using diagrammatic methods in refs. [3, 4] and functional approach in ref. [5] based on the one-dimensional auxiliary-field formalism [2, 6–9]; see ref. [5] for a review.

¹Throughout this work, our focus will be exclusively on off-lightcone Wilson lines and Wilson loops, which we will refer to simply as Wilson lines unless ambiguity arises.

Off-lightcone Wilson-line operators are constructed using local operators connected by time-like or space-like Wilson lines, which insure the gauge invariance of the non-local operators. Wilson-line operators have broad applications in various contexts. For instance, space-like Wilson-line operators play a crucial role in determining quasi-distribution function (quasi-PDFs) [10] and pseudo-distribution functions (pseudo-PDFs) [11], while time-like Wilson-line operators are essential for understanding quarkonium decay and production within the potential non-relativistic QCD (pNRQCD) effective theory [12–20].

Recently, there has been increasing interests in the study of the quasi-PDFs and pseudo-PDFs defined as the hadronic matrix elements of the Wilson-line operators of the type

$$\mathcal{O}_\Gamma(zv) = \bar{\psi}(zv)\Gamma W(zv,0)\psi(0), \quad (1.1)$$

$$\mathcal{O}^{g\mu\alpha\beta}(zv) = g^2 F^{\mu\nu}(zv) W(zv,0) F^{\alpha\beta}(0), \quad (1.2)$$

where ψ is a quark field, $F^{\mu\nu}$ is the gluon field strength tensor, Γ represents a Dirac structure, v is a four-vector satisfying $v^2 = -1$ (or $v^2 = 1$) for space-like (or time-like) Wilson-line operators in Minkowski spacetime², z is a real number and $W(zv,0)$ is a straight-line path-ordered Wilson line connecting the two fields,

$$W(zv,0) = \text{P exp} \left(ig \int_0^z ds v \cdot A(sv) \right), \quad (1.3)$$

in which the fundamental representation and the adjoint representation of the gauge group are assumed for the quark case, and the gluon case, respectively. Without loss of generality, we choose $z > 0$ throughout this work. Notably, quasi-PDFs and pseudo-PDFs can be factorized in relation to the standard light-cone parton distribution functions (PDFs) based on the short-distance operator product expansion (OPE) and the large momentum effective theory (LaMET) [21–23]. Employing hadronic matrix elements of off-lightcone Wilson-line operators as opposed to lightlike ones, offers the advantage of being considerably more suitable to lattice computations. However, this convenience comes with the trade-off of the highly non-trivial renormalization of the off-lightcone Wilson-line operators. For a comprehensive overview, we refer to ref. [24]. Furthermore, the vacuum expectation value of gluonic Wilson-line operators, such as the one defined in eq. (1.2), are related to the studies of quarkonium decay and production in the framework of pNRQCD, and the QCD vacuum structure (see ref. [25] for a review).

The gradient flow was originally introduced as a method to regularize and suppress the ultraviolet (UV) fluctuations [26, 27] in lattice calculations. It has proven useful in lattice QCD scale settings and in lattice calculation of local matrix elements and correlation functions [28–40]. Alongside its noise-reducing effect, gradient flow exhibits the notable property that composite operators at positive flow time remain finite even in the continuum limit, provided they are expressed using renormalized physical parameters and fields [29]. This property renders gradient flow suitable as a matching scheme between lattice and perturbative calculations. It is worth noting that gradient flow does not modify the infrared

²In this context and throughout subsequent discussions, we stay in Minkowski spacetime. However, a transition to Euclidean spacetime is warranted when introducing gradient flow, as it is defined in Euclidean spacetime, which indicates $v^2 = 1$ both for space-like and time-like Wilson-line operators.

behavior of operators in the small flow-time limit, allowing the use of dimensional regularization to regulate infrared (IR) divergences when matching from the gradient-flow scheme to $\overline{\text{MS}}$ -like schemes.

The initial proposal to explore quasi-PDFs with the incorporation of gradient flow was presented in ref. [41]. Subsequently, a one-loop matching calculation was performed, considering full flow time dependencies, for the hadronic matrix element defined using the operator in eq. (1.1) with zero external momenta, which is related to the quark quasi-PDF [42]. In this study, we present a systematic approach for calculating the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme for Wilson-line operators, such as those defined in eq. (1.1) and eq. (1.2), specifically in the limit of small flow time. Our approach is based on the one-dimensional auxiliary-field formalism, and we provide comprehensive details of the one-loop order matching calculations. For more recent applications of perturbative gradient flow, see refs. [43–53] and the references therein.

We structure the remaining content of the paper as follows. In section 2, we introduce the one-dimensional auxiliary-field formalism and elaborate on its application in the renormalization of Wilson-line operators. In section 3, we introduce the gradient-flow formalism and implement the small flow-time expansion to the local current operators, yielding the core formulas for our matching calculations. Section 4 is dedicated to the computation of the matching coefficients up to one-loop order. Our summary and conclusions can be found in section 5. In appendix A, we present the detailed matching calculation with full flow time dependence for the hadronic matrix element that is related to quasi-PDFs at zero external momenta. We compare these findings with existing results from ref. [42], as well as with the results of section 4 in the small flow-time limit.

2 One-dimensional auxiliary-field formalism and renormalization of the off-lightcone Wilson-line operators

2.1 The one-dimensional auxiliary-field formalism

The one-dimensional auxiliary-field formalism is defined by enlarging the QCD Lagrangian to include an extra term [5],

$$\mathcal{L}_{h_v} = \bar{h}_{v,0}(i\nabla \cdot D)h_{v,0}, \quad (2.1)$$

where the subscript “0” indicates a bare quantity, $h_{v,0}$ is an auxiliary “heavy” Grassmann (or complex) scalar field in either fundamental or adjoint representations of the SU(3) gauge group, $D_\mu^\alpha = \partial^\mu - ig_0 A_\mu^\alpha T^\alpha$ is the covariant derivative and T^α are the SU(3) generators in the appropriate representation. Note that, for $v = (1, 0, 0, 0)$ and $h_{v,0}$ being in the fundamental representation, the effective Lagrangian in eq. (2.1) essentially coincides with the Lagrangian of the heavy-quark effective theory (HQET) in the rest frame [54], except that we choose $h_{v,0}$ to be a scalar.

The effective Lagrangian $\mathcal{L}_{h_v}$ is amenable to renormalization through the redefinitions,

$$h_{v,0} = Z_{h_v}^{\frac{1}{2}} \hat{h}_v, \quad g_0 = Z_g g, \quad A_0 = Z_A^{\frac{1}{2}} A, \quad (2.2)$$

where Z_{h_v} is the wave-function renormalization constant of the “heavy” auxiliary field, Z_g and Z_A are the usual QCD renormalization constants of the strong coupling and the gluon field, respectively. There will be a “mass correction” $i\delta m$ for the “heavy” auxiliary field and the effective Lagrangian $\mathcal{L}_{h_v}$ expressed in terms of the renormalized fields and parameters is given by ³,

$$\mathcal{L}_{h_v} = Z_{h_v} \bar{h}_v (i v \cdot \partial - i\delta m) h_v + g Z_g Z_A^{\frac{1}{2}} Z_{h_v} \bar{h}_v v \cdot A^a T^a h_v, \quad (2.3)$$

where proper representations are implied for the fields and the generator T^a .

The “mass correction” $i\delta m$ is linearly divergent. The linear divergence vanishes in dimensional regularization, but persists, in lattice regularization and the gradient-flow formalism. Furthermore, the “mass correction” $i\delta m$ could incorporate contributions of $\mathcal{O}(\Lambda_{\text{QCD}})$, attributed to the renormalon ambiguities that are recognized to be present in HQET [55, 56].

Based on the one-dimensional auxiliary-field formalism, up to a constant normalization factor, the connected two-point function of the “heavy” auxiliary fields can be related to the Wilson lines through [5, 24, 57],

$$\langle h_{v,0}(x) \bar{h}_{v,0}(0) \rangle_{h_v} = W \left(\frac{v \cdot x}{v^2}, 0 \right) \theta \left(\frac{v \cdot x}{v^2} \right) \delta^{(d-1)}(x_\perp), \quad (2.4)$$

where $\langle \dots \rangle_{h_v}$ stands for integrating out the “heavy” auxiliary field h_v , and $x_\perp^\mu = x^\mu - v^\mu (v \cdot x)/v^2$. In this way, the Wilson-line operators can be substituted with products of local current operators. For the Wilson-line operators defined in eq. (1.1) and eq. (1.2), we have

$$\mathcal{O}_F^B(zv) = \int d^d x \, \delta \left(\frac{v \cdot x}{v^2} - z \right) \langle \bar{\psi}_0(x) h_{v,0}(x) \Gamma \bar{h}_{v,0}(0) \psi_0(0) \rangle_{h_v}, \quad (2.5)$$

$$\mathcal{O}^{\mu\nu\alpha\beta,B}(zv) = \int d^d x \, \delta \left(\frac{v \cdot x}{v^2} - z \right) \langle g^2 F_0^{\mu\nu}(x) h_{v,0}(x) \bar{h}_{v,0}(0) F_0^{\alpha\beta}(0) \rangle_{h_v}, \quad (2.6)$$

where the superscript “B” indicates bare composite operators, the bare “heavy” auxiliary fields $h_{v,0}$ in $\mathcal{O}_F^B(zv)$, $\mathcal{O}^{\mu\nu\alpha\beta,B}(zv)$ are in fundamental representation and adjoint representation, respectively.

2.2 Renormalization of the off-lightcone Wilson-line operators

The operator as defined in eq. (1.2) is not renormalized multiplicatively because, in the presence of the external four-vector v^μ , the components parallel and transverse to v^μ are renormalized differently [57]. We introduce the following projectors,

$$g_{\parallel}^{\mu\nu} = \frac{v_\mu v_\nu}{v^2}, \quad g_{\perp}^{\mu\nu} = g^{\mu\nu} - \frac{v_\mu v_\nu}{v^2}, \quad (2.7)$$

to project out the parallel and the transverse components of the field strength tensor $F^{\mu\nu}$ by defining,

$$F_{\perp\perp}^{\mu\nu} = g_{\parallel}^{\mu\alpha} g_{\perp}^{\nu\beta} F^{\alpha\beta}, \quad (2.8)$$

$$F_{\perp\perp}^{\mu\nu} = g_{\perp}^{\mu\alpha} g_{\perp}^{\nu\beta} F^{\alpha\beta}. \quad (2.9)$$

³The “mass correction” $i\delta m$ is imaginary for space-like v , but real for time-like v in Minkowski spacetime. In Euclidean spacetime, $i\delta m$ is always imaginary.

Notice that we ignore the mixing with gauge-noninvariant operators since they do not contribute to gauge-invariant observables. Nevertheless, we will verify the mixing patterns found in refs. [58–60] whenever they are relevant for our computations. Obviously, when $v = (1, 0, 0, 0)$ and $\mu \neq \nu$, $F_{\perp\perp}^{\mu\nu}$, $F_{\perp+}^{\mu\nu}$ are proportional to the chromoelectric field $\mathbf{E}$ and the chromomagnetic field $\mathbf{B}$, respectively. We introduce the following gluonic Wilson-line operators,

$$\mathcal{O}_{\perp\perp}^{\mu\alpha\beta}(zv) = g^2 F_{\perp\perp}^{\mu\nu}(zv) W(zv, 0) F_{\perp\perp}^{\alpha\beta}(0), \quad (2.10)$$

$$\mathcal{O}_{\perp+}^{\mu\alpha\beta}(zv) = g^2 F_{\perp\perp}^{\mu\nu}(zv) W(zv, 0) F_{\perp+}^{\alpha\beta}(0). \quad (2.11)$$

The off-lightcone Wilson-line operators defined in eq. (1.1), eq. (2.10), and eq. (2.11) undergo multiplicative renormalization within the auxiliary-field formalism, as proved in refs. [59, 61, 62], and shown diagrammatically in refs. [63, 64]. That is, the renormalized off-lightcone Wilson-line operators in eq. (1.1), eq. (2.10) and eq. (2.11) are related to the bare operators by the formulas,

$$\mathcal{O}_\Gamma^{\text{R}}(zv, \Lambda) = Z_q(\Lambda, \mu) e^{\delta m(\Lambda)z} \mathcal{O}_\Gamma^{\text{B}}(zv, \mu), \quad (2.12)$$

$$\mathcal{O}_{\perp\perp}^{\mu\alpha\beta, \text{R}}(zv, \Lambda) = Z_{\perp\perp}(\Lambda, \mu) e^{\delta m(\Lambda)z} \mathcal{O}_{\perp\perp}^{\mu\alpha\beta, \text{B}}(zv, \mu), \quad (2.13)$$

$$\mathcal{O}_{\perp+}^{\mu\alpha\beta, \text{R}}(zv, \Lambda) = Z_{\perp+}(\Lambda, \mu) e^{\delta m(\Lambda)z} \mathcal{O}_{\perp+}^{\mu\alpha\beta, \text{B}}(zv, \mu), \quad (2.14)$$

where the superscript “R” stands for renormalized composite operators, μ is the renormalization scale, Λ represents a UV cutoff (UV regulator), $\delta m(\Lambda)$ encodes the “mass correction” of the Wilson line, with its linear divergence, while $Z_q(\Lambda, \mu)$, $Z_{\perp\perp}(\Lambda, \mu)$ and $Z_{\perp+}(\Lambda, \mu)$ encode all logarithmic divergences originating from wave-function and vertex renormalizations.

Within the one-dimensional auxiliary-field formalism, after subtracting linear divergences, the renormalization of the off-lightcone Wilson-line operators defined in eq. (1.1), eq. (2.10), and eq. (2.11) is simplified into the renormalization of two local “heavy-to-light” or “heavy-to-gluon” currents⁴,

$$J_q(x) = \bar{\psi}(x) h_v(x), \quad (2.15)$$

$$J_{\perp\perp}^{\mu\nu}(x) = g F_{\perp\perp}^{\mu\nu}(x) h_v(x), \quad (2.16)$$

$$J_{\perp+}^{\mu\nu}(x) = g F_{\perp\perp}^{\mu\nu}(x) h_v(x). \quad (2.17)$$

The renormalization formulas for the above gauge-invariant local current operators are given by,

$$J_q^{\text{R}}(x, \Lambda) = Z_{J_q}(\Lambda, \mu) J_q^{\text{B}}(x, \mu), \quad (2.18)$$

$$J_{\perp\perp}^{\mu\nu, \text{R}}(x, \Lambda) = Z_{J_{\perp\perp}}(\Lambda, \mu) J_{\perp\perp}^{\mu\nu, \text{B}}(x, \mu), \quad (2.19)$$

$$J_{\perp+}^{\mu\nu, \text{R}}(x, \Lambda) = Z_{J_{\perp+}}(\Lambda, \mu) J_{\perp+}^{\mu\nu, \text{B}}(x, \mu), \quad (2.20)$$

⁴See refs. [65, 66] for the renormalization of off-lightcone Wilson-line operators in a different approach based on the Mandelstam formalism.

leading to the equivalence

$$Z_q(\Lambda, \mu) = Z_h^2(\Lambda, \mu), \quad (2.21)$$

$$Z_{\parallel\perp}(\Lambda, \mu) = Z_{h,\parallel\perp}^2(\Lambda, \mu), \quad (2.22)$$

$$Z_{\perp\perp}(\Lambda, \mu) = Z_{h,\perp\perp}^2(\Lambda, \mu). \quad (2.23)$$

In the following text, we may omit indicating the arguments Λ and μ in the renormalization constants.

The renormalization constants for the local current operators can be further decomposed as

$$Z_{J_h} = Z_h^{\frac{1}{2}} Z_{h_v}^{\frac{1}{2}} Z_J^V, \quad (2.24)$$

$$Z_{J,\parallel\perp} = Z_A^{\frac{1}{2}} Z_{h_v}^{\frac{1}{2}} Z_{\parallel\perp}^V, \quad (2.25)$$

$$Z_{J,\perp\perp} = Z_A^{\frac{1}{2}} Z_{h_v}^{\frac{1}{2}} Z_{\perp\perp}^V, \quad (2.26)$$

where Z_J^V , $Z_{\parallel\perp}^V$, and $Z_{\perp\perp}^V$ represent the corresponding renormalization factors arising from vertex corrections. Implicit in this notation is the dependence of the wave-function renormalization constant of the “heavy” auxiliary field from the color representations.

The Wilson-line operators that define the spin-dependent potentials in terms of chromomagnetic field insertions into a Wilson loop [67, 68] are also of interest in this study. Within the auxiliary-field formalism, each insertion of a chromomagnetic field into a Wilson loop can be related to the following current operator as,

$$J_F^{\mu\nu}(x) = \bar{h}_v(x) g F_{\perp\perp}^{\mu\nu}(x) h_v(x), \quad (2.27)$$

where $v = (1, 0, 0, 0)$ and the subscript F indicates that the “heavy” auxiliary fields are in the fundamental representation of the $SU(3)$ group. The renormalization formula for this operator is ,

$$J_F^{\mu\nu, R}(x) = Z_F J_F^{\mu\nu, R}(x) = Z_A^{1/2} Z_{h_v} Z_F^V J_F^{\mu\nu, R}(x), \quad (2.28)$$

where Z_F^V represents the renormalization factor stemming from the vertex corrections of the current $\bar{h}_v(x) g F_{\perp\perp}^{\mu\nu}(x) h_v(x)$.

3 Gradient flow and small flow-time expansion

3.1 The gradient-flow formalism

In the following, we work in d -dimensional Euclidean spacetime with $d = 4 - 2\epsilon$. The gradient flow extends the d -dimensional QCD to a $(d+1)$ -dimensional field theory by introducing an additional coordinate t (not to be confused with the time coordinate in Minkowski spacetime) of mass dimension -2 , which is called the flow time [28, 29]. In the gradient-flow formalism, the flow time dependent gluon field $B_\mu^a(x, t)$ and quark field

$\chi_\alpha^d(x, t)$ are determined by the flow equations [28–30],

$$\partial_t B_\mu = \mathcal{D}_\nu G_{\nu\mu} + \kappa \mathcal{D}_\mu \partial_\nu B_\nu, \quad (3.1)$$

$$\partial_t \chi = \mathcal{D}_\mu \mathcal{D}_\mu \chi - \kappa (\partial_\mu B_\mu) \chi, \quad (3.2)$$

$$\partial_t \bar{\chi} = \bar{\chi} \overleftarrow{\mathcal{D}}_\mu \overleftarrow{\mathcal{D}}_\mu + \kappa \bar{\chi} \partial_\mu B_\mu, \quad (3.3)$$

where κ is a gauge parameter that drops out in physical observables, and

$$G_{\mu\nu} = \partial_\nu B_\mu - \partial_\mu B_\nu + [B_\mu, B_\nu], \quad (3.4)$$

$$\mathcal{D}_\mu = \partial_\mu + [B_\mu, \cdot], \quad (3.5)$$

with the boundary conditions

$$B_\mu(x; t=0) = gA_\mu(x), \quad (3.6)$$

$$\chi(x; t=0) = \psi(x), \quad (3.7)$$

$$\bar{\chi}(x; t=0) = \bar{\psi}(x). \quad (3.8)$$

In perturbative gradient flow, one extends the regular QCD Feynman rules by adding flow time dependent exponentials to the propagators, introducing extra “flow lines” and “flow vertices” [29]; see ref. [43] for more details about the gradient flow Feynman rules and higher order calculation techniques. In the following perturbative calculation, we adopt the convenient choice $\kappa = 1$.

With renormalized physical parameters, the flowed field B_μ does not need further renormalization [28, 29], but the flowed quark fields χ need to be renormalized [30],

$$\chi_0(x; t) = Z_\chi^{\frac{1}{2}} \chi(x; t), \quad \bar{\chi}_0(x; t) = Z_\chi^{\frac{1}{2}} \bar{\chi}(x; t), \quad Z_\chi = 1 - \frac{\alpha_s}{4\pi} C_F \frac{3}{\epsilon_{UV}} + O(\alpha_s^2), \quad (3.9)$$

where $C_F = (N_c^2 - 1)/(2N_c)$ with $N_c = 3$ the number of colors. In order to avoid the complication due to the matching between Z_χ in dimensional regularization and that in the lattice regularization, the ringed fermion fields $\hat{\chi}(x, t)$, $\hat{\bar{\chi}}(x, t)$ were introduced in ref. [35],

$$\hat{\chi}(x, t) = \hat{Z}_\chi^{\frac{1}{2}} \chi_0(x, t), \quad \hat{\bar{\chi}}(x, t) = \hat{Z}_\chi^{\frac{1}{2}} \bar{\chi}_0(x, t), \quad (3.10)$$

with

$$\hat{Z}_\chi = \frac{-2N_c}{(4\pi)^2 t^2 \langle \hat{\bar{\chi}}_0(x, t) \overleftarrow{\mathcal{D}} \chi_0(x, t) \rangle|_{m=0}}, \quad (3.11)$$

for one quark flavor, where $\overleftarrow{\mathcal{D}}^\mu = \mathcal{D}^\mu - \overrightarrow{\mathcal{D}}^\mu$. The next-next-to-leading order (nnLO) result of $\hat{Z}_\chi$ can be found in ref. [43]. At next-to-leading order (NLO), the renormalization factor $\hat{Z}_\chi$ is given by [35]

$$\hat{Z}_\chi(t, \mu) = (8\pi t)^{-\epsilon} \left\{ 1 + \frac{\alpha_s}{4\pi} C_F \left[\frac{3}{\epsilon_{UV}} + 3 \log(2\mu^2 t e^{\gamma_E}) - \log(432) \right] \right\}, \quad (3.12)$$

where the factor $(8\pi t)^{-\epsilon}$ is introduced to balance the dimension of the ringed fermion field, which has dimension 3/2, while the flowed fermion field has dimension $(d-1)/2$.

3.2 The small flow-time expansion

In this work, we consider the local current operators defined in eqs. (2.15) and eq. (2.27), which are flowed at the flow time t . We use the same symbols for both the flowed and the un-flowed current operators as long as it does not lead to confusions. In the context of Wilson-line operators, we flow the external fields and the Wilson lines at the flow time t . Specifically, this means that all the gluon fields directly connecting to the “heavy” auxiliary field lines are flowed at the flow time t , while the “heavy” auxiliary fields themselves remain un-flowed. In the regime of small flow time, we employ a “short-distance” OPE for the flowed current operators. Our objective is to derive matching coefficients between renormalized flowed current operators and the $\overline{\text{MS}}$ renormalized un-flowed current operators,

$$\mathcal{O}^{\text{R}}(t) = c_{\mathcal{O}}(t, \mu) \mathcal{O}^{\overline{\text{MS}}}(\mu) + \mathcal{O}(t), \quad (3.13)$$

where we have neglected operator mixing.

We work in the massless limit for quark fields and set all the external scales to be zero whenever feasible. This strategic choice allows us to leverage established techniques from HQET for matching calculations. Specifically, we do not need to calculate the un-flowed operators because they yield scaleless integrals to all orders in perturbative calculations in dimensional regularization so that we can obtain the matching coefficients $c_{\mathcal{O}}(t, \mu)$ from the renormalized flowed operators simply by subtracting the IR poles.

Similarly to the renormalization of current operators, we can break down the matching coefficients $c_{\mathcal{O}}(t, \mu)$ for the local current operators into products of field matching coefficients and corresponding vertex matching coefficients. For the four local current operators $\bar{\psi} h_v$, $g F_{\perp\perp}^{\text{F}} h_v$, $g F_{\perp\perp}^{\text{AF}} h_v$, and $g \bar{h}_v F_{\perp\perp}^{\text{AF}} h_v$, the relationships between matching coefficients for the local current operators and those for the corresponding fields and vertices are given by

$$c_{\psi h_v}(t, \mu) = \hat{\zeta}_{\psi}(t, \mu) \zeta_{h_v}^{\text{F}}(t, \mu) \zeta_{\psi h_v}(t, \mu), \quad (3.14\text{a})$$

$$c_{\mathbb{Q}_{\perp}}(t, \mu) = \zeta_A(t, \mu) \zeta_{h_v}^{\text{A}}(t, \mu) \zeta_{\mathbb{Q}_{\perp}}(t, \mu), \quad (3.14\text{b})$$

$$c_{\perp\perp}(t, \mu) = \zeta_A(t, \mu) \zeta_{h_v}^{\text{A}}(t, \mu) \zeta_{\perp\perp}(t, \mu), \quad (3.14\text{c})$$

$$c_{\mathcal{F}}(t, \mu) = \zeta_A(t, \mu) (\zeta_{h_v}^{\text{F}}(t, \mu))^2 \zeta_{\mathcal{F}}(t, \mu). \quad (3.14\text{d})$$

Here $\hat{\zeta}_{\psi}(t, \mu)$, $\zeta_{h_v}^{\text{F}}(t, \mu)$ and $\zeta_A(t, \mu)$ are the matching coefficients of the ringed fermion field $\bar{\chi}$, the “heavy” auxiliary field h_v in fundamental (adjoint) representation and the flowed gluon field B , respectively. Additionally, $\zeta_{\psi h_v}(t, \mu)$, $\zeta_{\mathbb{Q}_{\perp}}(t, \mu)$, $\zeta_{\perp\perp}(t, \mu)$ and $\zeta_{\mathcal{F}}(t, \mu)$ denote matching coefficients stemming from vertex corrections for the corresponding flowed local current operators. We calculate these matching coefficients to one-loop order in section 4.

A key insight arising from our study in section 2 on the multiplicative renormalization of Wilson-line operators is that matching coefficients for Wilson-line operators in the transition from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme can be directly derived from matching coefficients for the corresponding local current operators. This occurs because in the one-dimensional auxiliary-field formalism, combinations of the renormalized local current operators at different spacetime points do not introduce additional UV divergences, apart from the subtracted linear divergence, and because of eq. (3.13) that relates local

operators in the small flow-time limit. Therefore, additional matching for these combined local current operators in different spacetime points is not needed in the small flow-time limit.

4 Computations and results

While the matching coefficients for the fields can be gauge-dependent, it is important to note that the combinations leading to the local current operators under consideration in this work are gauge invariant. Therefore, we have chosen to employ the Feynman gauge consistently throughout our calculations. Our computations are conducted in the context of Euclidean spacetime. In Euclidean spacetime, the renormalized Lagrangian $\mathcal{L}_{h_v}$ is given by

$$\mathcal{L}_{h_v}^E = Z_{h_v} \bar{h}_v (-i v \cdot \partial + i \delta m) h_v + g Z_g Z_A^{\frac{1}{2}} Z_{h_v} \bar{h}_v v \cdot A^a T^a h_v, \quad (4.1)$$

where the superscript “E” indicates Euclidean spacetime. Consequently, the Feynman rules for the “heavy” auxiliary field propagator, with momentum k flowing along the “heavy” auxiliary field line, and the interaction vertex are given by $\frac{1}{k^2 - i0}$ and $-g v^\mu T^a$, respectively. The $-i0$ prescription in Euclidean spacetime plays key role in obtaining correct linear divergences and “mass correction” $i \delta m$.

We decompose a vector into parallel and transverse components according to ($v^2 = 1$ in Euclidean spacetime in our settings)

$$q_{\parallel}^\mu = (v \cdot q) v^\mu, \quad q_{\perp}^\mu = q^\mu - v \cdot q v^\mu. \quad (4.2)$$

We use the following Schwinger parametrization (α -representation)

$$\frac{1}{A^n} = \frac{1}{\Gamma(n)} \int_0^{+\infty} d\alpha \alpha^{n-1} e^{-\alpha A}, \quad (4.3)$$

for the QCD propagator denominators. For the eikonal propagator, we use another version of Schwinger parametrization

$$\frac{1}{(k \cdot v - i0)^n} = \frac{(i)^n}{\Gamma(n)} \int_0^{+\infty} d\alpha \alpha^{n-1} e^{-i(k \cdot v)\alpha}, \quad (4.4)$$

with $k \cdot v$ real.

4.1 δm & ζ_{h_v}

The calculation of the self-energy of the “heavy” auxiliary field at one-loop in gradient flow is similar to that of the heavy quark field in HQET. Setting the external momentum p flowing from left to right, the one-loop self-energy diagram shown in figure 1 gives the amplitude

$$\mathcal{M}_{(a)} = g^2 \bar{\mu}^{2\epsilon} C_F \int \frac{d^4 k}{(2\pi)^4} \frac{1}{(k \cdot v + p \cdot v - i0) k^2} e^{-2tk^2}, \quad (4.5)$$

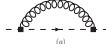


Figure 1. One-loop self-energy Feynman diagram for the “heavy” auxiliary field in gradient-flow formalism. The dashed arrow line denotes the “heavy” auxiliary field, the filled squares represent the flowed B_F field at flow time t .

where $\tilde{\mu}^{2\epsilon} = \left(\frac{\tilde{u}^2 \epsilon \bar{\kappa}}{4\pi} \right)^\epsilon$ and we have chosen the “heavy” auxiliary field to be in fundamental representation. For the adjoint representation case, we just need to replace C_F with C_A in the above expression.

Summing up geometric series of the one-loop self-energy diagrams of the “heavy” auxiliary field propagator gives $1/(p \cdot v - \mathcal{M}_{(a)})$, which indicates that the one-loop “mass correction” is given by $(-\mathcal{M}_{(a)})$. For simplicity, we set $p = 0$ in $\mathcal{M}_{(a)}$, yielding

$$-\mathcal{M}_{(a)}|_{p=0} = -g^2 \tilde{\mu}^{2\epsilon} C_F \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k \cdot v - i0) k^2} e^{-2\epsilon k^2}. \quad (4.6)$$

Note that the above expression is not odd under $k \cdot v \rightarrow -k \cdot v$ and the $k \cdot v$ integration does not give a vanishing result because the sign of the $i0$ prescription matters.

To derive the matching coefficient of the “heavy” auxiliary field at one-loop, we differentiate $\mathcal{M}_{(a)}$ with respect to $p \cdot v$, and set $p = 0$ for simplicity, resulting in

$$\frac{\partial \mathcal{M}_{(a)}}{\partial (p \cdot v)}|_{p=0} = -g^2 \tilde{\mu}^{2\epsilon} C_F \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k \cdot v - i0)^2 k^2} e^{-2\epsilon k^2}. \quad (4.7)$$

Now, we explicitly evaluate eq. (4.6) and eq. (4.7) using the Schwinger parameterization in eq. (4.3) and eq. (4.4). We find (we set $\epsilon = 0$ whenever the integral is finite)

$$-\mathcal{M}_{(a)}|_{p=0} = -ig^2 C_F \int \frac{d^4 k}{(2\pi)^4} \int_0^{+\infty} dx \int_0^{+\infty} dy e^{-(2t+x)k^2 - i(k \cdot v)y}, \quad (4.8)$$

$$\frac{\partial \mathcal{M}_{(a)}}{\partial (p \cdot v)}|_{p=0} = g^2 \tilde{\mu}^{2\epsilon} C_F \int \frac{d^4 k}{(2\pi)^4} \int_0^{+\infty} dx \int_0^{+\infty} dy y e^{-(2t+x)k^2 - i(k \cdot v)y}, \quad (4.9)$$

Performing the k momentum integration gives

$$-\mathcal{M}_{(a)}|_{p=0} = -i \frac{\alpha_s}{4\pi} C_F \int_0^{+\infty} dx \int_0^{+\infty} dy \frac{1}{(2t+x)^2} e^{-\frac{y^2}{4(2t+x)}}, \quad (4.10)$$

$$\frac{\partial \mathcal{M}_{(a)}}{\partial (p \cdot v)}|_{p=0} = \frac{g^2 \tilde{\mu}^{2\epsilon} C_F}{(4\pi)^{d/2}} \int_0^{+\infty} dx \int_0^{+\infty} dy y \frac{1}{(2t+x)^{d/2}} e^{-\frac{y^2}{4(2t+x)}}. \quad (4.11)$$

Integrating out x, y , we obtain the results

$$-\mathcal{M}_{(a)}|_{p=0} = -i \frac{\alpha_s}{4\pi} C_F \frac{\sqrt{2\pi}}{\sqrt{t}}, \quad (4.12)$$

$$\frac{\partial \mathcal{M}_{(a)}}{\partial (p \cdot v)}|_{p=0} = -2 \times \frac{\alpha_s C_F}{4\pi} \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) \right]. \quad (4.13)$$

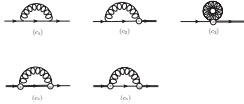


Figure 2 One-loop quark self-energy Feynman diagrams in the gradient-flow formalism (mirror diagrams are implicit). The solid line is the fermion line, the double solid line is the flowed fermion line, the open circle represents the flow vertex.

Repeating the same calculation using dimensional regularization with $t = 0$ gives vanishing $\mathcal{M}_{(a), t=0}|_{p=0}$ and

$$\frac{\partial \mathcal{M}_{(a), t=0}}{\partial(p \cdot v)}|_{p=0} = 2 \frac{\alpha_s C_F}{4\pi} \left[\frac{1}{\epsilon_{UV}} - \frac{1}{\epsilon_{IR}} \right]. \quad (4.14)$$

This confirms that we just need to subtract the IR poles from gradient flow results for the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme when external momenta are set to zero.

Therefore, we obtain the following results (notice that the “mass correction” in the renormalized Lagrangian is denoted as δm)

$$\delta m = -\frac{\alpha_s}{4\pi} C_R \frac{\sqrt{2\pi}}{\sqrt{t}} + O(\alpha_s^2), \quad (4.15)$$

$$\zeta_{h_v} = 1 - \frac{\alpha_s C_R}{4\pi} \log(2\mu^2 t e^{\gamma_E}) + O(\alpha_s^2), \quad (4.16)$$

where the $C_R = C_F$ for fundamental representation case and $C_R = C_A$ for adjoint representation case. The logarithmic dependence on t in the above result of ζ_{h_v} is consistent with the renormalization constant of h_v calculated in [4].

In the following matching calculations, we drop the calculations of un-flowed diagrams since they can be easily obtained from the amplitudes of the corresponding flowed diagrams by setting $t = 0$ from the beginning, which lead to results proportional to $\frac{1}{\epsilon_{UV}} - \frac{1}{\epsilon_{IR}}$ and obviously match the IR poles from flowed diagrams. We have explicitly checked that the coefficients of $-\log(2\mu^2 t e^{\gamma_E})$ in the matching coefficients calculated below and above align with those of the UV poles from the corresponding un-flowed operators at one-loop level, which can be served as a check point of our calculation.

4.2 $\hat{\zeta}_\psi$

The Feynman diagrams for the one-loop corrections of the quark self-energy in the gradient-flow formalism are displayed in figure 2. A similar calculation was performed in ref. [44].

here we repeat the calculation for completeness, adhering to our strategy of splitting the matching coefficients into several parts that can be calculated separately.

Diagram (c_1) is the same as the one in ordinary QCD in the small flow-time limit, which gives the contribution

$$\frac{1}{i} \frac{\partial \mathcal{M}_{(c_1)}}{\partial \beta} \Big|_{\beta=0} = -\frac{\alpha_s C_F}{4\pi} \left[\frac{1}{\epsilon_{UV}} - \frac{1}{\epsilon_{IR}} \right]. \quad (4.17)$$

For diagrams (c_2) and (c_3), we have

$$\begin{aligned} \mathcal{M}_{(c_2)} &= 2 \times (i)^2 g^2 \bar{\mu}^{2\epsilon} C_F \int_0^t ds \int \frac{d^d k}{(2\pi)^d} \frac{(-2\hat{k})\hat{k}}{(k^2)^2} e^{-ik^2} \\ &= 2 \times \frac{\alpha_s C_F}{4\pi} \left[\frac{1}{\epsilon_{UV}} + 1 + \log(2\mu^2 t e^{\gamma_E}) \right], \end{aligned} \quad (4.18)$$

and

$$\begin{aligned} \mathcal{M}_{(c_3)} &= 2 \times \frac{1}{2} g^2 \bar{\mu}^{2\epsilon} (-2dC_F) \int_0^t ds \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2} e^{-2ik^2} \\ &= -4 \times \frac{\alpha_s C_F}{4\pi} \left[\frac{1}{\epsilon_{UV}} + \frac{1}{2} + \log(2\mu^2 t e^{\gamma_E}) \right]. \end{aligned} \quad (4.19)$$

Diagram (c_4) and its corresponding mirror diagram contribute proportionally to the external fermion momentum (with the gradient flow gauge parameter $\kappa = 1$) and subsequently vanish upon setting the external momentum to zero. Diagram (c_5) yields a null contribution due to its inherent oddness under the gluon loop momentum transformation $k \rightarrow -k$.

Adding up $\frac{1}{i} \frac{\partial \mathcal{M}_{(c_1)}}{\partial \beta} \Big|_{\beta=0}$, $\mathcal{M}_{(c_2)}$ and $\mathcal{M}_{(c_3)}$, removing the UV divergences by introducing the ringed fermion field, and subtracting the IR divergences through matching procedure, we obtain the one-loop matching coefficient for the matching from the ringed fermion field $\tilde{\chi}$ to the $\overline{\text{MS}}$ renormalized fermion field ψ ,

$$\zeta_\psi = 1 + \frac{1}{2} \times \frac{\alpha_s}{4\pi} C_F \left[\log(2\mu^2 t e^{\gamma_E}) - \log(4\mathfrak{I}2) \right] + O(\alpha_s^2). \quad (4.20)$$

4.3 $\zeta_{\psi h_\kappa}$

The one-loop correction Feynman diagrams for the ψh_κ vertex in gradient-flow formalism are given in figure 3. Setting the quark external momentum to be zero, we have the amplitude for diagram (b_1),

$$\mathcal{M}_{(b_1)} = -g^2 \bar{\mu}^{2\epsilon} C_F \int \frac{d^d k}{(2\pi)^d} \frac{(\gamma \cdot v) \hat{k}}{(-k \cdot v - i0)(k^2)^2} e^{-2ik^2}. \quad (4.21)$$

Obviously, the transverse components of $\hat{k}$ give vanishing loop integration, thus we have

$$\begin{aligned} \mathcal{M}_{(b_1)} &= g^2 \bar{\mu}^{2\epsilon} C_F \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2)^2} e^{-2ik^2} \\ &= \frac{\alpha_s C_F}{4\pi} \left[-\frac{1}{\epsilon_{IR}} - \log(2\mu^2 t e^{\gamma_E}) - 1 \right]. \end{aligned} \quad (4.22)$$

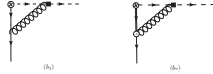


Figure 3. Feynman diagrams of the one-loop corrections for the $\bar{\psi}h_u$ vertex in gradient-flow formalism. The cross circle represents the $\bar{\psi}h_u$ vertex. The fermion field connected to the cross circle is flowed at flow time t .

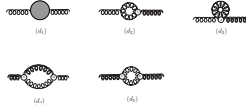


Figure 4. Feynman diagrams for the one-loop gluon self-energy in gradient-flow formalism (mirror diagrams are implicit). The blob in diagram (d_1) represents the gluon vacuum polarization. The single curly lines are gluon lines, while the double curly lines are the flowed gluon lines.

Diagram (b_2) gives contribution proportional to the external fermion momentum (with the gradient flow gauge parameter $\kappa = 1$) and subsequently vanishes upon setting the external momentum to zero. Therefore, we have the one-loop result of $\zeta_{\bar{\psi}h_u}$ after matching to the $\overline{\text{MS}}$ scheme,

$$\zeta_{\bar{\psi}h_u} = 1 - \frac{\alpha_s C_F}{4\pi} [\log(2\mu^2 t e^{\gamma_E}) + 1] + O(\alpha_s^2). \quad (4.23)$$

4.4 ζ_A

The Feynman diagrams for the one-loop corrections of gluon self-energy in the gradient-flow formalism are given by figure 4. For simplicity, we set the external momentum to be zero while extracting the corresponding contribution. Diagram (d_1) is just the conventional gluon vacuum polarization, which gives the contribution in the small flow-time limit,

$$\mathcal{M}_{(d_1)} = \frac{\alpha_s}{4\pi} \delta^{\mu\nu} \left(\frac{5}{3} C_A - \frac{2}{3} n_f \right) \left[\frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon_{\text{IR}}} \right]. \quad (4.24)$$

Diagrams (d_2) and (d_3) give the contributions

$$\begin{aligned} \mathcal{M}_{(d_2)} &= 4g^2 C_A \bar{\mu}^{2\epsilon} \int_0^t ds \int \frac{d^d k}{(2\pi)^d} \frac{(d-2)k^\mu k^\nu + k^2 \delta^{\mu\nu}}{k^2 k'^2} \exp(-2sk^2) \\ &= \frac{\alpha_s}{4\pi} C_A \delta^{\mu\nu} \left[\frac{3}{\epsilon_{\text{UV}}} + 3 \log(2\mu^2 t e^{\gamma_E}) + \frac{5}{2} \right], \end{aligned} \quad (4.25)$$

and

$$\begin{aligned}\mathcal{M}_{(d_3)} &= 2(1-d)g^2C_A\tilde{\mu}^{2\epsilon}\delta^{\mu\nu}\int_0^t ds\int\frac{d^dk}{(2\pi)^d}\frac{1}{k^2}\exp(-2sk^2) \\ &= -\frac{\alpha_s}{4\pi}C_A\delta^{\mu\nu}\left[\frac{3}{\epsilon_{\text{UV}}}+3\log(2\mu^2te^{\gamma_E})+1\right],\end{aligned}\quad (4.26)$$

from which, we see that the UV divergences in diagrams (d_2) and (d_3) cancel.

Diagram (d_4) gives the contribution

$$\begin{aligned}\mathcal{M}_{(d_4)} &= 4g^2C_A\tilde{\mu}^{2\epsilon}\int_0^t ds\int_0^s du\int\frac{d^dk}{(2\pi)^d}\frac{2(d-2)k^\mu k^\nu+\delta^{\mu\nu}k^2}{k^2}e^{-2sk^2} \\ &= \frac{\alpha_s}{4\pi}C_A\delta^{\mu\nu}\left[\frac{2}{\epsilon_{\text{UV}}}+2\log(2\mu^2te^{\gamma_E})-\frac{1}{2}\right].\end{aligned}\quad (4.27)$$

Diagram (d_5) does not contribute in the small flow-time limit because it gives contribution proportional to t in the small flow-time expansion. The results of $\mathcal{M}_{(d_2)}$, $\mathcal{M}_{(d_3)}$ and $\mathcal{M}_{(d_4)}$ are consistent with those given in ref. [35].

Adding up $\mathcal{M}_{(d_1)}$, $\mathcal{M}_{(d_2)}$, $\mathcal{M}_{(d_3)}$ and $\mathcal{M}_{(d_4)}$, we obtain

$$\begin{aligned}\mathcal{M}_{(d)} &= \mathcal{M}_{(d_1)} + \mathcal{M}_{(d_2)} + \mathcal{M}_{(d_3)} + \mathcal{M}_{(d_4)} \\ &= \frac{\alpha_s}{4\pi}\delta^{\mu\nu}\left[\frac{1}{\epsilon_{\text{UV}}}\beta_0 - \frac{1}{\epsilon_{\text{IR}}}\left(\frac{5}{3}C_A - \frac{2}{3}n_f\right) + 2C_A\log(2\mu^2te^{\gamma_E}) + C_A\right],\end{aligned}\quad (4.28)$$

with $\beta_0 = \frac{11}{3}C_A - \frac{2}{3}n_f$. The IR poles are removed through matching procedure and the UV poles are absorbed into the renormalization of the strong coupling in the $\overline{\text{MS}}$ scheme,

$$Z_g = 1 - \frac{\alpha_s}{4\pi}\frac{1}{2\epsilon_{\text{UV}}}\beta_0 + O(\alpha_s^2),\quad (4.29)$$

which is consistent with the general argument that the flowed gluon field in B_μ does not need renormalization while the strong coupling in B_μ needs to be renormalized [29].

Finally, we have the matching coefficient for the matching from the flowed B_μ field to the $\overline{\text{MS}}$ renormalized gluon field gA_μ ,

$$\zeta_A = 1 + \frac{\alpha_s}{4\pi}C_A\left[\log(2\mu^2te^{\gamma_E}) + \frac{1}{2}\right] + O(\alpha_s^2).\quad (4.30)$$

Note that the logarithmic dependence in ζ_A cancel that in ζ_{h_v} (see eq. (4.16)) in adjoint representation at one-loop level. However, the constant terms in the square brackets in eq. (4.30) are not canceled.

4.5 $\zeta_{\perp\perp}$ & ζ_F

The one-loop vertex type Feynman diagrams for the operators $gF_{\perp\perp}^{\mu\nu}h_v$, $gF_{\perp\perp}^{\mu\nu}h_v$ and $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$ in the gradient-flow formalism are shown in figure 5. Choosing the external gluon momentum to be q (flowing into the vertex), we have the leading-order amplitudes

$$\mathcal{M}_{F_{\perp\perp}h_v}^{\text{LO}} = ig\left(q_\perp^\mu A_\perp^\nu - q_\perp^\nu A_\perp^\mu\right),\quad (4.31)$$

$$\mathcal{M}_{F_{\perp\perp}h_v}^{\text{LO}} = ig\left(q_\perp^\mu A_\perp^\nu - q_\perp^\nu A_\perp^\mu\right),\quad (4.32)$$

$$\mathcal{M}_{\tilde{h}_vF_{\perp\perp}h_v}^{\text{LO}} = ig\left(q_\perp^\mu A_\perp^\nu - q_\perp^\nu A_\perp^\mu\right),\quad (4.33)$$

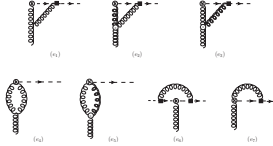


Figure 5. Vertex type Feynman diagrams that contribute to the one-loop matching for the operators $gF_{\perp\perp}^{\mu\nu}h_v$, $gF_{\perp\perp}^{\mu\nu}h_v$ and $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$ in the gradient-flow formalism (mirror diagrams are implicit). The gluon fields at the cross circles and the filled squares are flowed at flow time t . For $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$, the fields $\tilde{h}_v$ are not drawn in diagrams $(e_1) - (e_5)$ for simplicity. The “heavy” auxiliary fields in $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$ are in fundamental representation and the “heavy” auxiliary fields in $gF_{\perp\perp}^{\mu\nu}h_v$, $gF_{\perp\perp}^{\mu\nu}h_v$ are in adjoint representation. Diagrams (e_6) and (e_7) only give contributions to the operators $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$ and $gF_{\perp\perp}^{\mu\nu}h_v$, respectively.

where we have suppressed the color indices and dropped the external “heavy” fields but kept the external gluon fields for the convenience of matching procedure.

Since we are only interested in obtaining the matching coefficients, which are independent of the external momenta, we can set $q \cdot v = 0$ for the amplitudes of the operators $gF_{\perp\perp}^{\mu\nu}h_v$ and $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$, but not for the amplitudes of the operator $gF_{\perp\perp}^{\mu\nu}h_v$. The computations of the one-loop vertex corrections for the operators $gF_{\perp\perp}^{\mu\nu}h_v$ and $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$ are straightforward and similar. These three operators share the same diagrams $(e_1) - (e_6)$. We will only give the amplitudes of diagrams $(e_1) - (e_5)$ for the operators $gF_{\perp\perp}^{\mu\nu}h_v$ and $gF_{\perp\perp}^{\mu\nu}h_v$, supplied with the amplitude of diagram (e_6) for the operator $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$ and the amplitude of diagram (e_7) for the operator $g\tilde{h}_vF_{\perp\perp}^{\mu\nu}h_v$.

The calculation of the vertex corrections for $gF_{\perp\perp}^{\mu\nu}h_v$ is not so straightforward due to non-trivial mixing with the gauge non-invariant operator $iv^\mu A_\perp^\nu (iv \cdot D - i\delta m)h_v$. We will deal with $gF_{\perp\perp}^{\mu\nu}h_v$ in the next subsection in details.

The amplitudes of diagrams $(e_1) - (e_6)$ are given by

$$\begin{aligned} \mathcal{M}_{(e_1)} = & -g^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^4 k}{(2\pi)^4} \left\{ ik^\mu A^\nu \frac{\delta^{\mu\nu} (-q - k) \cdot v + v^\alpha (2q - k)^\alpha + v^\alpha (2k - q)^\alpha}{(k \cdot v - i0)k^2(k - q)^2} \right. \\ & \left. - (\mu \leftrightarrow \nu) \right\} e^{-t[k^2 + (k - q)^2]}, \end{aligned} \quad (4.34)$$

$$\mathcal{M}_{(e_1)} = g^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \int_0^t ds \left\{ i k^\mu \frac{v^\alpha (k-2q)^\nu + 2\delta^{\alpha\nu} q \cdot v - 2v^\nu (k-q)^\alpha}{(k \cdot v - i0)(k-q)^2} \right. \\ \left. - (\mu \leftrightarrow \nu) \right\} A^\alpha e^{-(s+t)(k-q)^2 - (t-s)k^2}, \quad (4.35)$$

$$\mathcal{M}_{(e_2)} = g^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \int_0^t ds \left\{ i k^\mu \frac{\delta^{\alpha\nu} (q+k) \cdot v + 2v^\nu (-k)^\alpha - 2v^\alpha q^\nu}{(k \cdot v - i0)k^2} \right. \\ \left. - (\mu \leftrightarrow \nu) \right\} A^\alpha e^{-(t+s)k^2 - (t-s)(k-q)^2}, \quad (4.36)$$

$$\mathcal{M}_{(e_3)} = -\frac{3}{2} g^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{i(q^\mu A^\nu - q^\nu A^\mu)}{k^2(k-q)^2} e^{-i[k^2 + (k-q)^2]}, \quad (4.37)$$

$$\mathcal{M}_{(e_4)} = -g^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \int_0^t ds \frac{i(k+3q)^\mu A^\nu - i(k+3q)^\nu A^\mu}{k^2} e^{-(t+s)k^2 - (t-s)(k-q)^2}, \quad (4.38)$$

$$\mathcal{M}_{(e_5)} = g^3 \left(C_F - \frac{C_A}{2} \right) \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \int_0^t ds \frac{i(q_\perp^\mu A_\perp^\nu - q_\perp^\nu A_\perp^\mu)}{(k \cdot v - i0)^2 k^2} e^{-2ik^2}. \quad (4.39)$$

Extracting the LO expressions and setting $q = 0$ thereafter, we obtain the contributions from each diagram for the operator $gF_{\perp\perp}^{\mu\nu} h_{\nu}$,

$$\zeta_{\perp\perp, (e_1)} = -\frac{\alpha_s C_A}{4\pi} \times \frac{1}{2} \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) + 1 \right], \quad (4.40)$$

$$\zeta_{\perp\perp, (e_2)} = 0, \quad (4.41)$$

$$\zeta_{\perp\perp, (e_3)} = \frac{\alpha_s C_A}{4\pi} \times \frac{1}{16}, \quad (4.42)$$

$$\zeta_{\perp\perp, (e_4)} = \frac{\alpha_s C_A}{4\pi} \times \frac{3}{2} \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) + 1 \right], \quad (4.43)$$

$$\zeta_{\perp\perp, (e_5)} = -\frac{\alpha_s C_A}{4\pi} \times \frac{25}{16}. \quad (4.44)$$

Summing up all the contributions from diagrams $(e_1) - (e_5)$ and discarding the IR poles through matching procedure, we obtain the vertex matching coefficient for the operator $gF_{\perp\perp}^{\mu\nu} h_{\nu}$,

$$\zeta_{\perp\perp} = 1 + \frac{\alpha_s C_A}{4\pi} \left[\log(2\mu^2 t e^{\gamma_E}) - \frac{1}{2} \right] + O(\alpha_s^2). \quad (4.45)$$

Similarly, we have the contributions from each diagram for the operator $g\bar{h}_e F_{\perp\perp}^{\mu\nu} h_e$,

$$\zeta_{F,(e_1)} = -\frac{\alpha_s}{4\pi} C_A \times \frac{1}{2} \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) + 1 \right], \quad (4.46)$$

$$\zeta_{F,(e_2)} = 0, \quad (4.47)$$

$$\zeta_{F,(e_3)} = \frac{\alpha_s}{4\pi} C_A \times \frac{1}{16}, \quad (4.48)$$

$$\zeta_{F,(e_4)} = \frac{\alpha_s}{4\pi} C_A \times \frac{3}{2} \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) + 1 \right], \quad (4.49)$$

$$\zeta_{F,(e_5)} = -\frac{\alpha_s}{4\pi} C_A \times \frac{25}{16}, \quad (4.50)$$

$$\zeta_{F,(e_6)} = \frac{\alpha_s}{4\pi} \left(C_F - \frac{C_A}{2} \right) \times 2 \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) \right], \quad (4.51)$$

which lead to

$$\zeta_{\perp\perp}^F = 1 + \frac{\alpha_s}{4\pi} \left[2C_F \log(2\mu^2 t e^{\gamma_E}) + \frac{1}{2} C_A \right] + O(\alpha_s^2). \quad (4.52)$$

4.6 $\zeta_{\perp\perp}$

Diagrams (e_4) , (e_5) have no mixing and give the same contributions for $gF_{\perp\perp}^{\mu\nu} h_e$ and $gF_{\perp\perp}^{\mu\nu} h_e$,

$$\zeta_{\parallel\perp,(e_4)} = \frac{\alpha_s C_A}{4\pi} \times \frac{3}{2} \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) + 1 \right], \quad (4.53)$$

$$\zeta_{\parallel\perp,(e_5)} = -\frac{\alpha_s C_A}{4\pi} \times \frac{25}{16}. \quad (4.54)$$

Complications arise from diagrams (e_1) , (e_2) , (e_3) and (e_7) , whose amplitudes are given by

$$\mathcal{M}_{(e_1)}^{\parallel\perp} = i g_s^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \left[\frac{(q+k)_{\parallel}^{\mu} A_{\perp}^{\nu} - (2q-k)_{\perp}^{\nu} A_{\parallel}^{\mu}}{k^2(k-q)^2} + \frac{v^{\mu} k_{\perp}^{\nu} (2k-q)_{\perp} \cdot A_{\perp}}{(k \cdot v - i0) k^2 (k-q)^2} \right] \times e^{-\epsilon[k^2 + (k-q)^2]}, \quad (4.55)$$

$$\mathcal{M}_{(e_2)}^{\parallel\perp} = i g_s^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \int_0^t ds \left[\frac{2(q_{\parallel}^{\mu} A_{\perp}^{\nu} - q_{\perp}^{\nu} A_{\parallel}^{\mu})}{(k-q)^2} + \frac{2v^{\mu} k_{\perp}^{\nu} (k-q) \cdot A}{(k \cdot v - i0)(k-q)^2} \right] \times e^{-(s+\epsilon)(k-q)^2 - (t-s)k^2}, \quad (4.56)$$

$$\mathcal{M}_{(e_3)}^{\parallel\perp} = i g_s^3 C_A \bar{\mu}^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \int_0^t ds \left[\frac{(k+q)_{\parallel}^{\mu} A_{\perp}^{\nu} - (k+q)_{\perp}^{\nu} A_{\parallel}^{\mu}}{k^2} + \frac{(A \cdot v) (q_{\parallel}^{\mu} k_{\perp}^{\nu} - q_{\perp}^{\nu} k_{\parallel}^{\mu}) + 2v^{\mu} k_{\perp}^{\nu} (k \cdot A)}{(k \cdot v - i0) k^2} \right] e^{-(s+\epsilon)k^2 - (t-s)(k-q)^2}, \quad (4.57)$$

$$\mathcal{M}_{(ev)}^{\parallel\perp} = ig^3 C_A \bar{\mu}^{-2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{-v^\mu A_\perp^\nu}{(k \cdot v - i0)(k - q)^2} e^{-2i\epsilon(k-q)^2}. \quad (4.58)$$

These amplitudes do not give expressions that are exactly proportional to the LO expression of the operator, $gF_{\perp\perp}^{\mu\nu} h_\nu$ even after tensor reduction for the loop integrations. They lead to mixing with the gauge non-invariant operators $iv^\mu A_\perp^\nu (iv \cdot D - i0m) h_\nu$, whose LO expression is proportional to $q_\parallel^\mu A_\perp^\nu$, and thus it is not straightforward to disentangle the mixing from those that give contributions to the operator $gF_{\perp\perp}^{\mu\nu} h_\nu$. Fortunately, we can extract the contributions to the operator $gF_{\perp\perp}^{\mu\nu} h_\nu$ through the results that are proportional to $q_\perp^\mu A_\parallel^\nu$. To check the mixing pattern and cancellation of linear divergences that were found in refs. [58–60], we will also keep the results that are proportional to $q_\parallel^\mu A_\perp^\nu$ and $v_\parallel^\mu A_\perp^\nu / \sqrt{t}$ in the small flow-time limit.

If we set $t = 0$ in $\mathcal{M}_{(e1)}^{\parallel\perp}$ and $\mathcal{M}_{(e2)}^{\parallel\perp}$, we can easily see that the linear divergences coming from $\mathcal{M}_{(e2)}^{\parallel\perp}$ and the second term of $\mathcal{M}_{(e1)}^{\parallel\perp}$ cancel with $d = 3 - 2\epsilon$. However, in gradient flow scheme, the cancellation of linear divergences are much more complicated. Apparently, the linearly divergences arising from $\mathcal{M}_{(ev)}^{\parallel\perp}$ and the second term of $\mathcal{M}_{(e1)}^{\parallel\perp}$ do not cancel with $d = 4 - 2\epsilon$ and $t > 0$. The cancellation of linear divergences in gradient-flow scheme will be restored when diagrams (e_2) and (e_3) are included.

Great simplification can be achieved for our matching calculation by adopting the well-developed method used in the matching calculation in HQET. That is, we focus on the large loop momentum region $k \gg q$ and expand the amplitudes up to linear order of the external momentum q before performing the loop momentum integration. This method is valid in our case because, in the small flow-time limit, the flowed operators share the same IR behavior with the corresponding un-flowed operators and contributions to the matching coefficients come from the UV region.

Expanding the amplitudes according to $k \gg q$ up to linear order of q and completing the loop integrations, we obtain

$$\mathcal{M}_{(e1)}^{\parallel\perp} = \frac{\alpha_s}{4\pi} g C_A \left\{ -\frac{2\sqrt{2\pi}}{3\sqrt{t}} v^\mu A_\perp^\nu - i\ell_1 \left[\left(\frac{3}{2} + \frac{4}{d} \right) q_\parallel^\mu A_\perp^\nu - \frac{3}{2} q_\perp^\mu A_\parallel^\nu \right] \right\}, \quad (4.59)$$

$$\mathcal{M}_{(e2)}^{\parallel\perp} = \frac{\alpha_s}{4\pi} g C_A \left\{ -\frac{1\sqrt{2\pi}}{6\sqrt{t}} v^\mu A_\perp^\nu + i \left[\frac{15}{8} q_\parallel^\mu A_\perp^\nu - \frac{1}{8} q_\perp^\mu A_\parallel^\nu \right] \right\}, \quad (4.60)$$

$$\mathcal{M}_{(e3)}^{\parallel\perp} = \frac{\alpha_s}{4\pi} g C_A \left\{ -\frac{1\sqrt{2\pi}}{6\sqrt{t}} v^\mu A_\perp^\nu + i \left[\frac{11}{16} q_\parallel^\mu A_\perp^\nu - \frac{15}{16} q_\perp^\mu A_\parallel^\nu \right] \right\}, \quad (4.61)$$

$$\mathcal{M}_{(ev)}^{\parallel\perp} = \frac{\alpha_s}{4\pi} g C_A \left\{ \frac{\sqrt{2\pi}}{\sqrt{t}} v^\mu A_\perp^\nu + \frac{i}{2} \ell_1 \frac{1}{2} q_\parallel^\mu A_\perp^\nu \right\}, \quad (4.62)$$

where

$$\ell_1 = \frac{1}{\epsilon_{\text{IR}}} + \log(2\mu^2 t e^{\gamma_E}) + 1. \quad (4.63)$$

From the above results, we can clearly see the cancellation of the linear divergences. After subtracting the contributions to the operator $gF_{\perp\perp}^{\mu\nu}h_\nu$, the leftover logarithmic dependence on t also matches the UV divergence of the mixing presented in refs. [58–60].

Summarizing all the contributions from diagrams $(e_1) - (e_5)$ and (e_7) , we obtain the matching coefficient contributed from the vertex correction of the operator $gF_{\perp\perp}^{\mu\nu}h_\nu$,

$$\zeta_{\perp\perp} = 1 - \frac{1}{2} \frac{\alpha_s}{4\pi} C_A + O(\alpha_s^2), \quad (4.64)$$

which cancels the constant term in the square brackets of ζ_A in eq. (4.30) and thus makes the one-loop contribution to the matching coefficient of the operator $gF_{\perp\perp}^{\mu\nu}h_\nu$ vanish ⁵.

4.7 Results and applications for Wilson-line operators

We are now ready to summarize the final results of the matching coefficients for the four local current operators: $\bar{\psi}h_\nu$, $gF_{\perp\perp}^{\mu\nu}h_\nu$, $gF_{\perp\perp}^{\mu\nu}h_\nu$ and $g\bar{h}_\nu F_{\perp\perp}^{\mu\nu}h_\nu$. The relationships between the matching coefficients of the local current operators and the matching coefficients of the fields and vertices are provided in eqs. (3.14). The one-loop results of ζ_{h_ν} , ζ_ψ , $\zeta_{\psi h_\nu}$, ζ_A , $\zeta_{\perp\perp}$, $\zeta_{\perp\perp}$ and ζ_F can be found in eq. (4.16), eq. (4.20), eq. (4.23), eq. (4.30), eq. (4.64), eq. (4.45) and eq. (4.52), respectively. Implementing those results, we finally obtain

$$c_{\psi h_\nu}(t, \mu) = 1 - \frac{\alpha_s}{4\pi} C_F \left[\frac{3}{2} \log(2\mu^2 t e^{\gamma_E}) + \frac{\log(432)}{2} + 1 \right] + O(\alpha_s^2), \quad (4.65)$$

$$c_{\perp\perp}(t, \mu) = 1 + O(\alpha_s^2), \quad (4.66)$$

$$c_{\perp\perp}(t, \mu) = 1 + \frac{\alpha_s}{4\pi} C_A \times \log(2\mu^2 t e^{\gamma_E}) + O(\alpha_s^2), \quad (4.67)$$

$$c_F(t, \mu) = 1 + \frac{\alpha_s}{4\pi} C_A \times \log(2\mu^2 t e^{\gamma_E}) + O(\alpha_s^2). \quad (4.68)$$

We have verified that the logarithmic dependences on t in eq. (4.65), eq. (4.66) and eq. (4.67) align with the anomalous dimensions given by refs. [5, 58, 70] for the current operators $\bar{\psi}h_\nu$, $gF_{\perp\perp}^{\mu\nu}h_\nu$ and $gF_{\perp\perp}^{\mu\nu}h_\nu$, respectively (see ref. [57] that uses similar notation for these operators for comparison). For the current operator $g\bar{h}_\nu F_{\perp\perp}^{\mu\nu}h_\nu$, its anomalous dimension was also calculated in refs. [71, 72] using background field method (see ref. [73] for a review of background field method), which is consistent with the logarithmic dependences on t in eq. (4.68). It is noteworthy that while gA does not contribute to the anomalous dimension of $g\bar{h}_\nu F_{\perp\perp}^{\mu\nu}h_\nu$ within the background field method, it does so using the method adopted in this work. The UV poles in the $\overline{\text{MS}}$ scheme for diagrams (e_1) and (e_4) differ depending on whether the calculations are conducted with or without the background field method, owing to the distinct Feynman rules of three-gluon vertex. We have checked that these differences compensate for those arising from the anomalous dimensions of gA with or without the background field method using Feynman gauge. With correct logarithmic dependences on t for the matching coefficients of local current operators, “heavy” auxiliary field h_ν , quark

⁵In ref. [60], the author calculated the color-electric field correlator G_E at one-loop level within the gradient-flow formalism. The matching calculation for G_E in the small flow-time limit can be reduced to the matching calculation for the local current operator $g\bar{h}_\nu F_{\perp\perp}^{\mu\nu}h_\nu$ with h_ν being in fundamental representation, which also has vanishing one-loop contribution to the matching coefficient.

field ψ and gA , the corresponding logarithmic dependences on t for the vertex matching coefficients are ensured to be correct. While the results of $c_{\perp\perp}(t, \mu)$ and $c_F(t, \mu)$ coincide at one-loop level, it is intriguing that the anomalous dimensions of $gF_{\perp\perp}^{\mu\nu}h_v$ and $gh_vF_{\perp\perp}^{\mu\nu}h_v$ differ by a term proportional to π^2 at two-loop level (see the two-loop anomalous dimensions given in refs. [57, 74, 75]). In addition, for the operator $gF_{\perp\perp}^{\mu\nu}h_v$, its anomalous dimension starts to contribute at two-loop level [57], the two-loop and three-loop anomalous dimensions of the operator $\bar{\psi}h_v$ can be found in refs. [76, 77] and ref. [78], respectively.

As mentioned in the end of section 3.2, the matching coefficients for the Wilson-line operators can be directly obtained from the corresponding matching coefficients of the local current operators in the small flow-time limit within the one-dimensional auxiliary-field formalism. In the small flow-time limit, expressing the matching equations for the Wilson-line operators defined in eq. (1.1), eq. (2.10) and eq. (2.11) as

$$\mathcal{O}_v^R(zv, t) = C_v(t, \mu) e^{\delta m z} \mathcal{O}_v^{\overline{\text{MS}}}(zv) + \mathcal{O}(t), \quad (4.69)$$

$$\mathcal{O}_{\perp\perp}^R(zv, t) = C_{\perp\perp}(t, \mu) e^{\delta m z} \mathcal{O}_{\perp\perp}^{\overline{\text{MS}}}(zv) + \mathcal{O}(t), \quad (4.70)$$

$$\mathcal{O}_{\perp\perp}^R(zv, t) = C_{\perp\perp}(t, \mu) e^{\delta m z} \mathcal{O}_{\perp\perp}^{\overline{\text{MS}}}(zv) + \mathcal{O}(t), \quad (4.71)$$

where we have omitted the Lorentz indices for simplicity, we establish relations

$$C_v(t, \mu) = c_{v\bar{v}0}^2(t, \mu), \quad (4.72)$$

$$C_{\perp\perp}(t, \mu) = c_{\perp\perp}^2(t, \mu), \quad (4.73)$$

$$C_{\perp\perp}(t, \mu) = c_{\perp\perp}^2(t, \mu). \quad (4.74)$$

These relations are independent of z and external states of the matrix elements constructed using these Wilson-line operators.

Regarding the spin-dependent potentials in terms of chromomagnetic field insertions into a Wilson loop, each such insertion will lead to a factor of the matching coefficient $c_F(t, \mu)$ for the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme.

5 Summary and outlook

We have developed a systematic approach for matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme for Wilson-line operators in the small flow-time limit. This approach extends to various applications, including quasi-PDFs, gliconic correlators that appear in quarkonium decay and production in the framework of pNRQCD and spin-dependent potentials in terms of chromomagnetic field insertions into a Wilson loop.

Within the one-dimensional auxiliary-field formalism, the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme for the Wilson-line operators discussed in this work can be reduced to the matching for the corresponding local current operators in the small flow-time limit. We match these local current operators from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme by comparing perturbative computations in these two schemes. Great simplification for the matching procedure can be achieved by using small flow-time expansion and setting external momenta to be zero whenever possible. In this way, the well-developed

method of matching calculation in HQET can be applied. After completing the matching for the local current operators, one automatically obtains the matching coefficients for the corresponding Wilson-line operators in the small flow-time limit, because the combinations of the renormalized local current operators at different spacetime do not introduce further UV divergences, apart from the subtracted linear divergences, due to the multiplicative renormalizability of the Wilson-line operators.

The matching for the local current operators can be further decomposed as the matching for the fields and vertices due to the multiplicative renormalizability of the local current operators. We thus compute the matching coefficients at one-loop level for auxiliary “heavy” auxiliary field h_v , massless quark field ψ , ψh_v vertex, gluon field gA and the vertices of the local current operators $gF_{\perp\perp}^{\mu\nu} h_v$, $gF_{\perp\perp}^{\mu\nu} h_v$ and $gh_v F_{\perp\perp}^{\mu\nu} h_v$. The one-loop result of the linearly divergent “mass” correction δm is also obtained. Provided with these results, we give the matching coefficients for the local current operators and the Wilson-line operators in the small flow-time limit.

It is worth mentioning that this method is also applicable for matching from lattice regularization schemes to the $\overline{\text{MS}}$ scheme in the small lattice spacing limit, albeit with added complexity due to UV and IR divergence treatment [79].

In the appendix, we have shown that, at one-loop level, for the case of hadronic matrix element defined in eq. (A.1) at zero external momenta, the finite flow time effect is very small as long as the flow radius is smaller than physical distance z , which is usually satisfied in lattice gradient flow computations.

With the systematic approach we have developed for the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme for the Wilson-line operators, it is straightforward to extend the current study to two-loop order, which would be more helpful for lattice gradient flow computations of matrix elements constructed using the off-lightcone Wilson-line operators. We leave it for future research.

Note added

Before submission of this work, we noticed that the one-loop result of the matching coefficient for the gluonic correlator G_B was recently published in ref. [80], which was the same as our one-loop result of $c_B^2(t, \mu)$, because the matching calculation for the gluonic correlator G_B could be reduced to the matching calculation for the local current operator $gh_v F_{\perp\perp}^{\mu\nu} h_v$.

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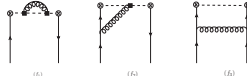


Figure 6. Feynman diagrams that contribute to the one-loop corrections for the hadronic matrix element defined as eq. (A.1) in gradient-flow formalism. The dashed line is the Wilson line, the filled squares represent the flowed B_μ field at flow time t , the cross circles represent the spacetime points that connecting the Wilson line and the fermion fields, which are all flowed at the flow time t . Diagrams that give contribution proportional to the external momentum and thus vanish in our calculation are dropped. The external quark one-loop self-energy diagrams are not listed here, they are shown in figure. 2

A Quark quasi-PDF operator in gradient flow

In this appendix, we consider the hadronic matrix element constructed using the operator defined in eq. (1.1),

$$h(z, P^z) = \langle P | \bar{\psi}(zv) \Gamma W(zv, 0) \psi(0) | P \rangle, \quad (\text{A.1})$$

where P is the hadronic state with momentum P^z , $\Gamma = \gamma \cdot v$ or $\Gamma = \gamma_4^a$. The one-loop matching calculation for $h(z, P^z)$ from the gradient-flow scheme to the $\overline{\text{MS}}$ -like schemes was done in ref. [42] with $P^z = 0$ and full flow time dependence.

However, the matching coefficient given by eq. (44) of ref. [42] does not lead to z independent result in the small flow-time limit, which contradicts with our argument that we only need to do matching calculations for the local current operators stemming from the Wilson-line operators based on the one-dimensional auxiliary-field formalism. Therefore, we re-calculate the matching coefficient in great detail and compare with the calculation done in ref. [42]⁶.

The matching relation for $h(z, P^z)$ can be expressed as

$$h^{\text{B}}(z, t) = C_q(t, z, \mu) h^{\overline{\text{MS}}}(z, \mu), \quad (\text{A.2})$$

where we have dropped the exponentiated “mass” correction δm in the matching relation and choose $P^z = 0$ for simplicity and comparison.

The Feynman diagrams that contribute to the one-loop corrections for $h(z, P^z)$ in gradient-flow formalism are shown in figure. 2 and figure. 6. The contribution coming from the quark self-energy diagrams in figure. 2 can be summarized as

$$\mathcal{M}_q^{\text{B}} = \frac{\alpha_s}{4\pi} C_F \left[\frac{1}{\epsilon_{\text{IR}}} + \log(2t\mu^2) + \gamma_E - \log(432) \right], \quad (\text{A.3})$$

where the subscript $\hat{\chi}$ indicates that we have used the ringed fermion field to remove the UV divergences.

⁶Our results correct and supersede the results of ref. [42], for which an erratum is in preparation.

A.1 Wilson line self-energy diagram (f_1)

The one-loop Wilson line self energy diagram is shown as diagram (f_1) in figure. 6 and its amplitude is given by

$$\mathcal{M}_{(f_1)} = -g^2 \bar{z}^{2\nu} C_F \int_0^z ds_1 \int_0^{s_1} ds_2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2} e^{-2ik^2} e^{i(k \cdot v)(s_2 - s_1)}, \quad (\text{A.4})$$

which is finite at $d = 4$. Therefore, we set $d = 4$ and obtain,

$$\begin{aligned} \mathcal{M}_{(f_1)} &= -g^2 C_F \int_0^z ds_1 \int_0^{s_1} ds_2 \int_0^{+\infty} dx \frac{1}{(2\pi)^4} e^{-\frac{(s_2 - s_1)^2}{4(\pi + x)}} \left(\frac{\pi}{2t + x} \right)^2 \\ &= \frac{\alpha_s}{4\pi} C_F \times 2 \left[\log(z^2) + \gamma_E + 2 \left(1 - e^{-z^2} \right) - \text{Ei}(-z^2) - 2\sqrt{\pi} z \text{erf}(z) \right], \quad (\text{A.5}) \end{aligned}$$

where $\bar{z} = \frac{z}{r_F}$ with $r_F = \sqrt{8t}$ representing the flow radius, $\text{Ei}(x)$ is the exponential integral

$$\text{Ei}(x) = - \int_{-x}^{+\infty} \frac{e^{-t}}{t} dt, \quad (\text{A.6})$$

and $\text{erf}(z)$ is the error function

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt. \quad (\text{A.7})$$

In the small flow-time limit, we have

$$\mathcal{M}_{(f_1), t \rightarrow 0} = \frac{\alpha_s}{4\pi} C_F \times 2 \left[\log(z^2) + \gamma_E + 2 - 2\sqrt{\pi} z \right], \quad (\text{A.8})$$

where the last term is consistent with the one-loop result of δm we have obtained in eq. (4.15). Notice that the $\overline{\text{MS}}$ renormalized result of the Wilson line self energy at one-loop is given by [23]

$$\mathcal{M}_{(f_1)}^{\overline{\text{MS}}} = \frac{\alpha_s}{4\pi} C_F \times 2 \left[2 + 2\gamma_E + \log\left(\frac{\mu^2 z^2}{4}\right) \right]. \quad (\text{A.9})$$

Dropping the linearly divergent term from $\mathcal{M}_{(f_1), t \rightarrow 0}$ and comparing with $\mathcal{M}_{(f_1)}^{\overline{\text{MS}}}$, gives the contribution from diagram (f_1) for the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme in the small flow-time limit,

$$\mathcal{C}_{(f_1), t \rightarrow 0} = \frac{\alpha_s C_F}{4\pi} \left[-2 \log(2\mu^2 t e^{\gamma_E}) \right], \quad (\text{A.10})$$

which is consistent with the result ζ_{h_v} we have obtained eq. (4.16).

A.2 Vertex diagram (f_2)

The amplitude of diagram (f_2) is given by (including its mirror diagram),

$$\mathcal{M}_{(f_2)} = 2ig C_F \int_0^z ds \int \frac{d^d k}{(2\pi)^d} \frac{-i\hat{k}}{(k^2)^2} (ig\gamma \cdot v) e^{-2ik^2} e^{-i(k \cdot v)s}. \quad (\text{A.11})$$

Since the transverse components of $\not{k}$ lead to vanishing momentum integration, we can replace $\not{k}$ with $(k \cdot v)(\gamma \cdot v)$ in $\mathcal{M}_{(f_2)}$, therefore, we have

$$\begin{aligned}\mathcal{M}_{(f_2)} &= 2g^2 C_F \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2)^2} \left(1 - e^{-i(k \cdot v)z}\right) e^{-2ik^2} \\ &= \frac{\alpha_s}{4\pi} C_F \times 2 \left[\log(z^2) + \gamma_E - 1 - \text{Ei}(-z^2) + \frac{1}{z^2} (1 - e^{-z^2}) \right].\end{aligned}\quad (\text{A.12})$$

In the small flow-time limit, we have

$$\mathcal{M}_{(f_2), t \rightarrow 0} = \frac{\alpha_s}{4\pi} C_F \times 2 \left[\log\left(\frac{z^2}{8t}\right) + \gamma_E - 1 \right]. \quad (\text{A.13})$$

Comparing the corresponding result in the $\overline{\text{MS}}$ scheme [23],

$$\mathcal{M}_{(f_2)}^{\overline{\text{MS}}} = \frac{\alpha_s}{4\pi} C_F \times 2 \left[2\gamma_E + \log\left(\frac{\mu^2 z^2}{4}\right) \right], \quad (\text{A.14})$$

we obtain the contribution from diagram (f_2) for the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme in the small flow-time limit,

$$C_{(f_2), t \rightarrow 0} = \frac{\alpha_s}{4\pi} C_F [-2 \log(2\mu^2 t e^{\gamma_E}) - 2], \quad (\text{A.15})$$

which is consistent with the result of ζ_{gh_s} we have obtained in eq. (4.23).

A.3 Diagram (f_3)

The amplitude of diagram (f_3) can be expressed as,

$$\begin{aligned}\mathcal{M}_{(f_3)} &= (ig)^2 \bar{\mu}^{2\omega} C_F \int \frac{d^d k}{(2\pi)^d} \frac{\delta^{\mu\nu} \gamma^\rho (-i\not{k}) \gamma^\alpha (-i\not{k}) \gamma^\nu}{(k^2)^3} e^{-2ik^2} e^{-i(k \cdot v)z} \\ &= g^2 C_F (d-2) \bar{\mu}^{2\omega} \int \frac{d^d k}{(2\pi)^d} \frac{k^2 \gamma^\alpha - 2k^\alpha \not{k}}{(k^2)^3} e^{-2ik^2} e^{-i(k \cdot v)z}.\end{aligned}\quad (\text{A.16})$$

If α is parallel to v , dropping the tree-level factor $v^\alpha \gamma \cdot v$, we have

$$\begin{aligned}\mathcal{M}_{(f_3), \parallel} &= g^2 C_F (d-2) \bar{\mu}^{2\omega} \int \frac{d^d k}{(2\pi)^d} \frac{[k^2 - 2(k \cdot v)^2]}{(k^2)^3} e^{-2ik^2} e^{-i(k \cdot v)z} \\ &= \frac{\alpha_s}{4\pi} C_F \left[-\frac{1}{\epsilon_{\text{IR}}} - \log\left(\frac{\mu^2 z^2}{4}\right) - 2\gamma_E + 3 \right. \\ &\quad \left. + \text{Ei}(-z^2) + \frac{3}{z^2} (1 - e^{-z^2}) - \frac{1}{z^2} (4 - e^{-z^2}) \right].\end{aligned}\quad (\text{A.17})$$

If α is transverse to v , dropping the tree-level factor $\gamma_\perp^\alpha$, we have

$$\begin{aligned}\mathcal{M}_{(f_3), \perp} &= g^2 C_F \frac{d-2}{d-1} \bar{\mu}^{2\omega} \int \frac{d^d k}{(2\pi)^d} \frac{(d-3)k^2 + 2(k \cdot v)^2}{(k^2)^3} e^{-2ik^2} e^{-i(k \cdot v)z} \\ &= \frac{\alpha_s}{4\pi} C_F \left[-\frac{1}{\epsilon_{\text{IR}}} - \log\left(\frac{\mu^2 z^2}{4}\right) - 2\gamma_E + 1 \right. \\ &\quad \left. + \text{Ei}(-z^2) - \frac{1}{z^2} (1 - e^{-z^2}) + \frac{1}{z^2} e^{-z^2} \right].\end{aligned}\quad (\text{A.18})$$

In the small flow-time limit, we have

$$\mathcal{M}_{(f_3),\parallel}^{t,\mu^0} = \frac{\alpha_s}{4\pi} C_F \left[-\frac{1}{\epsilon_{\text{IR}}} - \log\left(\frac{\mu^2 z^2}{4}\right) - 2\gamma_E + 3 \right], \quad (\text{A.19})$$

$$\mathcal{M}_{(f_3),\perp}^{t,\mu^0} = \frac{\alpha_s}{4\pi} C_F \left[-\frac{1}{\epsilon_{\text{IR}}} - \log\left(\frac{\mu^2 z^2}{4}\right) - 2\gamma_E + 1 \right]. \quad (\text{A.20})$$

Noticing the corresponding $\overline{\text{MS}}$ results are given by [23],

$$\mathcal{M}_{(f_3),\parallel}^{\overline{\text{MS}}} = \frac{\alpha_s}{4\pi} C_F \left[-\frac{1}{\epsilon_{\text{IR}}} - \log\left(\frac{\mu^2 z^2}{4}\right) - 2\gamma_E + 3 \right], \quad (\text{A.21})$$

$$\mathcal{M}_{(f_3),\perp}^{\overline{\text{MS}}} = \frac{\alpha_s}{4\pi} C_F \left[-\frac{1}{\epsilon_{\text{IR}}} - \log\left(\frac{\mu^2 z^2}{4}\right) - 2\gamma_E + 1 \right], \quad (\text{A.22})$$

which are exactly the same with the gradient flow results in the small flow-time limit. This indicates that diagram (f_3) does not contribute to the matching from the gradient-flow scheme to the $\overline{\text{MS}}$ scheme in the small flow-time limit, which supports our argument that we only need the matching for the local current operators stemming from the Wilson-line operators based on the one-dimensional auxiliary-field formalism.

A.4 Final results

Adding up the full result of each diagram in figure. 6 and the result of quark self-energy diagram in figure. 2, which are given by eq. (A.5), eq. (A.12), eq. (A.17), eq. (A.18) and eq. (A.3), dropping the linearly divergent term through “mass” renormalization, we obtain the normalized full result in the gradient-flow scheme

$$\mathcal{M}_q = \frac{\alpha_s}{4\pi} C_F \left[a_1 + 3 \log(z^2 e^{\gamma_E}) - \log(432) - 4e^{-z^2} - 3\text{Ei}(-z^2) - 4\sqrt{\pi} z \text{erf}(z) + 4\sqrt{\pi} z \right], \quad (\text{A.23})$$

where

$$a_{\gamma_v} = 5 + \frac{3}{z^4} \left(1 - e^{-z^2} \right) - \frac{1}{z^2} \left(2 + e^{-z^2} \right), \quad (\text{A.24})$$

$$a_{\gamma_L} = 3 - \frac{1}{z^4} \left(1 - e^{-z^2} \right) + \frac{1}{z^2} \left(2 - e^{-z^2} \right). \quad (\text{A.25})$$

The $\overline{\text{MS}}$ renormalized total result is

$$\mathcal{M}_q^{\overline{\text{MS}}} = \frac{\alpha_s}{4\pi} C_F \left[a_1' + 3 \log\left(\frac{z^2 \mu^2}{4}\right) + 6\gamma_E \right], \quad (\text{A.26})$$

with $a_{\gamma_v}' = 7$, $a_{\gamma_L}' = 5$.

Then, we have the full matching coefficient,

$$C_q(t, \mu, z) = 1 + \frac{\alpha_s}{4\pi} C_F \left[a_1' - a_1 - 3 \log(2\mu^2 t e^{\gamma_E}) - \log(432) - 4e^{-z^2} - 3\text{Ei}(-z^2) - 4\sqrt{\pi} z \text{erf}(z) + 4\sqrt{\pi} z \right] + O(\alpha_s^2), \quad (\text{A.27})$$

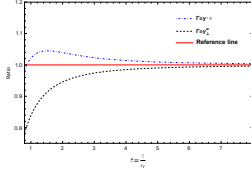


Figure 7. The ratio of the one-loop correction result with full flow time dependence and the one-loop correction result in the small flow-time limit: $\text{Ratio} = \frac{C_F(t, \mu, z) - 1}{C_F(t \rightarrow 0(t, \mu)) - 1}$. The renormalization scale μ is chosen so that $\log(2t\mu^2 e^{\gamma_E}) = 0$.

which gives the small flow-time limit,

$$C_{q, t \rightarrow 0}(t, \mu) = 1 - \frac{\alpha_s}{4\pi} C_F \left[3 \log(2t\mu^2 e^{\gamma_E}) + 2 + \log(432) \right], \quad (\text{A.28})$$

which is indeed independent of z and Γ . It is straightforward to check that $C_{q, t \rightarrow 0}(t, \mu) = c_{\psi h, \mu}^2$ as expected (see eq. (4.65)).

To see how large the finite flow time effect will be, we plot the ratio of the one-loop corrections with full flow time dependence and the one-loop corrections in the small flow-time limit in figure. 7. As it is shown in figure. 7, the finite flow time effect is very small as long as the flow radius $r_F = \sqrt{8t}$ is smaller than the distance z ($z > 1$), which is usually satisfied in lattice gradient flow computations.

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